



PISA

PISA for Development Assessment and Analytical Framework

READING, MATHEMATICS AND SCIENCE



Programme for International Student Assessment

PISA

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Foreword

The OECD Programme for International Student Assessment (PISA), created in 1997, represents a commitment by the governments of OECD and partner countries to monitor the outcomes of education systems, in terms of student achievement, within a common, internationally agreed framework. PISA is a collaborative effort, bringing together scientific expertise from the participating countries/economies and steered jointly by their governments on the basis of shared policy interests. Experts from participating countries also serve on working groups that are charged with linking the PISA policy objectives with the best available substantive and technical expertise in the field of internationally comparable assessments. Through involvement in these expert groups, countries ensure that the PISA assessment instruments are internationally valid and take into account the cultural and curricular context of the PISA-participating countries and economies.

Participation in PISA by non-OECD countries is growing and is combined with demand from these countries for innovations that will maximise their benefits from participation in the assessment. PISA for Development (PISA-D) is an initiative that has been developed in response to this demand and in the context of the Education Sustainable Development Goal that was adopted by the United Nations General Assembly in 2015 and which emphasises universal access to literacy and numeracy. This pilot project aims to make PISA more accessible and relevant to middle- and low-income countries. It does this by developing and piloting enhanced PISA survey instruments that are more relevant for the contexts found in middle- and low-income countries but which produce scores that are on the same scales as the main PISA assessment. The initiative also includes the development of an approach and methodology for including out-of-school youth in the survey. All of the instruments and approaches piloted in PISA-D will be mainstreamed in PISA from the 2021 edition of the assessment onwards.

This publication presents the guiding principles behind the PISA-D assessment for both the school-based and the out-of-school instruments. Sample tasks are also included. It assembles versions of the PISA assessment frameworks for reading, mathematical and scientific literacy that are based on the PISA 2012 and PISA 2015 frameworks, but extends these frameworks to allow for more relevant measurement in a broad range of middle- and low-income countries. Making the measurement more relevant to these countries requires more detail in the description of competencies of the most vulnerable students, those with the lowest levels of performance; which in turn requires including items that will enable the observation of these competencies in greater detail. Yet the relevance of PISA-D, and the aim of mainstreaming the outputs from the initiative in main PISA, also depends on comparability with international PISA results: the instruments therefore allow for students to demonstrate the full range of proficiency levels in PISA.

As with previous cycles of PISA, the PISA-D cognitive frameworks have been reviewed and updated by a network of international experts who have experience with PISA, the relevant domains and the contexts found in middle- and low-income countries. A group of Pearson content experts led by a chair, together with three experts suggested by the participating countries, reviewed existing versions of the PISA 2012 and 2015 assessment frameworks and prepared Chapters 2, 3 and 4 of this publication. The reading framework was led by Jean-François Rouet, the mathematics



framework by Zbigniew Marciniak and the science framework by Jonathan Osborne. Additionally, representatives of the participating countries have provided advice and guidance in the development and extension of the assessment and analytical framework for PISA-D.

This document also includes the framework for the PISA-D questionnaires in Chapter 5. This questionnaires framework was developed by J. Douglas Willms from The Learning Bar Inc. of Canada, with contributions from Lucia Tramonte and Robert Laurie. The questionnaire expert group included Servaas Van der Berg (South Africa, Chair), Andres Peri (Uruguay), Fons van de Vijver (Netherlands) and Wilima Wadhwa (India). Other experts who contributed to the development of the questionnaire framework include Danielle Durepos and Alma Lopez-Garcia.

The publication was prepared by the OECD Secretariat, principally by Michael Ward, Catalina Covacevich and Kelly Makowiecki, with contributions from Francesco Avvisati, Mario Piacentini and Alfonso Echazarra. A full list of all contributing experts and support staff is included at Annex B.



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Acronyms and Abbreviations

ASER	Annual Status of Education Report
CEFR	Common European Framework of Reference
DOK	Depth of knowledge
EGMA	Early Grade Mathematics Assessment
EGRA	Early Grade Reading Assessment
ESCS	Economic, social and cultural status
ETS	Educational Testing Service
IAG	International Advisory Group
IALS	International Adult Literacy Survey
ICT	Information and communications technology
LAMP	Literacy Assessment Monitoring Programme
LLECE	<i>Laboratorio Latinoamericano de Evaluación de la Calidad de la Educación</i> (Latin American Laboratory for Assessment of the Quality of Education)
NC	National Centre
NPM	National Project Manager
OECD	Organisation for Economic Co-operation and Development
PASEC	<i>Programme d'Analyse des Systèmes Éducatifs de la CONFEMEN</i> (CONFEMEN Education Systems Analysis Program)
PGB	PISA Governing Board
PIAAC	Programme for the International Assessment of Adult Competencies
PIRLS	Programme in International Reading Literacy Study
PISA	Programme for International Student Assessment
PISA-D	PISA for Development
SACMEQ	Southern and Eastern Africa Consortium for Monitoring Educational Quality
SDG	Sustainable Development Goal
SES	Socio-economic status
STEP	Skills Toward Employment and Productivity
TAG	Technical Advisory Group
TIMSS	Trends in International Mathematics and Science Study
UIS	UNESCO Institute of Statistics
UNESCO	United Nations Educational, Scientific and Cultural Organization
UNICEF	United Nations Children's Fund
WEI-SPS	World Education Indicators' Survey of Primary Schools



1

What are PISA and PISA for Development?

This Chapter provides an overview of the Programme for International Student Assessment (PISA) and explains how the PISA for Development (PISA-D) project was developed in response to a review of the experience of middle-income countries in PISA. It describes what PISA-D adds to PISA, including enhancements to the cognitive test and contextual questionnaires, an assessment of the out-of-school population, and support for building the capacity of participating countries to implement international large-scale assessments and use assessment results to support evidence-based policy making. The Chapter also discusses how PISA-D contributes to the monitoring and achievement of the Education Sustainable Development Goal (SDG), which emphasises quality and equity of learning outcomes for children, young people and adults.



“What is important for citizens to know and be able to do?” In response to that question and to the need for cross-nationally comparable evidence on student performance, the Organisation for Economic Co-operation and Development (OECD) launched the Programme for International Student Assessment (PISA) in 1997. PISA assesses the extent to which 15-year-old students, near the end of their compulsory education, have acquired key knowledge and skills that are essential for full participation in modern societies.

The triennial assessment focuses on the core school subjects of reading, mathematics and science. Students’ proficiency in an innovative domain is also assessed. The assessment does not just ascertain whether students can reproduce knowledge; it also examines how well students can extrapolate from what they have learnt and can apply that knowledge in unfamiliar settings, both in and outside of school. This approach reflects the fact that modern economies reward individuals not for what they know, but for what they can do with what they know.

Through questionnaires distributed to students, parents, school principals and teachers, PISA also gathers information about students’ home background, their approaches to learning and their learning environments.

In each round of PISA, one of the core domains is tested in detail, so a thorough analysis of achievement in each of the three core areas is presented every nine years and an analysis of trends is offered every three years. Combined with the information gathered through the various questionnaires, the PISA assessment provides three main types of outcomes:

- basic indicators that provide a baseline profile of the knowledge and skills of students
- indicators derived from the questionnaires that show how such skills relate to various demographic, social, economic and educational variables
- indicators on trends that show changes in outcome levels and distributions, and in relationships between student-level, school-level and system-level background variables and outcomes.

PISA is an ongoing programme that, over the longer term, will lead to the development of a body of information for monitoring trends in the knowledge and skills of students in various countries as well as in different demographic subgroups of each country. Policy makers around the world use PISA findings to gauge the knowledge and skills of students in their own country/economy in comparison with those in other participating countries/economies, establish benchmarks for improvements in the education provided and/or in learning outcomes, and understand the relative strengths and weaknesses of their own education systems.

THE EXPERIENCE OF MIDDLE-INCOME COUNTRIES IN PISA

Students representing more than 80 countries and economies that together make up over 80% of the world economy have participated in PISA since its launch, including 44 middle-income countries, 27 of which have been recipients of foreign aid. As more and more participants join it has become apparent that the design and implementation models for PISA need to evolve to successfully cater to a larger and more diverse set of countries, including the growing number of middle- and low-income countries who want to participate in the assessment (Lockheed, Prokic-Breuer and Shadrova, 2015). In particular, PISA needs to take more account of the marked differences between high- and middle-income countries in education quality and equity and their correlates.

The OECD’s analysis of the experience of middle-income countries in PISA has revealed the following three key results that have implications for the further development of the assessment and its framework:

- First, the overall performance of 15-year-old students in all the middle-income countries participating in PISA, except Viet Nam, is lower than that of students in OECD countries, and varies widely. Performance is also concentrated at the lower levels of the PISA proficiency scales.
- Second, some of the educational inputs as currently measured by PISA are unrelated to differences in performance across schools in the majority of the middle-income countries that participate in PISA. In addition, the measure of economic, social and cultural status currently used by PISA does not adequately capture lower levels of parental education, income and risk factors of poverty that are more frequent in low-income countries. Moreover, it has also become clear that the data captured on the context that surrounds students could be made more relevant, particularly in respect of policies, for middle- and low-income countries.
- Third, out-of-school rates for lower secondary school children are high in many middle- and low-income countries and, in addition, many 15-year-olds in these contexts are also enrolled in grades below those that are eligible for PISA (i.e. Grade 6 and below). The combination of these two exclusion mechanisms result in indices as low as 50% coverage



of the 15-year-old population in some PISA-participating countries, and limit the comparability of middle-income countries' results with other countries. It is also the case that PISA runs the risk of reinforcing policies of exclusion in middle-income countries, unless the assessment takes concrete steps to incorporate all the 15-year-olds in a country's population in the survey.

PISA FOR DEVELOPMENT

Building on the experience of middle-income countries in PISA, and in an effort to respond to the three results highlighted above, the OECD launched the PISA for Development (PISA-D) initiative in 2014. This is a one-off pilot project spanning six years that aims to make the assessment more accessible and relevant to a wider range of countries. The project is also a contribution to the monitoring of international educational targets related to the Education Sustainable Development Goal (SDG), adopted by the United Nations General Assembly in 2015 as part of the Agenda for Sustainable Development. The project has also been informed by analysis of the lessons and experiences from other regional and international large-scale assessments in education in middle- and low-income countries (Cresswell, Schwantner and Waters, 2015). To accomplish its aims, the project sets out to:

- increase the resolution of the PISA tests at the lower end of the student performance distribution
- capture a wider range of social and economic contexts
- incorporate an assessment of out-of-school 14-16 year-olds.

The highly collaborative PISA-D project is being carried out by the OECD, nine participating countries, international contractors, development partners and technical partners.

Eight countries are participating in the school-based implementation of PISA-D: Bhutan, Cambodia, Ecuador, Guatemala, Honduras, Paraguay, Senegal and Zambia. One of the main reasons for their participation is policy makers' wish to understand why students in their countries achieve certain levels of performance. Assessment results will provide these policy makers with data and evidence that can be used to determine what they can do to improve their educational systems and, ultimately, ensure that their students obtain the skills needed to succeed in tomorrow's world and as set out in the Education SDG Framework.

In addition to the school-based component of PISA-D, an out-of-school component is being piloted by six countries – Guatemala, Honduras, Senegal, Paraguay, Panama and Zambia – and focuses on the knowledge, skills and contextual factors of 14-16 year-old out-of-school youth. In PISA-D, the definition of out-of-school youth incorporates all those 14-16 year-olds that are not reflected in the school-based survey, including those who are out-of-school and those who are in school but enrolled at Grade 6 or below. This out-of-school component adopts the same framework used for the school-based component of PISA-D, as the description of competencies, particularly at lower levels of performance, will also apply to the out-of-school population. Through the out-of-school assessment, PISA-D will be able to report on what all 15-year-olds in a population know and can do. The analysis of these data should yield valuable insights for governments in middle- and low-income countries, in particular about the effectiveness of their education systems, and about the success of policies that aim to ensure inclusive and equitable quality education and learning opportunities for all. It will also serve to reinforce these policies of inclusion and contribute to the monitoring and achievement of the Education SDG with its emphasis on leaving no one behind.

Box 1.1 The out-of-school component

Across many middle- and low-income countries, relatively large proportions of 15-year-olds are not enrolled in school or are enrolled in school in grades below PISA's target grades (Grade 7 and above) and are therefore excluded from the PISA sample. In the PISA-D participating countries, between 10 and 50% of youth are in this situation. The PISA-D out-of-school component is establishing methods and approaches to include out-of-school youth aged 14 to 16 and also 14-16 year-old students that are in Grade 6 or below in the assessment. The sample range was expanded from 15-year-olds to 14- and 16-year-olds following the recommendations of Carr-Hill (2015), who highlighted the challenge of locating a single year age group in a household survey in middle- and low-income countries. The range of educational experiences in this out-of-school population is expected to vary substantially, from children with no experience in formal education to those who have recently left school or who are still in school but in Grade 6 or below.



The PISA-D instruments, once piloted and finalised, will be available for use in future PISA cycles (from PISA 2021 onwards) and will allow middle- and low-income countries to participate in PISA more meaningfully. The enhanced instruments will also support global measures of reading and mathematical skills as part of the Education SDG agenda, strengthening PISA's potential to provide a global metric for measuring progress towards the Education SDG targets and indicators.

The PISA-D framework maintains the concept of competency that was adopted by the PISA Governing Board as part of the long-term strategy for PISA in 2013, which seeks to go beyond the reproduction of subject-matter knowledge and focuses on the capacity of students to extrapolate from what they know and apply their knowledge. Furthermore, the PISA-D framework maintains the same design parameters that have guided all assessments from PISA 2000.

This publication presents the theory underlying the PISA-D assessment, which has been developed in the context of PISA. It includes frameworks for assessing the three core subjects – reading, mathematics and science (Chapters 2, 3 and 4, respectively), that build on the PISA 2012 and 2015 frameworks (OECD 2013 and 2016). The chapters outline the cognitive processes or competencies involved in the tasks of each testing domain, and the area of knowledge and contexts or situations in which these cognitive processes are applied. They also discuss how each domain is assessed. Chapter 5 explains the theory underlying the context questionnaires distributed to students, school principals and teachers, and the ones answered by the out-of-school youth, their parents (or the person most knowledgeable about the youth) and the interviewer.

WHAT PISA-D ADDS TO PISA

While PISA-D is being implemented within the overall PISA framework and in accordance with PISA's technical standards and usual practices, it includes new features and enhancements to make the assessment more accessible and relevant to middle- and low-income countries. These features and enhancements include:

- An equal treatment of the three major domains tested: reading, mathematics and science – unlike PISA, where one of the domains is given a particular focus in each cycle.
- Targeted test instruments that cover a wider range of performance at the lower levels of proficiency, while still providing scores that cover the whole of the PISA framework and are comparable to the main PISA results – unlike PISA where the tests are not targeted on particular levels of performance.
- Modified test instruments and questionnaires that have a reduced reading burden, in recognition of the lower levels of reading literacy capacity in middle- and low-income countries.
- Contextual questionnaires that have at their core items from PISA to facilitate international comparisons, but also include several distinct PISA-D items that are more relevant to middle- and low-income countries. These new items also respond to the policy priorities of the countries participating in PISA-D.
- An assessment of the out-of-school population: PISA assesses 15-year-olds that are in school in Grade 7 or above. PISA-D assesses this same population, but also has an out-of-school module aimed at 14-16 year-olds who are not in school, or are in school but in Grade 6 or below. The inclusion of out-of-school youth in the survey makes PISA-D unique in the landscape of international large-scale assessments. The project explores methodologies and data-collection tools regarding out-of-school youth, i) in terms of their skills, competencies and non-cognitive attributes; and ii) in terms of obtaining better actionable data on the characteristics of these children, the reasons for their not being in school and on the magnitudes and forms of exclusion and disparities.

Another feature unique to PISA-D is the learning and capacity-building opportunities that are built into each phase of project implementation. In preparing to implement the assessment, PISA-D countries undergo a capacity needs analysis based on PISA's technical standards and devise a capacity-building plan that is also relevant for strengthening their national assessment systems. The PISA-D countries are also assisted by the OECD to prepare a project implementation plan that guides their implementation of the survey and ensures that the necessary human and financial resources are in place. While PISA countries have not benefitted from similar support, the PISA-D project serves as the basis for developing a model of support within the core PISA survey which can be offered more widely to all participating countries from the 2021 cycle onwards.

PISA-D results will be published in national reports produced by the countries in collaboration with the OECD. As part of the report production process, the OECD and its contractors will provide inputs to the countries to strengthen their capacities for data analysis, interpretation of PISA results, report writing and the production of tailored communication products to support the dissemination of PISA results and policy messages. These national reports and other communication products will present results in the context of the international PISA scales and include relevant analyses and information



based on the policy priorities of each country. The reports will constitute a summary of key results and analysis designed to stimulate a constructive debate on improvement, building upon and enriching already existing data and evidence from national, regional or international sources. The national reports will be the culmination of an engagement and communication strategy that is being implemented by each country, another new feature introduced by PISA-D. These strategies involve key stakeholders in each country in the survey and the discussion of the results and implications for policy. Stakeholders include pupils, parents, teachers, teacher unions, school principals, academia, civil society, media, and central and local government.

Box 1.2 Key features of PISA-D

The content

The school-based survey assesses reading, mathematics and science, while the out-of-school survey includes reading and mathematics only. PISA-D assesses not only whether students can reproduce knowledge, but also whether they can extrapolate from what they have learnt and apply their knowledge in new situations. It emphasises the mastery of processes, the understanding of concepts, and the ability to function in various types of situations.

The students

Around 37 100 students will complete the school-based assessment, representing about 1 200 000 15-year-old students (in Grade 7 or above) in the schools of the seven participating countries. Furthermore, around 16 200 youth from six countries will participate in the out-of-school assessment, representing about 1 700 000 out-of-school youth between the ages of 14 and 16 and students aged 14 to 16 in Grade 6 or below.

The assessment

The school-based assessment is a paper-based test, lasting a total of two hours for each student. Test items are a mixture of multiple-choice questions and questions requiring students to construct their own responses. The items are organised in groups, each group based on a passage that sets out a real-life situation. The school-based assessment draws on about 195 test items, with different students taking different combinations of test items.

Students also answer a background questionnaire, which takes 35 minutes to complete. The questionnaire seeks information about the students themselves, their well-being, educational attainment and engagement, their homes, their families, and their school and learning experiences. School principals complete a school questionnaire that describes the school, its students and teachers, and the learning environment. Teachers also complete a questionnaire about themselves, the school's resources, their teaching practice and their students.

The out-of-school assessment is conducted on a tablet computer. The test takes 50 minutes and test items are a mixture of multiple-choice questions and questions requiring respondents to construct their own responses. The items are organised in groups, each group based on a passage that sets out a real-life situation. Youth participating in the out-of-school assessment will answer about 38 test items, with different respondents taking different combinations of test items.

The out-of-school respondents also answer a background questionnaire, which takes about 30 minutes to complete. The questionnaire seeks information about the youth themselves, their well-being, educational attainment and attitudes towards learning, their homes, and their school and learning experiences. Parents (or the most knowledgeable person) also answer a questionnaire about the youth's background and childhood experiences. A household observation questionnaire is completed by the interviewer and information about the location of the household is collected by PISA-D National Centres.

THE PISA-D TESTS: SCHOOL-BASED AND OUT-OF-SCHOOL ASSESSMENTS

The PISA-D school-based instrument is a paper-based assessment designed as a two-hour test. This test design includes four clusters from each of the domains of reading, mathematics and science to measure trends. There are 12 different test booklets, each containing PISA 2015 trend items from two of the three core PISA domains. Each booklet allocated to students comprises four 30-minute clusters of test material. In total, students spend 120 minutes on all three subjects, reading, mathematics and science.



Each test booklet is completed by a sufficient number of students to make appropriate estimates of the achievement levels on all items by students in each country and in relevant subgroups within a country (such as boys and girls, and students from different social and economic contexts). Comparability with PISA 2015, a computer-based assessment, is assured through trend items. In addition, each student answers a 35-minute background questionnaire, which gathers contextual information that is analysed with the test results to provide a broader picture of student performance.

The PISA-D out-of-school instrument is a tablet-based assessment designed as a 50-minute test. The computer-based household survey Programme for the International Assessment of Adult Competencies (PIACC) was used as a model for selecting the delivery mode, and tablets were chosen over laptops on account of cost, efficiency and user-friendliness. The test will include a ten-minute core module of basic reading and mathematics skills to ensure that respondents have an appropriate level of skills to proceed to the full assessment. An established minimum number of items answered correctly will determine the set of items that will be presented to respondents in the second stage of the cognitive assessment. The second stage was designed to take no longer than 40 minutes to complete. Respondents who pass the core module will be randomly assigned to one of the 30 forms measuring reading and mathematical literacy. Respondents who fail the core module will be directed to a 10-minute assessment of reading components followed by “Form 0”, a 30-minute assessment of basic reading and mathematical literacy tasks. In addition, participants answer a 30-minute questionnaire.

Box 1.3 Paper-based or computer-based – does it make a difference?

There is a great deal of research on paper- and computer-based test performance, but findings are mixed. Some early studies indicated that reading speed was slower in a computer-based environment (Dillon, 1994) and less accurate (Muter et al., 1982), although these studies were conducted on proofreading tasks, not in an assessment situation. Richardson et al. (2002) reported that students found computer-based problem-solving tasks engaging and motivating, often despite the unfamiliarity of the problem types and the challenging nature of the items. They were sometimes distracted by attractive graphics, and sometime used poor heuristics when attempting tasks.

There is a large body of more recent literature on paper- and computer-based tests’ equivalency (e.g. Macedo-Rouet et al., 2009; Paek, 2005); however these still reveal conflicting findings. In one of the largest comparisons of paper- and computer-based testing, Sandene et al. (2005) found that eighth-grade students’ mean score was four points higher on a computer-based mathematics test than on an equivalent paper-based test. Bennett et al. (2008) concluded from their research that computer familiarity affects performance on computer-based mathematics tests, while others have found that the range of functions available through computer-based tests can affect performance. For example, Mason, Patry and Berstein (2001) found that students’ performance was negatively affected in computer-based tests compared to paper-based tests when there was no opportunity on the computer version to review and check responses. Bennett (2003) found that screen size affected scores on verbal reasoning tests, possibly because smaller computer screens require scrolling.

By contrast, a meta-analysis of studies looking at kindergarten through Grade 12 (K-12) students’ mathematics and reading achievement (Wang et al., 2007) indicated that, overall, administration mode has no statistically significant effect on scores. A mode-effects study was conducted as part of the OECD Programme for the International Assessment of Adult Competencies (PIAAC) field trial. In this study, adults were randomly assigned to either a computer- or paper-based assessment of literacy and numeracy skills. The majority of the items used in the paper delivery mode was adapted for computer delivery and used in this study. Analyses of these data reveal that almost all of the item parameters were stable across the two modes, thus showing that responses could be measured along the same literacy and numeracy scales (OECD, 2014). Given this evidence, it was hypothesised that PISA 2009 reading items could be transposed onto a screen for PISA 2015 without affecting trend data. This evidence was also the basis for hypothesising that PISA-D reading and mathematics items could be transposed onto a tablet without affecting trend data.

An overview of what is assessed in each domain

Box 1.4 presents definitions of the three domains assessed in PISA-D, which are the same definitions used for PISA 2015. The definitions all emphasise functional knowledge and skills that allow one to participate fully in society. Such participation requires more than just being able to carry out tasks imposed externally by, for example, an employer; it



also means being able to participate in decision making. The more complex tasks in PISA-D require students to reflect on and evaluate material, not just to answer questions that have one correct answer.

Box 1.4 Definitions of the domains

Reading literacy: An individual's capacity to understand, use, reflect on and engage with written texts, in order to achieve one's goals, to develop one's knowledge and potential, and to participate in society.

Mathematical literacy: An individual's capacity to formulate, employ, and interpret mathematics in a variety of contexts. It includes reasoning mathematically and using mathematical concepts, procedures, facts and tools to describe, explain and predict phenomena. It assists individuals to recognise the role that mathematics plays in the world and to make the well-founded judgments and decisions needed by constructive, engaged and reflective citizens.

Scientific literacy: The ability to engage with science-related issues, and with the ideas of science, as a reflective citizen. A scientifically literate person is willing to engage in reasoned discourse about science and technology which requires the competencies to explain phenomena scientifically, evaluate and design scientific enquiry, and interpret data and evidence scientifically.

Reading literacy (Chapter 2) is defined as an individual's ability to understand, use, reflect on and engage with written texts to achieve their goals, develop their knowledge and potential, and participate in society.

PISA-D assesses students' performance in reading through questions related to three major task characteristics:

- processes, which refers to the cognitive approach that determines how readers engage with a text
- text, which refers to the range of material that is read
- situations, which refers to the range of broad contexts or purposes for which reading takes place.

Mathematical literacy (Chapter 3) is defined as an individual's capacity to formulate, employ and interpret mathematics in a variety of contexts. It includes reasoning mathematically and using mathematical concepts, procedures, facts and tools to describe, explain and predict phenomena. It assists individuals to recognise the role that mathematics plays in the world and to make the well-founded judgments and decisions needed by constructive, engaged and reflective citizens.

PISA-D assesses students' performance in mathematics through questions related to three inter-related aspects:

- processes, which describe what individuals do to connect the context of the problem with mathematics and thus solve the problem, and the capabilities that underlie those processes
- content, which is targeted for use in the assessment items
- contexts, in which the assessment items are located.

Scientific literacy (Chapter 4) is included in the school-based assessment only and is defined as the ability to engage with science-related issues, and with the ideas of science, as a reflective citizen. A scientifically literate person is willing to engage in reasoned discourse about science and technology, which requires the competencies to explain phenomena scientifically, evaluate and design scientific enquiry, and interpret data and evidence scientifically.

PISA assesses students' performance in science through questions related to:

- contexts, including personal, local/national and global issues, both current and historical, which demand some understanding of science and technology
- knowledge, which refers to an understanding of the major facts, concepts and explanatory theories that form the basis of scientific knowledge. Such knowledge includes knowledge of both the natural world and technological artefacts (content knowledge), knowledge of how such ideas are produced (procedural knowledge), and an understanding of the underlying rationale for these procedures and the justification for their use (epistemic knowledge).
- competencies, including the ability to explain phenomena scientifically, evaluate and design scientific enquiry, and interpret data and evidence scientifically.



The evolution of reporting student performance in PISA and PISA-D

Results from PISA are reported using scales. Initially, the OECD average score for all three subjects was 500 with a standard deviation of 100, which meant that two-thirds of students across OECD countries scored between 400 and 600 points. These scores represent degrees of proficiency in a particular domain. In subsequent cycles of PISA, the OECD average score has fluctuated slightly around the original. The evolution of reporting student performance in PISA and PISA-D in the three domains is summarised in the sections below.

Reading literacy

Reading literacy was the major domain in 2000, and the reading scales were divided into five levels of knowledge and skills. The main advantage of this approach is that it is useful for describing what substantial numbers of students can do with tasks at different levels of difficulty. Results were also presented through three “aspect” subscales of reading: accessing and retrieving information; integrating and interpreting texts; and reflecting and evaluating texts. A proficiency scale was also available for mathematics and science, though without described levels.

PISA 2003 built upon this approach by specifying six proficiency levels for the mathematics scale. There were four “content” subscales in mathematics: space and shape, change and relationships, quantity and uncertainty. Similarly, the reporting of science in PISA 2006 specified six proficiency levels. The three “competency” subscales in science related to identifying scientific issues, explaining phenomena scientifically and using scientific evidence. Country performance was compared on the bases of knowledge about science and knowledge of science. The three main areas of knowledge of science were physical systems, living systems, and earth and space systems.

PISA 2009 marked the first time that reading literacy was re-assessed as a major domain. Trend results were reported for all three domains. PISA 2009 added a Level 6 to the reading scale to describe very high levels of reading proficiency. The bottom level of proficiency, Level 1, was relabelled as Level 1a. Another level, Level 1b, was introduced to describe the performance of students who would previously have been rated as “below Level 1”, but who show proficiency in relation to new items that are easier than those included in previous PISA assessments. These changes allow countries to know more about what kinds of tasks students with very high and very low reading proficiency are capable of completing. To further extend the framework to the lower end of the scale of reading proficiency, PISA-D adds Level 1c to provide better coverage of basic processes, such as literal sentence and passage comprehension. Levels 1a and 1b have been modified for better alignment with the new descriptor for Level 1c.

Mathematical literacy

Mathematics was re-assessed as a major domain in PISA 2012. In addition to the “content” subscales (with the “uncertainty” scale renamed as “uncertainty and data” for improved clarity), three new subscales were developed to assess the three processes in which students, as active problem solvers, engage. These three “process” subscales are: formulating situations mathematically; employing mathematical concepts, facts, procedures and reasoning; and interpreting, applying and evaluating mathematical outcomes (known as “formulating”, “employing” and “interpreting”). To further extend the framework to the lower end of the scale of mathematical literacy proficiency, PISA-D renames Level 1 as 1a, and creates two new proficiency levels at the lower end of the scale, Levels 1b and 1c, to better measure basic processes; such as performing a simple calculation and selecting an appropriate strategy from a list.

Scientific literacy

Science, which was the main subject of assessment in PISA 2006, was again the main domain in PISA 2015. The assessment measures students’ ability to: explain phenomena scientifically; evaluate and design scientific enquiry; and interpret data and evidence scientifically. The science scale was also extended by the addition of Level 1b to better describe the proficiency of students at the lowest level of ability who demonstrate minimal scientific literacy and who would previously not have been included in the reporting scales. To further extend the framework to the lower end of the scale of scientific literacy proficiency PISA-D adds Level 1c to gather information on basic skills at the lowest performance levels, such as being able to recall appropriate scientific knowledge but not apply such knowledge, or to make a simple prediction but not justify it. Levels 2, 1a and 1b were modified to implement a clear line of progression in knowledge from Level 1c.

THE PISA-D CONTEXTUAL QUESTIONNAIRES

The focus of the PISA contextual questionnaires is on understanding how measures of student performance at age 15 are related to various aspects of school and classroom practice as well as other related factors, such as economic, social and cultural context. The PISA-D questionnaires include these aspects and also cover a broader set of well-being outcomes and



a wider range of risk and protective factors, taking into account differences in life experiences of children in developing countries, both of those who are in school and of those who are not.

The contextual framework for PISA-D

The PISA-D questionnaire framework uses the Education Prosperity model (Willms, 2015) as an overarching framework, while also taking into account the goals of PISA-D, lessons from past PISA cycles and other international studies, recommendations from research literature and the priorities of the participating countries. Education prosperity, as applied in PISA-D, is a life-course approach that includes a core set of metrics for success at six key stages of development, covering the period from conception to adolescence. It identifies a key set of outcomes called “Prosperity Outcomes” for six stages of development from conception to age 18, and a set of family, institutional and community factors, called “Foundations for Success”, which drive these outcomes. PISA-D focuses on the fifth stage of the Educational Prosperity framework, late primary and lower secondary (ages 10 to 15).

The framework places great emphasis on equality and equity, with equality referring to differences among sub-populations in the distribution of their educational outcomes and equity referring to differences among sub-populations in their access to the resources and schooling processes that affect schooling outcomes. The PISA-D contextual framework also focuses on the measurement of socio-economic status and poverty, with the purpose of exploring an international measure of poverty for youth in middle- and low- income countries; while also extending the measure of the PISA index of economic, social and cultural status (ESCS).

The framework for the PISA-D questionnaires focuses on 15 modules of content. These modules measure the four Prosperity Outcomes, the five Foundations for Success, and the six demographic factors relevant to assessing equality and equity that are listed below. In addition, the questionnaires include several teacher, school and system-level background measures that provide context for the Prosperity Outcomes. Chapter 5 presents the PISA-D questionnaire framework in detail.

Table 1.1 Modules assessed in the PISA-D questionnaires

1. Prosperity Outcomes	1.1 Academic performance (measured through the PISA-D tests)
	1.2 Educational attainment
	1.3 Health and well-being
	1.4 Attitudes towards school and learning
2. Foundations for Success	2.1 Inclusive environments
	2.2 Quality instruction
	2.3 Learning time
	2.4 Material resources
	2.5 Family and community support
3. Demographic factors for assessing equality and equity	3.1 Gender
	3.2 Socio-economic status and poverty
	3.3 Language spoken at home and language of instruction
	3.4 Urban/rural status
	3.5 Immigrant status
	3.6 Disability

PISA-D enhances the contextual questionnaires to better measure factors that are more strongly related to student performance in middle- and low-income countries, while maintaining comparability with PISA on a set of core indicators. For example, the questionnaires collect more detailed data on students’ language of instruction at school, language at home and their socio-economic status, as measured by home possessions and parents’ education, literacy skills and participation in the labour force. The questionnaires also identify additional indicators of educational success beyond performance on the PISA test. These indicators are measured through questions about educational attainment, health and well-being, and attitudes towards school and learning.

It is also important to note that the contextual information collected through the student, school and teacher questionnaires comprises only a part of the information available to PISA-D. System-level data describing the general structure of the education systems will be used in the PISA-D analysis and country reports. This system-level data includes information



on the structure of national programmes, national assessments and examinations, instruction time, teacher training and salaries, educational finance (including enrolment), national accounts and population data. Available data on all of these indicators have been reviewed for PISA-D countries, identifying the current status of system-level data collection and availability in terms of quality and completeness (UIS, 2016).

The school-based questionnaires

The school-based questionnaires for students, teachers and the principals of schools have been developed in accordance with the contextual framework. These questionnaires take about 35 minutes for the students to complete and about 25 minutes for teachers and the principals. The responses to the questionnaires are analysed with the assessment results to provide at once a broader and more nuanced picture of student, school and system performance. These questionnaires seek information about:

- students and their family backgrounds, including their economic, social and cultural capital, and the language they speak at home versus the language of instruction
- aspects of students' lives, such as their level of educational attainment, their health and well-being, and their engagement with school
- aspects of learning, including quality of instruction, inclusive environments, learning time, school material resources and family and community support
- contexts of learning, including teacher, school and system-level information.

The out-of-school questionnaires

The out-of-school component questionnaires for youth, parents and interviewers have been developed in accordance with the contextual framework. These questionnaires take between 15 and 30 minutes each for the youth, the person most knowledgeable about the youth (parent, guardian or other) and the interviewer to complete. These questionnaires seek information about:

- youths and their family backgrounds, including their economic, social and cultural capital, and the language they speak at home versus the language of instruction when they attended school
- aspects of youths' lives, such as their level of educational attainment, their attitudes towards learning, their employment status, their habits and life outside of school, and their health and well-being
- aspects of learning, including inclusive environments, family support, their perception of the inclusiveness of their school environment when they attended school, their reasons for being out of school and barriers preventing them from returning to school, and their family support and environment
- aspects of youths' early years, their educational experience and their parent/care-giver's educational expectations for the youth
- aspects of youths' households, including location and surrounding characteristics.

A COLLABORATIVE PROJECT

PISA-D is a highly collaborative effort carried by the OECD Secretariat, contractors and nine participating countries with the support of several development partners and institutional partners.

The OECD's Directorate for Education and Skills and the Development Co-operation Directorate share responsibility for the overall management of PISA-D, monitoring its implementation on a day-to-day basis and building consensus among countries. The OECD serves as the Secretariat and interlocutor between the PISA-D International Advisory Group (IAG), the PISA Governing Board (PGB), the Technical Advisory Group (TAG) and the PISA-D contractors. The OECD is also responsible for the capacity building of the participating countries, the production of the indicators, the analysis of results, and the preparation of the national reports and project publications in co-operation with the contractors and in close collaboration with the participating countries both at the policy level with the PGB and IAG, at the technical level with the TAG and at the implementation level with the National Project Managers (NPMs).

The IAG, which is specifically for PISA-D, meets annually and comprises government officials from participating countries, representatives of development partners supporting the initiative, representatives of institutional partners, such as UNESCO and UNICEF, invited experts and representatives of the OECD.



The PGB, representing all countries/economies with full PISA membership at senior policy levels, determines the policy priorities for PISA in the context of OECD objectives and oversees adherence to these priorities during the implementation of the programme. The PGB sets priorities for developing indicators, for establishing assessment instruments and for reporting results. Experts from participating countries/economies also serve on working groups to ensure that the instruments are internationally valid and take into account differences in the cultures and education systems.

The PISA-D TAG, managed by the OECD, explores technical issues that have policy or project implications and advises the OECD and its international contractors on these issues.

The PISA-D international contractors are responsible for survey operations and management and take the lead on supporting the countries to implement the programme. The contractors also take the lead on developing the enhanced assessment instruments, drawing on the technical expertise of the Subject Matter Expert Groups and Questionnaire Expert Groups that support PISA. The development of the PISA-D frameworks for reading, mathematics and science and the development of the PISA-D cognitive instruments are the responsibility of the contractor Educational Testing Service (ETS), while the design and development of the PISA-D questionnaires are the responsibility of the contractor The Learning Bar. Management and oversight of this survey, the development of the instruments, scaling and analysis are the responsibility of ETS, as is the development of the electronic platform. Other partners or subcontractors involved with ETS include Pearson for the development of the cognitive frameworks, cApStAn for linguistic quality assurance and control and Westat for survey operations and sampling.

Participating countries implement the survey at the national level through National Centres (NCs). Within the NCs, PISA is managed at the country level by NPMs, subject to the agreed administration procedures and in accordance with the PISA-D Technical Standards put in place by the OECD and its contractors. The NPMs play a vital role in ensuring that implementation is of high quality and help to shape and guide the project in accordance with the PISA-D Technical Standards. They also verify and evaluate the survey results, analyses, reports and publications. The co-operation of students, teachers and principals in participating schools is crucial to the success of PISA-D during all stages of development and implementation. National experts from the participating countries contribute to the preparation of the frameworks and instruments, and they also provide input for the design of analytical outputs. NCs collaborate with OECD on the analysis of PISA-D data for their countries and the production of national reports and other communication products.

From the outset of the project, the OECD has engaged the participation of the key international agencies and programmes concerned with student assessment and improving the quality of education in developing countries. These technical partners include UNESCO, UNESCO Institute of Statistics (UIS), the Global Education Monitoring Report team, UNICEF, the Global Partnership for Education and the following assessment programmes: ASER, EGRA, EGMA, SACMEQ, PASEC, Pre-PIRLS and PIRLS, TIMSS, LLECE, STEP, LAMP, Uwezo, and WEI-SPS.¹ Representatives of these agencies and programmes have been consulted on all aspects of project design and development.

The international and national costs of the project are funded through a combination of development partner support and financing from the PISA-D countries. The development partners that have provided financing or aid-in-kind are France (*Agence française de développement* / French Development Agency); Germany (*Bundesministerium für wirtschaftliche Zusammenarbeit und Entwicklung* / Federal Ministry for Cooperation and Development and *Deutsche Gesellschaft für Internationale Zusammenarbeit* / German Corporation for International Cooperation); Global Partnership for Education; Inter-American Development Bank; Ireland (Irish Aid); Japan (*独立行政法人国際協力機構* / Japan International Cooperation Agency); Korea; Microsoft Corporation; Positivo; Norway (*Norad er direktoratet for utviklingssamarbeid* / Norwegian Agency for Development Cooperation); Sunny Varkey Foundation; United Kingdom (Department for International Development); and the World Bank.

IMPLEMENTATION OF PISA-D

PISA-D is being implemented in five phases over the course of 2014 to 2019.

1. *Design, planning and co-ordination (2014-15)*: Producing expert papers to inform the work of enhancing the assessment instruments, selecting international contractors to conduct the work, and preparing participating countries, including Capacity Needs Analysis and developing a Capacity Building Plan and a Project Implementation Plan for each country. This phase also included the first and second annual meetings of the PISA-D IAG and the first and second annual meetings of the PISA-D TAG which were crucial for reaching agreements on the design of the initiative.



2. *Technical development (2015-16)*: Reviewing assessment frameworks and items, selecting items, designing enhancements, preparing materials, and planning for field trials, as well as the development of the project's Analysis and Reporting Plan.
3. *Field trials and in-country data collection (2016-18)*: Field trials in each country to test the enhanced instruments, reviewing and analysing the results of the field trial, preparing materials for the main study data collection, and conducting the main study data collection.
4. *Analysis and report writing (2018-19)*: Data cleaning and analysis, interpreting results, eight countries writing their national reports supported by the OECD and its contractors.
5. *Report production, dissemination and post-pilot governance (2018-19)*: Instruments finalised, an independent review of the project completed, national reports published, a project results report and a technical report published, a PISA-D international seminar, and PISA-D instruments incorporated in PISA from the 2021 cycle onwards.

STRENGTHENING CAPACITIES

Nine countries (Bhutan, Cambodia, Ecuador, Guatemala, Honduras, Panama, Paraguay, Senegal and Zambia) have partnered with the OECD to develop and test the enhanced PISA instruments. With the exception of Panama, these countries have never before participated in PISA, but they have experience with regional or international assessments and conduct national student assessments.

In addition to delivering the enhancements to PISA discussed above, PISA-D also builds capacity for managing large-scale student learning assessment and using the results to support policy dialogue and decision making in the participating countries. The OECD offers participating countries training on a variety of topics, including framework and item development, sampling, translation/adaptation of survey instruments, data management, coding of students' responses, data analysis and reporting.

Participating countries have each established an NC and nominated an NPM to ensure appropriate infrastructure and resources are in place to implement the assessment in accordance with the PISA Technical Standards. A three-stage process has been developed and implemented to prepare countries for PISA-D participation:

1. *Capacity Needs Analysis*: ensures there is a solid foundational capacity for implementing the project and identifies areas of potential growth for the country
2. *Capacity Building Plan*: addresses identified capacity needs and enhances the enabling environment for PISA, particularly the use of assessment results for national policy dialogue and evidence-based decision making
3. *Project Implementation Plan*: describes the actions to be carried out by the specific entities and agents that are named and commissioned for implementation by the authorities of the participating country, together with the necessary resources.

The project also promotes peer-to-peer learning by bringing together the countries already participating in PISA with PISA-D countries through individual country visits, staff exchanges, international meetings, technical training and workshops, and developing country case studies. These country partnerships allow for sharing information about the implementation of the study and also about working with education stakeholders, using PISA to inform a broader national discussion about the value and standards of assessment, and preparing national reports and disseminating the assessment results.

PISA-D AND THE SUSTAINABLE DEVELOPMENT GOALS (SDGS)

The SDG Education 2030 agenda (UNESCO, 2015) that is set within the framework of the Sustainable Development Goals (UN, 2015) emphasises the quality, equity and measurement of learning outcomes for young children through to working adults. The challenge now is to define global learning indicators that can be measured and tracked on a global scale over time. Through its enhancement of PISA, the PISA-D initiative is designed to inform and support the monitoring, reporting and achievement of the Education SDG and its related targets and indicators, particularly those related to learning outcomes.

The OECD has been a key partner of UNESCO and the other co-convening agencies in developing the Education SDG framework, and works closely with UIS in the development of indicators that will be used to measure progress towards SDG achievement. In turn, UNESCO, UIS and the World Bank have partnered with the OECD in support of the PISA-D initiative.



The OECD, UIS and the World Bank are working together and with other key practitioners, policy makers, researchers, representatives of governments, civil society organisations, funders, UN agencies, and other stakeholders committed to improving learning outcomes in all countries – particularly low- and middle-income countries. PISA-D and the OECD's plans for mainstreaming the outputs of the project in future cycles of PISA is a key contribution to these efforts, and an embodiment of international collaboration in support of the measurement and monitoring of learning outcomes in the context of the Education SDG.

Note

1. See the abbreviation and acronym section for the full names of these programmes.

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2

PISA for Development Reading Framework

This Chapter defines “reading literacy” as assessed in the Programme for International Student Assessment (PISA) and the extensions to the PISA reading framework that have been designed for the PISA for Development (PISA-D) project. It describes the processes involved in reading and the type of texts and response formats used in the PISA-D reading assessment and provides several sample items. The Chapter also discusses how student performance in reading is measured and reported.



WHAT IS NEW IN PISA-D? EXTENSIONS TO THE PISA READING LITERACY FRAMEWORK

The extensions made to the PISA frameworks for PISA-D are an attempt to gain more information about students at the bottom of the performance distribution, particularly for Level 1. The text in this chapter draws primarily on the PISA 2012 reading framework, with additions to facilitate the extensions of the framework and some modifications to address aspects particularly important to assessment for PISA-D. Some specific elements from the 2018 framework have also been included.¹ The extensions occur primarily in four locations: the literature review, descriptions of the reading processes, descriptions of the proficiency levels, and discussion on assessing the proficiencies. The rationale behind these changes is also provided.

Reading literacy was the major domain assessed in 2000 for the first PISA cycle (PISA 2000). For the fourth PISA cycle (PISA 2009), it was the first to be revisited as a major domain, requiring a full review of its framework and new development of the instruments that represent it. For the seventh PISA cycle (2018), the conceptual framework for reading literacy is again being revised. This chapter discusses the conceptual framework underlying the PISA 2012 assessment of students' reading competencies and its extension to PISA-D. The definition of the domain is the same as in PISA 2009 (when it was assessed as the major domain for a second time), apart from the enhanced descriptions of the levels of competencies that fall below the current PISA Level 1.

Starting in 2009, the PISA reading literacy frameworks took digital reading into account, and the assessment of digital reading was implemented only as a computer-based assessment. Much of the content related to paper-based reading remained consistent across the 2009, 2012 and 2015 frameworks. However, the 2015 framework was changed to make formulations for testing on computer. Because of this, the PISA-D framework is based on the 2012 framework. It must be stressed that both 2015 and 2018 offer a paper-based version that maintains its comparability with the computer-based version through the trend items. The use of trend items is the strategy used to ensure comparability between PISA-D and PISA 2015.

The PISA-D framework is designed for assessing the reading literacy of 15-year-old adolescents, who may be in or out of school. The 15-year-olds need to read proficiently in order to participate in school activities (Shanahan and Shanahan, 2008). But most of them also use reading in a wide range of out-of-school contexts; for instance, to communicate with their peers, to acquire information related to their personal interests, or to interact with institutions and businesses (IRA, 2012). Therefore, the framework must represent reading in a broad sense that encompasses basic as well as more advanced forms of reading, relevant for school as well as non-school situations. This includes not only the comprehension of a single given passage of text, but also an ability to find, select, interpret and evaluate information from the full range of texts associated with reading for school and out-of-school purposes.

The original reading literacy framework for PISA was developed through a consensus-building process involving reading experts selected by the participating countries to form the PISA 2000 reading expert group. The definition of reading literacy evolved in part from the International Association for the Evaluation of Educational Achievement's Reading Literacy Study (1992) and the International Adult Literacy Survey (IALS, 1994, 1997 and 1998). In particular, it reflected the IALS emphasis on the importance of reading skills for active participation in society. It was also influenced by contemporary – and still current – theories of reading, which emphasise the multiple cognitive processes involved in reading and their interactive nature (Britt, Goldman and Rouet, 2012; Dechant, 1991; Rayner and Reichle, 2010; Rumelhart, 1985), models of discourse comprehension (Kintsch, 1998; Zwaan and Singer, 2003) and theories of performance in solving information problems (Kirsch, 2001; Kirsch and Mosenthal, 1990; Rouet, 2006).

Changes in our concept of reading since 2000 have already led to an expanded definition of reading literacy, which recognises motivational and behavioural characteristics of reading alongside cognitive characteristics. In light of recent research, reading engagement and metacognition were featured more prominently in the PISA 2009 reading literacy framework as elements that can make an important contribution to policy makers' understanding of factors that can be developed, shaped and fostered as components of reading literacy.

The PISA-D reading literacy framework provides additional emphasis on the basic components of the cognitive processes that underlie reading skills. These components include being able to locate information that is explicitly stated in text, to access and comprehend the meaning of individual words, and to understand the literal meaning of information as it is expressed in sentences as well as across passages. As such, these components can provide information about what these students *can do* with respect to the building blocks of reading literacy proficiency.



This chapter is organised into three major sections. The first section, “Defining reading literacy”, explains the theoretical underpinnings of the PISA reading assessment, including the formal definition of the reading literacy construct. The second section, “Organising the domain of reading”, describes three elements: processes, which refers to the cognitive approach that determines how readers engage with a text; text, which refers to the range of material that is read; and situation, which refers to the range of broad contexts or purposes for which reading takes place. The third section, “Assessing reading literacy”, outlines the approach taken to apply the elements of the framework previously described, including factors affecting item difficulty, the response formats, coding and scoring, reporting proficiency, testing reading literacy among the out-of-school population and examples of items for addressing the extended PISA-D Framework.

DEFINING READING LITERACY

Definitions of reading and reading literacy have changed over time in parallel with changes in society, economy and culture. The concept of learning, particularly the concept of lifelong learning, has expanded the perception of reading literacy. Literacy is no longer considered to be an ability acquired only in childhood during the early years of schooling. Instead, it is viewed as an expanding set of knowledge, skills and strategies that individuals build on throughout life in various contexts, through interaction with their peers and the wider community.

Cognitively based theories of reading emphasise the constructive nature of comprehension, the diversity of cognitive processes involved in reading and their interactive nature (Binkley, Rust and Williams, 1997; Kintsch, 1998; McNamara and Magliano, 2009; Oakhill, Cain and Bryant, 2003; Snow, 2002; Zwaan and Singer, 2003). The reader generates meaning in response to text by using previous knowledge and a range of text and situational cues that are often socially and culturally derived. While constructing meaning, competent readers use various processes, skills and strategies to locate information, to monitor and maintain understanding (van den Broek et al., 2002), and to critically assess the relevance and validity of the information (Richter and Rapp, 2014). These processes and strategies are expected to vary with context and purpose as readers interact with multiple continuous and non-continuous texts both in print and when using digital technologies (Britt and Rouet, 2012; Coiro et al., 2008).

The PISA 2012 definition of reading literacy, the same used in PISA 2009 and 2015 and PISA-D, is shown in Box 2.1:

Box 2.1 The PISA 2012 definition of reading literacy

Reading literacy is understanding, using, reflecting on and engaging with written texts, in order to achieve one’s goals, develop one’s knowledge and potential, and participate in society.

While this definition is taken for PISA-D also, the project extends the PISA definition of reading literacy through the incorporation of the concept of reading components. Reading components are the sub-skills, or building blocks, that underlie reading literacy (Oakhill, Cain and Bryant, 2003). As they develop and integrate, they facilitate proficient reading comprehension. Conversely, if the components are under-developed or deployed inefficiently, they may hinder a person’s ability to comprehend texts (Perfetti, Landi and Oakhill, 2005). Although components can vary in their importance across languages (based on the structure of the language), there are several components that are generally agreed to be significant regardless of language family: word meaning (print vocabulary); sentence processing; and passage comprehension. An assessment of reading components was administered as part of the Programme for the International Assessment of Adult Competencies (PIAAC) and as an optional component of PISA 2012 (e.g. Sabatini and Bruce, 2009). The assessment of reading components can provide information on the component skills of students and out-of-school youth, particularly of those who fall into the lowest levels of literacy. They can also shed light on the kinds of educational/instructional programmes that improve their component skills, which will, in turn, improve their literacy. While word meaning is generally considered to be a proficiency that is already attained by 15 years, students in some countries may not have fully achieved this proficiency, most particularly when the established language of instruction is different from the student’s home language. Thus, the PISA-D reading literacy framework incorporates the reading components of word comprehension, sentence processing and passage comprehension.

It should be noted that there are other critical reading components, including the visual recognition of the printed elements of the alphabet, decoding words into sounds and basic oral comprehension. These are not included as part of the PISA-D framework as they are assumed to be skills attained by 15-year-olds who attend school at their regular grade level as well as out-of-school 15-year-olds who have mastered these basic levels of literacy.



Box 2.2 Foundational reading skills required for PISA-D

Successful performance on higher level reading tasks are dependent and built upon a foundation of component skills (e.g. Abadzi, 2003; Baer, Kutner and Sabatini, 2009; Curtis, 1980; Oakhill, Cain and Bryant, 2003; Perfetti, 2003; Rayner and Reichle, 2010; Sabatini and Bruce, 2009; Stine-Morrow, Miller and Hertzog, 2006). At the simplest view, reading consists of word recognition and linguistic comprehension, each being necessary but not sufficient for reading (e.g. Hoover and Tunmer, 1993). These components can be further elaborated to multiple foundational skills that are required to perform successfully at the lowest level of PISA. Below we outline five of these foundational skills. The first two are pre-conditions of the abilities needed to perform basic reading for PISA-D and are not assessed as part of the instrument. The remaining three are included as part of the assessment and are considered the basic skills that would be necessary to succeed at Level 1c.

1. Ability to relate characters (written symbols) to corresponding phonemes (acoustic sounds)

Reading requires mapping a system of printed visual symbols (individually and in combination) to the spoken form of the language (i.e. the phonetics, phonology) (e.g. Perfetti, 1985). However, there is a significant amount of variability in how the mapping is performed in different languages. For example, alphabetic writing systems map letters to phonemes, while other languages map characters at the syllable level and some at the level of individual words or morphemes (meaning-based units). Thus, the acquisition of this ability may vary by language.

2. Ability to recognise individual or groups of symbols as representing a word referring to objects and/or relationships between words

The printed forms of objects and concepts given the particular orthographic and morphological structure of the language must also be recognised as representing meaningful words (e.g. Anderson and Freebody, 1981; Hirsch, 2003; McCutchen, Green and Abbott, 2008; Nagy and Townsend, 2012; Ouellet, 2006). It should be noted that this ability can differ across languages due to the orthographic differences between languages, the degree of regularity of the relationship between the print and oral language forms; and how morphological and grammatical/syntactical features of the language are encoded in words. For these reasons, it is difficult to ensure cross-language comparability in assessment, as this requires evaluating how to match the sources of difficulty in acquiring these print skills for each language, and balancing them across stimuli and tasks.

3. Ability to literally understand relationships among groups of words at the sentence level

An individual sentence serves as a complete unit of one or more coherent ideas (e.g. Kintsch, 1998), and a student must be able to comprehend the literal meaning of sentences of varying lengths. Reading a sentence requires both the syntactic processes that interpret the order and function of the words, and the semantic processes of interpreting the meaning of words and propositions (e.g. Kintsch, 1998; Snow, 2002).

4. Ability to literally understand explicit relationships between sentences at the level of short texts

Beyond individual sentences, a reader must be able to understand the literal meaning of passages of text. This requires forming a representation of the information contained across multiple sentences, connecting the idea units and structuring them in memory.

5. Ability to make low-level inferences about relationships across sentences of short texts

Students must be able to represent the information from connected sentences and infer specific relationships. These relationships can include connecting simple referents between one sentence and the next, such as the use of a nominal phrase in one sentence and a pronoun in the next, or creating coherence between two related sentences.

Reading literacy...

The term “reading literacy” is preferred to “reading” because it is likely to convey to a non-expert audience more precisely what the survey is measuring. “Reading” is often understood as simply decoding, or even reading aloud, whereas the intention of this survey is to measure something broader and deeper. Reading literacy includes a wide range of cognitive competencies, from basic decoding, to knowledge of words, grammar and larger linguistic and textual structures and features, to knowledge about the world.



In this assessment, “reading literacy” is intended to express the active, purposeful and functional application of reading in a range of situations and for various purposes. According to Holloway (1999), reading skills are essential to the academic achievement of middle and high school students. PISA assesses a wide range of students. Some will go on to university; some will pursue further studies in preparation for joining the labour force; some will enter the workforce directly after completing compulsory education. Achievement in reading literacy is not only a foundation for achievement in other subject areas within the education system, but also a pre-requisite for successful participation in most areas of adult life (Cunningham and Stanovich, 1998; Smith et al., 2000). Indeed, regardless of their academic or labour-force aspirations, students’ reading literacy is important for their active participation in their community and economic and personal life.

Reading literacy skills matter not just for individuals, but for economies as a whole. Policy makers and others are coming to recognise that in modern societies, human capital – the sum of what the individuals in an economy know and can do – may be the most important form of capital. Economists have for many years developed models showing generally that a country’s education levels are a predictor of its economic growth potential (Coulombe, Tremblay and Marchand, 2004).

...is understanding, using, reflecting on...

The word “understanding” is readily connected with the widely accepted concept of “reading comprehension”, which emphasises that all reading involves some level of integrating information from the text with the reader’s knowledge structures. In order to achieve some degree of understanding, the reader must decode written words, comprehend the literal meaning of sentences and passages, but also elaborate and reason about the information. Even the most basic forms of understanding require readers to draw on symbolic knowledge to identify words and make meaning from them. However, this process of integration can also be much broader, such as developing mental models of how texts relate to the world. The word “using” refers to the notions of application and function – doing something with what we read. “Reflecting on” is added to “understanding” and “using” to emphasise the notion that reading is interactive: readers draw on their own thoughts and experiences when engaging with a text. Of course, every act of reading requires some reflection, drawing on information from outside the text. Even at the earliest stages, readers draw on symbolic knowledge to decode a text and require some vocabulary knowledge to construct meaning. As readers develop their stores of information, experience and beliefs, they constantly, often unconsciously, test what they read against outside knowledge, thereby continually reviewing and revising their sense of the text.

...and engaging with...

A reading literate person not only has the skills and knowledge to read well, but also values and uses reading for a variety of purposes. It is therefore a goal of education to cultivate not only proficiency but also engagement in reading. Engagement in this context implies the motivation to read and comprises a cluster of affective and behavioural characteristics that include an interest in and enjoyment of reading, a sense of control over what one reads, involvement in the social dimension of reading, and diverse and frequent reading practices.

...written texts...

The term “written texts” is meant to include all those coherent texts in which language is used in its graphic form, whether printed and digital. Instead of the word “information”, which is used in some other definitions of reading, the term “texts” was chosen because of its association with written language and because it more readily connotes literary as well as information-focused reading. The PISA-D reading literacy framework makes no assumption about the length or elaborateness of a written text. For example, a text could be a single word embedded within a graphic or short passage within a table.

These texts do not include aural language artefacts such as voice recordings; nor do they include film, TV, animated visuals or pictures without words. They do include visual displays such as diagrams, pictures, maps, tables, graphs and comic strips that include some written language (for example, captions). These visual texts can exist either independently or they can be embedded in larger texts. Digital texts are distinguished from printed texts in a number of respects, including physical readability; the amount of text visible to the reader at any one time; the way different parts of a text and different texts are connected with one another through hypertext links; and, given these text characteristics, the way that readers typically engage with digital texts. To a much greater extent than with printed or hand-written texts, readers need to construct their own pathways to complete any reading activity associated with a digital text.

...in order to achieve one’s goals, develop one’s knowledge and potential, and participate in society.

This phrase is meant to capture the full scope of situations in which reading literacy plays a role, from private to public, from school to work, from formal education to lifelong learning and active citizenship. “To achieve one’s goals and to develop one’s knowledge and potential” spells out the idea that reading literacy enables the fulfilment of individual aspirations – both defined ones, such as graduating or getting a job, and those less defined and less immediate that enrich and extend



personal life and lifelong education. The word “participate” is used because it implies that reading literacy allows people to contribute to society as well as to meet their own needs. “Participating” includes social, cultural and political engagement.

ORGANISING THE DOMAIN OF READING

This section describes how the domain is represented, a vital issue because the organisation and representation of the domain determines the test design and, ultimately, the evidence about student proficiencies that can be collected and reported.

Reading is a multidimensional domain. While many elements are part of the construct, not all can be taken into account in building the PISA assessment. Only those considered most important were selected.

The PISA reading literacy assessment is built on three major task characteristics to ensure a broad coverage of the domain:

- *processes*, which refers to the cognitive approach that determines how readers engage with a text
- *text*, which refers to the range of material that is read
- *situation*, which refers to the range of broad contexts or purposes for which reading takes place.

Note that the term “processes” – proposed as the term within PISA 2018 – is used in the PISA-D framework, although in PISA 2000 through PISA 2015, processes were referred to as “aspects”. This is because the term “processes” aligns better with the scholarly literature on reading comprehension and assessment. In addition, the task characteristics are introduced in a different order than in the 2012 framework, in order to highlight those characteristics that are directly construct-relevant, as opposed to characteristics such as text types or task contexts, which are included mainly for purposes of coverage.

In PISA assessments, features of the text and processes variables (but not of the situation variable) are manipulated to influence the difficulty of a task. The processes are manipulated through the goals set in tasks.

Reading is a complex activity. The elements of reading do not exist independently of one another in neat compartments. The assignment of texts and tasks to framework categories does not imply that the categories are strictly partitioned or that the materials exist in atomised cells determined by a theoretical structure. The framework scheme is provided to ensure coverage, to guide the development of the assessment and to set parameters for reporting, based on what are considered the marked features of each task.

Processes

Processes are the mental strategies, approaches or purposes that readers use to negotiate their way into, around and between texts. Five processes were defined for PISA 2009-15 to guide the development of the reading literacy assessment tasks:

- retrieving information
- forming a broad understanding
- developing an interpretation
- reflecting on and evaluating the content of a text
- reflecting on and evaluating the form of a text.

For PISA-D, an additional process titled “literal comprehension” has been added. Literal comprehension requires students to comprehend explicitly stated information that may be found in individual words, sentences or passages. In addition, the concept of “retrieving information” is broadened to range from locating explicitly stated individual pieces of information, such as individual words or phrases, up to finding information in large passages.

As it is not possible to include sufficient items in PISA to report on each of the six processes as a separate subscale, for reporting on reading literacy these six processes are organised into three broad categories of processes:

- access and retrieve
- integrate and interpret
- reflect and evaluate.

Generate literal comprehension, forming a broad understanding and developing an interpretation tasks focus the reader on relationships within a text. Tasks that focus on the whole text require readers to generate the literal meaning of words,



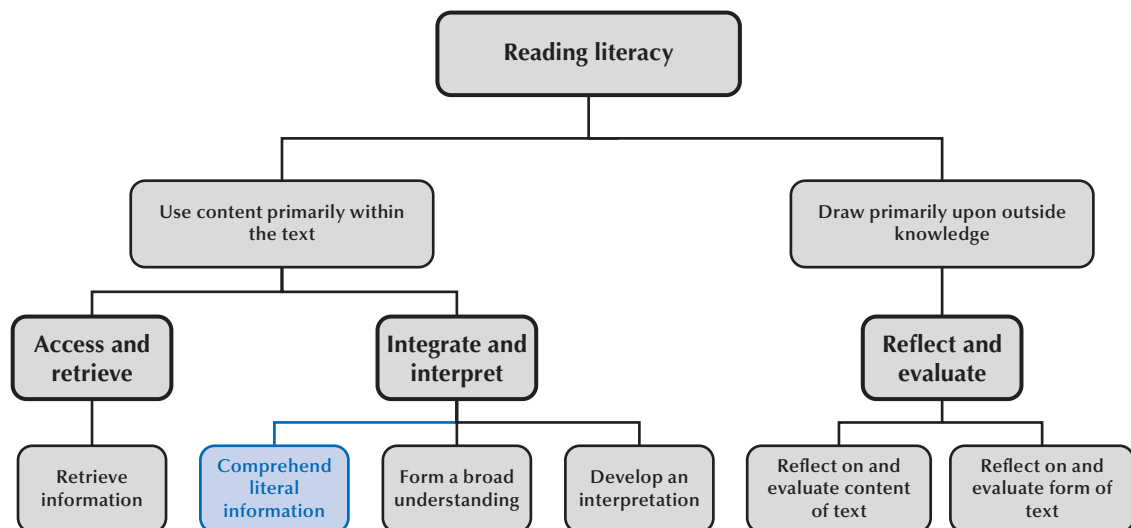
individual sentences, and short passages. They also require forming a broad understanding; tasks that focus on relationships between parts of the text require developing an interpretation. The three are grouped together under *integrate and interpret*.

Tasks related to the *retrieve information* process form the access and retrieve category.

Tasks addressing the last two processes, *reflecting on and evaluating the content of a text* and *reflecting on and evaluating the form of a text*, are grouped together into a single *reflect and evaluate* process category. Both require the reader to draw primarily on knowledge outside the text and relate it to what is being read. *Reflecting on and evaluating content* tasks are concerned with the notional substance of a text; *reflecting on and evaluating form* tasks are concerned with its structure or formal features.

Figure 2.1 shows the relationship between the five processes targeted in the test development for PISA in general and the additional process that will be assessed for PISA-D (in blue). The three broad categories reported on subscales in general PISA, when reading literacy is the major domain, are marked in bold. Because there is no major domain in PISA-D, reading literacy will be reported on a single overall scale only.

Figure 2.1 ■ **Processes targeted in reading literacy test development for PISA and PISA-D**



An elaboration of the three broad process categories, encompassing tasks in both print and digital media, is given below.

Access and retrieve

Accessing and retrieving involves going to the information space provided and navigating in that space to locate and retrieve one or more distinct pieces of information. Access and retrieve tasks can range from locating individual pieces of information, such as the details required by an employer from a job advertisement, to finding a telephone number with several prefix codes, to finding a particular fact to support or disprove a claim someone has made.

While *retrieving* describes the process of selecting the required information, *accessing* describes the process of getting to the place, the information space, where the required information is located (e.g. see sample item 4, question 12.1). Both processes are involved in most *access and retrieve* tasks in PISA. However, some items may require retrieving information only, especially in the print medium where the information is immediately visible and where the reader only has to select what is appropriate in a clearly specified information space.

Difficulty will be determined by several factors, including the number of paragraphs or pages that need to be used, the amount of information to be processed on any given place, and the specificity and explicitness of the task directions.

Integrate and interpret

Integrating and interpreting involves processing what is read to construct an internal representation of the meaning of the text.



At the most basic levels of comprehension, readers need to be able to identify in print the meaning of individual words that would occur in the everyday listening lexicon of average adult speakers of the language (e.g. Sabatini and Bruce, 2009). This would include the everyday words of the language that would be used in common social and commerce situations, but not those specialised to technical or academic areas. Beyond the word level, students must be able to combine words in order to parse sentences and to represent their literal meaning. This involves an ability to acknowledge when sentences are ill-structured or simply do not make sense (see the section with sample items at the end of the chapter). Readers also need to combine the meaning of small sets of sentences in order to form internal representations of simple descriptions or narrations. Processing the literal meaning of a text is a foundational competency that then allows additional deeper processes to be performed on the text. In order to better represent this basic comprehension level, in PISA-D the category “integrate and interpret” is extended so as to include the process of “comprehending the literal meaning of text”. Tasks that specifically require this process will be included in the assessment.

Integrating focuses on demonstrating an understanding of the coherence of the *text* and involves the processes to make internal sense of a text. *Integrating* involves connecting various pieces of information to make meaning, whether it be identifying similarities and differences, making comparisons of degree, or understanding cause and effect relationships.

Interpreting also requires going beyond the literal meaning and refers to the process of making meaning from something that is not stated. When interpreting, a reader is identifying the underlying assumptions or implications of part or all of the text.

Both *integrating* and *interpreting* are required to *form a broad understanding*. A reader must consider the text as a whole or in a broad perspective. Students may demonstrate initial understanding by identifying the main topic or message or by identifying the general purpose or use of the text.

Both *integrating* and *interpreting* are also involved in *developing an interpretation*, which requires readers to extend their initial broad impressions so that they develop a deeper, more specific or more complete understanding of what they have read. *Integrating* tasks include identifying and listing supporting evidence, and comparing and contrasting information in which the requirement is to draw together two or more pieces of information from the text. In order to process either explicit or implicit information from one or more sources in such tasks, the reader must often infer an intended relationship or category. *Interpreting* tasks may involve drawing an inference from a local context: for example, interpreting the meaning of a word or phrase that gives a particular nuance to the text. This process of comprehension is also assessed in tasks that require the student to make inferences about the author’s intention, and to identify the evidence used to infer that intention.

The relationship between the processes of integration and interpretation may therefore be seen as intimate and interactive. Integrating involves first inferring a relationship within the text (a kind of interpretation), and then bringing pieces of information together, therefore allowing an interpretation to be made that forms a new integrated whole.

Reflect and evaluate

Reflecting and evaluating involves drawing upon knowledge, ideas or attitudes beyond the text in order to relate the information provided within the text to one’s own conceptual and experiential frames of reference.

Reflect items may be thought of as those that require readers to consult their own experience or knowledge to compare, contrast or hypothesise. *Evaluate* items are those that ask readers to make a judgment drawing on standards beyond the text.

Reflecting on and evaluating the content of a text requires the reader to connect information in a text to knowledge from outside sources. Readers must also assess the claims made in the text against their own knowledge of the world. Often readers are asked to articulate and defend their own points of view. To do so, readers must be able to develop an understanding of what is said and intended in a text. They must then test that mental representation against what they know and believe on the basis of either prior information, or information found in other texts. Readers must call on supporting evidence from within the text and contrast it with other sources of information, using both general and specific knowledge as well as the ability to reason abstractly.

Reflecting on and evaluating the form of a text requires readers to stand apart from the text, to consider it objectively, and to evaluate its quality and appropriateness. Implicit knowledge of text structure, the style typical of different kinds of texts, can play an important role in these tasks. Evaluating how successful an author is in portraying some characteristic or persuading a reader depends not only on substantive knowledge but also on the ability to detect subtleties in language.



To some extent every critical judgment requires the reader to consult his or her own experience; some kinds of reflection, on the other hand, do not require evaluation (for example, comparing personal experience with something described in a text). Thus evaluation might be seen as a subset of reflection.

The processes of reading in print and digital media

The three broad processes defined for PISA reading literacy are not conceived of as entirely separate and independent, but rather as inter-related and interdependent. Indeed from a cognitive processing perspective, they can be considered semi-hierarchical: it is not possible to interpret or integrate information without having first retrieved it. And it is not possible to reflect on or evaluate information without having made some sort of interpretation. In PISA, however, the framework description of reading processes distinguishes approaches to reading that are demanded for different contexts and purposes; these are then reflected in assessment tasks that emphasise one or other process.

For PISA-D, the distribution of tasks across the major framework variables of situation and text should closely mirror the distributions used for the print items in PISA 2012, both for the school-based and the out-of-school tests. The distribution of process variables does have some differences.

Table 2.1 shows the approximate distribution of reading score points by the processes for the PISA 2012 assessment and the desired distribution of reading score points by the processes for PISA-D. Note that the distribution puts greater emphasis on *access and retrieve*, most particularly at the lower levels of proficiency, while also putting lower emphasis on *reflect and evaluate*. This enhances the sensitivity to competencies that will tend to fall at the lower levels of the PISA scale.

Table 2.1 Distribution of score points in reading, by processes, for PISA 2012 (approximate distribution) and PISA-D (desired distribution)

Processes (aspects)	Percentage of total score points in PISA 2012	Percentage of total score points in PISA-D
Access and retrieve	22	25-30% with 15% below Level 3
Integrate and interpret	56	45-55%
Reflect and evaluate	22	15-25%
Complex	0	0
Total	100	100

The desired distribution specifies the blueprint for selecting items according to important aspects of the domain frameworks. Item selection is based on the assessment design, as well as item characteristics related to a number of framework aspects – including coding requirement, process, situation, and text format, and consideration of the items' psychometric properties and appropriateness for this assessment. Following the assessment, the actual distributions of items across the framework aspects will be described in relation to the desired distributions. The extent to which the item pool for the assessment meets the framework specifications will be discussed in the technical report in the context of practical constraints in the item selection process.

Situation

The PISA-D situation variables remain the same as those for PISA 2012. They were adapted from the Common European Framework of Reference (CEFR) developed for the Council of Europe (Council of Europe, 1996). The four situation variables – personal, public, educational and occupational – are described in the following paragraphs.

The *personal* situation relates to texts that are intended to satisfy an individual's personal interests, both practical and intellectual. This category also includes texts that are intended to maintain or develop personal connections with other people. It includes personal letters, fiction, biography and informational texts that are intended to be read to satisfy curiosity, as a part of leisure or recreational activities. In the digital medium it includes personal emails, instant messages and diary-style blogs.

The *public* category describes the reading of texts that relate to activities and concerns of the larger society. The category includes official documents and information about public events. In general, the texts associated with this category assume a more or less anonymous contact with others; they also therefore include forum-style blogs, news websites, and public notices that are encountered both on line and in print.



The content of *educational* texts is usually designed specifically for the purpose of instruction. Printed text books and interactive learning software are typical examples of material generated for this kind of reading. Educational reading normally involves acquiring information as part of a larger learning task. The materials are often not chosen by the reader, but instead assigned by an instructor. The model tasks are those usually identified as “reading to learn” (Sticht, 1975; Stiggins, 1982).

Many 15-year-olds will move from school into the labour force within one to two years, and many out-of-school youth may already be part of the work force. A typical *occupational* reading task is one that involves the accomplishment of some immediate task. It might include searching for a job, either in a print newspaper’s classified advertisement section, or on line; or following workplace directions. The model tasks of this type are often referred to as “reading to do” (Sticht, 1975; Stiggins, 1982).

Situation is used in PISA reading literacy to define texts and their associated tasks, and refers to the contexts and uses for which the author constructed the text. The manner in which the situation variable is specified is therefore about supposed audience and purpose, and is not based simply on the place where the reading activity is carried out. Many texts used in classrooms are not specifically designed for classroom use. For example, a piece of literary text may typically be read by a 15-year-old in a mother-tongue language or literature class, yet the text was written (presumably) for readers’ personal enjoyment and appreciation. Given its original purpose, such a text is classified as *personal* in PISA. As Hubbard (1989) has shown, some kinds of reading usually associated with out-of-school settings for children, such as rules for clubs and records of games, often take place unofficially at school as well. These texts are classified as *public* in PISA. Conversely, textbooks are read both in schools and in homes, and the process and purpose probably differ little from one setting to another. Such texts are classified as *educational* in PISA.

It should be noted that the four categories overlap. In practice, for example, a text may be intended both to delight and to instruct (personal and educational); or to provide professional advice that is also general information (occupational and public). While content is not a variable that is specifically manipulated in this study, by sampling texts across a variety of situations the intent is to maximise the diversity of content that will be included in the PISA reading literacy survey.

Table 2.2 shows the approximate distribution of score points by situation for print reading tasks in PISA 2012 and the desired distribution for PISA-D. The distributions of situations used in PISA 2012 can be maintained at the same approximate values for PISA-D.

Table 2.2 Distribution of score points in reading, by situation, for PISA 2012 (approximate distribution) and PISA-D (desired distribution)

Situation	Percentage of total score points in PISA 2012	Percentage of total score points in PISA-D
Personal	36	25-45
Educational	33	25-45
Occupational	20	15-25
Public	11	5-15
Total	100	100

Text

The text dimensions for PISA-D remain the same as those used in PISA 2012. Reading requires material for the reader to read. In an assessment, that material – a text (or a set of texts) related to a particular task – must be coherent within itself. That is, the text must be able to stand alone without requiring additional material to make sense to the proficient reader. While it is obvious that there are many different kinds of texts and that any assessment should include a broad range, it is not so obvious that there is an ideal categorisation of kinds of texts. The addition of digital reading to the framework has made this issue still more complex. In 2009 and 2012, there have been four main text classifications:²

- medium: print and digital
- environment: authored and message-based



- text format: continuous, non-continuous, mixed and multiple
- text type: description, narration, exposition, argumentation, instruction and transaction.

The classification of medium – print and digital – is applied to each text as the broadest distinction. Below that classification, the text format and text type categories are applied to all texts, whether print or digital. The environment classification, on the other hand, is only applicable to digital texts.

Medium

Since PISA 2009, an important major categorisation of texts is the classification by medium: print or digital.

Print text usually appears on paper in forms such as single sheets, brochures, magazines and books. The physical status of the printed text encourages (though it does not compel) the reader to approach the content of the text in a particular sequence. In essence, printed texts have a fixed or static existence. Moreover, in real life and in the assessment context, the extent or amount of the text is immediately visible to the reader.

Digital text may be defined as the display of text through liquid crystal display (LCD), plasma, thin film transistor (TFT), and other electronic devices. For the purposes of PISA, however, digital text is synonymous with *hypertext*: a text or texts with navigation tools and features that make possible and indeed even require non-sequential reading. Each reader constructs a “customised” text from the information encountered at the links he or she follows. In essence, such digital texts have an unfixed, dynamic existence. In the digital medium, typically only a fraction of the available text can be seen at any one time, and often the extent of text available is unknown. The PISA-D instruments do not include hypertext, but digital text is mentioned here for completeness.

Text format

An important classification of texts is the distinction between continuous and non-continuous texts.

Texts in *continuous* and *non-continuous* format appear in both the print and digital media. *Mixed* and *multiple* format texts are also prevalent in both media, particularly so in the digital medium. Each of these four formats is elaborated as follow:

Continuous texts are formed by sentences organised into paragraphs. These may fit into even larger structures, such as sections, chapters, and books (e.g. newspaper reports, essays, novels, short stories, reviews and letters for the print medium, and reviews, blogs and reports in prose for the digital).

Non-continuous texts are organised differently to *continuous* texts, and therefore require a different kind of reading approach. *Non-continuous* texts are most frequently organised in matrix format, composed of a number of lists (Kirsch and Mosenthal, 1990) (e.g. lists, tables, graphs, diagrams, advertisements, schedules, catalogues, indexes and forms).

Many texts in both print and digital media are single, coherent artefacts consisting of a set of elements in both a *continuous* and *non-continuous* format. In well-constructed *mixed* texts, the constituents (e.g. a prose explanation, along with a graph or table) are mutually supportive through coherence and cohesion links at the local and global level. *Mixed* text in the print medium is a common format in magazines, reference books and reports. In the digital medium, authored web pages are typically mixed texts, with combinations of lists, paragraphs of prose, and often graphics. Message-based texts such as online forms, email messages and forums also combine texts that are *continuous* and *non-continuous* in format.

Multiple texts are defined as those that have been generated independently, and make sense independently; they are juxtaposed for a particular occasion or may be loosely linked together for the purposes of the assessment. The relationship between the texts may not be obvious; they may be complementary or may contradict one another. For example, with digital texts, a set of websites from different companies providing travel advice may or may not provide similar directions to tourists. For paper-based texts, multiple texts may include a bus time schedule, a map and a text explaining a set of tours around a town. Multiple texts may have a single “pure” format (for example, continuous), or may include both continuous and non-continuous texts.

Table 2.3 shows the approximate distributions of score points for print reading tasks by text format for PISA 2012, which should be maintained for PISA-D.

Table 2.3 Distribution of score points in reading, by text format, for PISA 2012 (approximate distribution) and PISA-D (desired distribution)

Text format	Percentage of total score points in PISA 2012	Percentage of total score points in PISA-D
Continuous	58	50-60
Non-continuous	31	25-35
Mixed	9	5-15
Multiple	2	0-10
Total	100	100

Text type

A different categorisation of text is by text type: description, narration, exposition, argumentation, instruction, and transaction. The text types are the same for PISA-D as they have been since PISA 2009.

Texts as they are found in the world typically resist categorisation; they are usually not written with rules in mind, and tend to cut across categories. That notwithstanding, in order to ensure that the reading instrument samples across a range of texts that represent different types of reading, PISA categorises texts based on their predominant characteristics.

The following classification of texts used in PISA is adapted from the work of Werlich (1976).

Description is the type of text where the information refers to properties of objects in space. The typical questions that descriptive texts provide an answer to are *what* questions (e.g. a depiction of a particular place in a travelogue or diary, a catalogue, a geographical map, an online flight schedule, or a description of a feature, function or process in a technical manual).

Narration is the type of text where the information refers to properties of objects in time. Narration typically answers questions relating to *when*, or *in what sequence*. Why characters in stories behave as they do is another important question that narration typically answers (e.g. a novel, a short story, a play, a biography, a comic strip, fictional texts and a newspaper report of an event). The proportion of narrative texts in the print medium in PISA 2012 was a little greater than that in the previous PISA cycles (2000-09), at about 20% (formerly about 15%).

Exposition is the type of text in which the information is presented as composite concepts or mental constructs, or those elements into which concepts or mental constructs can be analysed. The text provides an explanation of how the different elements interrelate in a meaningful whole and often answers questions about *how* (e.g. a scholarly essay, a diagram showing a model of memory, a graph of population trends, a concept map and an entry in an online encyclopaedia).

Argumentation is the type of text that presents the relationship among concepts or propositions. Argument texts often answer *why* questions. An important sub-classification of argument texts is persuasive and opinionative texts, referring to opinions and points of view. Examples of text in the text type category *argumentation* are a letter to the editor, a poster advertisement, the posts in an online forum, and a Web-based review of a book or film.

Instruction is the type of text that provides directions on what to do. The text presents directions for certain behaviours in order to complete a task (e.g. a recipe, a series of diagrams showing a procedure for giving first aid and guidelines for operating digital software).

Transaction represents the kind of text that aims to achieve a specific purpose outlined in the text, such as requesting that something is done, organising a meeting or making a social engagement with a friend. Before the spread of digital communication, this kind of text was a significant component of some kinds of letters and, as an oral exchange, the principal purpose of many phone calls. This text type was not included in Werlich's (1976) categorisation. It was used for the first time in the PISA 2009 framework because of its prevalence in the digital medium (e.g. everyday email and text message exchanges between colleagues or friends that request and confirm arrangements).

Strategy to extend the framework to provide better coverage of basic literacy levels

Two strategies are used in order to extend the framework to lower levels of reading proficiencies. Firstly, additional item types are included to assess word meaning, basic sentence and passage comprehension, and literal meaning (see above for the description of the process "integrate and interpret"). The purpose of the tasks is to measure the extent to which



students understand the literal and inferential meaning of words and connected text. Two tasks were defined: *sentence processing* and *passage comprehension*.

The sentence processing tasks assess the ability to comprehend written sentences of varying lengths. In the PISA reading components assessment, the construct is instantiated in a sensibility judgment task. Its purpose is to measure the extent to which students can comprehend sentences of increasing lengths. In the task, students see a set of sentences and decide if they make sense (“yes”) or do not make sense (“no”) with respect to general knowledge about the real world (as in the first item in the section on sample items), or the internal logic of the sentence itself (as in the second item).

The basic passage comprehension tasks assess the ability to understand the literal meaning or “gist” of connected text and to make low-level inferences across sentences in the text. In the PISA and PIAAC reading components assessments, the construct has been instantiated in an embedded cloze task, in which certain words are purposefully deleted from the text and replaced with blanks. The task is for the test-taker to complete the missing words. Its purpose is to measure the extent to which students understand the literal and inferential meaning of connected text. In the task, the participant sees a passage in which the sentences include an embedded cloze item (two word choices are given for a single blank). The participant reads the passage silently and circles the word that correctly completes each sentence. Sample item 2 shows an example of a passage comprehension task with multiple items embedded within it.

The second strategy consists in adapting existing PISA tasks to assess low-level comprehension and access and retrieve processes. Sample items 3, 4 and 5 illustrate this.

ASSESSING READING LITERACY

The previous section outlined the conceptual framework for reading literacy. The concepts in the framework must in turn be represented in tasks and questions in order to collect evidence of students’ proficiency in reading literacy.

The distribution of tasks across the major framework variables of process, situation and text was discussed in the previous section. In this section, the framework describes the distribution of tasks across the major framework variables as well as some of the other major issues in constructing and operationalising the assessment: factors affecting item difficulty and how difficulty can be manipulated; the choice of response formats; issues around coding and scoring; strategy to extend the framework to provide better coverage of basic literacy levels; reporting proficiency in reading in PISA-D; testing reading literacy among the out-of-school population; and examples of items for addressing the extended PISA-D framework.

Factors affecting item difficulty

The difficulty of any reading literacy task depends on an interaction among several variables. Drawing on Kirsch and Mosenthal’s work (e.g. Kirsch, 2001; Kirsch and Mosenthal, 1990), we can manipulate the difficulty of items by applying knowledge of the following process and text format variables.

In *access and retrieve* tasks, difficulty is conditioned by the number of pieces of information that the reader needs to locate, by the amount of inference required, by the amount and prominence of competing information, and by the length (e.g. number of words, sentences, paragraphs) and complexity of the text.

In *integrate and interpret* tasks, difficulty is affected by the type of interpretation required (for example, making a comparison is easier than finding a contrast, and comprehending a specified causal link is easier than inferring an implicit causal relationship); by the number of pieces of information to be considered; by the degree and prominence of competing information in the text; and by the nature of the text. The less familiar and the more abstract the content and the longer and more complex the text, the more difficult the task is likely to be.

In *reflect and evaluate* tasks, difficulty is affected by the type of reflection or evaluation required (from least to most difficult, the types of reflection are: connecting; explaining and comparing; hypothesising and evaluating); by the nature of the knowledge that the reader needs to bring to the text (a task is more difficult if the reader needs to draw on narrow, specialised knowledge rather than broad and common knowledge); by the relative abstraction and length of the text; and by the depth of understanding of the text required to complete the task.

In tasks relating to *continuous texts*, difficulty is influenced by the length of the text, the explicitness and transparency of its structure, how clearly the parts are related to the general theme, and whether there are text features, such as paragraphs or headings, and discourse markers, such as sequencing words.



In tasks relating to *non-continuous texts*, difficulty is influenced by the amount of information in the text; the list structure (simple lists are easier to negotiate than more complex lists); whether the components are ordered and explicitly organised, for example with labels or special formatting; and whether the information required is in the body of the text or in a separate part, such as a footnote.

Response formats

Coding requirements are shown in Table 2.4 for print score points in relation to the three processes of reading literacy and for digital score points in relation to the four processes. Items that require expert judgment consist of open-constructed and short-constructed responses that require expert coding. Items that do not require coder judgment consist of multiple-choice, complex multiple-choice and closed-constructed response items. The closed-constructed response items are those that require the student to generate a response, but require minimal judgment on the part of a coder.

Distribution of coding requirements for PISA-D should be kept comparable to mainstream PISA assessments. The distribution of item types in print reading does not vary much from one cycle/administration to the next. However, the selection for 2012 has a slightly higher proportion of items that do not require expert coding than in previous cycles: 58% non-expert coded and 42% expert coded in 2012 (compared with 55% and 45% respectively in previous administrations). The same ratio applies to print and to digital reading in PISA 2012.

Table 2.4 shows the approximate distribution of score points by coding requirement for each reading process in PISA 2012 and in the paper-based PISA-D test. Due to the extra testing time it would involve, the tablet-based test does not include items that require expert judgment in coding.

Table 2.4 Distribution of score points in reading, by coding requirement for each reading process, in PISA 2012 (approximate distribution) and PISA-D (desired distribution)

Process (aspect)	Percentage of total score points in PISA 2012: Print reading			Percentage of total score points in PISA-D		
	Expert judgment required	No expert judgment required	Total	Expert judgment required	No expert judgment required	Total
Access and retrieve	4	18	22	0-10	10-20	20-30
Integrate and interpret	20	36	56	15-30	30-40	45-60
Reflect and evaluate	18	4	22	15-25	0-10	20-30
Complex	0	0	0	0	0	0
TOTAL	42	58	100	35-50	45-65	100

Coding and scoring

Codes are applied to test items, either by a more or less automated process of capturing the alternative chosen by the student for a multiple-choice answer, or by a human judge (expert coder) selecting a code that best captures the kind of response given by a student to an item that requires a constructed response. The code is then converted to a score for the item. For multiple-choice or closed-response format items, the student has either chosen the designated correct answer or not, so the item is scored as 1 (full credit) or 0 (no credit) respectively. For more complex scoring of constructed response items, some answers, even though incomplete, indicate a higher level of reading literacy than inaccurate or incorrect answers, and receive partial credit.

Reporting proficiency in reading in PISA-D

PISA reports results in terms of proficiency scales that are interpretable for the purposes of policy. To capture the progression of complexity and difficulty, from 2009 and up to 2018, PISA has used seven levels based on the PISA 2009 combined print reading literacy scale. For PISA-D, an additional level has been added at the lowest level, so the combined print reading literacy scale is divided into eight levels. Figure 2.2 describes these eight levels of print reading proficiency. Level 6 is the highest described level of proficiency (Level 5 was the highest level before PISA 2009 reading assessments). Levels 2, 3, 4 and 5 remain the same as in PISA 2000. In the mainstream PISA, the lowest bottom level of measured proficiency is Level 1b, with Level 1a being the second lowest level. For PISA-D, Level 1c is added as the lowest level of proficiency with a focus on understanding words, short phrases and extracting literal meaning from sentences. These different levels of proficiency allow countries to know more about the kinds of tasks students with very high and very low reading proficiency are capable of performing. Levels 1a and 1b have been modified for better alignment with the new Level 1c.



Figure 2.2 ■ Summary description of the eight levels of reading proficiency in PISA-D

Level	Lower score limit	Percentage of students across OECD countries at each level, PISA 2012	Percentage of students across 18 middle- and low-income countries at each level, PISA 2012	Characteristics of tasks
6	698	1.1%	0.1%	Tasks at this level typically require the reader to make multiple inferences, comparisons and contrasts that are both detailed and precise. They require demonstration of a full and detailed understanding of one or more texts and may involve integrating information from more than one text. Tasks may require the reader to deal with unfamiliar ideas, in the presence of prominent competing information, and to generate abstract categories for interpretations. <i>Reflect and evaluate</i> tasks may require the reader to hypothesise about or critically evaluate a complex text on an unfamiliar topic, taking into account multiple criteria or perspectives, and applying sophisticated understandings from beyond the text. A salient condition for <i>access and retrieve</i> tasks at this level is precision of analysis and fine attention to detail that is inconspicuous in the texts.
5	626	7.3%	1.1%	Tasks at this level that involve retrieving information require the reader to locate and organise several pieces of deeply embedded information, inferring which information in the text is relevant. Reflective tasks require critical evaluation or hypothesis, drawing on specialised knowledge. Both interpretative and reflective tasks require a full and detailed understanding of a text whose content or form is unfamiliar. For all processes of reading, tasks at this level typically involve dealing with concepts that are contrary to expectations.
4	553	21.0%	6.7%	Tasks at this level that involve retrieving information require the reader to locate and organise several pieces of embedded information. Some tasks at this level require interpreting the meaning of nuances of language in a section of text by taking into account the text as a whole. Other interpretative tasks require understanding and applying categories in an unfamiliar context. Reflective tasks at this level require readers to use formal or public knowledge to hypothesise about or critically evaluate a text. Readers must demonstrate an accurate understanding of long or complex texts whose content or form may be unfamiliar.
3	480	29.1%	19.1%	Tasks at this level require the reader to locate, and in some cases recognise the relationship between, several pieces of information that must meet multiple conditions. Interpretative tasks at this level require the reader to integrate several parts of a text in order to identify a main idea, understand a relationship or construe the meaning of a word or phrase. They need to take into account many features in comparing, contrasting or categorising. Often the required information is not prominent or there is much competing information; or there are other text obstacles, such as ideas that are contrary to expectation or negatively worded. Reflective tasks at this level may require connections, comparisons, and explanations, or they may require the reader to evaluate a feature of the text. Some reflective tasks require readers to demonstrate a fine understanding of the text in relation to familiar, everyday knowledge. Other tasks do not require detailed text comprehension but require the reader to draw on less common knowledge.

Figure 2.2 [continued] ■ Summary description of the eight levels of reading proficiency in PISA-D

Level	Lower score limit	Percentage of students across OECD countries at each level, PISA 2012	Percentage of students across 18 middle- and low-income countries at each level, PISA 2012	Characteristics of tasks
2	407	23.5%	30.3%	Some tasks at this level require the reader to locate one or more pieces of information, which may need to be inferred and may need to meet several conditions. Others require recognising the main idea in a text, understanding relationships, or construing meaning within a limited part of the text when the information is not prominent and the reader must make low level inferences. Tasks at this level may involve comparisons or contrasts based on a single feature in the text. Typical reflective tasks at this level require readers to make a comparison or several connections between the text and outside knowledge, by drawing on personal experience and attitudes.
1a	335	12.3%	25.9%	Tasks at this level require the reader to understand the literal meaning of sentences or short passages. Most tasks require the reader to locate one or more independent pieces of information; to recognise the main theme or author's purpose in a text about a familiar topic, or to make a simple connection between information in the text and common, everyday knowledge. The reader is directed to consider relevant factors in the task and in the text. In tasks requiring interpretation, the reader may need to make simple connections between adjacent pieces of information.
1b	262	4.4%	12.6%	Tasks at this level require the reader to understand the literal meaning of sentences within single short passages. Some tasks require students to locate a piece of explicitly stated information in a single given text. The reader is explicitly directed to consider relevant factors in the task and in the text. Most texts at level 1b are short and they typically contain limited competing information.
1c	189	1.3% (percentage of students scoring below Level 1b, PISA 2012)	4.3% (percentage of students scoring below Level 1b, PISA 2012)	Tasks at this level require the reader to understand the literal meaning of individual written words and phrases within sentences or very short, syntactically simple passages with familiar contexts. Some tasks require students to locate a single word or phrase in a short list or text based on literal matching cues. Texts at level 1c are short and they include little if any competing information. Texts support students with a familiar structure, explicit pointers to the information, repetition and illustration.

Note: Descriptors 2 through 6 are the same as those used in PISA 2012 and 2015. Descriptors 1a and 1b have been revised for better alignment with the new descriptor for Level 1c.

Testing reading literacy among the out-of-school population

The extended PISA-D reading framework is appropriate for 15-year-old students whether in or out of school. The units and items are not directly based in the school context, and thus there is no particular requirement or change needed in the units that are categorised as relevant for educational activities, since educational activities also occur out of school. Therefore, the distribution and selection of units and items can be the same for PISA-D in-school and out-of-school populations.

The out-of-school component is assessed on a tablet computer, but only fixed-text items are used, so it is appropriate to use the same framework as for the paper-based test.



Box 2.3 Delivery mode

The PISA-D school-based assessment is paper-based, while the out-of-school assessment is conducted on a tablet computer. To ensure comparability between the tests, the tablet-based instruments for PISA-D are formed by a subgroup of the items used for the paper-based assessment. All these items were originally designed for a paper-based assessment, so when moving to a tablet-based delivery, care was taken to maintain comparability between the assessments. The PISA 2015 framework describes some factors that must be considered when transposing items from paper to computer mode. These elements were also taken into account when designing the out-of-school instrument for PISA-D.

Item types: The computer provides a range of opportunities for designers of test items, including new item formats (e.g. drag-and-drop, hotspots). Since the PISA-D tablet-based tests use a subgroup of items from the paper-based test, there is less opportunity to exploit innovative item types and the majority of response formats remains unchanged.

Stimulus presentation: A feature of fixed texts defined in the construct is that “the extent or amount of the text is immediately visible to the reader”. Clearly, it is impossible, both on paper and on a screen, to have long texts displayed on a single page or screen. To allow for this and still satisfy the construct of fixed texts, pagination is used for texts rather than scrolling. Texts that cover more than one page are presented in their entirety before the student sees the first question.

IT skills: Just as paper-based assessments rely on a set of fundamental skills for working with printed materials, so computer-based assessments rely on a set of fundamental information and communications technology skills for using computers. These include knowledge of basic hardware (e.g. keyboard and mouse) and basic conventions (e.g. arrows to move forward and specific buttons to press to execute commands). The intention is to keep such skills to a minimal core level in the tablet-based assessment.

Examples of items for addressing the extended PISA-D reading framework

The following six items illustrate the types of question that can be asked of students at Level 1a and below. The items either come from or are adapted from the PISA or PIAAC assessments.

Sample item 1

Directions: Circle **YES** if the sentence makes sense. Circle **NO** if the sentence does not make sense.

The red car had a flat tyre.	YES	NO
Airplanes are made of dogs.	YES	NO
The happy student read the book last night.	YES	NO
If the cat had stayed out all night, it would not have been in the house at 2 a.m.	YES	NO
The man who is taller than the woman and the boy is shorter than both of them.	YES	NO

Sample item 1 assesses sentence processing tasks and likely corresponds to proficiency Level 1c. In PISA-D, sentence processing tasks also included some short sentences with three options, with instructions to choose the word that makes the sentence make sense.

Sample item 2**PASSAGE COMPREHENSION**

In items assessing passage comprehension, respondents are asked to read a passage in which they are required at certain points to select the word that makes sense from the two alternatives provided.

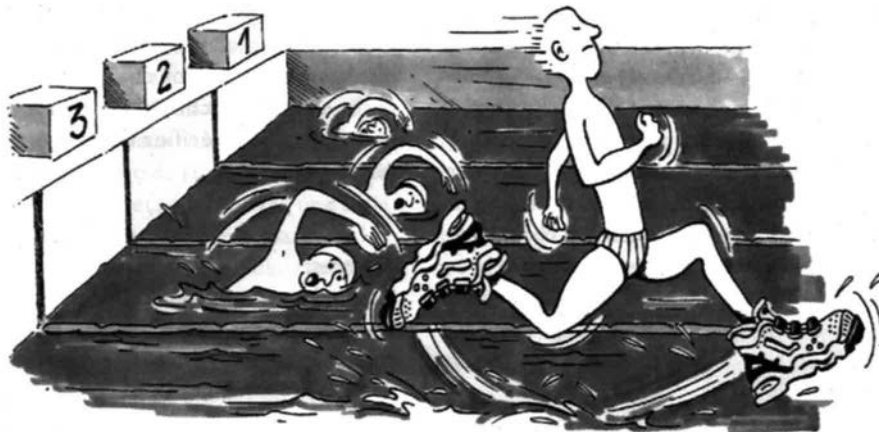
To the editor: Yesterday, it was announced that the cost of riding the bus will increase. The price will go up by twenty percent starting next wife / month. As someone who rides the bus every day, I am upset by this foot / increase. I understand that the cost of gasoline / student has risen. I also understand that riders have to pay a fair price / snake for bus service. I am willing to pay a little more because I rely on the bus to get to object / work. But an increase / uncle of twenty percent is too much.

This increase is especially difficult to accept when you see the city's plans to build a new sports stadium. The government will spend millions on this project even though we already have a science / stadium. If we delay the stadium, some of that money can be used to offset the increase in bus fares / views. Then, in a few years, we can decide if we really do need a new sports cloth / arena. Please let the city council know you care about this issue by attending the next public meeting / frames.

Sample item 2 assesses passage comprehension and likely corresponds to proficiency Level 1c. In PISA-D, the passage comprehension paragraphs (part of Reading Components) have been modified to have three options instead of two.

Sample item 3**FEEL GOOD IN YOUR RUNNERS**

For 14 years the Sports Medicine Centre of Lyon (France) has been studying the injuries of young sports players and sports professionals. The study has established that the best course is prevention ... and good shoes.





Knocks, falls, wear and tear...

Eighteen per cent of sports players aged 8 to 12 already have heel injuries. The cartilage of a footballer's ankle does not respond well to shocks, and 25% of professionals have discovered for themselves that it is an especially weak point. The cartilage of the delicate knee joint can also be irreparably damaged and if care is not taken right from childhood (10–12 years of age), this can cause premature osteoarthritis. The hip does not escape damage either and, particularly when tired, players run the risk of fractures as a result of falls or collisions.

According to the study, footballers who have been playing for more than ten years have bony

outgrowths either on the tibia or on the heel. This is what is known as “footballer’s foot”, a deformity caused by shoes with soles and ankle parts that are too flexible.

Protect, support, stabilise, absorb

If a shoe is too rigid, it restricts movement. If it is too flexible, it increases the risk of injuries and sprains. A good sports shoe should meet four criteria:

Firstly, it must *provide exterior protection*: resisting knocks from the ball or another player, coping with unevenness in the ground, and keeping the foot warm and dry even when it is freezing cold and raining.

It must *support the foot*, and in particular the ankle joint, to avoid sprains, swelling and

other problems, which may even affect the knee.

It must also provide players with good *stability* so that they do not slip on a wet ground or skid on a surface that is too dry.

Finally, it must *absorb shocks*, especially those suffered by volleyball and basketball players who are constantly jumping.

Dry feet

To avoid minor but painful conditions such as blisters or even splits or athlete’s foot (fungal infections), the shoe must allow evaporation of perspiration and must prevent outside dampness from getting in. The ideal material for this is leather, which can be water-proofed to prevent the shoe from getting soaked the first time it rains.

QUESTION 7.2

According to the article, why should sports shoes not be too rigid?

QUESTION 7.4

Look at this sentence from near the end of the article. It is presented here in two parts:

“To avoid minor but painful conditions such as blisters or even splits or athlete’s foot (fungal infections),...”

(first part)

“...the shoe must allow evaporation of perspiration and must prevent outside dampness from getting in.”

(second part)

What is the relationship between the first and second parts of the sentence?

The second part:

- A. contradicts the first part.
- B. repeats the first part.
- C. illustrates the problem described in the first part.
- D. gives the solution to the problem described in the first part.

Sample item 3 is a released PISA item that shows a basic informational text. Question 7.2 assesses a student’s literal comprehension from the text. Because of the amount of text students must read, it likely corresponds to Level 1a. Question 7.4, on the other hand, assesses integration and interpretation of information and thus is at a higher level of proficiency.



Sample item 4

The Moreland Library System gives new library members a bookmark showing its Hours of Opening. Refer to the bookmark to answer the questions which follow.

	HOURS OF OPENING					<i>Effective from February 1 1998</i>
	Brunswick Library	Campbell Turnbull Library	Coburg Library	Fawkner Library	Glenroy Library	
Sunday	1pm-5pm	Closed	2pm-5pm	Closed	2pm-5pm	
Monday	11am-8pm	11am-5.30pm	1pm-8pm	11am-5.30pm	10am-5.30pm	
Tuesday	11am-8pm	11am-8pm	11am-8pm	11am-8pm	10am-8pm	
Wednesday	11am-8pm	11am-5pm	10am-8pm	11am-5pm	10am-8pm	
Thursday	11am-8pm	11am-5.30pm	10am-8pm	11am-5.30pm	10am-8pm	
Friday	11am-5pm	11am-5pm	10am-8pm	11am-5pm	10am-5.30pm	
Saturday	10am-1pm	10am-1pm	9am-1pm	10am-1pm	9am-1pm	

QUESTION 12.1

What time does the Fawkner Library close on Wednesday?

.....

QUESTION 12.2

Which library is still open at 6 p.m. on Friday evening?

- A. Brunswick Library
- B. Campbell Turnbull Library
- C. Coburg Library
- D. Fawkner Library
- E. Glenroy Library

Sample item 4 is a released PISA item that assesses basic access and retrieve tasks in a simple non-continuous text. Question 12.1 requires accessing the information directly from a row in the table which is likely to be Levels 1a or 1b, while Question 12.2 requires combining multiple criteria in order to access the correct information which is more likely Level 2.



Sample item 5

SUPERMARKET NOTICE

Peanut Allergy Alert
Lemon Cream Biscuits

Date of alert: 04 February

Manufacturer's Name: Fine Foods Ltd

Product Information: 125 g Lemon Cream Biscuits (Best before 18 June and Best before 01 July)

Details: Some biscuits in these batches may contain pieces of peanut, which are not included in the ingredient list. People with an allergy to peanuts should not eat these biscuits.

Consumer action: If you have bought these biscuits you may return the product to the place of purchase for a full refund. Or call 1800 034 241 for further information.

QUESTION 3

What is the name of the company that made the biscuits?

.....

Sample item 5 is a released PISA item that assesses basic access and retrieve processes. Question 3, "What is the name of the company that makes the biscuits?" requires a small inference since the text says "manufacturer" rather than "company". Thus, as it stands, the item would likely be at Level 1b of proficiency. However, if it were modified to: "What is the name of the manufacturer that makes the biscuits?", then the item would require a literal match and would be considered as Level 1c.



Notes

1. The term “processes” from the 2018 framework is used instead of the term “aspects” used in previous versions.
2. In 2015 PISA was moved to computer-based delivery with additional consequences for the classification of text types. For more details the reader is referred to the PISA 2015 reading framework.

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3

PISA for Development Mathematics Framework

This chapter defines “mathematical literacy” as assessed in the Programme for International Student Assessment (PISA) and the extensions to the PISA mathematics framework that have been designed for the PISA for Development (PISA-D) project. It explains the processes, content knowledge and contexts reflected in PISA-D’s mathematics problems, and provides several sample items. The chapter also discusses how student performance in mathematics is measured and reported.



WHAT IS NEW IN PISA-D? EXTENSIONS TO THE PISA MATHEMATICAL LITERACY FRAMEWORK

The objective of the PISA-D mathematics framework is to extend the PISA framework, in order to measure mathematical skills of students who perform at or below the lowest level on the standard math PISA scale. The outcomes of such measurement should provide reliable data that could help to plan the most effective ways of improving those students' mathematical skills. This extended framework is applicable, not only for students, but also for 14-16 year-olds who are out of school or not enrolled in PISA's target grades (Grade 7 or above).

To achieve this objective of the framework, it would seem natural to concentrate on some very basic “numeracy skills”, such as fluency in performing simple arithmetical operations. However, it would not be an effective solution. While certainly some of these skills are needed to perform at the highest levels of PISA – such as arithmetical fluency, understanding of basic mathematical concepts, being able to recognise and identify graphs, and understanding of math vocabulary – they are not the focus of PISA.

The implementation of the PISA measurement of mathematics was preceded by a scientific discussion on the role of teaching mathematics. The references section of this chapter lists the most important scientific publications behind those discussions. The framework itself provides a comprehensive explanation of the most important conclusions, culminating in the concept of mathematical literacy. In short, it stresses the primary importance of the ability to use mathematics in a wide variety of contexts. The international success of PISA confirms this as a widely accepted way of understanding the primary goal of learning mathematics in today's world.

From this perspective, mastering the most basic technical skills is not enough. While it is important to be able to perform arithmetical operations, performing these operations it is not sufficient to get by mathematically in real life. To put this knowledge to use, one necessarily needs at least the basic skills of choosing the right model and selecting a strategy or an explanation. These skills constitute the core of the PISA understanding of mathematical literacy.

Identifying some of even the most basic technical skills as a measure of mathematical competencies would be, in this context, quite misleading. It could direct the attention of the users of the PISA results toward those skills as the primary education target, giving little chance to their students of becoming more mathematically literate.

The PISA-D mathematics framework adheres to the core idea of mathematical literacy, as defined by PISA. Therefore it is designed as an extension to the PISA 2015 mathematics framework, essentially measuring the same basic skills. The extensions aim to expand coverage at the lower ability levels in two ways: by using more straightforward, simply formulated items; and by suggesting a very careful analysis of students' attempts to solve the problem. The items used in PISA-D will also test the ability to choose the right model and select a strategy or an explanation. Thus PISA-D gains the potential to help improve students' mathematical literacy.

The extensions made to the PISA 2015 framework in PISA-D are an attempt to gain more information about students who currently perform below Level 1. In the mathematics framework, these extensions occur in three locations: descriptions of the proficiencies, where proficiency Level 1 was renamed as 1a and two new proficiency levels were added, 1b and 1c; adding five new activities to the process descriptors; and adding four new skills to the table relating the mathematical processes to the fundamental mathematical capabilities.

The PISA 2015 framework (OECD, 2016) continues the description and illustration of the PISA mathematics assessment as set out in the 2012 framework, when mathematics was re-examined and updated for use as the major domain in that cycle.

For PISA 2015, the computer was the primary mode of delivery for all domains, including mathematical literacy. The 2015 framework was updated to reflect the change in delivery mode, and includes a discussion of the considerations of transposing paper items to a screen and examples of what the results look like. The definition and constructs of mathematical literacy however, remain unchanged and consistent with those used in 2012. It is important to note that PISA-D includes a paper-based test for the in-school population and a tablet-based test for the out-of-school population. For this reason, therefore, the sections in this chapter dealing with computer-based assessment of mathematics only apply to the out-of-school assessment.

This chapter is organised into three major sections. The first section, “Defining mathematical literacy”, explains the theoretical underpinnings of the PISA mathematics assessment, including the formal definition of the mathematical literacy construct. The second section, “Organising the domain of mathematics”, describes three aspects: i) the mathematical *processes*



and the *fundamental mathematical capabilities* (in previous frameworks the “competencies”) underlying those processes; ii) the way mathematical *content* knowledge is organised in the PISA 2015 framework, and the content knowledge that is relevant to an assessment of 15-year-old students; and iii) the *contexts* in which students will face mathematical challenges. The third section, “Assessing mathematical literacy”, outlines the approach taken to apply the elements of the framework previously described, including the response formats, item scoring, reporting proficiency, testing mathematical literacy among the out-of-school population and examples of items for addressing the extended PISA-D framework.

DEFINING MATHEMATICAL LITERACY

An understanding of mathematics is central to a young person’s preparedness for life in modern society. A growing proportion of problems and situations encountered in daily life, including in professional contexts, require some level of understanding of mathematics, mathematical reasoning and mathematical tools, before they can be fully understood and addressed. Mathematics is a critical tool for young people as they confront issues and challenges in personal, occupational, societal, and scientific aspects of their lives. It is thus important to have an understanding of the degree to which young people emerging from school are adequately prepared to apply mathematics to understanding important issues and solving meaningful problems. An assessment at age 15 provides an early indication of how individuals may respond in later life to the diverse array of situations they will encounter that involve mathematics.

The construct of mathematical literacy used in this chapter is based on PISA 2015, and is intended to describe the capacities of individuals to reason mathematically and use mathematical concepts, procedures, facts and tools to describe, explain and predict phenomena. This conception of mathematical literacy supports the importance of students developing a strong understanding of concepts of pure mathematics and the benefits of being engaged in explorations in the abstract world of mathematics. The construct of mathematical literacy, as defined for PISA, strongly emphasises the need to develop students’ capacity to use mathematics in context, and it is important that they have rich experiences in their mathematics classrooms to accomplish this. For PISA 2012, mathematical literacy was defined as shown in Box 3.1. This is also the definition used in the PISA 2015 and PISA-D assessments.

Box 3.1 The PISA 2015 definition of mathematical literacy

Mathematical literacy is an individual’s capacity to formulate, employ and interpret mathematics in a variety of contexts. It includes reasoning mathematically and using mathematical concepts, procedures, facts and tools to describe, explain and predict phenomena. It assists individuals to recognise the role that mathematics plays in the world and to make the well-founded judgments and decisions needed by constructive, engaged and reflective citizens.

The focus of the language in the definition of mathematical literacy is on active engagement in mathematics, and is intended to encompass reasoning mathematically and using mathematical concepts, procedures, facts and tools in describing, explaining and predicting phenomena. In particular, the verbs “formulate”, “employ” and “interpret” point to the three processes in which students as active problem solvers will engage.

The language of the definition was also intended to integrate the notion of mathematical modelling, which has historically been a cornerstone of the PISA framework for mathematics (e.g. OECD, 2004), into the PISA 2015 definition of mathematical literacy. As individuals use mathematics and mathematical tools to solve problems in contexts, their work progresses through a series of stages (individually developed later in the document).

The modelling cycle is a central aspect of the PISA conception of students as active problem solvers; however, it is often not necessary to engage in every stage of the modelling cycle, especially in the context of an assessment (Niss et al., 2007). The problem solver frequently carries out some steps of the modelling cycle but not all of them, (e.g. when using graphs), or goes around the cycle several times to modify earlier decisions and assumptions.

The definition also acknowledges that mathematical literacy helps individuals to recognise the role that mathematics plays in the world and in helping them make the kinds of well-founded judgments and decisions required of constructive, engaged and reflective citizens.



Mathematical tools mentioned in the definition refer to a variety of physical and digital equipment, software and calculation devices. The 2015 computer-based survey, as well as the PISA-D tablet-based test, included an online calculator as part of the computer-based test material provided for some questions.

ORGANISING THE DOMAIN OF MATHEMATICS

The PISA mathematics framework defines the domain of mathematics for the PISA survey and describes an approach to the assessment of the mathematical literacy of 15-year-olds. That is, PISA assesses the extent to which 15-year-old students can handle mathematics adeptly when confronted with situations and problems – the majority of which are presented in real-world contexts.

For purposes of the assessment, the PISA 2015 definition of mathematical literacy can be analysed in terms of three inter-related aspects:

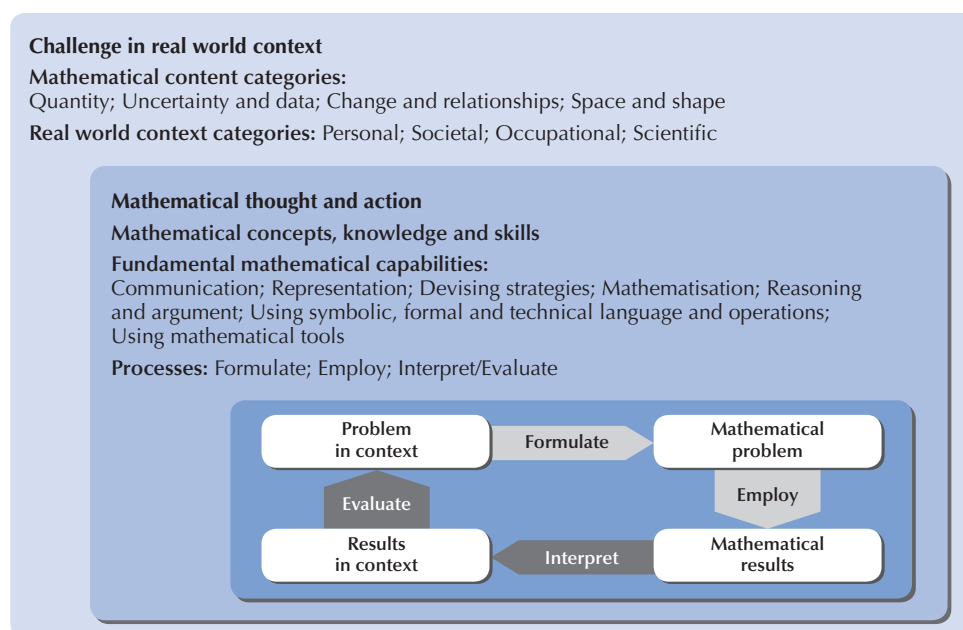
- the mathematical processes that describe what individuals do to connect the context of the problem with mathematics and thus solve the problem, and the capabilities that underlie those processes
- the mathematical content that is targeted for use in the assessment items
- the contexts in which the assessment items are located.

The following sections elaborate these aspects. In highlighting these aspects of the domain, the PISA 2012 mathematics framework, which was also used in PISA 2015 and PISA-D, helps ensure that assessment items developed for the survey reflect a range of processes, content, and contexts, so that, considered as a whole, the set of assessment items effectively operationalises what this framework defines as mathematical literacy. To illustrate the aspects of mathematical literacy, examples are available in the *PISA 2012 Assessment and Analytical Framework* (OECD, 2013) and on the PISA website (www.oecd.org/pisa).

Three questions, based on the PISA 2015 definition of mathematical literacy, lie behind the organisation of this section of the chapter. They are:

- What processes do individuals engage in when solving contextual mathematical problems, and what capabilities do we expect individuals to be able to demonstrate as their mathematical literacy grows?
- What mathematical content knowledge can we expect of individuals – and of 15-year-old students in particular?
- In what contexts can mathematical literacy be observed and assessed?

Figure 3.1 ■ **A model of mathematical literacy in practice**



Source: OECD (2013), *PISA 2012 Assessment and Analytical Framework*, <http://dx.doi.org/10.1787/9789264190511-en>.



Mathematical processes and the underlying mathematical capabilities

Mathematical processes

The definition of mathematical literacy refers to an individual's capacity to formulate, employ and interpret mathematics. These three words, formulate, employ and interpret, provide a useful and meaningful structure for organising the mathematical processes that describe what individuals do to connect the context of a problem with the mathematics and thus solve the problem. Items in the PISA 2015 and the PISA-D mathematics survey will be assigned to one of three mathematical processes:

- formulating situations mathematically
- employing mathematical concepts, facts, procedures and reasoning
- interpreting, applying and evaluating mathematical outcomes.

It is important for both policy makers and those engaged more closely in the day-to-day education of students to know how effectively students are able to engage in each of these processes. The *formulating* process indicates how effectively students are able to recognise and identify opportunities to use mathematics in problem situations and then provide the necessary mathematical structure needed to formulate that contextualised problem into a mathematical form. The *employing* process indicates how well students are able to perform computations and manipulations and apply the concepts and facts that they know to arrive at a mathematical solution to a problem formulated mathematically. The *interpreting* process indicates how effectively students are able to reflect upon mathematical solutions or conclusions, interpret them in the context of a real-world problem, and determine whether the results or conclusions are reasonable. Students' facility at applying mathematics to problems and situations is dependent on skills inherent in all three of these processes, and an understanding of their effectiveness in each category can help inform both policy-level discussions and decisions being made closer to the classroom level.

In an effort to better measure the capabilities of Level 1b and 1c students, specific extensions have been made in PISA-D to the descriptions of the processes formulate, employ and interpret. The five additions are intended to better describe students' attempts to apply mathematical processes. This approach acknowledges that before students may be fully capable of utilising processes, they must first be able to identify and select an appropriate model, strategy or argument.

Formulating situations mathematically

The word *formulate* in the mathematical literacy definition refers to individuals being able to recognise and identify opportunities to use mathematics and then provide mathematical structure to a problem presented in some contextualised form. In the process of formulating situations mathematically, individuals determine where they can extract the essential mathematics to analyse, set up and solve the problem. They translate from a real-world setting to the domain of mathematics and provide the real-world problem with mathematical structure, representations and specificity. They reason about and make sense of constraints and assumptions in the problem. Specifically, this process of formulating situations mathematically includes activities such as the following:

- identifying the mathematical aspects of a problem situated in a real-world context and identifying the significant variables
- recognising mathematical structure (including regularities, relationships and patterns) in problems or situations
- simplifying a situation or problem in order to make it amenable to mathematical analysis
- identifying constraints and assumptions behind any mathematical modelling and simplifications gleaned from the context
- representing a situation mathematically, using appropriate variables, symbols, diagrams and standard models
- representing a problem in a different way, including organising it according to mathematical concepts and making appropriate assumptions
- understanding and explaining the relationships between the context-specific language of a problem and the symbolic and formal language needed to represent it mathematically
- translating a problem into mathematical language or a representation
- recognising aspects of a problem that correspond with known problems or mathematical concepts, facts or procedures
- using technology (such as a spreadsheet or the list facility on a graphing calculator) to portray a mathematical relationship inherent in a contextualised problem.



In addition to the activities listed above, the following activity has been added to PISA-D:

- selecting an appropriate model from a list.

Employing mathematical concepts, facts, procedures and reasoning

The word *employ* in the mathematical literacy definition refers to individuals being able to apply mathematical concepts, facts, procedures and reasoning to solve mathematically formulated problems to obtain mathematical conclusions. In the process of employing mathematical concepts, facts, procedures and reasoning to solve problems, individuals perform the mathematical procedures needed to derive results and find a mathematical solution (e.g. performing arithmetic computations, solving equations, making logical deductions from mathematical assumptions, performing symbolic manipulations, extracting mathematical information from tables and graphs, representing and manipulating shapes in space, and analysing data). They work on a model of the problem situation, establish regularities, identify connections between mathematical entities and create mathematical arguments. Specifically, this process of employing mathematical concepts, facts, procedures and reasoning includes activities such as:

- devising and implementing strategies for finding mathematical solutions
- using mathematical tools, including technology, to help find exact or approximate solutions
- applying mathematical facts, rules, algorithms and structures when finding solutions
- manipulating numbers, graphical and statistical data and information, algebraic expressions and equations, and geometric representations
- making mathematical diagrams, graphs and constructions and extracting mathematical information from them
- using and switching between different representations in the process of finding solutions
- making generalisations based on the results of applying mathematical procedures to find solutions
- reflecting on mathematical arguments and explaining and justifying mathematical results.

In addition to the activities listed above, the following activities have been added to PISA-D:

- performing a simple calculation
- drawing a simple conclusion
- selecting an appropriate strategy from a list.

Interpreting, applying and evaluating mathematical outcomes

The word *interpret* used in the mathematical literacy definition focuses on the abilities of individuals to reflect upon mathematical solutions, results or conclusions and interpret them in the context of real-life problems. This involves translating mathematical solutions or reasoning back into the context of a problem and determining whether the results are reasonable and make sense in the context of the problem. This mathematical process category encompasses both the “interpret” and “evaluate” arrows noted in the previously defined model of mathematical literacy in practice (see Figure 3.1). Individuals engaged in this process may be called upon to construct and communicate explanations and arguments in the context of the problem, reflecting on both the modelling process and its results. Specifically, this process of interpreting, applying and evaluating mathematical outcomes includes activities such as:

- interpreting a mathematical result back into the real-world context
- evaluating the reasonableness of a mathematical solution in the context of a real-world problem
- understanding how the real-world impacts the outcomes and calculations of a mathematical procedure or model in order to make contextual judgments about how the results should be adjusted or applied
- explaining why a mathematical result or conclusion does, or does not, make sense given the context of a problem
- understanding the extent and limits of mathematical concepts and mathematical solutions
- critiquing and identifying the limits of the model used to solve a problem.

In addition to the activities listed above, the following activity has been added to PISA-D:

- evaluating a mathematical outcome in terms of the context.



Desired distribution of items by mathematical process

The goal in constructing the assessment is to achieve a balance that provides approximately equal weighting between the two processes that involve making a connection between the real world and the mathematical world and the process that calls for students to be able to work on a mathematically formulated problem. Table 3.1 shows the desired distribution of items by process for PISA 2015 and PISA-D (both for the in- and out-of-school instruments).

Table 3.1 Desired distribution of mathematics items, by process category

Process category	Percentage of items in PISA 2015	Percentage of items in PISA-D
Formulating situations mathematically	25	25
Employing mathematical concepts, facts, procedures and reasoning	50	50
Interpreting, applying and evaluating mathematical outcomes	25	25
Total	100	100

The desired distribution specifies the blueprint for selecting items according to important aspects of the domain frameworks. Item selection is based on the assessment design, as well as item characteristics related to a number of framework aspects – including process, content and context category, and consideration of the items’ psychometric properties and appropriateness for this assessment. Following the assessment, the actual distributions of items across the framework aspects will be described in relation to the desired distributions. The extent to which the item pool for the assessment meets the framework specifications will be discussed in the technical report in the context of practical constraints in the item selection process.

Fundamental mathematical capabilities underlying the mathematical processes

A decade of experience in developing PISA items and analysing the ways in which students respond to items has revealed that there is a set of fundamental mathematical capabilities that underpins each of these reported processes and mathematical literacy in practice. The work of Mogens Niss and his Danish colleagues (Niss, 2003; Niss and Jensen, 2002; Niss and Højgaard, 2011) identified eight capabilities – referred to as “competencies” by Niss and in the 2003 framework (OECD, 2004) – that are instrumental to mathematical behaviour.

The PISA 2015 and PISA-D frameworks uses a modified formulation of this set of capabilities, which condenses the number from eight to seven based on an investigation of the operation of the competencies through previously administered PISA items (Turner et al., 2013). There is wide recognition of the need to identify such a set of general mathematical capabilities to complement the role of specific mathematical content knowledge in mathematics learning. Prominent examples include the eight mathematical practices of the Common Core State Standards Initiative in the United States (CCSSI, 2010), the four key processes (representing, analysing, interpreting and evaluating, and communicating and reflecting) of England’s Mathematics National Curriculum (QCA, 2007), and the process standards in the United States’ National Council of Teachers of Mathematics’ Principles and Standards for School Mathematics (NCTM, 2000). These cognitive capabilities are available to or learnable by individuals in order to understand and engage with the world in a mathematical way, or to solve problems. As the level of mathematical literacy possessed by an individual increases, that individual is able to draw to an increasing degree on the fundamental mathematical capabilities (Turner and Adams, 2012). Thus, increasing activation of fundamental mathematical capabilities is associated with increasing item difficulty. This observation has been used as the basis of the descriptions of different proficiency levels of mathematical literacy reported in previous PISA surveys and discussed later in this framework.

The seven fundamental mathematical capabilities used in the PISA 2015 and PISA-D frameworks are as follows:

- **Communication:** Mathematical literacy involves communication. The individual perceives the existence of some challenge and is stimulated to recognise and understand a problem situation. Reading, decoding and interpreting statements, questions, tasks or objects enables the individual to form a mental model of the situation, which is an important step in understanding, clarifying and formulating a problem. During the solution process, intermediate results may need to be summarised and presented. Later on, once a solution has been found, the problem solver may need to present the solution, and perhaps an explanation or justification, to others.



- *Mathematising*: Mathematical literacy can involve transforming a problem defined in the real world to a strictly mathematical form (which can include structuring, conceptualising, making assumptions and/or formulating a model), or interpreting or evaluating a mathematical outcome or a mathematical model in relation to the original problem. The term mathematising is used to describe the fundamental mathematical activities involved.
- *Representation*: Mathematical literacy very frequently involves representations of mathematical objects and situations. This can entail selecting, interpreting, translating between, and using a variety of representations to capture a situation, interact with a problem, or to present one's work. The representations referred to include graphs, tables, diagrams, pictures, equations, formulae and concrete materials.
- *Reasoning and argument*: A mathematical ability that is called on throughout the different stages and activities associated with mathematical literacy is referred to as reasoning and argument. This capability involves logically rooted thought processes that explore and link problem elements so as to make inferences from them, check a justification that is given, or provide a justification of statements or solutions to problems.
- *Devising strategies for solving problems*: Mathematical literacy frequently requires devising strategies for solving problems mathematically. This involves a set of critical control processes that guide an individual to effectively recognise, formulate, and solve problems. This skill is characterised as selecting or devising a plan or strategy to use mathematics to solve problems arising from a task or context, as well as guiding its implementation. This mathematical capability can be demanded at any of the stages of the problem solving process.
- *Using symbolic, formal and technical language and operations*: Mathematical literacy requires using symbolic, formal, and technical language and operations. This involves understanding, interpreting, manipulating and making use of symbolic expressions within a mathematical context (including arithmetic expressions and operations) governed by mathematical conventions and rules. It also involves understanding and utilising formal constructs based on definitions, rules and formal systems and also using algorithms with these entities. The symbols, rules and systems used will vary according to what particular mathematical content knowledge is needed for a specific task to formulate, solve or interpret the mathematics.
- *Using mathematical tools*:¹ The final mathematical capability that underpins mathematical literacy in practice is using mathematical tools. Mathematical tools encompass physical tools such as measuring instruments, as well as calculators and computer-based tools that are becoming more widely available. This ability involves knowing about and being able to make use of various tools that may assist mathematical activity, and knowing about the limitations of such tools. Mathematical tools can also have an important role in communicating results.

These capabilities are evident to varying degrees in each of the three mathematical processes. The ways in which these capabilities manifest themselves within the three processes are described in Figure 3.2.

A good guide to the empirical difficulty of items can be obtained by considering which aspects of the fundamental mathematical capabilities are required for planning and executing a solution (Turner, 2012; Turner and Adams, 2012; Turner et al., 2013). The easiest items will require the activation of few capabilities and in a relatively straightforward way. The hardest items require complex activation of several capabilities. Predicting difficulty requires consideration of both the number of capabilities and the complexity of activation required. Based on the modifications to the proficiencies and processes for PISA-D, it was necessary to add particular skills to support these modifications. Four skills were added to the table in order to provide better understanding of the extensions of the mathematical process descriptions. These skills also support the capabilities delineated in the proficiencies 1b and 1c.



Figure 3.2 ■ **Relationship between mathematical processes (top row) and fundamental mathematical capabilities (left-most column)**

	Formulating situations mathematically	Employing mathematical concepts, facts, procedures and reasoning	Interpreting, applying and evaluating mathematical outcomes
Communicating	Read, decode and make sense of statements, questions, tasks, objects or images, in order to form a mental model of the situation	Articulate a solution, show the work involved in reaching a solution, and/or summarise and present intermediate mathematical results	Construct and communicate explanations and arguments in the context of the problem
Mathematising	Identify the underlying mathematical variables and structures in the real-world problem, and make assumptions so that they can be used For PISA-D, “Select a model appropriate to the context of real-world problems” has been added	Use an understanding of the context to guide or expedite the mathematical solving process, e.g. working to a context-appropriate level of accuracy	Understand the extent and limits of a mathematical solution that are a consequence of the mathematical model employed
Representation	Create a mathematical representation of real-world information For PISA-D, “Select a representation appropriate to the context” has been added	Make sense of, relate and use a variety of representations when interacting with a problem	Interpret mathematical outcomes in a variety of formats in relation to a situation or use; compare or evaluate two or more representations in relation to a situation
Reasoning and argument	Explain, defend or provide a justification for the identified or devised representation of a real-world situation	Explain, defend or provide a justification for the processes and procedures used to determine a mathematical result or solution Connect pieces of information to arrive at a mathematical solution, make generalisations or create a multi-step argument For PISA-D, “Select an appropriate justification” has been added	Reflect on mathematical solutions and create explanations and arguments that support, refute or qualify a mathematical solution to a contextualised problem
Devising strategies for solving problems	Select or devise a plan or strategy to mathematically reframe contextualised problems	Activate effective and sustained control mechanisms across a multi-step procedure leading to a mathematical solution, conclusion or generalisation	Devise and implement a strategy in order to interpret, evaluate and validate a mathematical solution to a contextualised problem For PISA-D, “Implement a given strategy” has been added
Using symbolic, formal, and technical language and operations	Use appropriate variables, symbols, diagrams and standard models in order to represent a real-world problem using symbolic/formal language	Understand and utilise formal constructs based on definitions, rules and formal systems as well as employing algorithms	Understand the relationship between the context of the problem and representation of the mathematical solution Use this understanding to help interpret the solution in context and gauge the feasibility and possible limitations of the solution
Using mathematical tools	Use mathematical tools in order to recognise mathematical structures or to portray mathematical relationships	Know about and be able to make appropriate use of various tools that may assist in implementing processes and procedures for determining mathematical solutions	Use mathematical tools to ascertain the reasonableness of a mathematical solution and any limits and constraints on that solution, given the context of the problem



Mathematical content knowledge

An understanding of mathematical content – and the ability to apply that knowledge to the solution of meaningful contextualised problems – is important for citizens in the modern world. That is, to solve problems and interpret situations in personal, occupational, societal and scientific contexts, there is a need to draw upon certain mathematical knowledge and understandings.

Mathematical structures have been developed over time as a means to understand and interpret natural and social phenomena. In schools, the mathematics curriculum is typically organised around content strands (e.g. number, algebra and geometry) and detailed topic lists that reflect historically well-established branches of mathematics and that help in defining a structured curriculum. However, outside the mathematics classroom, a challenge or situation that arises is usually not accompanied by a set of rules and prescriptions that shows how the challenge can be met. Rather it typically requires some creative thought in seeing the possibilities of bringing mathematics to bear on the situation and in formulating it mathematically. Often a situation can be addressed in different ways, drawing on different mathematical concepts, procedures, facts or tools.

Since the goal of PISA is to assess mathematical literacy, an organisational structure for mathematical content knowledge has been developed based on the mathematical phenomena that underlie broad classes of problems and which have motivated the development of specific mathematical concepts and procedures. Because national mathematics curricula are typically designed to equip students with knowledge and skills that address these same underlying mathematical phenomena, the outcome is that the range of content arising from organising content this way is closely aligned with that typically found in national mathematics curricula. This framework lists some content topics appropriate for assessing the mathematical literacy of 15-year-old students based on analyses of national standards from eleven countries.

To organise the domain of mathematics for purposes of assessing mathematical literacy, it is important to select a structure that grows out of historical developments in mathematics, that encompasses sufficient variety and depth to reveal the essentials of mathematics, and that also represents, or includes, the conventional mathematical strands in an acceptable way. Thus, a set of content categories that reflects the range of underlying mathematical phenomena was selected for the PISA 2015 framework, consistent with the categories used for previous PISA surveys.

The following list of content categories, therefore, are used in PISA 2015 and PISA-D to meet the requirements of historical development, coverage of the domain of mathematics and the underlying phenomena that motivate its development, and reflection of the major strands of school curricula. These four categories characterise the range of mathematical content that is central to the discipline and illustrate the broad areas of content used in the test items for PISA 2015 and PISA-D:

- Change and relationships
- Space and shape
- Quantity
- Uncertainty and data

With these four categories, the mathematical domain can be organised in a way that ensures a spread of items across the domain and focuses on important mathematical phenomena, but at the same time, avoids a too fine division that would work against a focus on rich and challenging mathematical problems based on real situations. While categorisation by content category is important for item development and selection, and for reporting of assessment results, it is important to note that some specific content topics may materialise in more than one content category. For example, a released PISA item called *Pizzas* involves determining which of two round pizzas, with different diameters and different costs but the same thickness, is the better value. This item draws on several areas of mathematics, including measurement, quantification (value for money, proportional reasoning and arithmetic calculations), and change and relationships (in terms of relationships among the variables and how relevant properties change from the smaller pizza to the larger one.) This item was ultimately categorised as a *change and relationships* item since the key to the problem lies in students being able to relate the change in areas of the two pizzas (given a change in diameter) and a corresponding change of price. Clearly, a different item involving circle area might be classified as a *space and shape* item. Connections between aspects of content that span these four content categories contribute to the coherence of mathematics as a discipline and are apparent in some of the assessment items selected for the PISA 2015 assessment.

The broad mathematical content categories and the more specific content topics appropriate for 15-year-old students described later in this section reflect the level and breadth of content that is eligible for inclusion on the PISA 2015 and PISA-D surveys. Narrative descriptions of each content category and the relevance of each to solving meaningful



problems are provided first, followed by more specific definitions of the kinds of content that are appropriate for inclusion in an assessment of mathematical literacy of 15-year-old students and out-of-school youth. These specific topics reflect commonalities found in the expectations set by a range of countries and educational jurisdictions. The standards examined to identify these content topics are viewed as evidence not only of what is taught in mathematics classrooms in these countries but also as indicators of what countries view as important knowledge and skills for preparing students of this age to become constructive, engaged and reflective citizens.

Descriptions of the mathematical content knowledge that characterise each of the four categories — *change and relationships*, *space and shape*, *quantity*, and *uncertainty and data* — are provided below.

Change and relationships

The natural and designed worlds display a multitude of temporary and permanent relationships among objects and circumstances, where changes occur within systems of inter-related objects or in circumstances where the elements influence one another. In many cases these changes occur over time, and in other cases changes in one object or quantity are related to changes in another. Some of these situations involve discrete change; others change continuously. Some relationships are of a permanent, or invariant, nature. Being more literate about change and relationships involves understanding fundamental types of change and recognising when they occur in order to use suitable mathematical models to describe and predict change. Mathematically this means modelling the change and the relationships with appropriate functions and equations, as well as creating, interpreting, and translating among symbolic and graphical representations of relationships.

Change and relationships is evident in such diverse settings as growth of organisms, music, and the cycle of seasons, weather patterns, employment levels, and economic conditions. Aspects of the traditional mathematical content of functions and algebra, including algebraic expressions, equations and inequalities, and tabular and graphical representations, are central in describing, modelling, and interpreting change phenomena. For example, the released PISA unit *Walking* (see the “Examples of items” section) contains two items that exemplify the *change and relationships* category since the focus is on the algebraic relationships between two variables, requiring students to activate their algebraic knowledge and skills. Students are required to employ a given formula for pacelength – a formula expressed in algebraic form – to determine pacelength in one item and walking speed in the other. Representations of data and relationships described using statistics also are often used to portray and interpret change and relationships, and a firm grounding in the basics of number and units is also essential to defining and interpreting *change and relationships*. Some interesting relationships arise from geometric measurement, such as the way that changes in perimeter of a family of shapes might relate to changes in area, or the relationships among lengths of the sides of triangles.

Space and shape

Space and shape encompasses a wide range of phenomena that are encountered everywhere in our visual and physical world: patterns, properties of objects, positions and orientations, representations of objects, decoding and encoding of visual information, and navigation and dynamic interaction with real shapes as well as with representations. Geometry serves as an essential foundation for *space and shape*, but the category extends beyond traditional geometry in content, meaning, and method, drawing on elements of other mathematical areas such as spatial visualisation, measurement and algebra. For instance, shapes can change, and a point can move along a locus, thus requiring function concepts. Measurement formulae are central in this area. The manipulation and interpretation of shapes in settings that call for tools ranging from dynamic geometry software to global positioning system (GPS) software are included in this content category.

PISA assumes that the understanding of a set of core concepts and skills is important to mathematical literacy relative to *space and shape*. Mathematical literacy in the area of *space and shape* involves a range of activities such as understanding perspective (for example in paintings), creating and reading maps, transforming shapes with and without technology, interpreting views of three-dimensional scenes from various perspectives and constructing representations of shapes.

Quantity

The notion of *quantity* may be the most pervasive and essential mathematical aspect of engaging with, and functioning in, our world. It incorporates the quantification of attributes of objects, relationships, situations and entities in the world, understanding various representations of those quantifications, and judging interpretations and arguments based on quantity. To engage with the quantification of the world involves understanding measurements, counts, magnitudes, units, indicators, relative size, and numerical trends and patterns. Aspects of quantitative reasoning—such as number



sense, multiple representations of numbers, elegance in computation, mental calculation, estimation and assessment of reasonableness of results—are the essence of mathematical literacy relative to *quantity*.

Quantification is a primary method for describing and measuring a vast set of attributes of aspects of the world. It allows for the modelling of situations, for the examination of change and relationships, for the description and manipulation of space and shape, for organising and interpreting data, and for the measurement and assessment of uncertainty. Thus mathematical literacy in the area of *quantity* applies knowledge of number and number operations in a wide variety of settings.

Uncertainty and data

In science, technology and everyday life, uncertainty is a given. Uncertainty is therefore a phenomenon at the heart of the mathematical analysis of many problem situations, and the theory of probability and statistics as well as techniques of data representation and description have been established to deal with it. The *uncertainty and data* content category includes recognising the place of variation in processes, having a sense of the quantification of that variation, acknowledging uncertainty and error in measurement, and knowing about chance. It also includes forming, interpreting and evaluating conclusions drawn in situations where uncertainty is central. The presentation and interpretation of data are key concepts in this category (Moore, 1997).

There is uncertainty in scientific predictions, poll results, weather forecasts and economic models. There is variation in manufacturing processes, test scores and survey findings, and chance is fundamental to many recreational activities enjoyed by individuals. The traditional curricular areas of probability and statistics provide formal means of describing, modelling, and interpreting a certain class of uncertainty phenomena, and for making inferences. In addition, knowledge of number and of aspects of algebra such as graphs and symbolic representation contribute to facility in engaging in problems in this content category. The focus on the interpretation and presentation of data is an important aspect of the *uncertainty and data* category.

Desired distribution of items by content category

The desired distribution of items selected for PISA 2015 and for PISA-D (both in- and out-of-school instruments) across the four content categories is shown in Table 3.2. The goal in constructing the survey is a balanced distribution of items with respect to content category, since all of these domains are important for constructive, engaged and reflective citizens.

Table 3.2 Desired distribution of mathematics items, by content category

Content category	Percentage of items in PISA 2015	Percentage of items in PISA-D
Change and relationships	25	25
Space and shape	25	25
Quantity	25	25
Uncertainty and data	25	25
Total	100	100

Content topics for guiding the assessment of mathematical literacy for 15-year-old students

To effectively understand and solve contextualised problems involving *change and relationships*, *space and shape*, *quantity*, and *uncertainty and data* requires drawing upon a variety of mathematical concepts, procedures, facts and tools at an appropriate level of depth and sophistication. As an assessment of mathematical literacy, PISA strives to assess the levels and types of mathematics that are appropriate for 15-year-old students on a trajectory to become constructive, engaged and reflective citizens able to make well-founded judgments and decisions. It is also the case that PISA, while not designed or intended to be a curriculum-driven assessment, strives to reflect the mathematics that students have likely had the opportunity to learn by the time they are 15 years old.

The content included in PISA-D and PISA 2015 is the same as that developed in PISA 2012. The four content categories of *change and relationships*, *space and shape*, *quantity*, and *uncertainty and data* serve as the foundation for identifying this range of content, yet there is not a one-to-one mapping of content topics to these categories. For example, proportional reasoning comes into play in such varied contexts as making measurement conversions, analysing linear relationships, calculating probabilities and examining the lengths of sides in similar shapes. The following content is intended to reflect



the centrality of many of these concepts to all four content categories and reinforce the coherence of mathematics as a discipline. It intends to be illustrative of the content topics included in PISA-D, rather than an exhaustive listing:

- *Functions*: the concept of function, emphasising but not limited to linear functions, their properties, and a variety of descriptions and representations of them. Commonly used representations are verbal, symbolic, tabular and graphical.
- *Algebraic expressions*: verbal interpretation of and manipulation with algebraic expressions, involving numbers, symbols, arithmetic operations, powers and simple roots.
- *Equations and inequalities*: linear and related equations and inequalities, simple second-degree equations, and analytic and non-analytic solution methods.
- *Co-ordinate systems*: representation and description of data, position and relationships.
- *Relationships within and among geometrical objects in two and three dimensions*: static relationships such as algebraic connections among elements of figures (e.g. the Pythagorean theorem as defining the relationship between the lengths of the sides of a right triangle), relative position, similarity and congruence, and dynamic relationships involving transformation and motion of objects, as well as correspondences between two- and three-dimensional objects.
- *Measurement*: quantification of features of and among shapes and objects, such as angle measures, distance, length, perimeter, circumference, area, and volume.
- *Numbers and units*: concepts; representations of numbers and number systems, including properties of integer and rational numbers; relevant aspects of irrational numbers; as well as quantities and units referring to phenomena such as time, money, weight, temperature, distance, area and volume, and derived quantities and their numerical description.
- *Arithmetic operations*: the nature and properties of these operations and related notational conventions.
- *Percents, ratios and proportions*: numerical description of relative magnitude and the application of proportions and proportional reasoning to solve problems.
- *Counting principles*: simple combinations and permutations.
- *Estimation*: purpose-driven approximation of quantities and numerical expressions, including significant digits and rounding.
- *Data collection, representation and interpretation*: nature, genesis, and collection of various types of data, and the different ways to represent and interpret them.
- *Data variability and its description*: concepts such as variability, distribution, and central tendency of data sets, and ways to describe and interpret these in quantitative terms.
- *Samples and sampling*: concepts of sampling and sampling from data populations, including simple inferences based on properties of samples.
- *Chance and probability*: notion of random events, random variation and its representation, chance and frequency of events, and basic aspects of the concept of probability.

Contexts

The choice of appropriate mathematical strategies and representations is often dependent on the context in which a problem arises. Being able to work within a context is widely appreciated to place additional demands on the problem solver (see Watson and Callingham, 2003, for findings about statistics). For the PISA survey, it is important that a wide variety of contexts are used. This offers the possibility of connecting with the broadest possible range of individual interests and with the range of situations in which individuals operate in the 21st century.

For purposes of the PISA-D mathematics framework, four context categories have been defined and are used to classify assessment items developed for the PISA survey:

- *Personal* – Problems classified in the personal context category focus on activities of one's self, one's family or one's peer group. The kinds of contexts that may be considered personal include (but are not limited to) those involving food preparation, shopping, games, personal health, personal transportation, sports, travel, personal scheduling and personal finance.
- *Occupational* – Problems classified in the occupational context category are centred on the world of work. Items categorised as occupational may involve (but are not limited to) such things as measuring, costing and ordering materials for building, payroll/accounting, quality control, scheduling/inventory, design/architecture and job-related decision making. Occupational contexts may relate to any level of the workforce, from unskilled work to the highest levels of professional work, although items in the PISA survey must be accessible to 15-year-old students.



- *Societal* – Problems classified in the societal context category focus on one's community (whether local, national or global). They may involve (but are not limited to) such things as voting systems, public transport, government, public policies, demographics, advertising, national statistics and economics. Although individuals are involved in all of these things in a personal way, in the societal context category, the focus of problems is on the community perspective.
- *Scientific* – Problems classified in the scientific category relate to the application of mathematics to the natural world and issues and topics related to science and technology. Particular contexts might include (but are not limited to) such areas as weather or climate, ecology, medicine, space science, genetics, measurement and the world of mathematics itself. Items that are intramathematical, where all the elements involved belong in the world of mathematics, fall within the scientific context.

PISA items were arranged in units that share stimulus material. It is therefore usually the case that all items in the same unit belonged to the same context category. Exceptions do arise; for example stimulus material may be examined from a personal point of view in one item and a societal point of view in another. When an item involved only mathematical constructs without reference to the contextual elements of the unit within which it is located, it was allocated to the context category of the unit. In the unusual case of a unit involving only mathematical constructs and being without reference to any context outside of mathematics, the unit was assigned to the scientific context category.

Using these context categories provided the basis for selecting a mix of item contexts and ensures that the assessment reflects a broad range of uses of mathematics, ranging from everyday personal uses to the scientific demands of global problems. Moreover it was important that each context category be populated with assessment items having a broad range of item difficulties. Given that the major purpose of these context categories is to challenge students in a broad range of problem contexts, each category was designed to contribute substantially to the measurement of mathematical literacy. It should not be the case that the difficulty level of assessment items representing one context category is systematically higher or lower than the difficulty level of assessment items in another category.

In identifying contexts that may be relevant, it is critical to keep in mind that a purpose of the assessment is to gauge the use of mathematical content knowledge, processes and capabilities that students have acquired by age 15. Contexts for assessment items, therefore, were selected in light of relevance to students' interests and lives, and the demands that will be placed upon them as they enter society as constructive, engaged and reflective citizens. National Project Managers from countries participating in the PISA-D survey were involved in judging the degree of such relevance.

Desired distribution of items by context category

The desired distribution of items selected for PISA 2015 and for PISA-D (both in- and out-of-school instruments) across the four content categories is shown in Table 3.3. With this balanced distribution, no single context type is allowed to dominate, providing students with items that span a broad range of individual interests and a range of situations that they might expect to encounter in their lives.

Table 3.3 Desired distribution of mathematics items, by context category

Context category	Percentage of items in PISA 2015	Percentage of items in PISA-D
Personal	25	25
Occupational	25	25
Societal	25	25
Scientific	25	25
Total	100	100

ASSESSING MATHEMATICAL LITERACY

This section outlines the approach taken to apply the elements of the framework described in previous sections to PISA 2015 and PISA-D. This includes the structure of the mathematics component of the PISA-D survey, arrangements for transferring the paper-based trend items to a computer-based delivery, and reporting mathematical proficiency.

Response formats

Three types of response format are used to assess mathematical literacy in PISA 2015 and PISA-D: open constructed-response, closed constructed-response and selected-response (simple and complex multiple-choice) items. Open constructed-response items require a somewhat extended written response from a student. Such items also may ask

the student to show the steps taken or to explain how the answer was reached. These items require trained experts to manually code student responses.

Closed constructed-response items provide a more structured setting for presenting problem solutions, and they produce a student response that can be easily judged to be either correct or incorrect. Often student responses to questions of this type can be keyed into data capture software, and coded automatically, but some must be manually coded by trained experts. The most frequent closed constructed-responses are single numbers.

Selected-response items require students to choose one or more responses from a number of response options. Responses to these questions can usually be automatically processed. About equal numbers of each of these response formats is used to construct the survey instruments.

Item scoring

Although the majority of the items were dichotomously scored (that is, responses are awarded either credit or no credit), the open constructed-response items can sometimes involve partial credit scoring, which allows responses to be assigned credit according to differing degrees of “correctness” of responses. For each such item, a detailed coding guide that allows for full credit, partial credit, or no credit was provided to persons trained in the coding of student responses across the range of participating countries to ensure coding of responses was done in a consistent and reliable way. To maximise the comparability between the paper-based and computer-based assessment, careful attention is given to the scoring guides in order to ensure that the important elements were included.

Reporting proficiency in mathematics

The outcomes of the PISA mathematics survey are reported in a number of ways. Estimates of overall mathematical proficiency are obtained for sampled students in each participating country, and a number of proficiency levels are defined. Descriptions of the degree of mathematical literacy typical of students in each level are also developed. For PISA 2003, scales based on the four broad content categories were developed. In Figure 3.3, descriptions for the six proficiency levels reported for the overall PISA mathematics scale in 2012 are presented. These form the basis for the PISA 2015 mathematics scale and the PISA-D mathematics scale. The finalised 2012 scale is used to report the PISA 2015 outcomes. For PISA-D, in addition, the existing Level 1 was renamed Level 1a, and the table describing the proficiencies has been extended to include Levels 1b and 1c.

Fundamental mathematical capabilities play a central role in defining what it means to be at different levels of the scales for mathematical literacy overall and for each of the reported processes. For example, in the proficiency scale description for Level 4 (see Figure 3.3), the second sentence highlights aspects of mathematising and representation that are evident at this level. The final sentence highlights the characteristic communication, reasoning and argument of Level 4, providing a contrast with the short communications and lack of argument of Level 3 and the additional reflection of Level 5. In an earlier section of this framework and in Figure 3.2, each of the mathematical processes was described in terms of the fundamental mathematical capabilities that individuals might activate when engaging in that process.

Figure 3.3 ■ Summary description of the eight levels of mathematics proficiency in PISA-D

Level	Lower score limit	Percentage of students across OECD countries at each level, PISA 2015	Percentage of students across 23 middle- and low-income countries at each level, PISA 2015	Descriptor
6	669	2.3%	0.3%	At Level 6, students can conceptualise, generalise and utilise information based on their investigations and modelling of complex problem situations, and can use their knowledge in relatively non-standard contexts. They can link different information sources and representations and flexibly translate among them. Students at this level are capable of advanced mathematical thinking and reasoning. These students can apply this insight and understanding, along with a mastery of symbolic and formal mathematical operations and relationships, to develop new approaches and strategies for attacking novel situations. Students at this level can reflect on their actions, and can formulate and precisely communicate their actions and reflections regarding their findings, interpretations, arguments and the appropriateness of these to the original situation.



Figure 3.3 [continued] ■ Summary description of the eight levels of mathematics proficiency in PISA-D

Level	Lower score limit	Percentage of students across OECD countries at each level, PISA 2015	Percentage of students across 23 middle- and low-income countries at each level, PISA 2015	Descriptor
5	607	8.4%	1.5%	At Level 5, students can develop and work with models for complex situations, identifying constraints and specifying assumptions. They can select, compare and evaluate appropriate problem-solving strategies for dealing with complex problems related to these models. Students at this level can work strategically using broad, well-developed thinking and reasoning skills, appropriate linked representations, symbolic and formal characterisations, and insight pertaining to these situations. They begin to reflect on their work and can formulate and communicate their interpretations and reasoning.
4	545	18.6%	5.3%	At Level 4, students can work effectively with explicit models for complex concrete situations that may involve constraints or call for making assumptions. They can select and integrate different representations, including symbolic, linking them directly to aspects of real-world situations. Students at this level can utilise their limited range of skills and can reason with some insight, in straightforward contexts. They can construct and communicate explanations and arguments based on their interpretations, arguments and actions.
3	482	24.8%	12.6%	At Level 3, students can execute clearly described procedures, including those that require sequential decisions. Their interpretations are sufficiently sound to be a base for building a simple model or for selecting and applying simple problem-solving strategies. Students at this level can interpret and use representations based on different information sources and reason directly from them. They typically show some ability to handle percentages, fractions and decimal numbers, and to work with proportional relationships. Their solutions reflect that they have engaged in basic interpretation and reasoning.
2	420	22.5%	21.6%	At Level 2, students can interpret and recognise situations in contexts that require no more than direct inference. They can extract relevant information from a single source and make use of a single representational mode. Students at this level can employ basic algorithms, formulae, procedures or conventions to solve problems involving whole numbers. They are capable of making literal interpretations of the results.
1a	358	14.9%	26.3%	At Level 1a, students can answer questions involving familiar contexts where all relevant information is present and the questions are clearly defined. They are able to identify information and to carry out routine procedures according to direct instructions in explicit situations. They can perform actions that are almost always obvious and follow immediately from the given stimuli.
1b	295	8.5% (percentage of students scoring below Level 1, PISA 2015)	32.4% (percentage of students scoring below Level 1, PISA 2015)	At Level 1b, students can respond to questions involving easy to understand contexts where all relevant information is clearly given in a simple representation (for example tabular or graphic) and defined in a short syntactically simple text. They are able to follow clearly prescribed instructions.
1c	236			At Level 1c, students can respond to questions involving easy to understand contexts where all relevant information is clearly given in a simple, familiar format (for example a small table or picture) and defined in a very short syntactically simple text. They are able to follow a clear instruction describing a single step or operation.

Note: Descriptors 2 through 6 are the same as those used in PISA 2012, and Level 1 was renamed Level 1a.



In order to gain useful information for these new levels, 1b and 1c, it is vital that context and language do not interfere with the mathematics being assessed. To this end, the context and language must be carefully considered.

The context for both 1b and 1c should be situations that students encounter on a daily basis. Examples of these contexts may include money, temperature, food, time, date, weight, size and distance. All items should be concrete and not abstract. The focus of the item should be mathematical only. The understanding of the context should not interfere with the performance of the item.

Equally important is to have all items formulated in the simplest possible terms. Sentences should be short and direct. Compound sentences, compound nouns and conditional sentences should be avoided. Vocabulary used in the items must be carefully examined to ensure that students will have a clear understanding of what is being required. In addition, special care will be given to ensure that no extra difficulty is added due to a heavy text load or by a context that is unfamiliar to students based on their cultural background.

Items designed for Level 1c should only ask for a single step or operation. However, it is important to note that a single step or operation is not limited to an arithmetical step. This step might be demonstrated by making a selection or identifying some information. All parts of the modelling cycle can and should be used to measure the mathematical abilities of students at Levels 1b and 1c.

Testing mathematical literacy among the out-of-school population

For the out-of-school population, item selection focused on the scale at or below Level 2 with an emphasis on the lower end of the scale in terms of item distribution. The selection process was similar to that used for the in-school population: coverage of all processes was maintained and contexts of the items were reviewed to ensure appropriateness for what individuals would encounter in an out-of-school context.

Box 3.2 Delivery mode

The PISA-D school-based assessment is paper-based, while the out-of-school assessment is conducted on a tablet computer. To ensure comparability between the tests, the tablet-based instruments for PISA-D are formed by a subgroup of the items used for the paper-based assessment. All these items were originally designed for a paper-based assessment, so when moving to a tablet-based delivery, care was taken to maintain comparability between the assessments. The PISA 2015 framework describes some factors that must be considered when transposing items from paper to computer mode. These elements were also taken into account when designing the out-of-school instrument for PISA-D.

Item types: The computer provides a range of opportunities for designers of test items, including new item formats (e.g. drag-and-drop, hotspots). Since the PISA-D tablet-based tests use a subgroup of items from the paper-based test, there is less opportunity to exploit innovative item types and the majority of response formats remains unchanged.

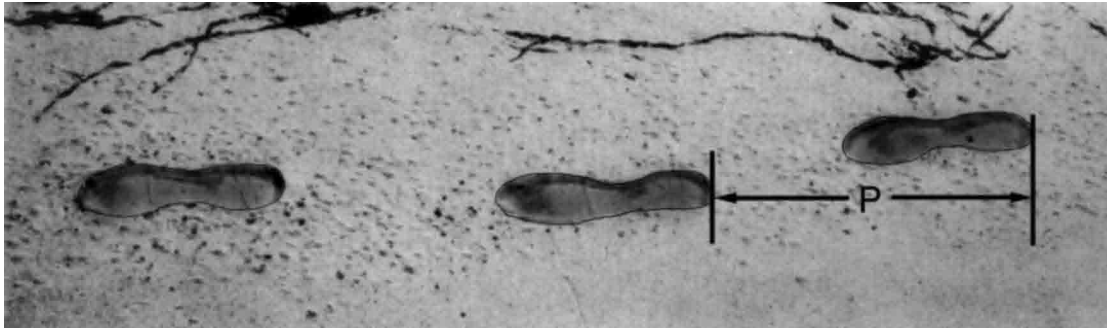
Stimulus presentation: A feature of fixed texts defined in the construct is that “the extent or amount of the text is immediately visible to the reader”. Clearly, it is impossible, both on paper and on a screen, to have long texts displayed on a single page or screen. To allow for this and still satisfy the construct of fixed texts, pagination is used for texts rather than scrolling. Texts that cover more than one page are presented in their entirety before the student sees the first question.

IT skills: Just as paper-based assessments rely on a set of fundamental skills for working with printed materials, so computer-based assessments rely on a set of fundamental information and communications technology skills for using computers. These include knowledge of basic hardware (e.g. keyboard and mouse) and basic conventions (e.g. arrows to move forward and specific buttons to press to execute commands). The intention is to keep such skills to a minimal core level in the tablet-based assessment.

Examples of items for addressing the extended PISA-D mathematics framework

The following items illustrate proficiency Levels 1a, 1b and 1c. In each case the tasks involved are described and an explanation is given about why the item fits a certain proficiency level. The items either come from or are adapted from PISA or work carried out by the Institute of Educational Research in Poland.

Sample item 1



The picture shows the footprints of a man walking. The pacelength P is the distance between the rear of two consecutive footprints.

For men, the formula $\frac{n}{P} = 140$ gives an approximate relationship between n and P where

n = number of steps per minute, and

P = pacelength in metres.

Heiko has a pacelength that is 0.5 metres. Using this formula, how many steps per minute, n , does Heiko take each minute?

For this item, the student simply needs to substitute the values into the formula and solve it. Since the pacelength is given (0.5 metres), the student only needs to complete a single operational step. Multiplying both sides of the equation by 0.5 gives a value for n of 70. This addresses the added process “performing a simple calculation.” A student who only substitutes the values correctly would meet the requirements of proficiency 1a. The abstract format of a formula involving two variables does not meet the requirements of 1b or 1c.

Sample item 2

Mei-Ling found out that the exchange rate between Singapore dollars and South African rand was

$$1 \text{ SGD} = 4.2 \text{ ZAR}$$

Mei-Ling changed 3 000 Singapore dollars into South African rand at this exchange rate. Choose a correct method from those listed. Then calculate n , the amount of South African rand Mei-Ling received after the exchange.

$$\frac{1}{4.2} = \frac{n}{3000} \quad \frac{1}{3000} = \frac{4.2}{n} \quad 4.2n = 3000 \quad n = 3000(4.2)$$

For this item, the student is given four methods to solve for n . Two of these methods will result in a correct value for n . The expectation is that a student will be able to select one of the correct methods and then solve for the value of n . This addresses the added process, “selecting an appropriate model from a list.” If a student is able to choose one of the correct methods, the requirements for proficiency 1b are met. If a student is also able to solve for n correctly, the requirements for proficiency 1a are met.



Sample item 3

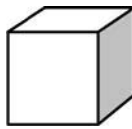
Nick wants to pave the rectangular patio of his new house with bricks. The patio has a length of 5.25 metres and a width of 3.00 metres. One box of bricks can pave 2 square metres.

Calculate the number of boxes of bricks Nick needs for the entire patio.

For this item, the student must perform two steps to arrive at a correct solution. First, the student must find the area of the patio. For the second step, the student must divide the number of square metres by two in order to find the total number of boxes of bricks. This addresses multiple processes in understanding what must be done, devising a strategy and performing the calculations. If a student is only able to successfully find the area, proficiency 1b is addressed. Proficiency 1a is addressed if a student does both steps correctly.

Sample item 4

Susan likes to build blocks using small cubes like the one shown in the following diagram:



Small cube

Susan will build a block as shown in Diagram A below:

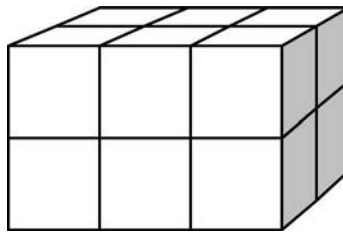


Diagram A

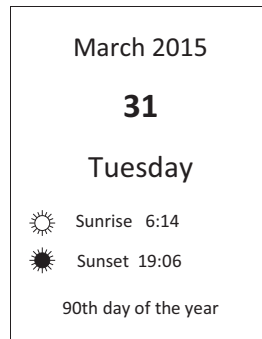
How many small cubes will Susan need to build the block shown in Diagram A?

For this item, the student needs to determine the number of small cubes needed to build the larger block. In doing this, the process “applying mathematical facts, rules, algorithms and structures when finding solutions” is addressed. Since this is a simple, one-step problem, it meets the requirements of proficiency 1c.



Sample item 5

The picture represents one page of a calendar.



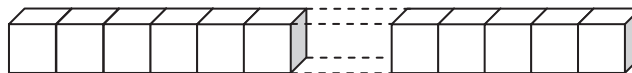
How much time will pass on that day from sunrise until sunset?

- A. 12 hours and 52 minutes
- B. 13 hours and 8 minutes
- C. 13 hours and 32 minutes
- D. 13 hours and 52 minutes

For this item, the student must determine the elapsed time. To solve this successfully, students must recognise the modification to the normal subtraction algorithm when regrouping. Because this recognition is required, even though it is one single operational step, the thought process involved makes this a 1a item rather than 1b or 1c.

Sample item 6

A cube of volume 1 m^3 has been cut off into small cubes of edge length 1 cm. If we put those small cubes one after another, as shown in the picture, we would get a square prism.



Which of the following statements are true? Mark T when the statement is true or F when it is false.

The volume of this square prism is 100 times larger than the volume of the original cubes.	T	F
One of the edges of this square prism has length 10 km.	T	F

For this item, the student must demonstrate an understanding of the concept of volume. The first statement requires no calculations at all, only simple reasoning. It is proficiency Level 1a, because there is modelling hidden here. We do not see the large cube in the picture which prevents this item from being 1b. The second statement requires a calculation in order to answer correctly. Students who go by “common sense” usually choose the wrong answer. The student has to ignore his or her intuitive judgment and pick the mathematical way of dealing with the problem by determining the number of small cubes and recognising the change in units. This statement is proficiency Level 2.



Notes

1. In some countries, “mathematical tools” can also refer to established mathematical procedures such as algorithms. For the purposes of the PISA framework, “mathematical tools” refers only to the physical and digital tools described in this section.

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4

PISA for Development Science Framework

This Chapter defines “scientific literacy” as assessed in the Programme for International Student Assessment (PISA) and the extensions to the PISA science framework that have been designed for the PISA for Development (PISA-D) project. It describes the types of contexts, knowledge and competencies reflected in PISA-D’s science problems, and provides several sample items. The Chapter also discusses how student performance in science is measured and reported.



WHAT IS NEW IN PISA-D? EXTENSIONS TO THE PISA SCIENTIFIC LITERACY FRAMEWORK

This chapter presents the assessment framework for science in PISA-D and shows how it specifically addresses the needs and contexts of using PISA for assessing student competency in a wider range of countries. This chapter explains how the PISA 2015 science framework has been extended to provide more information regarding student performance at the lower levels of proficiency. While PISA-D's out-of-school component does not include the science domain, this framework is applicable for students who are in school as well as 15-year-olds who are out of school.

PISA establishes a baseline level – proficiency Level 2, on a scale with 6 as the highest level and 1b the lowest – at which individuals begin to demonstrate the competencies that will enable them to participate effectively and productively in life as students, workers and citizens. The extensions made to the PISA-D science framework are an attempt to gain more information about students who currently perform at or below Level 1. PISA-D builds on the PISA 2015 science framework, extending it to yet a lower level of performance (1c) to gather precise data on the science skills of the lowest performers. These extensions have been achieved by describing how the expectations of the three competencies – “Explain phenomena scientifically”, “Evaluate and design scientific enquiry”, and “Interpret data and evidence scientifically” – can help distinguish the differences between Level 1a, 1b and 1c students, based on increasing, but limited, cognitive demands. In general, all Level 1 items make less extensive demands on students’ knowledge, and require less cognitive processing. In order to provide greater clarity, the document also explains what kinds of competencies and displays of understanding are not expected.

This chapter adds elements to indicate what it is reasonable to assess and what is expected of students who might perform at Levels 1 and 2 on the PISA scales, suggesting that assessment at these levels should be restricted, wherever possible, to items that make the lowest level of cognitive demand. In addition, to reduce the linguistic demands and cognitive load of any item, careful attention should be paid to simplifying the language of any item and removing extraneous text.

The PISA-D science framework adheres to the core idea of scientific literacy, as defined by PISA. The 1999, 2004 and 2006 PISA frameworks have elaborated a conception of scientific literacy as the central construct for science assessment. These documents have established a broad consensus among science educators of the concept of scientific literacy. The framework for PISA 2015 refines and extends the previous construct, in particular by drawing on the PISA 2006 framework that was used as the basis for assessment in 2006, 2009 and 2012. In 2015 science was the main domain, and PISA-D has no main domains. So those sections that are not relevant to the PISA-D framework – and hence much of the discussion on attitudes – have been omitted or made briefer in this framework.

Scientific literacy matters at both the national and international levels as humanity faces major challenges in providing sufficient water and food, controlling diseases, generating sufficient energy and adapting to climate change (UNEP, 2012). Many of these issues arise, however, at the local level where individuals may be faced with decisions about practices that affect their own health and food supplies, the appropriate use of materials and new technologies, and decisions about energy use. Dealing with all of these challenges will require a major contribution from science and technology. Yet, as argued by the European Commission, the solutions to political and ethical dilemmas involving science and technology “cannot be the subject of informed debate unless young people possess certain scientific awareness” (EC, 1995, p.28). Moreover, “this does not mean turning everyone into a scientific expert, but enabling them to fulfil an enlightened role in making choices which affect their environment and to understand in broad terms the social implications of debates between experts” (ibid. p.28). Given that knowledge of science and science-based technology contributes significantly to individuals’ personal, social and professional lives, an understanding of science and technology is thus central to a young person’s “preparedness for life”.

The concept of scientific literacy in this framework refers to knowledge of both science and science-based technology, even though science and technology do differ in their purposes, processes and products. Technology seeks the optimal solution to a human problem, and there may be more than one optimal solution. In contrast, science seeks the answer to a specific question about the natural, material world. Nevertheless, the two are closely related. For instance, new scientific knowledge leads to the development of new technologies (think of the advances in material science that led to the development of the transistor in 1948). Likewise, new technologies can lead to new scientific knowledge (think of how knowledge of the universe has been transformed through the development of better telescopes). Individuals make decisions and choices that influence the directions of new technologies (consider the decision to drive a smaller, more fuel-efficient car). Scientifically literate individuals should therefore be able to make more informed choices. They should also be able to recognise that, while science and technology are often a source of solutions, paradoxically, they can also be seen as a source of risk, generating new problems that can only be solved through the use of science and technology. Therefore, individuals need to be able to weigh the potential benefits and risks of applying scientific knowledge to themselves and society.



Scientific literacy also requires not just knowledge of the concepts and theories of science but also knowledge of the common procedures and practices associated with scientific enquiry and how these enable science to advance. Therefore, individuals who are scientifically literate have a knowledge of the major concepts and ideas that form the foundation of scientific and technological thought; how such knowledge has been derived; and the degree to which such knowledge is proved by evidence or theoretical explanations.

Undoubtedly, many of the challenges of the 21st century will require innovative solutions that have a basis in scientific thinking and scientific discovery. Societies will require a cadre of well-educated scientists to undertake the research and nurture the innovation that will be essential to meet the economic, social and environmental challenges that the world faces.

For all of these reasons, scientific literacy is perceived to be a key competency (Rychen and Salganik, 2003) and defined in terms of the ability to use knowledge and information interactively – that is “an understanding of how it [a knowledge of science] changes the way one can interact with the world and how it can be used to accomplish broader goals” (ibid.:10). As such, it represents a major goal for science education for all students. Therefore, the view of scientific literacy that forms the basis for the 2015 international assessment of 15-year-old students is a response to the question: What is important for young people to know, value and be able to do in situations involving science and technology?

This chapter is organised into the following sections. The first section, “Defining scientific literacy”, explains the theoretical underpinnings of the PISA science assessment, including the formal definition of the scientific literacy construct and describing the three competencies required for scientific literacy. The second section, “Organising the domain of science”, describes the four inter-related aspects that form the definition of scientific literacy: contexts, competencies, knowledge and attitudes. The third section, “Assessing scientific literacy”, outlines the approach taken to apply the elements of the framework previously described, including cognitive demand, test characteristics, reporting proficiency, testing scientific literacy among the out-of-school population and examples of items for addressing the extended PISA-D Framework.

DEFINING SCIENTIFIC LITERACY

Current thinking about the desired outcomes of science education is rooted strongly in a belief that an understanding of science is so important that it should be a feature of every young person’s education (AAAS, 1989; COSCE, 2011; Fensham, 1985; Millar and Osborne, 1998; National Research Council, 2012; KMK, 2005; Taiwan Ministry of Education, 1999). Indeed, in many countries science is an obligatory element of the school curriculum from kindergarten until the completion of compulsory education.

Many of the documents and policy statements cited above give pre-eminence to an education for citizenship. However, many of the curricula for school science across the world are based on a view that the primary goal of science education should be the preparation of the next generation of scientists (Millar and Osborne, 1998). These two goals are not always compatible. Attempts to resolve the tension between the needs of the majority of students who will not become scientists and the needs of the minority who will have led to an emphasis on teaching science through enquiry (National Academy of Science, 1995; National Research Council, 2000), and new curriculum models (Millar, 2006) that address the needs of both groups. The emphasis in these frameworks and their associated curricula lies not on producing individuals who will be “producers” of scientific knowledge, i.e. the future scientists; rather, it is on educating all young people to become informed, critical users of scientific knowledge.

To understand and engage in critical discussions about issues that involve science and technology requires three domain-specific competencies. The first is the ability to provide explanatory accounts of natural phenomena, technical artefacts and technologies and their implications for society. Such ability requires knowledge of the fundamental ideas of science and the questions that frame the practice and goals of science. The second is the knowledge and understanding of scientific enquiry to: identify questions that can be answered by scientific enquiry; identify whether appropriate procedures have been used; and propose ways in which such questions might be answered. The third is the competency to interpret and evaluate data and evidence scientifically and evaluate whether the conclusions are justified. Thus, scientific literacy in PISA 2015 and PISA-D is defined by the three competencies to:

- explain phenomena scientifically
- interpret data and evidence scientifically
- evaluate and design scientific enquiry.



All of these competencies require knowledge. Explaining scientific and technological phenomena, for instance, demands knowledge of the content of science (hereafter, content knowledge). The second and third competencies, however, require more than knowledge of what is known; they depend on an understanding of how scientific knowledge is established and the degree of confidence with which it is held. Some have argued for teaching what has variously been called “the nature of science” (Lederman, 2006), “ideas about science” (Millar and Osborne, 1998) or “scientific practices” (National Research Council, 2012). Recognising and identifying the features that characterise scientific enquiry require knowledge of the standard procedures that underlie the diverse methods and practices used to establish scientific knowledge (hereafter, procedural knowledge). Finally, the competencies require epistemic knowledge – an understanding of the rationale for the common practices of scientific enquiry, the status of the knowledge claims that are generated, and the meaning of foundational terms, such as theory, hypothesis and data.

Both procedural and epistemic knowledge are necessary to identify questions that are amenable to scientific enquiry, to judge whether appropriate procedures have been used to ensure that the claims are justified, and to distinguish scientific issues from matters of values or economic considerations. This definition of scientific literacy assumes that, throughout their lives, individuals will need to acquire knowledge, not through scientific investigations, but through the use of resources such as libraries and the Internet. Procedural and epistemic knowledge are essential to decide whether the many claims of knowledge and understanding that pervade contemporary media are based on the use of appropriate procedures and are justified.

Box 4.1 **Scientific knowledge: PISA 2015 terminology**

This document is based upon a view of scientific knowledge as consisting of three distinguishable but related elements. The first of these and the most familiar is knowledge of the facts, concepts, ideas and theories about the natural world that science has established. For instance, how plants synthesise complex molecules using light and carbon dioxide or the particulate nature of matter. This kind of knowledge is referred to as “**content knowledge**” or “knowledge of the content of science”.

Knowledge of the procedures that scientists use to establish scientific knowledge is referred to as “**procedural knowledge**”. This is a knowledge of the practices and concepts on which empirical enquiry is based such as repeating measurements to minimise error and reduce uncertainty, the control of variables, and standard procedures for representing and communicating data (Millar et al., 1995). More recently these have been elaborated as a set of “concepts of evidence” (Gott, Duggan, and Roberts, 2008).

Furthermore, understanding science as a practice also requires “**epistemic knowledge**” which refers to an understanding of the role of specific constructs and defining features essential to the process of knowledge building in science (Duschl, 2007). Epistemic knowledge includes an understanding of the function that questions, observations, theories, hypotheses, models and arguments play in science; recognition of the variety of forms of scientific enquiry; and the role peer review plays in establishing knowledge that can be trusted.

A more detailed discussion of these three forms of knowledge is provided in the later section on scientific knowledge and in Figures 4.5, 4.6 and 4.7.

People need all three forms of scientific knowledge to perform the three competencies of scientific literacy. PISA 2015 and PISA-D focus on assessing the extent to which 15-year-olds are capable of displaying the three aforementioned competencies appropriately within a range of personal, local/national (grouped in one category) and global contexts. (For the purposes of the PISA assessment, these competencies are only tested using the knowledge that 15-year-old students can reasonably be expected to have already acquired.) This perspective differs from that of many school science programmes that are dominated by content knowledge. Instead, the framework is based on a broader view of the kind of knowledge of science required of fully engaged citizens.

In addition, the competency-based perspective also recognises that there is an affective element to a student’s display of these competencies: students’ attitudes or disposition towards science will determine their level of interest, sustain their engagement, and may motivate them to take action (Schibeci, 1984). Thus, the scientifically literate person would typically have an interest in scientific topics; engage with science-related issues; have a concern for issues of technology,



resources and the environment; and reflect on the importance of science from a personal and social perspective. This requirement does not mean that such individuals are necessarily disposed towards becoming scientists themselves rather, such individuals recognise that science, technology and research in this domain are an essential element of contemporary culture that frames much of our thinking.

These considerations led to the definition of scientific literacy used in PISA 2015 and PISA-D (see Box 4.2). The use of the term “scientific literacy”, rather than “science”, underscores the importance that the PISA science assessment places on the application of scientific knowledge in the context of real-life situations.

Box 4.2 **The 2015 definition of scientific literacy**

Scientific literacy is the ability to engage with science-related issues, and with the ideas of science, as a reflective citizen.

A scientifically literate person is willing to engage in reasoned discourse about science and technology, which requires the competencies to:

- **Explain phenomena scientifically** – recognise, offer and evaluate explanations for a range of natural and technological phenomena.
- **Interpret data and evidence scientifically** – analyse and evaluate data, claims and arguments in a variety of representations and draw appropriate scientific conclusions.
- **Evaluate and design scientific enquiry** – describe and appraise scientific investigations and propose ways of addressing questions scientifically.

The competencies required for scientific literacy

Competency 1: Explain phenomena scientifically

The cultural achievement of science has been to develop a set of explanatory theories that have transformed our understanding of the natural world (in this document, “natural world” refers to phenomena associated with any object or activity occurring in the living or the material world), such as the idea that day and night is caused by a rotating Earth, or the idea that diseases can be caused by invisible micro-organisms. Moreover, such knowledge has enabled us to develop technologies that support human life by, for example, preventing disease or enabling rapid human communication across the globe. The competency to explain scientific and technological phenomena is thus dependent on knowledge of these major explanatory ideas of science.

Explaining scientific phenomena, however, requires more than the ability to recall and use theories, explanatory ideas, information and facts (content knowledge). Offering scientific explanations also requires an understanding of how such knowledge has been derived and the level of confidence we might hold about any scientific claims. For this competency, the individual requires a knowledge of the standard forms and procedures used in scientific enquiry to obtain such knowledge (procedural knowledge) and an understanding of their role and function in justifying the knowledge produced by science (epistemic knowledge).

Competency 2: Interpret data and evidence scientifically

Interpreting data is such a core activity of all scientists that some rudimentary understanding of the process is essential for scientific literacy. Initially, data interpretation begins with looking for patterns, constructing simple tables and graphical visualisations, such as pie charts, bar graphs, scatterplots or Venn diagrams. At a higher level, it requires the use of more complex data sets and the use of the analytical tools offered by spreadsheets and statistical packages. It would be wrong, however, to look at this competency as merely an ability to use these tools. A substantial body of knowledge is required to recognise what constitutes reliable and valid evidence and how to present data appropriately.

Scientists make choices about how to represent the data in graphs, charts or, increasingly, in complex simulations or 3D visualisations. Any relationships or patterns must then be read using knowledge of standard patterns. Whether uncertainty has been minimised by standard statistical techniques must also be considered. All of this draws on a body of procedural knowledge. The scientifically literate individual can also be expected to understand that uncertainty is an inherent feature of all measurement, and that one criterion for expressing confidence in a finding is determining the probability that the finding might have occurred by chance.



It is not sufficient, however, to understand the procedures that have been applied to obtain any data set. The scientifically literate individual needs to be able to judge whether they are appropriate and the ensuing claims are justified (epistemic knowledge). For instance, many sets of data can be interpreted in multiple ways. Argumentation and critique are essential to determining which is the most appropriate conclusion.

Whether it is new theories, novel ways of collecting data, or fresh interpretations of old data, argumentation is the means that scientists and technologists use to make their case for new ideas. Disagreement among scientists is normal, not extraordinary. Determining which interpretation is the best requires knowledge of science (content knowledge). Consensus on key scientific ideas and concepts has been achieved through this process of critique and argumentation (Longino, 1990). Indeed, it is a critical and sceptical disposition towards all empirical evidence that many would see as the hallmark of the professional scientist. The scientifically literate individual understands the function and purpose of argument and critique and why they are essential to the construction of knowledge in science. In addition, they should be able both to construct claims that are justified by data and to identify any flaws in the arguments of others.

Competency 3: Evaluate and design scientific enquiry

Scientific literacy implies that students have some understanding of the goal of scientific enquiry, which is to generate reliable knowledge about the natural world (Ziman, 1979). Data collected and obtained by observation and experiment, either in the laboratory or in the field, lead to the development of models and explanatory hypotheses that enable predictions that can then be tested experimentally. New ideas, however, commonly build on previous knowledge. Scientists themselves rarely work in isolation; they are members of research groups or teams that engage, nationally and internationally, in extensive collaboration with colleagues. New knowledge claims are always perceived to be provisional and may lack justification when subjected to critical peer review – the mechanism through which the scientific community ensures the objectivity of scientific knowledge (Longino, 1990). Hence, scientists have a commitment to publish or report their findings and the methods used in obtaining their evidence. Doing so enables empirical studies, at least in principle, to be replicated and results confirmed or challenged. However, measurements can never be absolutely precise; they all contain a degree of error. Much of the work of the experimental scientist is thus devoted to resolving uncertainty by repeating measurements, collecting larger samples, building instruments that are more accurate, and using statistical techniques that assess the degree of confidence in any result.

In addition, science has well-established procedures that are the foundations of any experiment to establish cause and effect. The use of controls enables the scientist to claim that any change in a perceived outcome can be attributed to a change in one specific feature. Failure to use such techniques leads to results where effects are confounded and cannot be trusted. Likewise, double-blind trials enable scientists to claim that the results have not been influenced either by the subjects of the experiment, or by the experimenter themselves. Other scientists, such as taxonomists and ecologists, are engaged in the process of identifying underlying patterns and interactions in the natural world that warrant a search for an explanation. In other cases, such as evolution, plate tectonics or climate change, scientists examine a range of hypotheses and eliminate those that do not fit with the evidence.

Facility with this competency draws on content knowledge, a knowledge of the common procedures used in science (procedural knowledge), and the function of these procedures in justifying any claims advanced by science (epistemic knowledge). Procedural and epistemic knowledge serve two functions. First, such knowledge is required by individuals to appraise scientific investigations and decide whether they have followed appropriate procedures and whether the conclusions are justified. Second, individuals who have this knowledge should be able to propose, at least in broad terms, how a scientific question might be investigated appropriately.

The evolution of the definition of scientific literacy in PISA

In PISA 2000 and 2003, scientific literacy was defined as:

“...the capacity to use scientific knowledge, to identify questions and to draw evidence-based conclusions in order to understand and help make decisions about the natural world and the changes made to it through human activity.” (OECD, 2000, 2003)



In 2000 and 2003, the definition embedded knowledge *of* science and understandings *about* science within the one term “scientific knowledge”. The 2006 definition separated and elaborated the term “scientific knowledge” by dividing it into two components: “knowledge *of* science” and “knowledge *about* science” (OECD, 2006). Both definitions referred to the application of scientific knowledge to understanding, and making informed decisions about, the natural world. In PISA 2006, the definition was enhanced by the addition of knowledge of the relationship between science and technology – an aspect that was assumed but not elaborated in the 2003 definition.

“For the purposes of PISA, scientific literacy refers to an individual’s:

- Scientific knowledge and use of that knowledge to identify questions, acquire new knowledge, explain scientific phenomena and draw evidence-based conclusions about science-related issues.
- Understanding of the characteristic features of science as a form of human knowledge and enquiry.
- Awareness of how science and technology shape our material, intellectual and cultural environments.
- Willingness to engage in science-related issues, and with the ideas of science, as a reflective citizen.” (OECD, 2006)

These ideas have evolved further in the PISA 2015 definition of scientific literacy. The major difference is that the notion of “knowledge *about* science” has been specified more clearly and split into two components – procedural knowledge and epistemic knowledge.

In 2006, the PISA framework was also expanded to include attitudinal aspects of students’ responses to scientific and technological issues within the construct of scientific literacy. In 2006, attitudes were measured in two ways: through the student questionnaire and through items embedded in the student test. Discrepancies were found between the results from the embedded questions and those from the background questionnaire with respect to “interest in science” for all students and gender differences in these issues (OECD, 2009; see also: Drechsel, Carstensen and Prenzel, 2011). More important, embedded items extended the length of the test. Hence, in PISA 2015, attitudinal aspects are only measured through the student questionnaire; there are no embedded attitudinal items. PISA-D does not include the measurement of attitudes towards learning science.

As for the constructs measured within this domain, the first (“interest in science”) and third (“environmental awareness”) remain the same as in 2006. The second (“support for scientific enquiry”) has been changed to a measure of “valuing scientific approaches to enquiry”, which is essentially a change in terminology to better reflect what is measured.

In addition, the contexts in PISA 2015 have been changed from “personal, social and global” in the 2006 assessment to “personal, local/national and global” to make the headings more coherent.

ORGANISING THE DOMAIN OF SCIENCE

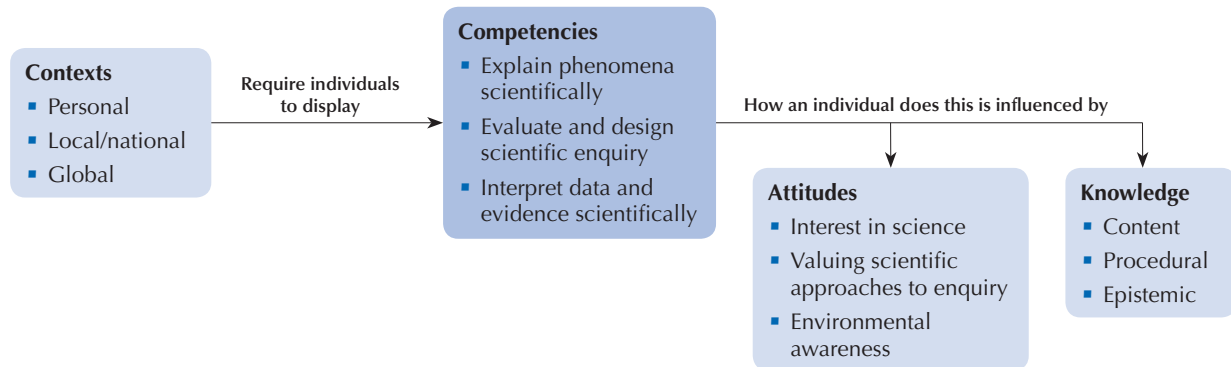
The PISA 2015 definition of scientific literacy used in PISA-D consists of four inter-related aspects (see Figures 4.1 and 4.2).

Figure 4.1 ■ **Aspects of the scientific literacy assessment framework for PISA 2015**

Contexts	Personal, local/national and global issues, both current and historical, which demand some understanding of science and technology.
Knowledge	An understanding of the major facts, concepts and explanatory theories that form the basis of scientific knowledge. Such knowledge includes knowledge of both the natural world and technological artefacts (content knowledge), knowledge of how such ideas are produced (procedural knowledge), and an understanding of the underlying rationale for these procedures and the justification for their use (epistemic knowledge).
Competencies	The ability to explain phenomena scientifically, interpret data and evidence scientifically, and evaluate and design scientific enquiry.
Attitudes	A set of attitudes towards science indicated by an interest in science and technology, valuing scientific approaches to enquiry where appropriate, and a perception and awareness of environmental issues.

Source: OECD (2016a), *PISA 2015 Assessment and Analytical Framework*, <http://dx.doi.org/10.1787/9789264255425-en>.

Figure 4.2 ■ Inter-relations between the four aspects



Source: OECD (2016a), *PISA 2015 Assessment and Analytical Framework*, <http://dx.doi.org/10.1787/9789264255425-en>.

Contexts of assessment items

PISA 2015 and PISA-D assess scientific knowledge in contexts that are relevant to the science curricula of participating countries. Such contexts are not, however, restricted to the common aspects of participants' national curricula. Rather, the assessment requires evidence of the successful use of the three competencies required for scientific literacy in situations set in personal, local/national and global contexts.

Assessment items are not limited to school science contexts. In the PISA 2015 and PISA-D scientific literacy assessments, the items focus on situations relating to the self, family and peer groups (personal), to the community (local and national), and to life across the world (global). Technology-based topics may be used as a common context. Some topics may be set in historical contexts, which are used to assess students' understanding of the processes and practices involved in advancing scientific knowledge.

Figure 4.3 shows how science and technology issues are applied within personal, local/national and global settings. The contexts are chosen in light of their relevance to students' interests and lives. The areas of application are: health and disease, natural resources, environmental quality, hazards, and the frontiers of science and technology. They are the areas in which scientific literacy has particular value for individuals and communities in enhancing and sustaining quality of life, and in developing public policy.

Figure 4.3 ■ Contexts in the PISA 2015 and PISA-D scientific literacy assessment

	Personal	Local/National	Global
Health and disease	Maintenance of health, accidents, nutrition	Control of disease, social transmission, food choices, community health	Epidemics, spread of infectious diseases
Natural resources	Personal consumption of materials and energy	Maintenance of human populations, quality of life, security, production and distribution of food, energy supply	Renewable and non-renewable natural systems, population growth, sustainable use of species
Environmental quality	Environmentally friendly actions, use and disposal of materials and devices	Population distribution, disposal of waste, environmental impact	Biodiversity, ecological sustainability, control of pollution, production and loss of soil/biomass
Hazards	Risk assessments of lifestyle choices	Rapid changes (e.g. earthquakes, severe weather), slow and progressive changes (e.g. coastal erosion, sedimentation), risk assessment	Climate change, impact of modern communication
Frontiers of science and technology	Scientific aspects of hobbies, personal technology, music and sporting activities	New materials, devices and processes, genetic modifications, health technology, transport	Extinction of species, exploration of space, origin and structure of the universe



The PISA science assessment is *not* an assessment of contexts. Rather, it assesses competencies and knowledge *in* specific contexts. These contexts are chosen on the basis of the knowledge and understanding that students are likely to have acquired by the age of 15.

Sensitivity to linguistic and cultural differences is a priority in item development and selection, not only for the sake of the validity of the assessment, but also to respect these differences among participating countries.

Scientific competencies

Figures 4.4a, 4.4c and 4.4e provide a detailed description of how students may display the three competencies required for scientific literacy. The set of scientific competencies in Figures 4.4a, 4.4c and 4.4e reflects a view that science is best seen as an ensemble of social and epistemic practices that are common across all sciences (National Research Council, 2012). Hence, all these competencies are framed as actions. They are written in this manner to convey the idea of what the scientifically literate person both understands and is capable of doing. Fluency with these practices is, in part, what distinguishes the expert scientist from the novice. While it would be unreasonable to expect a 15-year-old student to have the expertise of a scientist, a scientifically literate student can be expected to appreciate the role and significance of these practices and try to use them.

Figure 4.4a ■ **PISA 2015 and PISA-D scientific competencies: Explain phenomena scientifically**

Explain phenomena scientifically

Recognise, offer and evaluate explanations for a range of natural and technological phenomena demonstrating the ability to:

- recall and apply appropriate scientific knowledge
- identify, use and generate explanatory models and representations
- make and justify appropriate predictions
- offer explanatory hypotheses
- explain the potential implications of scientific knowledge for society.

Demonstrating the competency of *explaining phenomena scientifically* requires students to recall the appropriate content knowledge in a given situation and use it to interpret and explain the phenomenon of interest. Such knowledge can also be used to generate tentative explanatory hypotheses in contexts where there is a lack of knowledge or data. A scientifically literate person is expected to be able to draw on standard scientific models to construct simple representations to explain everyday phenomena, such as why antibiotics do not kill viruses, how a microwave oven works, or why gases are compressible but liquids are not, and use these to make predictions. This competency includes the ability to describe or interpret phenomena and predict possible changes. In addition, it may involve recognising or identifying appropriate descriptions, explanations and predictions.

For the purposes of assessing students who perform at Level 1, a more detailed description of this competency is defined beneath for PISA-D. All Level 1 students should be able to demonstrate some ability to explain phenomena scientifically.

Figure 4.4b ■ **PISA-D Levels 1a, 1b and 1c for scientific competency “Explain phenomena scientifically”**

Explain phenomena scientifically for Level 1c

Recognise explanations for a limited range of the most simple natural and technological phenomena demonstrating the ability to:

- recall appropriate scientific knowledge.

Explain phenomena scientifically for Level 1b

Recognise explanations for a range of simple or familiar natural and technological phenomena demonstrating the ability to:

- identify an explanatory model or representation
- recognise the potential implications of scientific knowledge for society and individuals.

Explain phenomena scientifically for Level 1a

Recognise explanations for a range of simple or familiar natural and technological phenomena demonstrating the ability to:

- make appropriate predictions
- recognise an appropriate explanatory hypothesis
- recognise simple causal or correlational relationships.



At Level 1c students can be required to:

- identify what the elements are of a standard representation used in science. For instance, a question might present an unlabelled diagram of an object and students could be asked to add the appropriate labels from a list provided by the question.
- recall appropriate scientific knowledge but not apply such knowledge. For instance, a student might be asked to identify which scientific phenomenon is being described in an item.

At Level 1b students can be required to:

- recall appropriate scientific knowledge but not to apply such knowledge. For instance, a question might ask which one of several familiar scientific concepts from a list would explain a simple phenomenon described at the beginning of the question.
- use a familiar piece of scientific knowledge. For instance, a question about the freezing point of water might ask students to determine whether water will freeze in a given context.

At Level 1a students can be required to:

- make a simple prediction but not justify it. For instance, a question might ask which of several predictions might be correct, or students could be asked to predict the reading of an ammeter on a simple circuit where the reading on one ammeter is provided and the other is not.
- to identify from a list which evidence supports a particular claim, e.g. that a rock is a sedimentary rock or that a whale is a mammal rather than a fish.
- provide descriptive explanations of the properties of objects or substances – for instance that a rock must be sedimentary because it can be easily scratched.

The following requirements, however, would be considered too advanced and beyond the scope of a Level 1 competency; students would only be expected to attain partial credit on an item. Thus competency at this level would not require students to:

- offer explanatory hypotheses, or explain the potential implications of scientific knowledge for society
- construct an explanation for why a given explanation might be flawed
- offer explanatory hypotheses that require students to recall knowledge and draw an appropriate inference
- provide a causal explanation for how a device works
- identify an explanatory model in a question that requires the recall of more than two pieces of knowledge
- provide explanations of unfamiliar phenomena.

Figure 4.4c ■ **PISA 2015 and PISA-D scientific competencies: Interpret data and evidence scientifically**

Interpret data and evidence scientifically
Analyse and evaluate scientific data, claims and arguments in a variety of representations and draw appropriate conclusions, demonstrating the ability to: ▪ transform data from one representation to another ▪ analyse and interpret data and draw appropriate conclusions ▪ identify the assumptions, evidence and reasoning in science-related texts ▪ distinguish between arguments that are based on scientific evidence and theory and those based on other considerations ▪ evaluate scientific arguments and evidence from different sources (e.g. newspapers, the Internet, journals).

A scientifically literate person should be able to interpret and make sense of basic forms of scientific data and evidence that are used to make claims and draw conclusions. Displaying this competency may require all three forms of scientific knowledge.

Those who possess this competency should be able to interpret the meaning of scientific evidence and its implications to a specified audience in their own words, using diagrams or other representations as appropriate. This competency requires the use of mathematical tools to analyse or summarise data, and the ability to use standard methods to transform data into different representations.



This competency also includes accessing scientific information and producing and evaluating arguments and conclusions based on scientific evidence (Kuhn, 2010; Osborne, 2010). It may also involve evaluating alternative conclusions using evidence; giving reasons for or against a given conclusion using procedural or epistemic knowledge; and identifying the assumptions made in reaching a conclusion. In short, the scientifically literate individual should be able to identify logical or flawed connections between evidence and conclusions.

The higher cognitive demand required to interpret data and evidence scientifically means that this competency is generally above Level 1c. More detailed descriptions of this competency for Levels 1a and 1b are provided below for PISA-D.

Figure 4.4d ■ **PISA-D Levels 1a and 1b for scientific competency**
“Interpret data and evidence scientifically”

Interpret data and evidence scientifically for Level 1b

Recognise a specific scientific claim, justification or data set in a simple or familiar context, demonstrating the ability to:

- identify the evidence, claim or justification in a science-related text
- identify simple patterns in data.

Interpret data and evidence scientifically for Level 1a

Recognise specific scientific data, claims and justifications in simple or familiar contexts, and identify an appropriate conclusion, demonstrating the ability to:

- recognise an appropriate conclusion that can be drawn from a simple set of data
- extract a specific piece of information from a scientific text
- interpret graphical and visual data
- identify simple causal or correlational relationships.

At Level 1b students can be required to:

- describe a simple trend in data, but not be asked to draw a conclusion based on the data. For instance, a question might be asked to identify how temperatures have changed over a period of time when provided data in a graph or table.
- identify a claim, evidence or a reason in a science-related text. Alternatively, students could be asked to identify which is the claim, evidence or reasoning in a science text from a list that is provided.

At Level 1a students can be required to:

- state which one of several conclusions about a simple phenomenon drawn from a data set is the most appropriate, using a deduction requiring one step
- given a simple table, graph or other form of data representation, identify which conclusion or prediction is correct, e.g. identifying trends in a graph where there is no extraneous information
- extract meaning from simple scientific texts, for instance, asking students to identify the states through which matter moves, e.g. about solids, liquids and gases
- identify whether the conclusion drawn from a table of results, graph or other form of data is justified or not, e.g. whether the interpretation drawn from a table of materials and the effect of a magnet on the material is correct.

However, the following requirements would be considered too advanced and beyond the scope of a Level 1 competency; students would only be expected to attain partial credit on an item. Thus competency at this level would not require students to:

- distinguish between arguments that are based on scientific evidence or scientific theories and those that are based on other considerations
- evaluate two competing arguments from different sources (e.g. newspaper, Internet, journals)
- analyse or interpret more than one data set in any question
- consider multiple pieces of evidence or multiple theories and whether the information supports one or more theories.

Table 4.1 shows the desired distribution of items, by competency, for the PISA 2015 science assessment and for PISA-D. For science, the desired distributions for PISA-D are for the school-based instrument only, as science is not included in the out-of-school assessment.



Figure 4.4e ■ **PISA 2015 and PISA-D scientific competencies: Evaluate and design scientific enquiry**

Evaluate and design scientific enquiry

Describe and appraise scientific investigations and propose ways of addressing questions scientifically demonstrating the ability to:

- identify the question explored in a given scientific study
- distinguish questions that could be investigated scientifically
- propose a way of exploring a given question scientifically
- evaluate ways of exploring a given question scientifically
- describe and evaluate how scientists ensure the reliability of data, and the objectivity and generalisability of explanations.

The competency of *evaluating and designing scientific enquiry* is required to evaluate reports of scientific findings and investigations critically. It relies on the ability to distinguish scientific questions from other forms of enquiry or recognise questions that could be investigated scientifically in a given context. This competency requires a knowledge of the key features of a scientific investigation – for example, what things should be measured, what variables should be changed or controlled, or what action should be taken so that accurate and precise data can be collected. It requires an ability to evaluate the quality of data, which, in turn, depends on recognising that data are not always completely accurate. It also requires the ability to determine whether an investigation is driven by an underlying theoretical premise or, alternatively, whether it seeks to determine patterns.

A scientifically literate person should also be able to recognise the significance of previous research when judging the value of any given scientific enquiry. Such knowledge is needed to situate the work and judge the importance of any possible outcomes. For example, knowing that the search for a malaria vaccine has been an ongoing programme of scientific research for several decades, and given the number of people who are killed by malarial infections, any findings that suggested a vaccine would be achievable would be of substantial significance.

Moreover, students need to understand the importance of developing a sceptical attitude towards all media reports in science. They need to recognise that all research builds on previous work, that the findings of any one study are always subject to uncertainty, and that the study may be biased by the sources of funding. This competency requires students to possess both procedural and epistemic knowledge but may also draw on their content knowledge of science, to varying degrees.

The higher cognitive demand required to evaluate and design scientific enquiry means that this competency is generally above Level 1c and attained only to a limited extent by Level 1b students. More detailed descriptions of this competency for Levels 1a and 1b are provided below for PISA-D.

Figure 4.4f ■ **PISA-D Levels 1a and 1b for scientific competency “Evaluate and design scientific enquiry”**

Evaluate and design scientific enquiry for Level 1b

Appraise simple scientific investigations, demonstrating the ability to:

- carry out a simple scientific procedure when provided explicit instructions
- determine which of several variables is the dependent variable in an investigation.
- recognise appropriate measures for a quantity (units appropriate for measuring).

Evaluate and design scientific enquiry for Level 1a

Appraise simple scientific investigations and recognise ways of addressing questions scientifically, demonstrating the ability to:

- identify the question explored in a simple scientific study
- distinguish a question that is possible to investigate scientifically from one that is not
- evaluate if one way of exploring a given question is scientifically appropriate
- identify a source of error in a measurement or a flaw in an experimental design.

At Level 1b students can be required to:

- determine which variables were changed, measured, or held constant when provided with a description of a scientific investigation
- identify the appropriate instrument or units to measure a quantity from a selection of different instruments or units.



At Level 1a students can be required to:

- identify the question that is being answered in a simple scientific investigation in which only one factor is varied at a time; for instance, by describing a study and then asking the student to explain what question is being answered
- identify which of several actions it might be best to undertake to answer a simple scientific question. For instance, the question “Where do woodlice live?” is best answered by pattern seeking, identification using criteria or fair testing.
- propose specific measurements that might be needed to answer a simple scientific question. For instance, a question might ask which of several variables should be measured to investigate whether the length of a pendulum affects the time of swing. Alternatively, a question might ask which of several variables should be controlled when conducting a simple investigation.
- from a list of several actions, identify which actions would reduce the error in an experiment. Such questions should be assessed using partial credit scoring.
- identify a variable (dependent and independent variables and controlled variable) in a simple scientific enquiry that should be controlled or should be varied to answer a given question.
- identify a simple flaw in an experimental design, e.g. a failure to control variables, taking a single measurement or measuring the wrong factor.

However, the following requirements would be considered too advanced and beyond the scope of a Level 1 competency; students would only be expected to attain partial credit on an item. Thus, competency at this level would not require students to:

- evaluate multiple ways of exploring a given question scientifically
- evaluate multiple ways that are proposed to ensure the reliability of data in an investigation
- explain why some data might be anomalous
- given a phenomenon, generate questions for investigation.

Table 4.1 Desired distribution of science items, by competency

Competency	Percentage of items in PISA 2015	Percentage of items in PISA-D
Explain phenomena scientifically	40-50	40-50
Interpret data and evidence scientifically	30-40	30-40
Evaluate and design scientific enquiry	20-30	20-30

The desired distribution specifies the blueprint for selecting items according to important aspects of the domain frameworks. Item selection is based on the assessment design as well as item characteristics related to a number of framework aspects, including competency, content, type of knowledge and response formats, as well as consideration of the items’ psychometric properties and appropriateness for this assessment. Following the assessment, the actual distributions of items across the framework aspects will be described in relation to the desired distributions. The extent to which the item pool for the assessment meets the framework specifications will be discussed in the technical report in the context of practical constraints in the item selection process.

Scientific knowledge

Content knowledge

Given that only a sample of the content domain of science can be assessed in the PISA 2015 and PISA-D scientific literacy assessments, clear criteria are used to guide the selection of the knowledge that is assessed. The criteria are applied to knowledge from the major fields of physics, chemistry, biology, earth and space sciences, and require that the knowledge:

- has relevance to real-life situations
- represents an important scientific concept or major explanatory theory that has enduring utility
- is appropriate to the developmental level of 15-year-olds.

It is thus assumed that students have some knowledge and understanding of the major explanatory ideas and theories of science, including an understanding of the history and scale of the universe, the particle model of matter and the theory of evolution by natural selection. These examples of major explanatory ideas are provided for illustrative purposes; there has been no attempt to list comprehensively all the ideas and theories that might be considered fundamental for a scientifically literate individual.



Figure 4.5 shows the content knowledge categories and examples selected by applying these criteria. Such knowledge is required for understanding the natural world and for making sense of experiences in personal, local/national, and global contexts. The framework uses the term “systems” instead of “sciences” in the descriptors of content knowledge. The intention is to convey the idea that citizens have to understand concepts from the physical and life sciences, and earth and space sciences, and how they apply in contexts where the elements of knowledge are interdependent or interdisciplinary. Things viewed as subsystems at one scale may be viewed as whole systems at a smaller scale. For example, the circulatory system can be seen as an entity in itself or as a subsystem of the human body; a molecule can be studied as a stable configuration of atoms but also as a subsystem of a cell or a gas. Thus, applying scientific knowledge and exhibiting scientific competencies requires a determination of which system and which boundaries apply in any particular context.

Figure 4.5 ■ **Knowledge of the content of science**

Physical systems that require knowledge of:
- Structure of matter (e.g. particle model, bonds) - Properties of matter (e.g. changes of state, thermal and electrical conductivity) - Chemical changes of matter (e.g. chemical reactions, energy transfer, acids/bases) - Motion and forces (e.g. velocity, friction) and action at a distance (e.g. magnetic, gravitational and electrostatic forces) - Energy and its transformation (e.g. conservation, dissipation, chemical reactions) - Interactions between energy and matter (e.g. light and radio waves, sound and seismic waves)
Living systems that require knowledge of:
- Cells (e.g. structures and function, DNA, plant and animal) - The concept of an organism (e.g. unicellular and multicellular) - Humans (e.g. health, nutrition, subsystems such as digestion, respiration, circulation, excretion, reproduction and their relationship) - Populations (e.g. species, evolution, biodiversity, genetic variation) - Ecosystems (e.g. food chains, matter and energy flow) - Biosphere (e.g. ecosystem services, sustainability)
Earth and space systems that require knowledge of:
- Structures of the Earth systems (e.g. lithosphere, atmosphere, hydrosphere) - Energy in the Earth systems (e.g. sources, global climate) - Change in Earth systems (e.g. plate tectonics, geochemical cycles, constructive and destructive forces) - Earth's history (e.g. fossils, origin and evolution) - Earth in space (e.g. gravity, solar systems, galaxies) - The history and scale of the universe and its history (e.g. light year, Big Bang theory)

Source: OECD (2016a), *PISA 2015 Assessment and Analytical Framework*, <http://dx.doi.org/10.1787/9789264255425-en>.

Table 4.2 shows the desired distribution of items, by content of science, for PISA 2015 and PISA-D.

Table 4.2 Desired distribution of science items, by content

System	Percentage of items in PISA 2015	Percentage of items in PISA-D
Physical	36	36
Living	36	36
Earth and space	28	28
Total	100	100



Procedural knowledge

A fundamental goal of science is to generate explanatory accounts of the material world. Tentative explanatory accounts are first developed and then tested through empirical enquiry. Empirical enquiry relies on certain well-established concepts, such as the notion of dependent and independent variables, the control of variables, types of measurement, forms of error, methods of minimising error, common patterns observed in data and methods of presenting data.

It is this knowledge of the concepts and procedures that are essential for scientific enquiry that underpins the collection, analysis and interpretation of scientific data. Such ideas form a body of procedural knowledge that has also been called “concepts of evidence” (Gott, Duggan and Roberts, 2008; Millar et al., 1995). One can think of procedural knowledge as knowledge of the standard procedures scientists use to obtain reliable and valid data. Such knowledge is needed both to undertake scientific enquiry and engage in critical reviews of the evidence that might be used to support particular claims. It is expected, for instance, that students will know that scientific knowledge has differing degrees of certainty associated with it, and so can explain why there is a difference between the confidence associated with measurements of the speed of light (which has been measured many times with ever more accurate instrumentation) and measurements of fish stocks in the North Atlantic or the mountain lion population in California. The examples listed in Figure 4.6 convey the general features of procedural knowledge that may be tested.

Figure 4.6 ■ PISA 2015 and PISA-D procedural knowledge

Procedural knowledge

- The concept of variables, including dependent, independent and control variables
- Concepts of measurement e.g. quantitative (measurements), qualitative (observations), the use of a scale, categorical and continuous variables
- Ways of assessing and minimising uncertainty, such as repeating and averaging measurements
- Mechanisms to ensure the replicability (closeness of agreement between repeated measures of the same quantity) and accuracy of data (the closeness of agreement between a measured quantity and a true value of the measure)
- Common ways of abstracting and representing data using tables, graphs and charts, and using them appropriately
- The control-of-variables strategy and its role in experimental design or the use of randomised controlled trials to avoid confounded findings and identify possible causal mechanisms
- The nature of an appropriate design for a given scientific question, e.g. experimental, field-based or pattern-seeking.

Epistemic knowledge

Epistemic knowledge refers to an understanding of the role of specific constructs and defining features essential to the process of knowledge building in science (Duschl, 2007). Those who have such knowledge can explain, with examples, the distinction between a scientific theory and a hypothesis or a scientific fact and an observation. They know that models, whether representational, abstract or mathematical, are a key feature of science, and that such models are like maps rather than accurate pictures of the material world. These students can recognise that any particle model of matter is an idealised representation of matter and can explain how the Bohr model is a limited model of what we know about the atom and its constituent parts. They recognise that the concept of a “theory” as used in science is not the same as the notion of a “theory” in everyday language, where it is used as a synonym for a “guess” or a “hunch”. Procedural knowledge is required to explain what is meant by the control-of-variables strategy; epistemic knowledge is required to explain *why* the use of the control-of-variables strategy or the replication of measurements is central to establishing knowledge in science.

Scientifically literate individuals also understand that scientists draw on data to advance claims to knowledge, and that argument is a commonplace feature of science. In particular, they know that some arguments in science are hypothetico-deductive (e.g. Copernicus’ argument for the heliocentric system), some are inductive (the conservation of energy), and some are an inference to the best explanation (Darwin’s theory of evolution or Wegener’s argument for moving continents). They also understand the role and significance of peer review as the mechanism that the scientific community has established for testing claims to new knowledge. As such, epistemic knowledge provides a rationale for the procedures and practices in which scientists engage, knowledge of the structures and defining features that guide scientific enquiry, and the foundation for the basis of belief in the claims that science makes about the natural world.



Figure 4.7 represents what are considered to be the major features of epistemic knowledge necessary for scientific literacy.

Figure 4.7 ■ PISA 2015 and PISA-D epistemic knowledge

Epistemic knowledge

The constructs and defining features of science. That is:

- The nature of scientific observations, facts, hypotheses, models and theories
- The purpose and goals of science (to produce explanations of the natural world) as distinguished from technology (to produce an optimal solution to human need), and what constitutes a scientific or technological question and appropriate data
- The values of science, e.g. a commitment to publication, objectivity and the elimination of bias
- The nature of reasoning used in science, e.g. deductive, inductive, inference to the best explanation (abductive), analogical and model-based.

The role of these constructs and features in justifying the knowledge produced by science. That is:

- How scientific claims are supported by data and reasoning in science
- The function of different forms of empirical enquiry in establishing knowledge, their goal (to test explanatory hypotheses or identify patterns) and their design (observation, controlled experiments, correlational studies)
- How measurement error affects the degree of confidence in scientific knowledge
- The use and role of physical, system and abstract models and their limits
- The role of collaboration and critique, and how peer review helps to establish confidence in scientific claims
- The role of scientific knowledge, along with other forms of knowledge, in identifying and addressing societal and technological issues.

Epistemic knowledge is most likely to be tested pragmatically in a context where a student is required to interpret and answer a question that requires some of this type of knowledge rather than assessing directly whether they understand the features detailed in Figure 4.7. For example, students may be asked to identify whether the conclusions are justified by the data, or what piece of evidence best supports the hypothesis advanced in an item and explain why.

Table 4.3 describes the desired distribution of items by type of knowledge for PISA 2015 and PISA-D.

Table 4.3 Desired distribution of science items, by type of knowledge

Knowledge	Percentage of items in PISA 2015	Percentage of items in PISA-D
Content	54-66	54-66
Procedural	19-31	19-31
Epistemic	10-22	10-22

The desired balance, by percentage of items, among the three knowledge components – content, procedural and epistemic – for PISA 2015 and PISA-D is shown in Table 4.4.¹

Table 4.4 Desired distribution of science items for knowledge

Knowledge types	Percentage of items in PISA 2015	Percentage of items in PISA-D
Total over systems (physical, living, earth and space)		
Content	54-66	54-66
Procedural	19- 31	19- 31
Epistemic	10-22	10-22
Total over knowledge types	100	100

ASSESSING SCIENTIFIC LITERACY

Cognitive demand

A key new feature of the PISA 2015 framework that will also be used in PISA-D is the definition of levels of cognitive demand within the assessment of scientific literacy and across all three competencies of the framework. In assessment frameworks, item difficulty, which is empirically derived, is often confused with cognitive demand. Empirical item difficulty is estimated from the proportion of test-takers who solve the item correctly, and thus assesses the amount of knowledge held by the test-taker population, whereas cognitive demand refers to the type of mental processes required (Davis and



Buckendahl, 2011). Care needs to be taken to ensure that the depth of knowledge required, i.e. the cognitive demand test items, is understood explicitly by the item developers and users of the PISA framework. For instance, an item can have high difficulty because the knowledge it is testing is not well known, but the cognitive demand is simply recall. Conversely, an item can be cognitively demanding because it requires the individual to relate and evaluate many items of knowledge – each of which is easily recalled. Thus, not only should the PISA test instrument discriminate in terms of performance between easier and harder test items, the test also needs to provide information on how students across the ability range can deal with problems at different levels of cognitive demand (Brookhart and Nitko, 2011).

The competencies are articulated using a range of terms defining cognitive demand through the use of verbs such as “recognise”, “interpret”, “analyse” and “evaluate”. However, in themselves these verbs do not necessarily indicate a hierarchical order of difficulty that is dependent on the level of knowledge required to answer any item. Various classifications of cognitive demand schemes have been developed and evaluated since Bloom’s Taxonomy was first published (Bloom, 1956). These have been largely based on categorisations of knowledge types and associated cognitive processes that are used to describe educational objectives or assessment tasks.

Bloom’s revised Taxonomy (Anderson and Krathwohl, 2001) identifies four categories of knowledge – factual, conceptual, procedural and meta-cognitive. This categorisation considers these forms of knowledge to be hierarchical and distinct from the six categories of performance used in Bloom’s first taxonomy – remembering, understanding, applying, analysing, evaluating and creating. In Anderson and Krathwohl’s framework, these two dimensions are now seen to be independent of each other, allowing for lower levels of knowledge to be crossed with higher-order skills, and vice versa.

A similar framework is offered by Marzano and Kendall’s Taxonomy (2007), which also provides a two-dimensional framework based on the relationship between how mental processes are ordered and the type of knowledge required. The use of mental processes is seen as a consequence of a need to engage with a task with meta-cognitive strategies that define potential approaches to solving problems. The cognitive system then uses either retrieval, comprehension, analysis or knowledge utilisation. Marzano and Kendall divide the knowledge domain into three types of knowledge, information, mental procedures and psychomotor, compared to the four categories in Bloom’s revised Taxonomy. Marzano and Kendall argue that their taxonomy is an improvement upon Bloom’s Taxonomy because it offers a model of how humans actually think rather than simply an organising framework.

A different approach is offered by Ford and Wargo (2012), who offer a framework for scaffolding dialogue as a way of considering cognitive demand. Their framework uses four levels that build on each other: recall, explain, juxtapose and evaluate. Although this framework has not been specifically designed for assessment purposes, it has many similarities to the PISA 2015 definition of scientific literacy and the need to make more explicit references to such demands in the knowledge and competencies.

Another schema can be found in the framework based on Depth of Knowledge developed by Webb (1997) specifically to address the disparity between assessments and the expectations of student learning. For Webb, levels of depth can be determined by taking into account the complexity of both the content and the task required. His schema consists of four major categories: level 1 (recall), level 2 (using skills and/or conceptual knowledge), level 3 (strategic thinking) and level 4 (extended thinking). Each category is populated with a large number of verbs that can be used to describe cognitive processes. Some of these appear at more than one level. This framework offers a more holistic view of learning and assessment tasks and requires an analysis of both the content and cognitive process demanded by any task. Webb’s Depth of Knowledge (DOK) approach is a simpler but more operational version of the SOLO Taxonomy (Biggs and Collis, 1982) which describes a continuum of student understanding through five distinct stages of pre-structural, unistructural, multistructural, relational and extended abstract understanding.

All the frameworks described briefly above have served to develop the knowledge and competencies in the PISA 2015 and PISA-D Frameworks. In drawing up such a framework, it is recognised that there are challenges in developing test items based on a cognitive hierarchy. The three main challenges are that:

- a) Too much effort is made to fit test items into particular cognitive frameworks, which can lead to poorly developed items.
- b) Intended items (with frameworks defining rigorous, cognitively demanding goals) may differ from actual items (which may operationalise the standard in a much less cognitively demanding way).
- c) Without a well-defined and understood cognitive framework, item writing and development often focuses on item difficulty and uses a limited range of cognitive processes and knowledge types, which are then only described and interpreted post-hoc, rather than building from a theory of increasing competency.



The approach taken in this framework is to use an adapted version of Webb's Depth of Knowledge grid (Webb, 1997) alongside the desired knowledge and competencies. As the competencies are the central feature of the framework, the cognitive framework needs to assess and report on them across the student ability range. Webb's Depth of Knowledge Levels offer a taxonomy for cognitive demand that requires items to identify both the cognitive demand from the verbal cues that are used, e.g. analyse, arrange, compare, and the expectations of the depth of knowledge required.

Figure 4.8 ■ PISA 2015 and PISA-D Framework for Cognitive Demand

		Competencies			Depth of Knowledge		
		Explain phenomena scientifically	Evaluate and design scientific enquiry	Interpret data and evidence scientifically	Low	Medium	High
Knowledge	Content knowledge						
	Procedural knowledge						
	Epistemic knowledge						

The grid in Figure 4.8 provides a framework for mapping items against the two dimensions of knowledge and competencies. In addition, each item can also be mapped using a third dimension based on a depth-of-knowledge taxonomy. This provides a means of operationalising cognitive demand as each item can be categorised as making demands that are:

■ **Low**

Carry out a one-step procedure, for example recall a fact, term, principle or concept, or locate a single point of information from a graph or table.

■ **Medium**

Use and apply conceptual knowledge to describe or explain phenomena, select appropriate procedures involving two or more steps, organise/display data, interpret or use simple data sets or graphs.

■ **High**

Analyse complex information or data; synthesise or evaluate evidence; justify; reason, given various sources; develop a plan or sequence of steps to approach a problem.

Table 4.5 shows the real distribution of items by depth of knowledge for PISA 2015 (there was no desired distribution specified for the depth of knowledge categories). Since the PISA-D assessment design calls for a greater proportion of items measuring the lower end of the scale, this criterion will presumably affect the distribution of items across the three categories of depth of knowledge. Compared to the distribution of items in the PISA 2015 assessment, there will likely be a greater proportion of items classified as “low” or “medium” depth of knowledge than in the “high” category in the PISA-D assessment.

Table 4.5 Distribution of science items, by depth of knowledge

Depth of knowledge	Percentage of items in PISA 2015	Percentage of items in PISA-D
Low	8	Not yet available
Medium	30	Not yet available
High	61	Not yet available
Total	100	100

Items that merely require recall of one piece of information make low cognitive demands, even if the knowledge itself might be quite complex. In contrast, items that require recall of more than one piece of knowledge, and require a comparison and evaluation of the competing merits of their relevance would be seen as having high cognitive demand. The difficulty of any item, therefore, is a combination both of the degree of complexity and range of knowledge it requires, and the cognitive operations that are required to process the item.



Therefore, the factors that determine the demand of items assessing science achievement include:

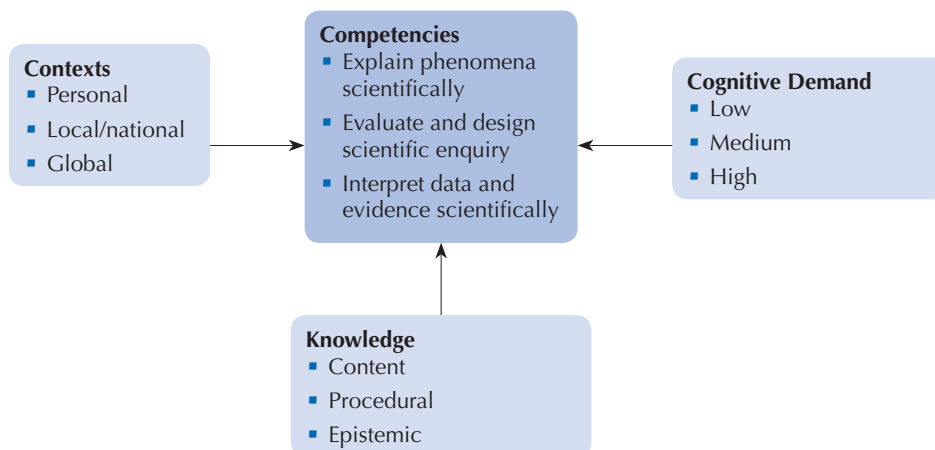
- The number and degree of complexity of elements of knowledge demanded by the item.
- The level of familiarity and prior knowledge that students may have of the content, procedural and epistemic knowledge involved.
- The cognitive operation required by the item, e.g. recall, analysis, evaluation.
- The extent to which forming a response is dependent on models or abstract scientific ideas.

This four-factor approach allows for a broader measure of scientific literacy across a wider range of student ability. Categorising the cognitive processes required for the competencies that form the basis of scientific literacy together with a consideration of the depth of knowledge required offers a model for assessing the level of demand of individual items. In addition, the relative simplicity of the approach offers a way to minimise the problems encountered in applying such frameworks. The use of this cognitive framework also facilitates the development of an a priori definition of the descriptive parameters of the reporting proficiency scales (see Figure 4.10).

Test characteristics

Figure 4.9 is a variation of Figure 4.2 that presents the basic components of the PISA framework for the 2015 scientific literacy assessment in a way that can be used to relate the framework with the structure and the content of assessment units. This may be used as a tool both to plan assessment exercises and to study the results of standard assessment exercises. As a starting point to construct assessment units, it shows the need to consider the contexts that will serve as stimulus material, the competencies required to respond to the questions or issues, the knowledge central to the exercise, and the cognitive demand.

Figure 4.9 ■ **A tool for constructing and analysing assessment units and items**



Source: OECD (2016a), *PISA 2015 Assessment and Analytical Framework*, <http://dx.doi.org/10.1787/9789264255425-en>.

A test unit is defined by specific stimulus material, which may be a brief written passage, or text accompanying a table, chart, graph or diagram. In units created for PISA 2015, the stimulus material may also include non-static stimulus material, such as animations and interactive simulations. The items are a set of independently scored questions of various types, as illustrated by the examples already discussed. Further examples can be found at the PISA website (www.oecd.org/pisa).

PISA uses this unit structure to facilitate the use of contexts that are as realistic as possible, reflecting the complexity of real-life situations, while making efficient use of testing time. Using situations about which several questions can be posed, rather than asking separate questions about a larger number of different situations, reduces the overall time required for a student to become familiar with the material in each question. However, the need to make each score point independent of others within a unit needs to be taken into account. It is also necessary to recognise that, because this approach reduces the number of different assessment contexts, it is important to ensure that there is an adequate range of contexts so that bias due to the choice of contexts is minimised.

PISA 2015 test units require the use of all three scientific competencies and draw on all three forms of science knowledge. In most cases, each test unit assesses multiple competencies and knowledge categories. Individual items, however, assess only one form of knowledge and one competency.



The need for students to read texts in order to understand and answer written questions on scientific literacy raises an issue of the level of reading literacy that are required. Stimulus material and questions use language that is as clear, simple and brief, and as syntactically simplified as possible while still conveying the appropriate meaning. The number of concepts introduced per paragraph are limited. Questions within the domain of science that assess reading or mathematical literacy are avoided.

In PISA-D, for a better measurement of items at proficiency Levels 1 and 2, items should only make the lower-level cognitive demands of recalling or recognising appropriate knowledge, understanding the meaning of texts, applying that knowledge, and very simple data analysis drawing on either factual knowledge or foundational concepts (Anderson and Krathwohl, 2001; Webb, 1997). In addition, whatever proficiency level they measure, items and language should wherever possible be simplified to reduce the cognitive load demanded of students (Sweller, 1994).

Response formats

Three classes of items will be used to assess the competencies and scientific knowledge identified in the framework. About one-third of the items are in each of the three classes in PISA 2015 and PISA-D:

- simple multiple choice: items calling for
 - selection of a single response from four options
 - selection of a “hot spot”, an answer that is a selectable element within a graphic or text.
- complex multiple choice: items calling for
 - responses to a series of related “Yes/No” questions that are treated for scoring as a single item (the typical format in 2006)
 - selection of more than one response from a list.
- constructed response: items calling for written or drawn responses.
 - Constructed-response items in scientific literacy typically call for a written response ranging from a phrase to a short paragraph (e.g. two to four sentences of explanation). A small number of constructed-response items call for drawing (e.g. a graph or diagram). In a computer-based assessment, any such items is supported by simple drawing editors that are specific to the response required.

Reporting proficiency in science in PISA-D

To achieve the aims of PISA, scales must be developed to measure student proficiency. A descriptive scale of levels of competence needs to be based on a theory of how the competence develops, not just on a post-hoc interpretation of what items of increasing difficulty seem to be measuring. The 2015 draft framework therefore defined explicitly the parameters of increasing competence and progression, allowing item developers to design items representing this growth in ability (Kane, 2006; Mislevy and Haertel, 2006). Although comparability with the 2006 scale descriptors (OECD, 2007) has been maximised in order to enable trend analyses, the new elements of the 2015 framework, such as depth of knowledge, have also been incorporated. The scales have also been extended by the addition of a Level 1b to specifically address and provide a description of students at the lowest level of ability who demonstrate minimal scientific literacy and would previously not have been included in the reporting scales. The scales for the 2015 Framework therefore propose more detailed and more specific descriptors of the levels of scientific literacy, and not an entirely different model as shown in Figure 4.10. For PISA-D, the table describing the performance level expectations has been extended to include a new Level 1c.

The proposed level descriptors are based on the *PISA 2015 Results Volume I* (OECD, 2016b) and offer a qualitative description of the differences between levels of performance. Proficiency Levels 2, 1a and 1b were modified to implement a clear line of progression in knowledge from 1c. The factors used to determine the demand of items assessing science achievement that have been incorporated into this outline of the proficiency scales include:

- the number and degree of complexity of elements of knowledge demanded by the item
- the level of familiarity and prior knowledge that students may have of the content, procedural and epistemic knowledge involved
- the cognitive operation required by the item, e.g. recall, analysis, evaluation
- the extent to which forming a response is dependent on models or abstract scientific ideas.



Figure 4.10 ■ Summary description of the eight levels of science proficiency in PISA-D

Level	Lower score limit	Percentage of students across OECD countries at each level, PISA 2015	Percentage of students across 23 middle- and low-income countries at each level, PISA 2015	Descriptor
6	708	1.1%	0.1%	At Level 6, students can draw on a range of inter-related scientific ideas and concepts from the physical, life and earth and space sciences and use content, procedural and epistemic knowledge in order to offer explanatory hypotheses of novel scientific phenomena, events and processes or to make predictions. In interpreting data and evidence, they are able to discriminate between relevant and irrelevant information and can draw on knowledge external to the normal school curriculum. They can distinguish between arguments that are based on scientific evidence and theory and those based on other considerations. Level 6 students can evaluate competing designs of complex experiments, field studies or simulations and justify their choices.
5	633	6.7%	0.8%	At Level 5, students can use abstract scientific ideas or concepts to explain unfamiliar and more complex phenomena, events and processes involving multiple causal links. They are able to apply more sophisticated epistemic knowledge to evaluate alternative experimental designs and justify their choices and use theoretical knowledge to interpret information or make predictions. Level 5 students can evaluate ways of exploring a given question scientifically and identify limitations in interpretations of data sets including sources and the effects of uncertainty in scientific data.
4	559	19.0%	5.0%	At Level 4, students can use more complex or more abstract content knowledge, which is either provided or recalled, to construct explanations of more complex or less familiar events and processes. They can conduct experiments involving two or more independent variables in a constrained context. They are able to justify an experimental design, drawing on elements of procedural and epistemic knowledge. Level 4 students can interpret data drawn from a moderately complex data set or less familiar context, draw appropriate conclusions that go beyond the data and provide justifications for their choices.
3	484	27.2%	15.5%	At Level 3, students can draw upon moderately complex content knowledge to identify or construct explanations of familiar phenomena. In less familiar or more complex situations, they can construct explanations with relevant cueing or support. They can draw on elements of procedural or epistemic knowledge to carry out a simple experiment in a constrained context. Level 3 students are able to distinguish between scientific and non-scientific issues and identify the evidence supporting a scientific claim.
2	410	24.8%	28.3%	At Level 2, students are able to draw on scientific content knowledge or procedural knowledge to identify an appropriate scientific explanation, interpret data, and identify the question being addressed in a simple experimental design. They can use basic or everyday scientific knowledge to identify a valid conclusion from a simple data set. Level 2 students demonstrate basic epistemic knowledge by being able to identify questions that could be investigated scientifically.

Figure 4.10 [continued] ■ **Summary description of the eight levels of science proficiency in PISA-D**

Level	Lower score limit	Percentage of students across OECD countries at each level, PISA 2015	Percentage of students across 23 middle- and low-income countries at each level, PISA 2015	Descriptor
1a	335	15.7%	31.5%	At Level 1a, students are able to draw on basic scientific content or procedural knowledge to recognise or identify explanations of simple scientific phenomenon presented using scientific language. With support, they can undertake structured scientific enquiries with no more than two variables. They are able to identify simple causal or correlational relationships and interpret graphical and visual data that require a low level of cognitive demand. Level 1a students can select the best scientific explanation for given data in familiar personal, local and global contexts. When presented with multiple factors of varying complexity requiring a low level of content knowledge or cognitive demand, students can select the best scientific explanations or procedures in a question in most but not all instances.
1b	260	4.9%	15.7%	At Level 1b, students can draw on everyday scientific knowledge to recognise aspects of familiar or simple phenomena presented using minimal scientific language. They are able to identify simple patterns in data, recognise basic scientific terms, identify the real-world features represented by simple models, and follow explicit instructions to carry out a scientific procedure.
1c	186	0.6% (percentage of students scoring below Level 1b, PISA 2015)	3.1% (percentage of students scoring below Level 1b, PISA 2015)	At Level 1c, students can recall an element of everyday scientific information or observations of common macroscopic phenomena to identify a correct scientific explanation or conclusion which has been communicated using non-technical or non-academic language and supported by illustrations.

Note: Descriptors 3 through 6 are the same as those used in PISA 2015, while descriptors 2, 1a and 1b have been revised for a better progression in knowledge from 1c. While the description of Level 1c was added, it has not been populated with any items, so PISA-D will not report student results in this level. Level 1c and the progression in knowledge from 1c will be further developed in PISA 2024, when science will be the main domain.

Items at the newly created Level 1c should be familiar to students' everyday lives or draw on ideas that permeate contemporary culture. All items should, whenever possible, attempt to draw on macroscopic phenomena that students may have experienced or observed or learnt in the curriculum. Equally important is to have all items formulated in the simplest possible language. Sentences should be short and direct. Lengthy sentences, compound nouns and complex phrasing should be avoided. Vocabulary used in the items must be carefully examined to avoid the use of academic language and, wherever possible, simplify the scientific language. Wherever possible, the cognitive processing should only require one-step reasoning and use simple data or descriptions.

In order to enter Level 1c performance, a student must have the foundational skills required to:

- read and comprehend simple sentences
- use numeracy and basic computation
- understand the basic components of tables and graphs
- apply the basic procedures of scientific enquiry
- interpret simple data sets.

Testing scientific literacy in the out-of-school population

The scientific literacy domain is not included in the out-of-school PISA-D assessment due to practical reasons related with the instrument. On one hand, the total test allows a maximum of 50 minutes, which is not enough time to include an assessment of three domains, so it became necessary to choose only two. In deciding, it was taken into account that reading and mathematics literacy are considered as foundational skills and necessary for the development of scientific



literacy skills. In addition, the target population was also considered. Science is the domain with the strongest link to school, so the least appropriate for a group that by definition has been exposed to less formal schooling. Thus, it was decided that reading and mathematics were the only domains that should be included in the assessment for out-of-school 14-16 year-olds.

Examples of items for addressing the extended PISA-D science framework

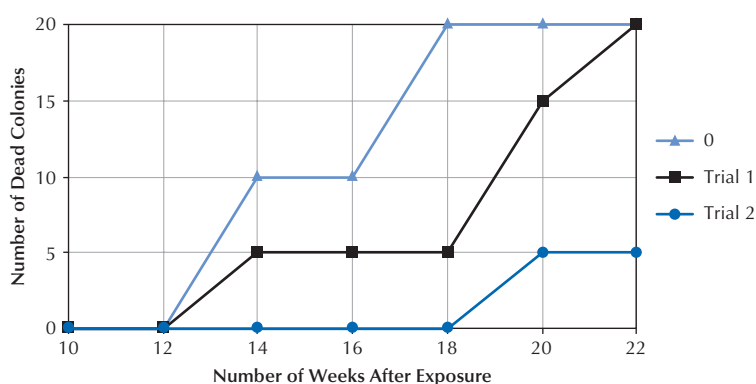
The following items illustrate the types of questions that can be asked of students at Level 1.

Sample item 1: Death of Bee Colonies – Level 1a

Scientists believe there are many causes why bee colonies die. One possibility is an insecticide that may cause bees to lose their sense of direction outside the hive.

Researchers tested whether this insecticide leads to the death of bee colonies. In a number of hives, they added the insecticide to the food of the bees for three weeks. All of the hives were given the same amount of food but the food had different amounts of insecticide in. Some hives were not given any insecticide.

None of the colonies died immediately. However, by week 14, some of the hives were empty. The following graph records the results:



What did the experiment test? Choose one of the responses below:

- A. The experiment tested the effect of insecticide on the resistance of bees over time.
- B. The experiment tested the effect of varying amounts of insecticide on the number of empty hives found over time.
- C. The experiment tested the effect of the death of bee colonies on the resistance of bees to insecticide.
- D. The experiment tested the effect of the death of bee colonies on the concentration of the insecticide.

Framework categories	2015 Framework extension
Competency	Evaluate and design scientific enquiry
Full description of competency	Students must identify a question being asked in a simple scientific enquiry where only one factor is being varied at a time
Knowledge	Evaluate and design scientific enquiry
Context	Local/national-environmental quality
Cognitive demand	Low
Item format	Multiple choice

Sample item 2: Fossil fuels – Level 1a

Many power plants burn fuel that gives off carbon dioxide. Adding more carbon dioxide into the air has a negative impact on the climate. There are different strategies to reduce the amount of carbon dioxide added to the air.

One such strategy is to burn biofuels instead of fossil fuels.

Another strategy involves trapping some of the carbon dioxide emitted by power plants and storing it deep underground or in the ocean. This strategy is called carbon capture.

Using biofuels does not have the same effect on levels of carbon dioxide in the air as using fossil fuels. Which of the statements below best explains why?

- A. Biofuels do not release carbon dioxide when they burn.
- B. Plants used for biofuels absorb carbon dioxide from the air as they grow.
- C. As they burn, biofuels take in carbon dioxide from the air.
- D. The carbon dioxide released by power plants using biofuels has different chemical properties than that released by power plants using fossil fuels.

Framework categories	2015 Framework extension
Competency	Explain phenomena scientifically
Full description of competency	Identify scientific concept appropriate to explain a phenomenon
Knowledge	Physical sciences
Context	Global/Natural resources
Cognitive demand	Low
Item format	Multiple choice

Sample item 3: Meteoroids and craters – Level 1b

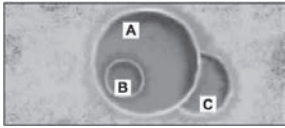
PISA 2015

Meteoroids and Craters

Question 3 / 3

Refer to "Meteoroids and Craters" on the right. Use drag and drop to answer the question.

Consider the following three craters.



Put the craters in order by the size of the meteoroids that caused them, from largest to smallest.

A

B

C

Largest
→
Smallest

Put the craters in order by when they were formed, from oldest to newest

A

B

C

Oldest
→
Newest

METEORIDS AND CRATERS

Rocks in space that enter Earth's atmosphere are called meteoroids. Meteoroids heat up, and glow as they fall through Earth's atmosphere. Most meteoroids burn up before they hit Earth's surface. When a meteoroid hits Earth it can make a hole called a crater.

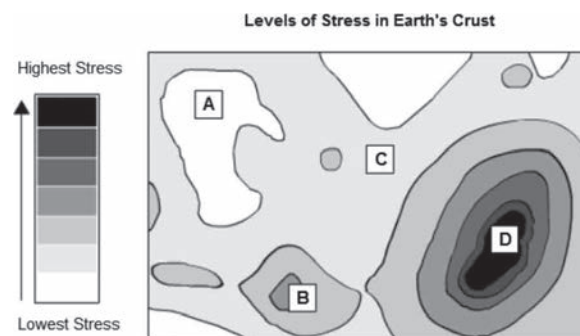




Framework categories	2015 Framework extension
Competency	Interpret data and evidence scientifically
Full description of competency	Given a simple set of observations, draw a correct inference
Knowledge	Earth and space sciences
Context	Global
Cognitive demand	Low
Item format	Multiple choice

Sample item 4: Groundwater extraction and earthquakes – Level 1b

GROUNDWATER EXTRACTION AND EARTHQUAKES Stress in Earth's Crust



The map above shows the levels of stress in Earth's crust in a region. Four locations within the region are identified as A, B, C, and D. Each location is on or near a fault that runs through the region.

Which of the following correctly rank risk of earthquake from lowest to highest? Choose one of the answers below:

- A. D, B, A, C
- B. A, C, B, D
- C. D, B, C, A
- D. A, D, C, B

Framework categories	2015 Framework extension
Competency	Interpret data and evidence scientifically
Full description of competency	Given a simple chart, identify which conclusion is correct
Knowledge	Procedural
Context	Local/National hazards
Cognitive demand	Low
Item format	Multiple choice



Notes

1. Because science was the main domain in PISA 2015 this was reported separately for the three systems (physical, living and earth and space). Since there are no subscales reported for PISA-D, the desired distribution for knowledge types is the total over all systems.

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5

PISA for Development Contextual Questionnaires Framework

This Chapter describes the framework and core content for the PISA for Development (PISA-D) contextual questionnaires, for both the school-based assessment and the out-of-school assessment. The Chapter presents the content and aims of the instruments for students who were in school and in Grade 7 or higher at the time of the assessment; who were in school but in a grade lower than Grade 7; and also for youths who were out of school. The Chapter also describes the teacher and school questionnaires that are used for the school-based assessment and the instruments used for the out-of-school population: a questionnaire for the parents or the person most knowledgeable about the youths, and a household observation questionnaire.



As noted in Chapter 1, the focus of the PISA contextual questionnaires is on understanding how measures of student performance at age 15 are related to various aspects of school and classroom practice as well as to other related factors, such as economic, social and cultural context. The PISA-D questionnaires include these aspects and also cover a broader set of well-being outcomes and risk and protective factors, taking into account differences in life experiences of children in middle- and low-income countries, both of those who are in school and those who are not.

The PISA-D questionnaire framework uses the Educational Prosperity model (Willms, 2015) as an over-arching framework. It incorporates lessons from other international studies, inputs from the participating countries and many elements from the PISA questionnaires.

A review of the experience of middle- and low-income countries participating in PISA 2000 to 2015 shows that the PISA questionnaires do not always capture the most relevant contextual factors for these countries. For example, questions about school infrastructure and teaching and learning materials are related to student performance in high-income countries, but are often unrelated to differences in performance in middle-income countries (Lockheed, Prokic-Breuer and Shadrova, 2015). In addition, the measure of economic, social and cultural status used by PISA does not adequately capture lower levels of parental education and income or the risk factors associated with poverty that are more frequent in low-income countries.

PISA-D enhances the contextual questionnaires to better measure factors that are more strongly related to student performance in middle- and low-income countries, while maintaining comparability with PISA on a set of core indicators. For example, the questionnaires collect more detailed data on students' language of instruction at school, language at home and their socio-economic status, as measured by home possessions and parents' education, literacy skills and participation in the labour force. The questionnaires also identify additional indicators of educational success beyond performance on the PISA test. These indicators comprise, for example, questions about educational attainment, health and well-being, and attitudes towards school and learning.

In addition to assessing student performance, PISA-D introduces an out-of-school assessment to collect data on youth who have not been eligible to sit the PISA school-based test. The out-of-school instruments gather much of the same data as the school-based instruments, as well as data on barriers to school attendance and factors that may impede students' progress through school.

The PISA-D school-based instruments include student, teacher and school questionnaires. In contrast, PISA distributes questionnaires to students and schools and offers countries four optional questionnaires, including a computer familiarity questionnaire, an educational career questionnaire, a parent questionnaire and a teacher questionnaire. The PISA-D instruments for the out-of-school population include a youth questionnaire (which is administered by an interviewer), a questionnaire for their parents or the person most knowledgeable about the youth, and a household observation questionnaire which is completed by the interviewer.

The Questionnaire Framework for PISA 2012 established the core questionnaire content that should be kept comparable across cycles to allow for continuous monitoring of education systems and identified four broad areas: outcomes, student background, teaching and learning processes, and school policies and educational governance (OECD, 2013). These four areas are all included in the PISA-D framework.

This chapter presents the framework for the PISA-D contextual questionnaires. The first section defines the core of the PISA-D contextual assessment, explaining 1) the Education Prosperity framework that shaped the enhancements made to the contextual questionnaires for PISA-D; 2) the approach to including the out-of-school population; and 3) the selection and organisation of the core content of the PISA-D instruments. The second section of this chapter explores the full breadth of policy issues to be covered, structured in 15 modules and one complementary category, and explains how the modules have been implemented in PISA-D. The second section also includes a comparison of the policy issues covered by PISA-D and PISA, highlighting similarities and differences.

DEFINING THE CORE OF CONTEXTUAL ASSESSMENT IN PISA-D

The PISA-D framework is an adapted version of the Education Prosperity approach. It takes into account the goals of PISA-D, lessons from past PISA cycles and other international studies, recommendations from research literature and the priorities of the participating countries. This overarching framework maintains that policy makers in middle- and low-income countries need to be informed principally on the Prosperity Outcomes, Foundations for Success and student-level demographic factors for monitoring performance of their education system and assessing equality and equity of outcomes that are described in this document. In addition, the questionnaires include several teacher, school and system-level



background measures that provide a context for the Prosperity Outcomes. The framework also proposes an approach for equality and equity for both the in-school and out-of-school populations.

This framework is discussed below, specifying the constructs and measures and providing arguments that support the choice of core content for PISA-D.

Educational Prosperity

The PISA-D questionnaire framework draws on the Educational Prosperity model (Willms, 2015), which follows a life-course approach to assessing children's outcomes, considering the effects of several factors over a student or youth's lifetime. The capacity of a society to develop young peoples' literacy skills and well-being depends on its ability to provide the right kinds of human and material resources to support healthy development from conception to childhood and beyond. Educational Prosperity refers to the success of the education system in developing children's cognitive skills and their social, emotional, physical and spiritual well-being. The term "prosperity" simply refers to the condition of experiencing success or thriving (Willms, 2015).

Educational Prosperity, as applied in PISA-D, considers development from conception to adolescence as the result of individuals' personal characteristics, their actions, their culture and the contexts in which they live (Mayer, 2009). It identifies a set of key outcomes, called "Prosperity Outcomes", for each of six stages of development, and a set of family, institutional and community factors, called "Foundations for Success", which drive these outcomes. The stages, which are described in Annex 5.A1, are prenatal, early development (ages 0 to 2), pre-primary (ages 3 to 5), early primary (ages 6 to 9), late primary and lower secondary (ages 10 to 15), and upper secondary (ages 16 to 18). Both the school-based and out-of-school components of PISA-D focus on the Prosperity Outcomes and the Foundations for Success for the fifth stage of the Educational Prosperity framework, while the out-of-school component also collects some data on earlier stages.

The approach has three explicit links to national and local policy and practice. First, it allows countries to align data collection with explicit goals at all levels of the system, from the minister and his or her staff to the front-line educators, students and parents. The challenge for countries is to maintain a focus on the alignment between data and policy goals. Second, the data collected with this approach has immediate implications for educational policies that involve decisions about the allocation of resources and its implications for equity. Countries will have reliable data on a wide set of Prosperity Outcomes as well as the foundation factors. With reliable data on differences across groups in outcomes and access to foundations, countries will be able to determine whether poor and marginalised populations are given equal opportunities to succeed at school and beyond. Third, the data collected will enable countries to set targets consistent with the SDG Education 2030 framework and monitor progress towards them. Many policy issues in middle- and low-income countries concern long-standing structural features of schools, such as the incidence of grade repetition or the choice of the language of instruction for minority groups. Making progress in reaching the SDG Education 2030 targets will require confronting these issues on the basis of solid evidence on how these structural features of education systems relate to the achievement and well-being of students on average and of specific groups at risk. Comparable data from other countries facing similar policy changes can also facilitate the identification of policy options that can yield the desired results. PISA-D provides an infrastructure for analysing relationships between trends in outcomes and policy changes. The descriptive evidence from PISA usefully complements experimental policy evaluations and more qualitative assessments of the implementation of policy reforms.

Prosperity Outcomes

The framework for PISA-D conceptualises success as something cumulative, emphasising that development at age 15 is a product of children's environments and experiences since birth. The PISA contextual questionnaires framework emphasises understanding how measures of student performance, attitudes and beliefs at age 15 are related to various aspects of student background and school and classroom practice. The PISA-D framework proposes a wider set of cognitive and non-cognitive outcomes and foundation factors to better measure the life experiences of in- and out-of-school of children in middle- and low-income countries. The Prosperity Outcomes include measures of academic performance, educational attainment, attitudes towards school and learning, and health and well-being.

The Educational Prosperity model was adapted to fit with the needs of the PISA-D participating countries, taking account of analysis of the results of middle- and low-income countries in PISA questionnaires, reviews of relevant international and regional studies and consultation with representatives of the participating countries.



The Educational Prosperity model distinguishes four processes that determine how success accumulates from one stage to the next: biological embedding, Foundations for Success, cumulative development and institutional selection. The Foundations for Success are described below, while a description of the other processes and the effects associated with each of the processes along the Educational Prosperity pathway are available in Annex 5.A1.

Foundations for Success

The Foundations for Success are factors that affect children's outcomes at each stage of development. For example, from age 2 to age 5, children's development is affected by parents' engagement with the child and intra-family relations, as well as by the quality of care at home and in early childhood centres. They are considered to be universal in that they are necessary conditions for success at each stage of development. The selection of the foundation factors was based on theory and a large body of research that provides evidence of the effects of each factor on student outcomes.

Three additional criteria were considered in determining which factors to include as Foundations for Success: the factors must be potent, proximal *and* pervasive. A "potent" factor is one that has a strong correlation with an outcome or set of outcomes. For example, the quality of classroom instruction is arguably the most important driver of student outcomes during the schooling period (Anderson, 2004; Rosenshine, 2010; Kyriakides, Christoforou and Charalambous, 2013; Creemers and Kyriakides, 2006).

A "proximal" factor is close to the outcome in the sense that its relationship with the outcome is not mediated through some other factor. For example, the quality of classroom instruction has a direct, positive relationship on student outcomes, without any intervening factors. "Principal leadership" is also an important factor and several studies have shown that it is correlated with student outcomes. However, it is not proximal because the "effects" of principal leadership are mediated through the school-related foundations factors, namely inclusive context, quality instruction, learning time and material resources. Thus, a jurisdiction may allocate resources to improving principal leadership, but this would only result in improved outcomes if it leads to improvements in quality instruction, increased learning time and so on.

A "pervasive" factor is positively correlated with a wide range of outcomes, although the strength of the correlation may vary with each outcome. For example, the effects associated with an "inclusive school context" not only affect student's academic performance, but also their educational attainment, their health and well-being, and their social, institutional and intellectual engagement.

Equality and equity

The terms "equality" and "equity" have been used by researchers and policy makers to denote several different concepts. These include, for example, the achievement gap between low- and high-status groups, differences in access to schooling, and the segregation of students into different types of schools and school programmes. Willms (2011) argued in the OECD's *Education at a Glance* (OECD, 2011), that equality and equity should be defined as separate concepts and measured with a consistent approach.

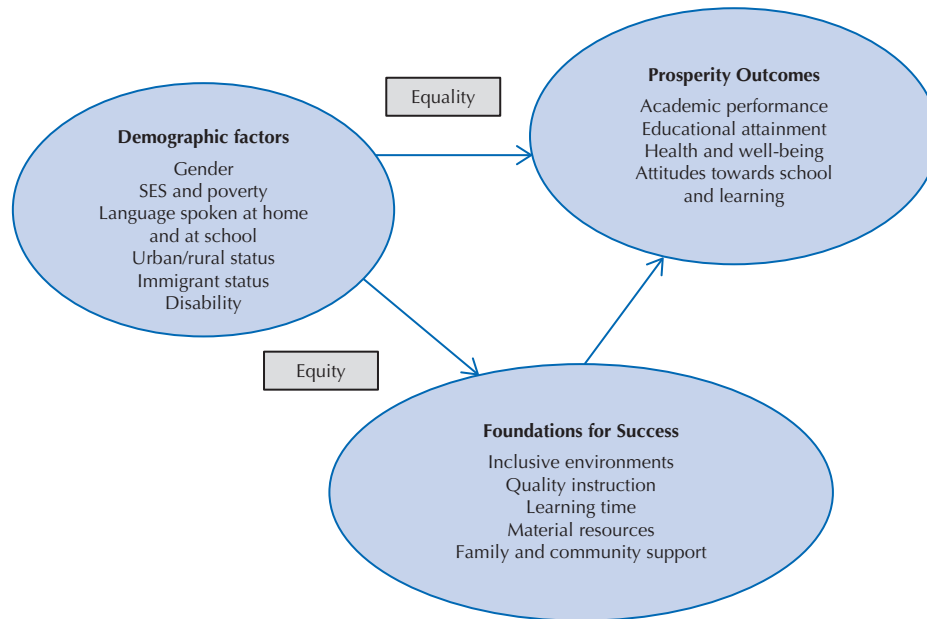
PISA-D defines inequality as differences among sub-populations in the distribution of their educational outcomes, while the measure of equity, a normative concept, requires also an assessment of fairness based on the observed differences among sub-populations in their access to the resources and schooling processes that affect schooling outcomes. Equality is therefore measured by the differences among groups in the distribution of Prosperity Outcomes, which are performance, attainment, health and well-being, and attitudes towards school and learning. Equity, on the other hand, also has to do with ensuring that all children benefit in the same way from school, and requires measures of whether children from different groups have fair access to the five Foundations for Success, which are inclusive environments, quality instruction, learning time, material resources, and family and community support. Unfair access to the foundation factors increases inequalities in outcomes. For example, not providing the right level of support to disadvantaged children to attend schools regularly will inevitably result in socio-economic inequalities in attainment and performance.

If we consider equality and equity in reading performance for students from differing socio-economic backgrounds, for example, equality is assessed by examining the relationship between reading performance and socio-economic status (SES), while equity is assessed by also examining the relationship between SES and the foundation factors that are considered core to learning how to read.



The PISA-D questionnaires collect information on several demographic factors that impact equality and equity and are relevant to both the in- and out-of-school populations. The framework focuses in particular on gender, SES and poverty, language, urban/rural status, immigrant background and disability. This model is characterised in Figure 5.1 (modified from Willms et al., 2012).

Figure 5.1 ■ **A model for assessing equality and equity**



Source: Modified from Willms, J.D. et al. (2012), "Assessing Educational Equality and Equity with Large-Scale Assessment Data: Brazil as a case study", Technical Notes No. IDB-TN-389, Inter-American Development Bank.

When discussing equity it is important to always consider fairness and equality together with the need for quality. For example, a risk of policies focusing on equality without quality would be an education system where students from all social backgrounds have access to equally low quality education and perform equally poorly.

Assessing the out-of-school population

The first benchmark to assess equality and equity is whether all children are given the same opportunities to be in school and stay in school to acquire the skills they need for life. One of the unique features of PISA-D is that it gathers information on how many youths are in school at ages 14 to 16 and the reasons why some youths have left school at that age. It also allows for the combination of data for the in-school and out-of-school populations.

PISA is aimed at 15-year-old students that are in Grade 7 or above, leaving a large population of middle- and low-income countries out of the assessment. PISA-D incorporates these youth in the assessment through the out-of-school component which is conducted through a household survey. This component assesses 14-16 year-olds that are either not in school or in school but in Grade 6 or below, which represents approximately one-third of youth in the participating countries. Through its two components PISA-D includes students who are in school (those in Grade 7 or above through the school-based component, those who are in Grade 6 or below through the out-of-school component) and youth who are not in school. The youth who are out of school include those that have never enrolled and those with some schooling, ranging from a few months to several years.



Box 5.1 Definition of access to schooling

The term “access” in education generally refers to whether schooling is freely available to children in a jurisdiction. The emphasis is on the *provision* of schooling, and it is incumbent upon governments and educational institutions to ensure that schools are available locally and that educational policies do not create barriers for attending school. In practical terms, however, access is gauged simply by measures of school attendance (e.g. UIS, 2006). This approach takes into account not only the supply of schooling, but also the cultural, social, religious, political and economic factors that affect the demand for schooling. In striving to improve school attendance, several governments have turned to demand-side initiatives, such as providing free meals, cash transfers to families which are conditional on their child’s attendance, and vouchers designed to increase school choice (Patrinos, 2007). Some definitions of access also incorporate the *quality* of school provision and in some cases are attached to a desired outcome. For example, the UN Sustainable Development Goal 4.1 states: “By 2030, ensure that all girls and boys complete free, equitable and quality primary and secondary education leading to relevant and effective learning outcomes” (UN, 2016). The statement calls not only for equal opportunities to attend school, but also equality of outcomes (relevant and effective learning outcomes) and equity of school provision (quality primary and secondary education).

The Educational Prosperity model and the approach taken in PISA-D identify two types of access: access as an outcome, which depends on both demand and supply and measured by attainment and learning outcomes; and access as a condition for success, which depends on supply and is measured by the Foundations for Success related to the quality of schooling.

The PISA-D framework includes four key aspects of school quality: inclusive environments, quality instruction, learning time and material resources. A pre-requisite to benefit from all school-related Foundations for Success is to be in school, therefore, access pertains to equity: do children from various sub-populations differ in their access to inclusive environments, quality instruction, learning time and material resources?

The PISA-D framework also includes academic performance and educational attainment, which refers to the extent to which children participate in school at various stages of their life-course. Thus, access also has to do with equality: do children from various sub-populations differ in their distribution of attainment and performance?

PISA-D also includes measures pertaining to the barriers to attending school for out-of-school children, which can help discern the extent to which access is predominantly a supply- or demand-side issue for each country.

In general, out-of-school youth tend to be poorer than those attending school – many of them are in the lowest quintile. They are mainly from rural settings, and more likely to be girls. Youth with disabilities and those belonging to minority ethnic, linguistic or religious groups are also more likely to be out-of-school. All of these factors are usually confounded with poverty (Carr-Hill, 2015).

PISA-D’s approach to measuring economic, social and cultural status (ESCS) and poverty is especially important for the out-of-school population, as poverty is one of the main reasons, if not the main reason, for being out of school. Both the students and out-of-school youth are asked an extensive set of questions relevant to poverty and language spoken at home. The youth questionnaire also includes several questions relevant to the demand for schooling, including questions about their work experience and support for their family as well as questions concerning perceived barriers to schooling.

The life-course approach assumes children’s attainment is determined by various events and family circumstances that begin at conception and continue through to age 15. The questionnaire completed by the person most knowledgeable about the youth asks about some elements of their early life-course foundations, such as the nutrition and health of the biological mother during pregnancy, and the engagement of the family during the preschool years. This provides data about the first four stages of development relevant to the accumulated effects of these factors on school attendance at ages 14 to 16.



Selecting and organising the core content

The instruments

The questionnaires for the school students include: a student questionnaire with 49 questions, a teacher questionnaire with 33 questions administered to the majority of teachers in each school, and a school questionnaire with 28 questions administered to the school principal.

The questionnaires for the out-of-school youth include: a questionnaire for the youth with 77 questions that is administered by an interviewer, a questionnaire for the person most knowledgeable about the youth with 19 questions, and a household observation questionnaire completed by the interviewer with 14 questions.

The distribution of questions across the elements of the Educational Prosperity model is shown in Figure 5.2, with black stars indicating questions that can be linked with PISA 2015 and blue stars indicating questions that are new to PISA-D.

Figure 5.2 ■ Questions in the PISA-D contextual questionnaires

	School-based assessment			Out-of-school assessment		
	Student	Teacher	School	Youth	Person most knowledgeable about the youth	Household
Prosperity Outcomes						
Educational attainment	****			***** *****	***	
Health and well-being	****			****		
Attitudes towards school and learning	*			**	*	
Foundations for Success						
Inclusive environments	****	**	****	***		
Quality instruction	***	*	*			
Learning time	*****	*	**			
Material resources		*****	*****			
Family and community support	*	*	**	*	*	
Demographic factors to assess equity and equality						
Gender	*			*		
Socio-economic status and poverty	***** ***** **		**	***** ***** ***** ***** **	*****	*****
Language spoken at home	****	***		****		
Urban/rural status			*			
Immigrant status	*			***		
Disability	*			**		
Context factors		***** *****	***** ****		*****	*****
Total	49	33	28	77	19	14

Note: Black stars indicate questions that come from the main PISA assessment; blue stars indicate questions new to PISA-D.

The questions for the PISA-D questionnaires were drawn mainly from PISA, and complemented with questions from other international and regional studies or developed in consultation with the PISA-D participating countries. The criteria for selecting and developing items included: their fit with the Educational Prosperity model; relevance, as confirmed through analysis of the results of middle- and low-income countries in PISA questionnaires; reviews of relevant international and regional studies; consultation with representatives of the participating countries; and maintaining links with PISA 2015.



All the items in the questionnaires were tested in a field trial. Questions were not retained for the main study if their psychometric properties (e.g. reliability, unidimensionality, completion of items and consistency across cultures) were inadequate. When there were two versions of a particular question, only one question was usually retained. To be retained for the main study, questions also had to meet at least one of the following conditions:

1. Relevant to the measurement of ESCS common to PISA 2015 or new measures required to extend the scale to lower values of ESCS and to collect information on poverty
2. Required for a measure of material resources
3. Relevant for comparability with PISA 2015
4. Required for coverage of all domains of the Educational Prosperity framework
5. Relevant to the classification of students into the six key sub-populations.

CORE CONTENT FOR ASSESSING EDUCATIONAL PROSPERITY

As noted above, the conceptual framework for the PISA-D questionnaires includes 15 modules. These include four Prosperity Outcomes, five Foundations for Success, and six student demographic background factors relevant to assessing equality and equity (see Figure 5.2 above). It also includes a set of teacher, school and system-level questions grouped under the category of context information that complement the variables included in the Educational Prosperity approach.

The content of these modules is discussed below.

Assessing Prosperity Outcomes

As noted above, Prosperity Outcomes include: academic performance, educational attainment, health and well-being, and attitudes towards school and learning. These are each described briefly below.

Academic performance

The measures of academic performance in PISA-D are based on the assessments of performance in reading, mathematics and science. The frameworks for these assessments are described in Chapters 2, 3 and 4.

Educational attainment

Educational attainment – how far students go in school – is a key outcome for middle- and low-income countries that sits alongside measures of academic performance. Many of the key policy questions of middle- and low-income countries pertain to students' and families' demand for education, which depends on students' early learning experiences and their perceptions of its relevance, quality and long-term benefits. In many middle- and low-income countries, students do not attend school beyond the primary level.

A salient feature of middle- and low-income countries is that the distribution of 15-year-old students stretches well below Grades 9 or 10. Another salient feature is that even though formally education might be compulsory, a large proportion of 15-year-old youth have dropped out. The primary aim in measuring attainment is to gain a better understanding of students' pathways to their current level of attainment and to also understand the reason for abandoning school, when this has happened.

Information about attainment is collected through all PISA-D questionnaires except for the teacher questionnaire and the household observations questionnaire, which is answered by the out-of-school youth interviewer. Like PISA, the PISA-D student and out-of-school youth questionnaires include questions about grade, grade repetition and early childhood education attendance (this is asked in the parent's questionnaire for the out-of-school component); and the school questionnaire asks about grade retention policies and academic support services. PISA-D further investigates the experience of students and out-of-school youth with questions about long-term absenteeism and the reasons for it, such as being sick or having to look after others. Also unique to PISA-D, the parent questionnaire asks about factors that could hinder the youth's completion of compulsory education.¹

The approach used by PISA-D to assess educational attainment is inspired by the framework set out by UIS and UNICEF (2005), which have been used to characterise the entire school-age population. In PISA-D they will be used to describe the levels of attainment of age 15 youth who are in school and 14-16 year-olds who are out of school.



Data from students' current grade level, or in the case of out-of-school students the last grade completed, as well as data on students' birthdate, will be used to construct an ordinal variable describing five levels of attainment (the first three levels are used for the in-school component and the last two for the out-of-school component):

1. **On-track.** Students are in their expected grade, given their birthdate; that is, they started school on schedule and have not repeated a grade. In most cases, this would be Grades 9 or 10.
2. **One year below expected grade.** These students have usually repeated a grade or were out of school for a prolonged period. They would typically be in Grades 8 or 9.
3. **Two or three years below expected grade.** In most cases, these students have repeated two or three grades, but some may have started late or simply faded in and out of school for a year. They would typically be in Grades 7 or 8.
4. **Enrolled in school but are four or five grades below the expected grade.** In most cases, these students will have repeated more than three times, but some may have started late or simply faded in and out of school for one year or more. They would typically be in Grades 5 or 6.
5. **Not attending school.** These students are not attending school, and the highest grade they attained will be described.

Health and well-being

The concept of well-being is very broad, and typically refers to the quality of people's life. Diener (2006) defines subjective well-being as "an umbrella term for the different valuations people make regarding their lives, the events happening to them, their bodies and minds, and the circumstances in which they live" (p. 400). PISA 2015 uses the following definition of well-being that extends beyond students' subjective appraisal of their life quality: "Students' well-being refers to the psychological, cognitive, social and physical functioning and capabilities that students need to live a happy and fulfilling life" (OECD, 2017).

This module is based on the New South Wales Department of Education and Communities framework for student well-being (New South Wales Department of Education and Communities, 2015) that considers the following five domains: emotional, physical, social, cognitive and spiritual well-being. The health and well-being module focuses on the first two of these domains, while social and cognitive well-being are included in other modules. Spiritual well-being is not included in the PISA-D framework.

Emotional well-being is the affective component of well-being – people's reactions to their experiences. It can be positive, such as people's overall rating of their happiness as used in the World Happiness Report (Helliwell, Layard and Sachs, 2012); or negative, such as people's feelings of anxiety, depression or fear. As in PISA, the PISA-D student and out-of-school youth questionnaires ask about general life satisfaction. PISA-D includes measures of anxiety and depression and questions about their physical and mental health during the past year.

Children's physical health is the key element of physical well-being. It is particularly important in middle- and low-income countries, as children's health in these countries is more often compromised in ways that affect their educational outcomes – due to hunger; physical and emotional abuse; chronic illnesses such as asthma, bronchitis, diabetes or epilepsy; and acute illnesses that cause children to miss school and fall behind. While PISA collects information on students' nutrition and physical activity, PISA-D asks about respondents' overall perception of their health and about their mental health during the past year. The parental questionnaire for the out-of-school youth also asks about the youth's prenatal and early experiences; about the mother's health during pregnancy, the conditions and any complications of the mother's birth experience, how the youth was fed during the first six months of life, and whether the youth had any health problems during the first five years of life.

Social well-being pertains to students' sense of belonging and their connectedness to others. In this framework, it is covered by elements of the attitudes towards school and learning and inclusive environment modules, primarily focusing on students' sense of belonging at school and their connectedness to others at school.

One of the elements of cognitive well-being is academic performance. In this framework this is covered as a Prosperity Outcome in its own right.



Box 5.2 Well-being in PISA 2015 and PISA-D

According to the Framework for the Analysis of Student Well-Being in the PISA 2015 Study (Borgonovi and Pál, 2016), the five dimensions of students' well-being captured in PISA 2015 are:

Cognitive well-being: The cognitive dimension of students' well-being refers to the skills and foundations students have to participate effectively in today's society, as lifelong learners, effective workers and engaged citizens. It comprises students' proficiency in academic subjects, their ability to collaborate with others to solve problems and their sense of mastery of school subjects. In PISA 2015 it is assessed as the level of subject-specific skills and competencies students have acquired, measured through the PISA tests; and their self-belief in those subjects, measured through the questionnaires.

In PISA-D this is assessed through the tests and included in the academic performance module, which is considered a Prosperity Outcome.

Psychological well-being: The psychological dimension of students' well-being includes students' evaluations and views about life, their engagement with school, and the goals and ambitions they have for their future. In PISA 2015 it includes students' self-reported psychological functioning, and covers life satisfaction – students' overall evaluation about life in general – and three aspects of education related to psychological functioning: 1) the goal setting and emotions related to *students' career and educational expectations*; 2) *achievement motivation* related to students' appreciation of the educational opportunities they have, an engagement with learning and an interest in acquiring knowledge; and 3) *test and learning anxiety*.

As in PISA, the PISA-D student and out-of-school youth questionnaires ask about general life satisfaction and PISA-D includes a measure of emotional distress (severe anxiety and depression) and questions about their physical and mental health during the past year in the health and well-being module. Concerning education related elements, PISA-D explores educational attainment in greater depth than PISA by asking students and out-of-school youth about long-term absenteeism and the reasons for it. PISA-D further investigates the experience of out-of-school youth with questions about whether they work, their profession, hours worked per week, and wage or salary. The out-of-school component also gathers information about youth's engagement in reading and writing literacy activities, such as how often they read a newspaper, magazine or book, write a text or email, etc.

Physical well-being: The physical dimension of students' well-being refers to students' health status, engagement in physical exercise and the adoption of healthy eating habits. PISA 2015 covers two aspects of students' lifestyles: physical activity and eating habits.

PISA-D focuses on physical health and assesses it as part of the module on health and well-being.

Social well-being: The social dimension of students' well-being refers to the quality of their social lives including their relationship with their family, their peers and their teachers (positive or negative), and how they perceive their social life in school (positive or negative), and how they perceive their social life in school. PISA 2015 measures students' sense of belonging at school and their relationships with their teachers, their peers and their parents.

PISA-D assesses social well-being within the attitudes towards school and learning and inclusive environment modules, primarily focusing on students' sense of belonging at school and their connectedness to others at school.

Material well-being: Material resources make it possible for families to better provide for their children's needs and for schools to support students' learning and healthy development. Households who live in poverty find it difficult to ensure that their children have access to the educational and cultural resources they need to thrive in school and to realise their potential. Research shows a strong link between material well-being in childhood and different dimensions of well-being in adulthood. Providing adequate resources to children is important, not only because it is a pre-requisite for successful development, but also because teenagers in poverty do not have adequate living and learning conditions to fulfil their personal goals. PISA contains a rich set of information on the types of resources students have at home and, most importantly, at school: human resources, material resources and extracurricular activities.



Box 5.2 [continued] **Well-being in PISA 2015 and PISA-D**

PISA-D assesses material well-being through the socio-economic status and poverty module, and also through questions on schools' material resources. In both cases it creates new questions to better address the needs of middle-income countries.

The PISA 2015 questionnaire design does not attempt to clearly articulate and identify input and outcome indicators for the five well-being dimensions, so some dimensions focus on well-being inputs and others on outcomes, without an integrated measurement approach. This is coherent with PISA-D classifying some of the PISA 2015 well-being factors as Prosperity Outcomes, others as Foundations for Success, and others as demographic factors related with equity and equality.

Attitudes towards school and learning

The PISA studies have examined students' interest and motivation in reading, mathematics and science, and their participation in activities related to the subject. For example, the OECD report, *Reading for Change: Performance and Engagement across Countries* examined students' motivation and interest in reading, and the time students spend reading for pleasure and reading diverse material (OECD, 2002). PISA has also considered "engagement" more broadly, to refer to students' attitudes towards schooling and their participation in school activities (Willms, 2003).

Several studies have considered engagement to be a predictor of educational performance and attainment, and there is strong evidence that engagement is correlated with both performance and attainment (Willms, 2003). However, in PISA-D it is considered an important outcome in its own right, a Prosperity Outcome alongside performance and attainment. A strong case can be made that the direction of causation is reversed (from performance to engagement) at certain stages of the school system. For example, children who do not make a successful transition from learning-to-read to reading-to-learn are more likely to become disaffected with school during the late primary and lower secondary years. Moreover, engagement is seen "as a *disposition* towards learning, working with others and functioning in a social institution" (Willms, 2003, p. 8), and as such is a key Prosperity Outcome that leads to lifelong learning and the likelihood of becoming a productive member of society.

Like PISA, the PISA-D student and out-of-school youth questionnaires include a measure of institutional engagement, providing information on general attitudes towards school and learning outcomes, as well as attitudes towards learning activities. Out-of-school youth are asked about attitudes towards school and learning based on their experience when attending school; and their parents are asked about their attitudes towards education as well.

The out-of-school component also gathers information about youth's engagement in reading and writing literacy activities, such as how often they read a newspaper, magazine or book, write a text or email, etc.

Assessing the Foundations for Success

As noted earlier in this chapter, the Foundations for Success are factors that affect children's outcomes at each stage of development. They are considered to be universal in that they are necessary conditions for success at each stage of development. The selection of the foundation factors for PISA-D was based on theory and a large body of research that provides evidence of the effects of each factor on student outcomes. The factors selected for PISA-D are: inclusive environments, quality instruction, learning time, material resources, and family and community support. These factors are each described briefly below. Some of the elements included in each factor are core to the factor, while other elements can be considered as supporting content.

Inclusive environments

Inclusive environments are classrooms, schools and broader communities that value and support inclusion. "Inclusion is a process of addressing and responding to the diversity of needs of all learners through increasing participation in learning, cultures and communities, and reducing exclusion within and from education. It involves changes and modifications in content, approaches, structures and strategies, with a common vision which covers all children of the appropriate age range and a conviction that it is the responsibility of the regular system to educate all children" (UNESCO, 2005, p. 13). UNESCO's (2009) policy guidelines provide a schema for measuring aspects of inclusion relevant to teachers' and principals' attitudes and values.



Inclusive environments are places in which all students can succeed. *All* means learners across the categorical boundaries of disability, social class, gender, ethnicity, national origin, sexual orientation and religion. *Succeed* means succeeding in terms of learning, as well as in terms of physical, social, emotional and spiritual outcomes (Willms, 2009). The provision of inclusive environments is a foundation for Educational Prosperity in middle- and low-income countries as it concerns the opportunities for children with disabilities; children from ethnic, linguistic and religious minorities; and girls to have equal access to schooling and a complete school experience, including opportunities to learn, engage in the social life of the school, and feel accepted by their peers and teachers.

Inclusive classroom and school practices affect students' sense of belonging at school, their participation in the social life of the school and their opportunities to learn. A reason for abandoning school is negative attitudes and responses to diversity as well as a failure to provide necessary accommodations to meet special learning needs. The metrics also need to capture the attitudes and practices of teachers and principals. Inclusion requires teachers to be ambassadors for inclusion in their community by embracing and celebrating diversity, becoming skilled at meeting the needs of students with special needs and using new approaches to assessing learning (Riehl, 2000). At the system level, inclusion is concerned with the extent to which students from different sub-populations or ability are segregated into different schools or school programmes.

For the school-based component, PISA-D collects information on inclusion from students, teachers and school principals. For the out-of-school component, it collects this information from the youth questionnaire, asking youth to describe their experience when they attended school.

As in PISA, PISA-D asks students to report on their sense of belonging at school. PISA-D further explores school climate with questions to students about the safety of their school, whether they feel safe at school and whether they have been sexually harassed at school; out-of-school youth are asked these same questions based on their experiences when they attended school.

PISA-D asks teachers a set of questions about their attitudes and practices towards teaching students with low literacy levels. Both PISA and PISA-D ask school principals about school policies concerning how students are admitted to the school and grouped for instruction as well as about the diversity of the school. PISA-D also asks about their attitude towards grade retention.

Quality instruction

Quality of instruction is the most important driver of student performance, but arguably the most difficult to define and measure. Anderson (2004) defined effective teachers as “those who achieve the goals they set for themselves or which they have set for them by others (e.g. ministries of education, legislators and other government officials, school principals).” (p. 22). His model assumes that teachers are aware of, understand and actively pursue goals; that they teach with a purpose – to facilitate learning – material which they consider worthwhile; and that their goals are concerned directly or indirectly with student learning. This perspective, that effective teachers are goal-oriented, is evident in virtually all of the contemporary models of effective instruction (Coe et al., 2014).

The “delivery of the lesson” and “interacting with students” are at the centre of Anderson’s (2004) conceptual framework of teacher effectiveness. Four other elements of his framework – standards, learning units, classroom climate and culture, and classroom organisation and management – have effects that are mediated through lesson delivery and teacher-student interactions. All six elements have direct effects on student learning and engagement. Teachers’ characteristics – including their professional knowledge, expectations and leadership, and students’ characteristics, including their aptitudes, prior knowledge, and attitudes and values – are positioned behind the six core elements of the framework. In the language of Educational Prosperity presented above, they are distal factors that have their effects through the proximal core elements, and thus are included as contextual factors and not as a foundation for success in the PISA-D framework.

This module is assessed through questions to students, teachers and school principals and is not assessed for the out-of-school component. Similar to PISA, the PISA-D student questionnaire includes measures on student-teacher interactions and assesses the classroom learning climate. PISA-D adds new questions on lesson delivery to gather information on the structure of lessons and teaching practices in mathematics lessons. PISA-D adds questions about their practices for teaching less able students. PISA-D also asks school principals about teachers’ behaviours that could negatively impact on classroom climate.



Learning time

Learning time in middle- and low-income countries differs from that of high-income countries in several ways. In many cases, children of middle- and low-income families start school at a later age, they miss many days of school during the primary school period, and they are more likely to repeat grades. Many children work in part-time jobs outside the home from an early age. Moreover, there appears to be considerable variation in class time devoted to the three core PISA subjects, and curriculum coverage is not as deep. How learning time is measured in main PISA has changed through the cycles.

The school-based component of PISA-D captures learning time in and out of school. Similar to PISA, the PISA-D student questionnaire asks about reasons for loss in learning time due to student truancy. PISA-D also asks students for information on reasons for reduced teaching time. PISA-D asks teachers about the reasons they are absent and school principals about their policies regarding teacher absenteeism. PISA-D also asks school principals about the reasons for and amount of instructional time lost during the last year, as does PISA. This module is not assessed for the out-of-school component.

Material resources

Studies based on the Latin American Laboratory for Assessment of the Quality of Education (LLECE) data by Murillo and Román (2011) and Willms and Somers (2001) suggest that in middle- and low-income countries school resources have substantial effects, even after taking into account the socio-economic characteristics of students. PISA-D's school-based component used a schema set out by Murillo and Román (2011), which distinguishes between basic services, didactic facilities and didactic resources:

- *Basic services* at the school include factors such as potable water, sewage services, bathrooms, electricity and telephones.
- *Didactic facilities* refer to places other than the classroom for teaching and learning. These include, for example, school libraries, gymnasiums, art and music rooms, science laboratories, computer rooms and sports fields.
- *Didactic resources* can include very basic materials, such as textbooks and blackboards, through to computers in the school, laptop computers for students and teachers, and quality books in the library.

Whereas PISA asks about principals' perceptions of school resources (lack of or inadequate physical infrastructure and educational material) and collects information on the availability of information and communication technology resources and Internet connectivity, PISA-D questions to school principals focus on the availability and condition of specific elements of school infrastructure (such as walls, windows, etc.) and facilities (such as toilets, classrooms, etc.) as well as the availability of textbooks. The questions in PISA-D also distinguish between the availability of didactic facilities and resources and teachers' use of didactic facilities and resources.

Family and community support

For children, few relationships in life are as significant and enduring as the relationship with their parents or the adults who raised them. The nature and extent of family and community support differs among countries, not only because of cultural differences, but also due to the large number of children living in poverty in many of the partner countries. But families – whether small, nuclear families, or extended families – invariably are the first social unit in which children learn and develop. And while good parenting can take different forms and be shaped by various social and cultural forces, it always involves providing children with the support, care, love, guidance and protection that set the conditions for healthy physical, mental and social development.

PISA questionnaires include questions pertaining to family's static cultural capital, about the relations between parents and children, and between parents and other parents.

In consultations with countries participating in PISA-D there was a demand for questions about community support. Small and Newman (2001) describe two over-arching connections between community and families that are relevant for developing measures of community support. One considers the socialisation of children, with neighbourhoods moulding children into certain behavioural patterns. The other pertains to the access of resources that support parents in raising their children. This could include, for example, literacy programmes, recreation facilities and programmes, or interventions to combat drug use and violence. In PISA there have been few questions traditionally about the communities or neighbourhoods of students, though PISA 2015 includes a question to school principals about whether they identify and integrate resources from the community to strengthen school programmes, family practices, and student learning and development.



For PISA-D, the community comprises local neighbourhoods nested within the school catchment area, which is defined with geographic boundaries or by the area from which the school draws its students.

PISA-D asks both students and out-of-school youth about the types of communication they have with their parents or someone in their family, whereas PISA asks about the student-parent relationship in terms of emotional warmth, stimulation, etc.

PISA-D asks teachers about families' involvement at school and asks school principals about how parent and community members or organisations contribute to the school, whereas PISA gathers information about school policies for parental involvement. PISA-D also asks parents of out-of-school youth about the type of support they provided to the youth in their early years, while PISA countries that distribute the optional parent questionnaire ask parents of PISA students about the support they provided to their children at the beginning of primary education and at age 15. PISA also asks parents about their participation in school-related activities and whether there are factors that have prevented them from participating in the activities.

Student-level demographic factors for assessing equality and equity

PISA-D focuses on the following measures pertaining to students' and families' backgrounds that are particularly relevant for low- and middle-income countries: gender, socio-economic status and poverty, the language spoken at home and the language of instruction at school, urban/rural status, immigrant status and disability. Though ethnicity is a variable related with being out-of-school, it was not included as one of the five demographic factors because it is embedded within poverty, immigrant status, language spoken at home and language of instruction.

Gender

Like PISA, the PISA-D question on gender simply asks students and youth whether they are male or female. PISA-D does not capture data about gender identity or sexual orientation.

Socio-economic status and poverty

Socio-economic status (SES) refers to the position of a person or family in a hierarchical social structure, based on their access to, or control over, wealth, prestige and power (Mueller and Parcel, 1981; Dutton and Levine, 1989). Numerous studies have shown that a person's position on an SES hierarchy is related to a wide range of outcomes pertaining to their physical, economic and social well-being. SES affects people's educational opportunities, their access to certain labour markets, their exposure to health and crime risks, and their lifetime earnings.

The literature on child development in middle- and low-income countries focuses mainly on the risk factors associated with poverty, especially during the prenatal period and the early years. These include, for example, poor nutrition during pregnancy, a lack of stimulation during the early years, and stressful living conditions.

The measure of SES in PISA, called the index of economic, social and cultural status (ESCS) does not adequately capture lower levels of education and lower levels of income and wealth for the majority of students in middle- and low-income countries. PISA-D extends this measure to lower levels of SES, keeping the link with the PISA measure. PISA-D also gathers specific information on poverty, and explores the possibility of developing a separate measure of it relevant to Educational Prosperity.

The PISA-D student and out-of-school youth questionnaires include a large number of questions pertaining to family SES and poverty, including the long-standing questions used in PISA which assess the highest educational level of the parents, the highest occupational status of parents, and an index of home possessions, which has been adjusted to middle- and low-income countries. The questionnaires also include new questions designed to capture youth's experience of poverty, including questions about material possessions, parents' education and engagement in literacy activities, and more detailed information about their parents' occupation. PISA-D further investigates the experience of out-of-school youth with questions about whether they work, their profession, hours worked per week, and wage or salary. Information about school meals is collected through the PISA-D school questionnaire.

Poverty is expected to be a fundamental characteristic of the out-of-school population. Unique to PISA-D, parents provide information about the out-of-school youth's food security during the youth's first two years of life and whether the youth has received any government support for schooling. The household observation questionnaire includes questions about the youth's housing and neighbourhood.



Language spoken at home and language of instruction

In several middle- and low-income countries, the students' first language differs from the language of instruction at school. Also, in some countries, the language of instruction during the primary grades, when children are learning to read, differs from the official language of instruction after the third or fourth grade. A further issue, which is more difficult to capture with a survey, is that in some schools, the teachers use the students' native language, or some combination of the native language and the official language of instruction.

PISA asks students, "What language do you speak at home most of the time?" This construct is extended for PISA-D in both the school-based and out-of-school components to include questions about the language of instruction at school and the language they first learnt to read. In addition, teachers are asked about which language they use during their lessons, as well as which language they use when talking with students.

Urban/rural status

The school questionnaire includes a variable pertaining to the size of the community, which can be used to determine the school's rural status. Living in a rural area versus a larger community is sometimes confounded with other student-level demographic factors and the analyses will enable to discern this. For the out-of-school component the urban/rural status information was collected during the sampling process using each country's administrative classification of regions as urban or rural.

Immigrant status

The measure of immigrant status is based on a long-standing approach used in PISA which is based on questions of students and youth about where they and their parents were born.

Disability

PISA-D is the first PISA study to include self-reported measures pertaining to disability. The questions follow contemporary approaches to disability, which emphasise the extent to which a disability limits people in doing certain activities in a particular environment. For example, students are asked about whether a disability limited their participation in school activities, while out-of-school youth are asked about whether they have a disability or medical condition that limits their daily activities. Out-of-school youth who report having a disability are also asked about the nature of the disability.

Context factors

The school and teacher questionnaires of the in-school component of PISA-D also gather data on other teacher, school and system-level background variables that are expected to help explain student outcomes but are not included in one of the previous modules. Some of the questions used to assess these variables come from PISA 2015 and others were created to fit the needs of middle- and low-income countries.

Like PISA, PISA-D asks teachers about their age and sex, qualification, employment status, educational background, years of experience and professional development activities. PISA-D gathers new information about whether the teacher teaches multi-grade classrooms, holds multiple teaching jobs or works other jobs in addition to teaching, and a number of factors relevant to their pre-service training, SES, and health and well-being. Also unique to PISA-D, the teacher questionnaire asks about the proportion of students in their class that lack the literacy and numeracy skills to learn the curriculum.

As policy makers have limited direct impact on teaching and learning processes, information on school-level factors that help to improve schools, and thus indirectly improve student learning, have high priority. To meet policy requests directly, PISA also needs to address issues related to governance at the system level. School principals in both PISA and PISA-D are asked numerous questions on resources and management, including type of school (public vs. private, distinguishing between types of private schools), number of students, average class size, school management and funding, as well as how many full- and part-time teachers work at their school. PISA-D adds questions on school location and nearby hazards. Like PISA, the PISA-D teacher questionnaire asks about school leadership at the school where they work and job satisfaction. PISA-D adds a question about their satisfaction with specific aspects of their job, such as benefits and pay.

As in PISA, the contextual information collected through the PISA-D questionnaires is complemented by system-level data on contextual variables in educational systems. The system-level questionnaire used in PISA was adapted for use by PISA-D countries, and both versions capture data on the structure of national programmes, national assessments and examinations, instructional time, teacher training and salaries, educational finance (including enrolment), national accounts and population data (UIS, 2016).



For the out-of-school component there is no data collected on teachers and school-level background variables, however the system-level data can provide relevant contextual information. Some background variables are also collected about the person who answers the “person most knowledgeable about the youth” questionnaire.

Note

1. While PISA offers an optional parent questionnaire, it is distributed to the parents of students sitting the PISA test in school and focuses on the parents’ perceptions of and involvement in their child’s school, their support for learning at home and school choice; and it acquires information on basic characteristics of the early childhood education and care arrangements of PISA participants, and reasons for attending or not attending early childhood education and care.

Annex 5.A1

Annex 5.A1 presents descriptions of the key elements of the six stages of development and the processes that determine how success accumulates from one stage to the next.

Key elements of each of the six stages of development

Prenatal period

The Prosperity Outcomes at this stage are a healthy pregnancy and delivery. The Foundations for Success include four family factors: nutrition, no exposure to toxins, the mother’s physical health and the mother’s emotional health. A number of studies have shown that poor nutrition during the prenatal period modifies the development of the unborn child, leading to low birth weight and a greater susceptibility to coronary heart disease, obesity and diabetes later in life (Barker, 1994; Barker and Sultan, 1995). The exposure of the foetus to environmental toxins or to alcohol or drugs can also compromise the healthy development of the unborn child (Nelson and Panksepp, 1998). The mental and physical well-being of the mother also plays a key role (Liu et al., 1997). The prevalence of healthy pregnancies and deliveries can be increased through the provision of prenatal care and primary health care.

At this stage and in subsequent stages, the framework includes social capital and resources. The term “social capital” is intended to capture positive socialising forces, such as trust among community members, social ties and networks connecting people, the presence of strong adult role models, and actively engaged citizens.

Early development (birth to age 2)

The Prosperity Outcomes at this stage include language, cognitive and physical development. The key family foundations include: breast-feeding and nutrition, the mother’s physical and emotional health, parenting skills and intra-family relations. These factors can be supported with positive post-natal care and primary health care.

Pre-primary (ages 3 to 5)

The Prosperity Outcomes at this stage includes skills in five domains: awareness of self and environment, cognitive development, language and communication, physical development, and social skills and approaches to learning. These outcomes are consistent with frameworks set out by UNICEF and the United States Congress (Shepard, Kagan and Wurtz, 1998).

Three of the most important family factors affecting children’s development in these domains are family involvement, especially reading to the child, positive intra-family relations and parenting styles (Willms, 2002). Children whose parents adopt an “authoritative” parenting style, which incorporates being responsive to the child’s needs but also involves setting expectations for positive behaviour, tend to have better outcomes in these domains (Tramonte, Willms and Gauthier, 2013).

Attendance in high-quality early childhood and care programmes has positive short-term outcomes and enduring long-term benefits (Burchinal, Howes and Kontos, 2002; Currie, 2001; Peisner-Feinberg et al., 2001; Ramey and Ramey, 1998; Schweinhart and Weikart, 1998), especially for children from less advantaged backgrounds (Burchinal et al., 2000). Several large national studies and many smaller local studies suggest that high-quality child care experiences are related, albeit modestly, to child outcomes, even after adjusting for factors such as socio-economic status and parental child-rearing attitudes and practices (Howes, Phillips and Whitebook, 1992; Peisner-Feinberg and Burchinal, 1997; Zill, 1999). Interventions are more effective when early learning programmes take place within a general framework of anti-poverty



and community development (Kagan and Zigler, 1987) and when programmes promote family engagement alongside high-quality learning experiences for children (Bertram et al., 2002). Programmes for children should be intensive, year-long and conducted by appropriately trained professionals (Leseman, 2002).

Early primary (ages 6 to 9)

After children enter school, there is another critical period that has a dramatic effect on their attainment and performance at age 15. The timely transition from learning-to-read to reading-to-learn, which for most children occurs at about age 8 or 9, is essential to academic success, school attainment and well-being (Snow, Burns and Griffin, 1998). During the primary school years, from kindergarten to Grade 3, considerable emphasis is placed on the development of reading skills. Of course, children learn subject-matter content and acquire a wide range of skills while they are learning to read. But after Grade 3 there is a tacit assumption that children can read fluently and comprehend curricular materials in subject domains such as health, social studies and science. The curriculum changes: students are expected to learn the languages of subject domains and use that language to think critically, solve problems and create new knowledge. The demands for strong reading skills increase as students make their way into the higher grades. Students who lack fundamental reading skills fall further and further behind.

Late primary and lower secondary (ages 10 to 15)

After age 10, during the late primary and lower secondary years, the relationship between early reading skills and future literacy skills is solidified (Francis et al., 1996; Kozminsky and Kozminsky, 2001), as is the relationship between early literacy and social and emotional outcomes (Coleman and Vaughn, 2000). This is the “reading-to-learn” period, during which students require strong literacy skills in all subjects in order to make inferences, monitor comprehension and use higher-order skills, such as previewing, predicting and summarising (O’Reilly and McNamara, 2007). Students who have not made the transition from learning-to-read to reading-to-learn are unable to handle the demands for understanding increasingly complex subject-matter content (Morris, Bloodgood and Perney, 2003).

Upper secondary (ages 16 to 18)

Completing secondary school is a key outcome at this stage. Longitudinal studies that have followed students through to the school-leaving age have identified a number of demographic and school-related factors related to completion (Barrington and Hendricks, 1989; Crane, 1991; Ensminger and Slusarcick, 1992; Fagan and Pabon, 1990; Gilbert et al., 1993; Janosz et al., 1997; Rumberger, 1983; Wehlage and Rutter, 1986). Literacy skills, grade repetition, attendance, engagement and positive behaviours are among the most important determinants, and nearly all studies emphasise the role of family socio-economic status and parental engagement.

Processes that determine how success accumulates from one stage to the next

Biological embedding

Children’s potential for success at school is affected by factors during the prenatal period that contribute to a healthy pregnancy and a healthy delivery. Recent advances in neurobiology, molecular biology and genomics have provided compelling evidence that children’s early experiences interact with their genetic disposition in ways that affect brain development as well as other neurological and biological systems associated with healthy child development (Boyce, Sokolowski and Robinson, 2012). Some of these biological processes are “biologically embedded” during the prenatal period through epigenetic processes in which chemical signatures are attached to genes that predispose the child to either vulnerability or resilience (Boyce and Kobor, 2015).

At birth, children have billions of neurons; during the course of early development the neurons form connections called synapses in response to environmental stimuli. As this occurs, many of the neurons that are not being used are pruned away. This process of synapse formation and neuron pruning – the sculpting of the brain – is more rapid during certain *critical periods* of the first two or three years of life (McEwan and Schmeck, 1994; Cynader and Frost, 1999). The notion that children’s early experiences are biologically embedded is gaining further support from research showing that the development of children’s endocrine and immune systems are also influenced by children’s environments during the early years (Barr, Beek and Caloiniu, 1999; Gunnar, 1998; McEwan, 1998).

Foundations for Success

See the “Foundations for Success” section of Chapter 5.

Cumulative development

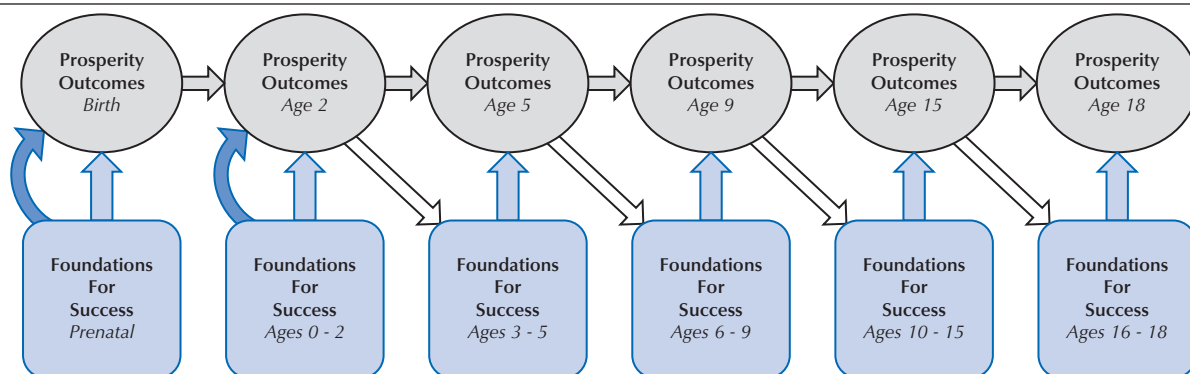
Children's development and success is cumulative. For example, children develop their literacy skills in a cumulative way as they move from one stage to the next. The rate at which they develop these skills depends on the strength and duration of their exposure to the family, institution and community factors that comprise the Foundations for Success in the Educational Prosperity model. For example, a child's literacy skills at age 15 depends on his or her literacy skills at age 8, which is strongly affected by the quality of instruction the child received during the primary grades. The increase in the child's literacy skills from ages 9 to 18 depends on the quality of instruction he or she received during the late primary and secondary school years.

Institutional selection

When students are successful at one stage of development, their life-course may be altered if they are selected into certain classes, school programmes or schools. For example, children who have strong reading and language skills are more likely to be streamed into classes or school programmes where they benefit from positive peer interactions, a higher quality of instruction and other factors that enable them to develop their skills at a faster pace. On the other hand, children who experience learning difficulties at a particular stage are more likely to be streamed into lower ability classes and have less access to the factors that improve their skills.

Figure 5.A1.1 shows the effects associated with each of the four processes along the Educational Prosperity pathway. The outcomes at birth are affected by the Foundations for Success (light blue arrows), which to some extent are biologically embedded (dark blue arrows) through epigenetic processes. The age 2 outcomes are determined by a cumulative effect (grey arrows) and the Foundations for Success associated with that stage, which include a foundation effect (light blue arrows) and an effect that is biologically embedded through the sculpting of the brain during critical periods (dark blue arrows). We assume there are no institutional selection effects at this stage. The age 5 outcomes are also determined by cumulative effects, foundation effects and prior biologically embedded effects. In addition, there can be an institution selection effect (white arrows) if children's outcomes at age 5 are to some extent determined by their access to preschools with varying quality. The outcomes at age 10 and age 18 are affected by the same factors. We assume that the "hard-wired" effects of biological embedding have diminished, although for some outcomes the process of biological embedding continues through to adolescence.

Figure 5.A1.1 ■ **Four ways in which success accumulates**
*Biological embedding (dark blue arrows), Foundations for Success (light blue arrows),
 cumulative effects (grey arrows), and institutional selection effects (white arrows)*





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Annex A

PISA FOR DEVELOPMENT CONTEXTUAL QUESTIONNAIRES

Annex A presents the contextual questionnaires used in PISA for Development. These are the **school questionnaire** distributed to school principals; the **student questionnaire** distributed to all participating students; the **teacher questionnaire** distributed to school teachers; the **out-of-school youth questionnaire** administered to all participating out-of-school 14-16 year-olds; the **parent questionnaire** distributed to the parent, or person most knowledgeable, of the out-of-school youth; and the **household observation questionnaire** distributed to the interviewers conducting the out-of-school assessment.

SCHOOL QUESTIONNAIRE

Main survey version

The school principal completes the school questionnaire. The questionnaire takes about 25 minutes to complete and covers information about the school and in particular its structure, organisation, student intake and social environment.

Technical terms are given in <brackets> and are adapted to the national context by the national data collection centre of the participating country or economy.

SC001 SC001Q01TA	Which of the following definitions best describes the community in which your school is located? (Please tick only one box.)	
	A village, hamlet or rural area (fewer than 3 000 people)	☐ ₁
	A small town (3 000 to about 15 000 people)	☐ ₂
	A town (15 000 to about 100 000 people)	☐ ₃
	A city (100 000 to about 1 000 000 people)	☐ ₄
	A large city (with over 1 000 000 people)	☐ ₅

SC002	Which of the following levels of education does your school provide? (Please select all that apply.)	
SC002Q01NA	<ISCED0>	☐ ₁
SC002Q01NB	< ISCED1>	☐ ₁
SC002Q01NC	< ISCED2>	☐ ₁
SC002Q01ND	< ISCED3>	☐ ₁

SC003	As of <August 1, 2017>, what was the total school enrolment (number of students)? (Please write a number on each line. Write "0" (zero) if there are none.)	
SC003Q01TA	Number of boys:	_____
SC003Q02TA	Number of girls:	_____

SC004	How many full-time and part-time teachers are on the staff of your school? <i>A full-time teacher is employed at least 90% of the time as a teacher for the full school year. All other teachers should be considered part-time.</i> (Please write a number in each space provided. Write "0" (zero) if there are none.)	
SC004Q01NA	Full-time teachers in TOTAL	_____
SC004Q02NA	Part-time teachers in TOTAL	_____



SC005
SC005Q01TA

What is the average size of <test language> classes in <national modal grade for 15-year-olds> in your school?

If your school does not teach this grade then answer this question for an adjacent grade where most of your 15 year-old students are enrolled.

(Please tick only one box.)

15 students or fewer	☐ ₁
16-20 students	☐ ₂
21-25 students	☐ ₃
26-30 students	☐ ₄
31-35 students	☐ ₅
36-40 students	☐ ₆
41-45 students	☐ ₇
46-50 students	☐ ₈
More than 50 students	☐ ₉

SC006
SC006Q01TA

Is your school a public or a private school?

(Please tick only one box.)

A public school <i>(This is a school managed directly or indirectly by a public education authority, government agency, or governing board appointed by government or elected by public franchise.)</i>	☐ ₁
A private school <i>(This is a school managed directly or indirectly by a non-government organisation; e.g. a church, trade union, business, or other private institution.)</i>	☐ ₂

SC007
C007Q01TA

What kind of organisation runs your school?

(Please tick only one box.)

A church or other religious organisation	☐ ₁
Another not-for-profit organisation	☐ ₂
A for-profit organisation	☐ ₃
The government	☐ ₄

SC008

About what percentage of your total funding for a typical school year comes from the following sources?

(Please write a number on each line. Write "0" (zero) if no funding comes from that source.)

		%
SC008Q01TA	Government (includes departments, local, regional, state and national)	_____
SC008Q02TA	Student fees or school charges paid by parents	_____
SC008Q03TA	Benefactors, donations, bequests, sponsorships, parent fundraising	_____
SC008Q04TA	Other	_____
	<i>Total</i>	<i>100%</i>

SC009
SC009Q01NA

In which of the following <type of building> is your school located?

(Please tick only one box.)

A school building	☐ ₁
Rooms of a government building	☐ ₂
Rooms of a church	☐ ₃
Rooms of an NGO	☐ ₄
A private home	☐ ₅
Rooms of a multi-purpose building	☐ ₆
A university	☐ ₇

SC010

Is your school located near one of the following <settings>?

(Please tick one box in each row.)

		Yes	No
SC010Q01NA	<Highway or freeway>	☐ ₁	☐ ₂
SC010Q02NA	Busy roads or intersections	☐ ₁	☐ ₂
SC010Q03NA	A neighbourhood with a high rate of crime	☐ ₁	☐ ₂
SC010Q04NA	A dump or waste land	☐ ₁	☐ ₂
SC010Q05NA	A geologically unstable area	☐ ₁	☐ ₂
SC010Q06NA	An industrial district	☐ ₁	☐ ₂
SC010Q07NA	Factories	☐ ₁	☐ ₂

SC011

Does your school have the following features, and if so, in what condition are they?

(Please tick one box in each row.)

		No, not available	Yes, but in poor condition	Yes, but in need of minor repairs	Yes, in good condition
SC011Q01NA	Roof	☐ ₁	☐ ₂	☐ ₃	☐ ₄
SC011Q02NA	Walls	☐ ₁	☐ ₂	☐ ₃	☐ ₄
SC011Q03NA	Floors	☐ ₁	☐ ₂	☐ ₃	☐ ₄
SC011Q04NA	Building entrance	☐ ₁	☐ ₂	☐ ₃	☐ ₄
SC011Q05NA	Doors	☐ ₁	☐ ₂	☐ ₃	☐ ₄
SC011Q06NA	Windows	☐ ₁	☐ ₂	☐ ₃	☐ ₄
SC011Q07NA	Hallways	☐ ₁	☐ ₂	☐ ₃	☐ ₄
SC011Q08NA	Classrooms	☐ ₁	☐ ₂	☐ ₃	☐ ₄

**SC012****Does your school have the following features, and if so, in what condition are they?***(Please tick one box in each row.)*

		<i>No, not available</i>	<i>Yes, but in poor condition</i>	<i>Yes, but in need of minor repairs</i>	<i>Yes, in good condition</i>
SC012Q01NA	Flush toilets	☐ ₁	☐ ₂	☐ ₃	☐ ₄
SC012Q02NA	Other types of toilets (e.g., <latrines>, <squat holes>, <pit toilets>)	☐ ₁	☐ ₂	☐ ₃	☐ ₄
SC012Q03NA	Kitchen	☐ ₁	☐ ₂	☐ ₃	☐ ₄
SC012Q04NA	Place with drinkable water	☐ ₁	☐ ₂	☐ ₃	☐ ₄
SC012Q05NA	Running water	☐ ₁	☐ ₂	☐ ₃	☐ ₄
SC012Q06NA	Electricity	☐ ₁	☐ ₂	☐ ₃	☐ ₄
SC012Q07NA	Indoor plumbing	☐ ₁	☐ ₂	☐ ₃	☐ ₄
SC012Q08NA	<First aid room>	☐ ₁	☐ ₂	☐ ₃	☐ ₄
SC012Q09NA	Immunisation or health care room	☐ ₁	☐ ₂	☐ ₃	☐ ₄
SC012Q10NA	Cafeteria	☐ ₁	☐ ₂	☐ ₃	☐ ₄
SC012Q11NA	Sports area or playground	☐ ₁	☐ ₂	☐ ₃	☐ ₄
SC012Q12NA	Fence or hedge on the school borders	☐ ₁	☐ ₂	☐ ₃	☐ ₄
SC012Q13NA	<Access ramp>	☐ ₁	☐ ₂	☐ ₃	☐ ₄
SC012Q14NA	<Fans>	☐ ₁	☐ ₂	☐ ₃	☐ ₄
SC012Q15NA	<Lighting>	☐ ₁	☐ ₂	☐ ₃	☐ ₄

SC013**Regarding toilets for students and staff, does your school have the following?***(Please tick one box in each row.)*

		<i>No, not available or inaccessible</i>	<i>Yes, but in poor condition</i>	<i>Yes, but in need of minor repairs</i>	<i>Yes, in good condition</i>
SC013Q01NA	Separate toilets for girls and boys	☐ ₁	☐ ₂	☐ ₃	☐ ₄
SC013Q02NA	Separate toilets for school staff and students	☐ ₁	☐ ₂	☐ ₃	☐ ₄

SC014

SC014Q01NA

Do students in your school have textbooks for instruction in <test language>?*(Please tick only one box.)*

Yes, every student has at least one.	☐ ₁
Yes, but not enough. Sometimes two students need to share a textbook.	☐ ₂
Yes, but so few that sometimes more than two students need to share a textbook.	☐ ₃
No, there are no textbooks.	☐ ₄

SC015

SC015Q01NA

Do students in your school have textbooks for mathematics instruction?*(Please tick only one box.)*

Yes, every student has at least one.	☐ ₁
Yes, but not enough. Sometimes two students need to share a textbook.	☐ ₂
Yes, but so few that sometimes more than two students need to share a textbook.	☐ ₃
No, there are no textbooks.	☐ ₄

SC016**Are the following meals or <snacks> offered in your school?***(Please tick only one box.)*

		Yes, free for all students	Yes, free for some students	Yes, for a fee	No, not offered
SC016Q01NA	<breakfast>	☐ ₁	☐ ₂	☐ ₃	☐ ₄
SC016Q02NA	<snack>	☐ ₁	☐ ₂	☐ ₃	☐ ₄
SC016Q03NA	<lunch>	☐ ₁	☐ ₂	☐ ₃	☐ ₄

SC017**Do the following statements regarding students' meals apply to your school?***(Please tick one box in each row.)*

		Yes	No
SC017Q01NA	Our school provides free meals for all students.	☐ ₁	☐ ₂
SC017Q02NA	Our school provides free meals for students of certain grades.	☐ ₁	☐ ₂
SC017Q03NA	Our school provides free meals only for students in need.	☐ ₁	☐ ₂
SC017Q04NA	Our school provides meals to students for a charge.	☐ ₁	☐ ₂

SC018**Do community members or organisations contribute to the school by doing the following?***(Please tick one box in each row.)*

		Yes	No
SC018Q01NA	Build school facilities such as classrooms or teacher houses.	☐ ₁	☐ ₂
SC018Q02NA	Maintain school facilities such as classrooms or teachers' houses.	☐ ₁	☐ ₂
SC018Q03NA	Maintain school grounds and fences or hedges around them.	☐ ₁	☐ ₂
SC018Q04NA	Construct, maintain or repair furniture or equipment.	☐ ₁	☐ ₂
SC018Q05NA	Teach when teachers are absent.	☐ ₁	☐ ₂
SC018Q06NA	Assist teachers in addressing the needs of students with disabilities.	☐ ₁	☐ ₂
SC018Q07NA	Organise sport activities or school trips.	☐ ₁	☐ ₂
SC018Q08NA	Assist with preparation and distribution of school meals.	☐ ₁	☐ ₂

SC019**Do parents or parental organisations contribute to the school by doing the following?***(Please tick one box in each row.)*

		Yes	No
SC019Q01NA	Build school facilities such as classrooms or teacher houses.	☐ ₁	☐ ₂
SC019Q02NA	Maintain school facilities such as classrooms or teachers' houses.	☐ ₁	☐ ₂
SC019Q03NA	Maintain school grounds and fences or hedges around them.	☐ ₁	☐ ₂
SC019Q04NA	Construct, maintain or repair furniture or equipment.	☐ ₁	☐ ₂
SC019Q05NA	Teach when teachers are absent.	☐ ₁	☐ ₂
SC019Q06NA	Assist teachers in addressing the needs of students with disabilities.	☐ ₁	☐ ₂
SC019Q07NA	Organise sport activities or school trips.	☐ ₁	☐ ₂
SC019Q08NA	Assist with preparation and distribution of school meals.	☐ ₁	☐ ₂

**SC025**

During the last month of school, how often was the school confronted with the following teacher behaviours?

(Please tick one box in each row.)

		Never	Once or twice	Every week
SC025Q01NA	Arriving late at school	☐ ₁	☐ ₂	☐ ₃
SC025Q02NA	Absenteeism (i.e., unjustified absence)	☐ ₁	☐ ₂	☐ ₃
SC025Q03NA	Skipping classes	☐ ₁	☐ ₂	☐ ₃
SC025Q04NA	Intimidation or bullying of students	☐ ₁	☐ ₂	☐ ₃
SC025Q05NA	Sexual harassment of other teachers	☐ ₁	☐ ₂	☐ ₃
SC025Q06NA	Sexual harassment of students	☐ ₁	☐ ₂	☐ ₃
SC025Q07NA	Use of abusive language	☐ ₁	☐ ₂	☐ ₃
SC025Q08NA	Drug use or possession	☐ ₁	☐ ₂	☐ ₃
SC025Q09NA	Alcohol use or possession	☐ ₁	☐ ₂	☐ ₃
SC025Q10NA	Health problems	☐ ₁	☐ ₂	☐ ₃
SC025Q11NA	Physical aggression against colleagues	☐ ₁	☐ ₂	☐ ₃
SC025Q12NA	Physical aggression against students	☐ ₁	☐ ₂	☐ ₃

SC024

On approximately how many instructional days in the last 12 months was your school closed or deviated from the regular curriculum for the following reasons?

(Please write a number on each line. Write "0" (zero) if there are none.)

		Days
SC024Q01NA	Vacations and holidays	_____
SC024Q02NA	National or local elections	_____
SC024Q03NA	Teacher conferences or professional development activities	_____
SC024Q04NA	School events and activities such as field trips, school dance, sporting events	_____
SC024Q05NA	Regional or local problems such as weather (e.g. storm, heat) or hazards (e. g. epidemics, fire, floods, or landslides)	_____
SC024Q06NA	Regional or local demonstrations such as manifestations, strikes, riots, protests or aggressive conflicts	_____
SC024Q07NA	Safety concerns related to school infrastructural issues	_____
SC024Q08NA	Safety concerns related to violence or delinquency in the school	_____

SC027**When a teacher is absent for a week or more, how often do you take the following actions in your school?***(Please tick one box in each row.)*

		<i>Never</i>	<i>Sometimes</i>	<i>Always</i>
SC027Q01NA	Send the pupils home.	☐ ₁	☐ ₂	☐ ₃
SC027Q02NA	Leave the pupils to learn on their own.	☐ ₁	☐ ₂	☐ ₃
SC027Q03NA	Combine the class with another class.	☐ ₁	☐ ₂	☐ ₃
SC027Q04NA	Reallocate pupils to several other classes.	☐ ₁	☐ ₂	☐ ₃
SC027Q05NA	Assign a senior pupil to supervise the class.	☐ ₁	☐ ₂	☐ ₃
SC027Q06NA	Substitute the absent teacher with a parent.	☐ ₁	☐ ₂	☐ ₃
SC027Q07NA	Substitute the absent teacher with a community member.	☐ ₁	☐ ₂	☐ ₃
SC027Q08NA	Substitute the absent teacher with another qualified teacher.	☐ ₁	☐ ₂	☐ ₃
SC027Q09NA	Substitute the absent teacher with a qualified <relief teacher>.	☐ ₁	☐ ₂	☐ ₃
SC027Q10NA	Substitute the absent teacher with an unqualified <relief teacher>.	☐ ₁	☐ ₂	☐ ₃
SC027Q11NA	Substitute the absent teacher with the <School Head>.	☐ ₁	☐ ₂	☐ ₃

SC026**Which policies regarding <grade repetition> are implemented in your school?***(Please tick one box in each row.)*

		<i>Yes</i>	<i>No</i>
SC026Q01NA	<Grade repetition> is possible on a voluntary basis, i.e. with request or permission from the parents.	☐ ₁	☐ ₂
SC026Q02NA	If a student fails the minimum achievement standards at the end of the school year, he or she has to repeat the grade.	☐ ₁	☐ ₂
SC026Q03NA	Individual courses can be repeated without repeating a whole grade.	☐ ₁	☐ ₂
SC026Q04NA	After repeating grades a certain number of times, students are expected to leave the school.	☐ ₁	☐ ₂
SC026Q05NA	<Grade repetition> is prohibited by <state or district> regulations.	☐ ₁	☐ ₂
SC026Q06NA	<Grade repetition> is prohibited by school policies.	☐ ₁	☐ ₂

SC028**Which of the following measures are implemented in your school to support students at risk of academic failure?***(Please tick one box in each row.)*

		<i>Yes</i>	<i>No</i>
SC028Q01NA	Our school has a professional counsellor for students at risk of academic failure.	☐ ₁	☐ ₂
SC028Q02NA	Our school provides mandatory <remedial courses> during the school year.	☐ ₁	☐ ₂
SC028Q03NA	Our school offers <remedial courses> during the school year.	☐ ₁	☐ ₂
SC028Q04NA	Our school offers <remedial courses> during the <summer>.	☐ ₁	☐ ₂
SC028Q05NA	Our school offers students <remedial education> while repeating.	☐ ₁	☐ ₂
SC028Q06NA	Our school offers students specific counselling during repeating.	☐ ₁	☐ ₂
SC028Q07NA	Our school has <guide teachers> assigned to students deemed to be at risk.	☐ ₁	☐ ₂

**SC020****How often are the following factors considered when students are admitted to your school?***(Please tick one box in each row.)*

		<i>Never</i>	<i>Sometimes</i>	<i>Always</i>
SC020Q01TA	Student's record of academic performance (including placement tests).	☐ ₁	☐ ₂	☐ ₃
SC020Q02TA	Recommendation of feeder schools.	☐ ₁	☐ ₂	☐ ₃
SC020Q03TA	Parents' endorsement of the instructional or religious philosophy of the school.	☐ ₁	☐ ₂	☐ ₃
SC020Q04TA	Whether the student requires or is interested in a special programme.	☐ ₁	☐ ₂	☐ ₃
SC020Q05TA	Preference given to family members of current or former students.	☐ ₁	☐ ₂	☐ ₃
SC020Q06TA	Residence in a particular area.	☐ ₁	☐ ₂	☐ ₃
SC020Q07TA	Student's disciplinary record in this or another school.	☐ ₁	☐ ₂	☐ ₃
SC020Q08TA	Student's criminal record.	☐ ₁	☐ ₂	☐ ₃
SC020Q09TA	Student's parental status or pregnancy.	☐ ₁	☐ ₂	☐ ₃
SC020Q10TA	Student's working status.	☐ ₁	☐ ₂	☐ ₃
SC020Q11TA	Student's cultural or ethnical background.	☐ ₁	☐ ₂	☐ ₃

SC021**Some schools organise instruction differently for students with different abilities.****What is your school's policy about this for students in <national modal grade for 15-year-olds>?***(Please tick one box in each row.)*

		<i>For all subjects</i>	<i>For some subjects</i>	<i>Not for any subjects</i>
SC021Q01TA	Students are grouped by ability into different classes.	☐ ₁	☐ ₂	☐ ₃
SC021Q02TA	Students are grouped by ability within their classes.	☐ ₁	☐ ₂	☐ ₃

SC022**Please estimate the percentage of students in your school who have the following characteristics.***(Please consider that students may fall into multiple categories.)**(Please tick one box for the estimated percentage in each line.)*

		<i>Less than 1%</i>	<i>Between 1 and 5%</i>	<i>Between 6 and 10%</i>	<i>Between 11 and 20%</i>	<i>Between 21 and 30%</i>	<i>More than 30%</i>
SC022Q01NA	Students whose <heritage language> is different from <test language>	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₅	☐ ₆
SC022Q02NA	Students from very poor homes	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₅	☐ ₆
SC022Q03NA	Students who are orphans	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₅	☐ ₆
SC022Q04NA	Students who are not living with their parents	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₅	☐ ₆
SC022Q05NA	Students who are pregnant	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₅	☐ ₆
SC022Q06NA	Students who have children	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₅	☐ ₆
SC022Q07NA	Students who have disciplinary problems or show delinquent behaviours	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₅	☐ ₆
SC022Q08NA	Students who work for <formal pay>	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₅	☐ ₆
SC022Q09NA	Students who work informally or for their families	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₅	☐ ₆
SC022Q10NA	Students with physical disabilities (e.g. serious hearing or sight difficulties, or impaired mobility)	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₅	☐ ₆
SC022Q11NA	Students with learning disabilities	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₅	☐ ₆

SC023**Think about what happens in your school when students need extra support. To what extent do you agree with the following statements?***(Please tick one box in each row.)*

		<i>Strongly Disagree</i>	<i>Disagree</i>	<i>Agree</i>	<i>Strongly Agree</i>
SC023Q01NA	Many of the students who are behind should have been <held back>.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
SC023Q02NA	Students who repeat one or more grades will eventually leave school before they graduate.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
SC023Q03NA	After students have repeated a grade, they are still unprepared for the next grade.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
SC023Q04NA	Students who have repeated a grade learn that they must try harder to succeed.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
SC023Q05NA	Teachers should try to teach the curriculum, even to students who do not have the basic reading and numeracy skills.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
SC023Q06NA	Students with disabilities should be taught in <special schools>.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
SC023Q07NA	We need more special classes for students who lag behind.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
SC023Q08NA	Teachers are able to teach classes with students with differing levels of ability.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
SC023Q09NA	A student should never have to repeat more than one grade.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
SC023Q10NA	A student should never have to repeat the same grade twice.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
SC023Q11NA	Because students learn at different rates, school programs should be <ungraded>.	☐ ₁	☐ ₂	☐ ₃	☐ ₄



STUDENT QUESTIONNAIRE

Main survey version

Students complete the student questionnaire after the literacy assessment. The questionnaire takes about 35 minutes to complete and covers information about the student, their school experience, and their family and life at home.

Technical terms are given in <brackets> and are adapted to the national context by the national data collection centre of the participating country or economy.

About the student

ST001 ST001Q01TA	What <grade> are you in?

	<grade>

ST003	On what date were you born?	
	(Please write in the day, month and year you were born.)	
ST003Q01TA	Day	_____
ST003Q02TA	Month	_____
ST003Q03TA	Year	_____

ST004 ST004Q01TA	Are you female or male?	
	(Please tick only one box.)	
	<i>Female</i>	<i>Male</i>
	☐ ₁	☐ ₂

ST021 ST021Q01TA	What language do you speak at home most of the time?	
	(Please tick only one box.)	
	<Language 1>	☐ ₁
	<Language 2>	☐ ₂
	<Language 3>	☐ ₃
	< ...etc. >	☐ ₄
	Other language	☐ ₅

ST026	In what country were you and your parents born?		
	(Please tick only one box in each column.)		
	<i>You</i> ST026Q01TA	<i>Mother</i> ST026Q02TA	<i>Father</i> ST026Q03TA
<Country A>	☐ ₁	☐ ₁	☐ ₁
<Country B>	☐ ₂	☐ ₂	☐ ₂
<Country C>	☐ ₃	☐ ₃	☐ ₃
<Country D>	☐ ₄	☐ ₄	☐ ₄
<...etc.>	☐ ₅	☐ ₅	☐ ₅
Other country	☐ ₆	☐ ₆	☐ ₆

ST015
ST015Q01TA**Overall, how satisfied are you with your life as a whole these days?**

(Please tick only one box.)

☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
0	1	2	3	4	5	6	7	8	9	10	
Not at all satisfied											Completely satisfied

ST016

The next five statements are about your health.

(Please tick one box in each row.)

		Yes	No
ST016Q01NA	I can see what is written on the board without difficulty.	☐ ₁	☐ ₂
ST016Q02NA	I can hear the teacher's voice clearly when he or she is giving a lesson.	☐ ₁	☐ ₂
ST016Q03NA	I have a physical disability that makes it difficult for me to walk or use stairs.	☐ ₁	☐ ₂
ST016Q04NA	I have a physical disability that makes it difficult for me to grasp small objects like a pencil or scissors.	☐ ₁	☐ ₂
ST016Q05NA	I often get so sick I cannot play, work or go to school.	☐ ₁	☐ ₂

ST017

We would like to know about certain feelings you may have at home or at school. For each statement below, please tell us how often you feel this way.

(Please tick one box in each row.)

		Never or almost never	About once a week	2 to 3 times a week	Almost every day
ST017Q01NA	I am too fearful or nervous.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
ST017Q02NA	I am afraid that other students think I am stupid.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
ST017Q04NA	I worry about a teacher asking me a question.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
ST017Q05NA	I worry about what other students think of me.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
ST017Q06NA	I cry without a good reason.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
ST017Q07NA	I feel lonely.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
ST017Q08NA	Other students seem to have more fun than me.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
ST017Q09NA	I feel sad or depressed.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
ST017Q10NA	I have trouble falling asleep at night.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
ST017Q11NA	A lot of things seem to bother me.	☐ ₁	☐ ₂	☐ ₃	☐ ₄

ST018

ST018Q01NA

In general, would you say your health is:

(Please tick only one box.)

☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
0	1	2	3	4	5	6	7	8	9	10
Poor		Fair			Good		Very Good		Excellent	

**ST019****During the past year, have you had any of the following health problems?***(Please tick one box in each row.)*

		Yes	No
ST019Q01NA	A chronic disease (e.g., heart disease, lung or other respiratory problems, cancer, diabetes)	☐ ₁	☐ ₂
ST019Q02NA	An infectious disease (e.g., cholera, malaria, tuberculosis)	☐ ₁	☐ ₂
ST019Q03NA	Gastrointestinal problems (e.g., heartburn, stomach pain, constipation, diarrhoea)	☐ ₁	☐ ₂
ST019Q04NA	A cold or flu	☐ ₁	☐ ₂
ST019Q05NA	An injury that needed treatment	☐ ₁	☐ ₂
ST019Q06NA	Pain that was long-lasting or recurring	☐ ₁	☐ ₂
ST019Q07NA	Depression	☐ ₁	☐ ₂
ST019Q08NA	Panic and anxiety attacks	☐ ₁	☐ ₂
ST019Q09NA	Insomnia	☐ ₁	☐ ₂
ST019Q10NA	Fatigue that was long-lasting or recurring	☐ ₁	☐ ₂

About the student's school experience

ST005

ST005Q01TA

Did you attend <ISCED 0>?*(Please tick only one box.)*

No	☐ ₁
Yes, for one year or less	☐ ₂
Yes, for more than one year	☐ ₃

ST007

ST007Q01NA

What language did most of your teachers use for instruction in <ISCED 1>?*(Please tick only one box.)*

<Language 1>	☐ ₁
<Language 2>	☐ ₂
<Language 3>	☐ ₃
< ...etc. >	☐ ₄
Other language	☐ ₅

ST023

ST023Q01NA

When did you begin learning <language of instruction>?*(Please tick only one box.)*

At home before I started school.	☐ ₁
In <ISCED 0>.	☐ ₂
When I started <ISCED 1>.	☐ ₃
When I started <ISCED 2>.	☐ ₄

ST024 Which language did you first learn to read?

(Please tick only one box.)

<Language 1>	☐ ₁
<Language 2>	☐ ₂
<Language 3>	☐ ₃
< ...etc. >	☐ ₄
Other language	☐ ₅

ST009 Have you ever repeated a <grade>?

(Please tick one box in each row.)

		No, never	Yes, once	Yes, twice or more
ST009Q01TA	At <ISCED 1>	☐ ₁	☐ ₂	☐ ₃
ST009Q02TA	At <ISCED 2>	☐ ₁	☐ ₂	☐ ₃
ST009Q03TA	At <ISCED 3>	☐ ₁	☐ ₂	☐ ₃

ST011 Have you ever missed school for more than three months in a row?

(Please tick only one box.)

No	☐ ₁	Please go to Question 18.
Yes, once	☐ ₂	Please go to Question 17.
Yes, twice or more	☐ ₃	Please go to Question 17.

ST012 Why did you miss school for more than three months in a row?

(Please tick one box in each row.)

		Yes	No
ST012Q01NA	I was bored.	☐ ₁	☐ ₂
ST012Q02NA	I was suspended for something I did (e.g., violence, aggression, use of drugs, drug dealing).	☐ ₁	☐ ₂
ST012Q03NA	I was pregnant.	☐ ₁	☐ ₂
ST012Q04NA	I could not reach school because of transportation problems.	☐ ₁	☐ ₂
ST012Q05NA	I did not have a teacher.	☐ ₁	☐ ₂
ST012Q06NA	I could not understand the language in which the lessons were given.	☐ ₁	☐ ₂
ST012Q07NA	I had to take care of a family member.	☐ ₁	☐ ₂
ST012Q08NA	I had to help with work at home or on the family land.	☐ ₁	☐ ₂
ST012Q09NA	I had to get work to bring money home.	☐ ₁	☐ ₂
ST012Q10NA	I was sick.	☐ ₁	☐ ₂
ST012Q11NA	I did not feel safe at school.	☐ ₁	☐ ₂
ST012Q12NA	I had to take care of sick parents or relatives.	☐ ₁	☐ ₂
ST012Q13NA	I was no longer interested in school.	☐ ₁	☐ ₂
ST012Q14NA	I could not pay <school fees>.	☐ ₁	☐ ₂
ST012Q15NA	School was closed because of a natural disaster (e.g., flood, earthquake).	☐ ₁	☐ ₂

**ST067****Thinking about your school: to what extent do you agree with the following statements?***(Please tick one box in each row.)*

		Strongly agree	Agree	Disagree	Strongly disagree
ST067Q03TA	School has helped give me confidence to make decisions.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
ST067Q04TA	School has taught me things which could be useful in a job.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
ST067Q05TA	Trying hard at school will help me get a good job.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
ST067Q06TA	Trying hard at school will help me get into a good <university>.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
ST067Q07TA	I enjoy receiving good <grades>.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
ST067Q08TA	Trying hard at school is important.	☐ ₁	☐ ₂	☐ ₃	☐ ₄

ST068**Thinking about your school: to what extent do you agree with the following statements?***(Please tick one box in each row.)*

		Strongly agree	Agree	Disagree	Strongly disagree
ST068Q01TA	I feel like an outsider (or left out of things) at school.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
ST068Q02TA	I make friends easily at school.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
ST068Q03TA	I feel like I belong at school.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
ST068Q04TA	I feel awkward and out of place in my school.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
ST068Q05TA	Other students seem to like me.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
ST068Q06TA	I feel lonely at school.	☐ ₁	☐ ₂	☐ ₃	☐ ₄

ST069**Thinking about your school: to what extent do you agree with the following statements?***(Please tick one box in each row.)*

		Strongly agree	Agree	Disagree	Strongly disagree
ST069Q01NA	I feel safe at our school.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
ST069Q02NA	I feel safe on my way to school.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
ST069Q03NA	I feel safe on my way home from school.	☐ ₁	☐ ₂	☐ ₃	☐ ₄

ST070**During the past four weeks, did any of the following events occur?***(Please tick one box in each row.)*

		Yes	No
ST070Q01NA	I was in a physical fight on school property.	☐ ₁	☐ ₂
ST070Q02NA	I stayed away from home from school because I felt unsafe.	☐ ₁	☐ ₂
ST070Q03NA	Our school was vandalised.	☐ ₁	☐ ₂
ST070Q04NA	I gave money to someone at school because they have threatened to hurt me.	☐ ₁	☐ ₂
ST070Q05NA	I witnessed a fight on school property in which someone got hurt.	☐ ₁	☐ ₂
ST070Q06NA	I saw gangs in and around the school.	☐ ₁	☐ ₂
ST070Q07NA	I heard a student threaten to hurt another student.	☐ ₁	☐ ₂
ST070Q08NA	Someone stole something of mine at school.	☐ ₁	☐ ₂
ST070Q09NA	I saw a student carrying a gun or a knife at school.	☐ ₁	☐ ₂

ST072**Thinking about the teachers at your school: to what extent do you agree or disagree with the following statements?***(Please tick one box in each row.)*

		<i>Strongly agree</i>	<i>Agree</i>	<i>Disagree</i>	<i>Strongly disagree</i>
ST072Q01NA	I get along well with most of my teachers.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
ST072Q02NA	Most of my teachers are interested in my well-being.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
ST072Q03NA	Most of my teachers listen to what I have to say.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
ST072Q04NA	If I need extra help, I will receive it from my teachers.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
ST072Q05NA	Most of my teachers treat me fairly.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
ST072Q06NA	The teachers show an interest in every student's learning.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
ST072Q07NA	The teachers give students an opportunity to express opinions.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
ST072Q08NA	Our teachers expect us to work hard.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
ST072Q09NA	Our teachers encourage students to do their best work.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
ST072Q10NA	Our teachers expect us to do our homework on time.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
ST072Q11NA	Students understand what is expected of them for their <courses>.	☐ ₁	☐ ₂	☐ ₃	☐ ₄

ST073**Sexual harassment is any unwanted or inappropriate language or touching of a sexual nature that makes you feel upset, hurt or angry.**

It can be verbal, such as comments about your body, sexual remarks, or the spreading of rumours about a person. It can be physical, such as touching, rubbing, pinching or hugging in a sexual way. It can be a request for a sexual favour in return for something else. It can happen to both boys and girls.

(Please tick one box in each row.)

		<i>Yes</i>	<i>No</i>
ST073Q01NA	In the past 4 weeks, have you felt sexually harassed at school by a student?	☐ ₁	☐ ₂
ST073Q02NA	In the past 4 weeks, have you felt sexually harassed at school by a teacher or other staff member?	☐ ₁	☐ ₂

ST074**How often do these things happen in your classroom?***(Please tick one box in each row.)*

		<i>Every lesson</i>	<i>Most lessons</i>	<i>Some lessons</i>	<i>Never or hardly ever</i>
ST074Q01TA	Students don't listen to what the teacher says.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
ST074Q02TA	There is noise and disorder.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
ST074Q03TA	The teacher has to wait a long time for students to quiet down.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
ST074Q04TA	Students cannot work well.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
ST074Q05TA	Students don't start working for a long time after the lesson begins.	☐ ₁	☐ ₂	☐ ₃	☐ ₄

**ST075****How often do these things happen in your lessons in mathematics?***(Please tick one box in each row.)*

		<i>Every lesson</i>	<i>Most lessons</i>	<i>Some lessons</i>	<i>Never or hardly ever</i>
At the beginning of a lesson:					
ST075Q01NA	The teacher explains the purpose of the lesson.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
ST075Q02NA	The teacher reviews what we learned in previous lessons.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
During a lesson:					
ST075Q03NA	The teacher shows us how to solve problems.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
ST075Q04NA	The teacher provides examples of successful work.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
ST075Q05NA	The teacher gives clear answers to students' questions.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
ST075Q06NA	The teacher gives a formal lecture on the topic.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
ST075Q07NA	The teacher explains mathematical concepts.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
ST075Q08NA	The teacher gives us work to do at our desk.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
ST075Q09NA	The teacher talks with students about their work.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
At the end of the lesson:					
ST075Q11NA	The teacher summarises what we have done that day.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
ST075Q12NA	The teacher gives us homework to practise what we have learned.	☐ ₁	☐ ₂	☐ ₃	☐ ₄

ST078**In the last two weeks at school, how often did the following things occur?***(Please tick one box in each row.)*

		<i>Never</i>	<i>One or two times</i>	<i>Three or four times</i>	<i>Five or more times</i>
ST078Q01TA	I <skipped> a whole school day.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
ST078Q02TA	I <skipped> some classes.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
ST078Q03TA	I arrived late for school.	☐ ₁	☐ ₂	☐ ₃	☐ ₄

ST061
ST061Q01NA**How long does it usually take you to get from your home to school?***(Please tick only one box.)*

	15 minutes or less	☐ ₁
	More than 15 minutes but less than 30 minutes	☐ ₂
	30 minutes or more, but less than 60 minutes	☐ ₃
	60 minutes or more, but less than 90 minutes	☐ ₄
	More than 90 minutes	☐ ₅

ST079

In the last two weeks at school, did any of these happen?*(Please tick one box in each row.)*

		Yes	No
ST079Q01NA	One or more classes were cancelled.	☐ ₁	☐ ₂
ST079Q02NA	School was cancelled.	☐ ₁	☐ ₂
ST079Q03NA	One of my teachers was late for class.	☐ ₁	☐ ₂
ST079Q04NA	One of my teachers did not come for class.	☐ ₁	☐ ₂
ST079Q05NA	There was a teacher strike.	☐ ₁	☐ ₂
ST079Q06NA	My teacher worked at the computer during class time.	☐ ₁	☐ ₂
ST079Q07NA	My teacher answered personal calls during class time.	☐ ₁	☐ ₂
ST079Q08NA	My teacher attended a meeting during class time.	☐ ₁	☐ ₂

About the student's family and life at home

ST083

In general, how often do your parents or someone in your family do the following things with you?*(Please tick one box in each row.)*

		Never or hardly ever	A few times a year	About once a month	Several times a month	Several times a week
ST083Q03NA	Discuss how well you are doing at school.	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₅
ST083Q04NA	Eat <the main meal> with you.	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₅
ST083Q05NA	Spend time just talking with you.	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₅
ST083Q06NA	Talk to you about the importance of completing <secondary school>.	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₅
ST083Q07NA	Talk to you about any problems you might have at school.	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₅
ST083Q08NA	Ask you about how well you are getting along with kids at school.	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₅
ST083Q09NA	Encourage you to get good <grades>.	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₅
ST083Q11NA	Take an interest in what you are learning at school.	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₅
ST083Q12NA	Talk to you about your future education.	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₅
ST083Q13NA	Ask you what you did in school that day.	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₅

ST020

Think of the work you did in the past week.*(Please tick one box in each row.)*

		Yes	No
ST020Q01NA	I worked for payment for someone who is not a member of my household.	☐ ₁	☐ ₂
ST020Q02NA	I fetched water for household use.	☐ ₁	☐ ₂
ST020Q03NA	I collected firewood for household use.	☐ ₁	☐ ₂
ST020Q04NA	I worked on our family farm, in our family business, or selling goods on the street.	☐ ₁	☐ ₂
ST020Q05NA	I helped in the care of children, or an elderly or sick person.	☐ ₁	☐ ₂
ST020Q06NA	I helped with other household chores such as shopping, cleaning, washing clothes, or cooking.	☐ ₁	☐ ₂

**ST029****Who do you live with at home?***(Please tick one box in each row.)*

		Yes	No
ST029Q01NA	Your mother	☐ ₁	☐ ₂
ST029Q02NA	Your father	☐ ₁	☐ ₂
ST029Q03NA	Your grandmother	☐ ₁	☐ ₂
ST029Q04NA	Your grandfather	☐ ₁	☐ ₂
ST029Q05NA	Your siblings (including step-sisters or step-brothers)	☐ ₁	☐ ₂
ST029Q06NA	Other relatives (e.g., aunts, uncles, cousins)	☐ ₁	☐ ₂
ST029Q10NA	Your child/children	☐ ₁	☐ ₂

**ST062
ST064****Which of the following are in your home?***(Please tick one box in each row.)*

		Yes	No
ST062Q01TA	A desk to study at	☐ ₁	☐ ₂
ST062Q02TA	A room of your own	☐ ₁	☐ ₂
ST062Q03TA	A quiet place to study	☐ ₁	☐ ₂
ST062Q04TA	A computer you can use for school work	☐ ₁	☐ ₂
ST062Q05TA	Educational software	☐ ₁	☐ ₂
ST062Q06TA	A link to the internet	☐ ₁	☐ ₂
ST062Q10TA	Books to help with your school work	☐ ₁	☐ ₂
ST062Q12TA	A dictionary	☐ ₁	☐ ₂
ST064Q01NA	A table to have meals	☐ ₁	☐ ₂
ST064Q03NA	A washer	☐ ₁	☐ ₂
ST064Q04NA	A refrigerator or freezer	☐ ₁	☐ ₂
ST064Q06NA	A stove or burner for cooking	☐ ₁	☐ ₂

ST063**How many of these are there in your home?***(Please tick one box in each row.)*

		None	One	Two	Three or more
ST063Q01TA	Televisions	☐	☐	☐	☐
ST063Q02TA	Cars, vans or trucks	☐	☐	☐	☐
ST063Q03TA	Rooms with a bath or shower	☐	☐	☐	☐
ST063Q04TA	<Cell phones> with internet access (e.g., smartphones)	☐	☐	☐	☐
ST063Q05TA	Computers (desktop computer, portable laptop, or notebook)	☐	☐	☐	☐
ST063Q06TA	Musical instruments (e.g., guitar, piano, <country-specific example>)	☐	☐	☐	☐

ST066
ST066Q01NA**How many books are there in your home?**

*Do not include magazines, newspapers, or your schoolbooks.
(Please tick only one box.)*

There are no books.	☐ ₁
There are fewer than 10 books.	☐ ₂
There are 10 - 50 books.	☐ ₃
There are more than 50 books.	☐ ₄

ST049
ST049Q01NA**Do you share a toilet facility with others who are not members of your family?**Yes ☐ ₁ No ☐ ₂ST051
ST051Q01NA**What is the floor of your home mostly made of?**

(Please tick only one box.)

Earth, sand or dung	☐ ₁
Wood planks, palm, or bamboo	☐ ₂
Parquet, polished wood	☐ ₃
Vinyl or asphalt strips	☐ ₄
Ceramic tiles	☐ ₅
Cement	☐ ₆
Stone	☐ ₇

ST048
ST048Q01NA**Do you have a <flush toilet> at your house?**Yes ☐ ₁ No ☐ ₂ST057
ST057Q01NA**Does any member of your household have a bank account?**Yes ☐ ₁ No ☐ ₂ST059
ST059Q01NA**In the past 30 days, how often were you hungry because there was not enough food?**

(Please tick only one box.)

<i>Never or almost never</i>	<i>About once a week</i>	<i>2 to 3 times a week</i>	<i>Almost every day</i>
☐ ₁	☐ ₂	☐ ₃	☐ ₄

ST031
ST031Q01TA**What is the <highest level of schooling> completed by your mother?**

*If you are not sure which box to choose, please ask the <test administrator> for help.
(Please tick only one box.)*

<ISCED level 3A>	☐ ₁
<ISCED level 3B, 3C>	☐ ₂
<ISCED level 2>	☐ ₃
<ISCED level 1>	☐ ₄
She did not complete <ISCED level 1>	☐ ₅



ST032
ST032Q01NA

Can your mother read and write?

(Please tick only one box.)

My mother cannot read or write.	☐ ₁
My mother can read but not write.	☐ ₂
My mother can read and write well.	☐ ₃
I do not know.	☐ ₄

ST034

Does your mother have any of the following qualifications?

If you are not sure how to answer this question, please ask the <test administrator> for help.

(Please tick one box in each row.)

		Yes	No
ST034Q01TA	<ISCED level 6>	☐ ₁	☐ ₂
ST034Q02TA	<ISCED level 5A>	☐ ₁	☐ ₂
ST034Q03TA	<ISCED level 5B>	☐ ₁	☐ ₂
ST034Q04TA	<ISCED level 4>	☐ ₁	☐ ₂

ST037
ST037Q01NA

What kind of job does your mother have?

(Please tick only one box.)

Stay at home mother (e.g., housewife)	☐ ₁
No job, but looking for work	☐ ₂
Armed forces (e.g., captain, lieutenant, sergeant, corporal, private)	☐ ₃
Labourer (e.g., hotel or office cleaner, farm labourer, factory or mining labourer, kitchen helper, newspaper vendor, mail carrier, janitor)	☐ ₄
Machine Operator (e.g., dry-cleaner, worker in clothing or shoe factory, sewing machine operator, paper products machine operator, crane operator, bus driver, truck driver)	☐ ₅
Craft and Trades Worker (e.g., house builder, bricklayer, carpenter, handicraft maker, dress maker, jewelry maker, traditional food maker, mechanic)	☐ ₆
Skilled Worker (e.g., cattle or dairy farmer, fisher, gardener, child care aide, senior support worker, painter, baker)	☐ ₇
Services and Sales Worker (e.g., cook, waitress, hairdresser, taxi driver, street food vendor, grocer, store attendant, security guard, car dealer, cashier)	☐ ₈
Clerical Worker (e.g., secretary, data entry clerk, bank teller, hotel receptionist, tourist agent, assistant accountant)	☐ ₉
Technical Worker (e.g., building inspector, nursing aide, bookkeeper, chef, electrician, plumber)	☐ ₁₀
Professional (e.g., doctor, school teacher, nurse, engineer, accountant, computer programmer, lawyer, architect, psychologist, social worker)	☐ ₁₁
Manager (e.g., government official, human resource manager in a large company, bank director)	☐ ₁₂

ST036**The following two questions concern your mother's job.***(If she is not working now, please tell us her last main job.)*

ST036Q01TA

What is your mother's main job?
(e.g., school teacher, kitchen-hand, sales manager)*Please write the job title.* _____

ST036Q02TA

What does your mother do in her main job?
(e.g., teaches high school students, helps the cook prepare meals in a restaurant, manages a sales team)*Please use a sentence to describe the kind of work she does or did in that job.*
_____**ST038**

ST038Q01TA

What is the <highest level of schooling> completed by your father?*If you are not sure which box to choose, please ask the <test administrator> for help.**(Please tick only one box.)*

<ISCED level 3A>

☐ ₁

<ISCED level 3B, 3C>

☐ ₂

<ISCED level 2>

☐ ₃

<ISCED level 1>

☐ ₄

He did not complete <ISCED level 1>

☐ ₅**ST039**

ST039Q01NA

Can your father read and write?*(Please tick only one box.)*

My father cannot read or write.

☐ ₁

My father can read but not write.

☐ ₂

My father can read and write well.

☐ ₃

I do not know.

☐ ₄**ST041****Does your father have any of the following qualifications?***If you are not sure how to answer this question, please ask the <test administrator> for help.**(Please tick one box in each row.)*

		Yes	No
ST041Q01TA	<ISCED level 6>	☐ ₁	☐ ₂
ST041Q02TA	<ISCED level 5A>	☐ ₁	☐ ₂
ST041Q03TA	<ISCED level 5B>	☐ ₁	☐ ₂
ST041Q04TA	<ISCED level 4>	☐ ₁	☐ ₂



ST044
ST044Q01NA

What kind of job does your father have?

(Please tick only one box.)

Stay at home father	☐ ₁
No job, but looking for work	☐ ₂
Armed forces (e.g., captain, lieutenant, sergeant, corporal, private)	☐ ₃
Labourer (e.g., hotel or office cleaner, farm labourer, factory or mining labourer, kitchen helper, newspaper vendor, mail carrier, janitor)	☐ ₄
Machine Operator (e.g., dry-cleaner, worker in clothing or shoe factory, sewing machine operator, paper products machine operator, crane operator, bus driver, truck driver)	☐ ₅
Craft and Trades Worker (e.g., house builder, bricklayer, carpenter, handicraft maker, dress maker, jewelry maker, traditional food maker, mechanic)	☐ ₆
Skilled Worker (e.g., cattle or dairy farmer, fisher, gardener, child care aide, senior support worker, painter, baker)	☐ ₇
Services and Sales Worker (e.g., cook, waiter, hairdresser, taxi driver, street food vendor, grocer, store attendant, security guard, car dealer, cashier)	☐ ₈
Clerical Worker (e.g., secretary, data entry clerk, bank teller, hotel receptionist, tourist agent, assistant accountant)	☐ ₉
Technical Worker (e.g., building inspector, nursing aide, bookkeeper, chef, electrician, plumber)	☐ ₁₀
Professional (e.g., doctor, school teacher, nurse, engineer, accountant, computer programmer, lawyer, architect, psychologist, social worker)	☐ ₁₁
Manager (e.g., government official, human resource manager in a large company, bank director)	☐ ₁₂

ST043

The following two questions concern your father's job.

(If he is not working now, please tell us his last main job.)

ST043Q01TA	What is your father's main job? (e.g., school teacher, kitchen-hand, sales manager) Please write the job title. _____
ST043Q02TA	What does your father do in his main job? (e.g., teaches high school students, helps the cook prepare meals in a restaurant, manages a sales team) Please use a sentence to describe the kind of work he does or did in that job. _____

TEACHER QUESTIONNAIRE

Main survey version

Teachers complete the teacher questionnaire. The questionnaire takes about 25 minutes to complete and covers information about the teacher's professional background, the environment, practices and available resources at their school, and their personal life.

Technical terms are given in <brackets> and are adapted to the national context by the national data collection centre of the participating country or economy.

TC001 TC001Q01TA	Are you female or male? (Please tick only one box.)
Female	☐ ₁
Male	☐ ₂

TC002 TC002Q01NA	How old are you? (Please write a number on the line.)
	_____ Years

TC003 TC003Q01NA	What is the highest level of formal education you have completed? (Please tick only one box.)
<ISCED Level 2>	☐ ₁
<ISCED Level 3>	☐ ₂
<ISCED Level 4>	☐ ₃
<ISCED Level 5B>	☐ ₄
<ISCED Level 5A Bachelor degree>	☐ ₅
<ISCED Level 5A Masters degree>	☐ ₆
<ISCED Level 6>	☐ ₇

TC004	What <grades> do you teach in this school? (Please select all that apply.)
TC004Q01NA	<Kindergarten> ☐ ₁
TC004Q01NB	<Grade 1> ☐ ₁
TC004Q01NC	<Grade 2> ☐ ₁
TC004Q01ND	<Grade 3> ☐ ₁
TC004Q01NE	<Grade 4> ☐ ₁
TC004Q01NF	<Grade 5> ☐ ₁
TC004Q01NG	<Grade 6> ☐ ₁
TC004Q01NH	<Grade 7> ☐ ₁
TC004Q01NI	<Grade 8> ☐ ₁
TC004Q01NJ	<Grade 9> ☐ ₁
TC004Q01NK	<Grade 10> ☐ ₁
TC004Q01NL	<Grade 11> ☐ ₁
TC004Q01NM	<Grade 12> ☐ ₁



TC005 TC005Q01NA	Do you teach <multi-grade classrooms> in this school?	
	<i>(Please tick only one box.)</i>	
	Yes	☐ ₁
	No	☐ ₂

TC017	Do you have the following resources in your school, and if so, in what condition are they?				
	<i>(Please tick one box in each row.)</i>				
		No, not available	Yes, but in poor condition	Yes, but in need of minor repairs	Yes, in good condition
TC017Q01NA	Chairs for students	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC017Q02NA	Desks for students	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC017Q03NA	Writing board (black, white, green)	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC017Q04NA	Chalk (or other markers)	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC017Q05NA	A wall chart, map or diagram	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC017Q06NA	One or more bookshelves	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC017Q07NA	Workbooks	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC017Q08NA	Work sheets	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC017Q09NA	Dictionary	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC017Q10NA	Reading, mathematics, or science textbooks	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC017Q11NA	Reference books for teachers	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC017Q12NA	Teacher's guide	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC017Q13NA	School library	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC017Q14NA	Gym	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC017Q15NA	Music room	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC017Q16NA	Art room	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC017Q17NA	Teacher table and chair	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC017Q18NA	Room for student guidance or counselling	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC017Q19NA	<Education resource centre>	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC017Q20NA	<Area for productive projects>	☐ ₁	☐ ₂	☐ ₃	☐ ₄

TC018**How often do you use the following resources in your lessons?***(Please tick one box in each row.)*

		<i>I have never used it</i>	<i>About one to three times a year</i>	<i>About once or twice a month</i>	<i>About once or twice a week</i>	<i>Daily</i>
TC018Q01NA	Writing board (black, white, green)	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₅
TC018Q02NA	Chalk (or other markers)	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₅
TC018Q03NA	A wall chart, map or diagram	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₅
TC018Q04NA	Workbooks	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₅
TC018Q05NA	Work sheets	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₅
TC018Q06NA	Dictionary	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₅
TC018Q07NA	Reading, mathematics, or science textbooks	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₅
TC018Q08NA	Reference books for teachers	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₅
TC018Q09NA	Teacher's guide	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₅
TC018Q10NA	School library	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₅
TC018Q11NA	Gym	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₅
TC018Q12NA	Science lab	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₅
TC018Q13NA	Music room	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₅
TC018Q14NA	Art room	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₅

TC019

TC019Q01NA

Do students in your school have textbooks for instruction in <test language>?*(Please tick only one box.)*

Yes, every student has at least one.	☐ ₁
Yes, but not enough. Sometimes two students need to share a textbook.	☐ ₂
Yes, but so few that sometimes more than two students need to share a textbook.	☐ ₃
No, there are no textbooks.	☐ ₄

TC020

TC020Q01NA

Do students in your school have textbooks for mathematics instruction?*(Please tick only one box.)*

Yes, every student has at least one.	☐ ₁
Yes, but not enough. Sometimes two students have to share a textbook.	☐ ₂
Yes, but so few that sometimes more than two students have to share a textbook.	☐ ₃
No, there are no textbooks.	☐ ₄

**TC035****Do you have the following resources in this school, and if so, in what condition are they?***(Please tick one box in each row.)*

		<i>No, not available</i>	<i>Yes, but in poor condition</i>	<i>Yes, but in need of minor repairs</i>	<i>Yes, in good condition</i>
TC035Q01NA	Computers for students	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC035Q02NA	Internet connection for students	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC035Q03NA	Computers for teachers	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC035Q04NA	Internet connection for teachers	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC035Q05NA	Computers for administrative use	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC035Q06NA	Computer room	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC035Q07NA	Science lab	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC035Q08NA	Photocopier	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC035Q09NA	Overhead or slide projector	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC035Q10NA	Audio or video disk players (e.g., CD, DVD, or VCD)	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC035Q11NA	Radio	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC035Q12NA	TV or screens	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC035Q13NA	Telephone line	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC035Q14NA	Teacher staff room	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC035Q15NA	School administrative office	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC035Q16NA	Storage room	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC035Q17NA	<Education resource centre>	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC035Q18NA	<Area for productive projects>	☐ ₁	☐ ₂	☐ ₃	☐ ₄

TC036**How often do you use the following resources in your lessons?***(Please tick one box in each row.)*

		<i>I have never used it</i>	<i>About one to three times a year</i>	<i>About once or twice a month</i>	<i>About once or twice a week</i>	<i>Daily</i>
TC036Q01NA	Computers for students	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₅
TC036Q02NA	Internet connection for students	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₅
TC036Q03NA	Computers for teachers	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₅
TC036Q04NA	Internet connection for teachers	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₅
TC036Q05NA	Photocopier	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₅
TC036Q06NA	Overhead or slide projector	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₅
TC036Q07NA	Audio and video disk players (e.g., CD, DVD, or VCD)	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₅
TC036Q08NA	Radio	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₅
TC036Q09NA	TV or screens	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₅
TC036Q10NA	Computer room	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₅
TC036Q11NA	<Area for productive projects>	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₅

TC031

How often do family members of students in your class do the following things?*(Please tick one box in each row.)*

		<i>Never or almost never</i>	<i>Sometimes</i>	<i>Often</i>	<i>Always or almost always</i>
TC031Q01NA	They attend parent-teacher meetings.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC031Q02NA	They ask for personal meetings to discuss the progress of their child.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC031Q03NA	They ask for personal meetings to discuss other school matters.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC031Q04NA	They participate in school fundraising events or campaigns.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC031Q05NA	They help in your class.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC031Q06NA	They volunteer after school with helping students do their homework.	☐ ₁	☐ ₂	☐ ₃	☐ ₄

TC006

TC006Q01NA

Are you a <permanent> or <non-permanent> teacher?*(Please tick only one box.)*

<Permanent> teacher who is paid by the government (<permanent> and pensionable, paid by public institutions, such as <local or regional governments or national government>).	☐ ₁
<Permanent> teacher who is not paid by the government (<permanent> and pensionable, paid by private institutions such as <churches, private companies and organisations, or NGOs>).	☐ ₂
<Non-permanent> teacher (temporary, contract, or student teacher) who is paid by the government.	☐ ₃
<Non-permanent> teacher (temporary, contract, or student teacher) who is not paid by the government.	☐ ₄

TC007

TC007Q01NA

In how many schools do you currently teach?*(Please tick only one box.)*

In this school only	☐ ₁
In this school and in one other school	☐ ₂
In this school and in two other schools	☐ ₃
In this school and in more than two other schools	☐ ₄

TC008

TC008Q01NA

In addition to teaching in school, how many hours per week do you work as <private tutor>?*(Please tick only one box.)*

None	☐ ₁
Up to 10 hours per week	☐ ₂
Between 10 and 19 hours per week	☐ ₃
Between 20 and 30 hours per week	☐ ₄
More than 30 hours per week	☐ ₅



TC009
TC009Q01NA

In addition to teaching in school, how many hours per week do you work in another job that is not related to teaching?

(Please tick only one box.)

None	☐ ₁
Up to 10 hours per week	☐ ₂
Between 10 and 19 hours per week	☐ ₃
Between 20 and 30 hours per week	☐ ₄
More than 30 hours per week	☐ ₅

TC010
TC010Q01NA

Which of the following most accurately reflects the condition of your living accommodation?

(Please tick only one box.)

Generally in a poor state	☐ ₁
Some parts require major repairs	☐ ₂
Some parts require minor repairs	☐ ₃
Generally in good condition	☐ ₄

TC012

How many years of work experience do you have?

(Please round up to whole years no matter whether you worked part-time or full-time and write the appropriate number of years on each line. If any option does not apply to you write "0" (zero).)

TC012Q01TA	Year(s) working as a teacher <u>at this school.</u>	_____
TC012Q02TA	Year(s) working as a teacher <u>in total.</u>	_____
TC012Q03TA	Year(s) working in other education roles (do not include years working as a teacher or principal).	_____
TC012Q04TA	Year(s) working in <u>other jobs.</u>	_____

TC013
TC013Q01NA

Did you complete any pre-teaching service training?

(Please tick only one box.)

No	☐ ₁
Yes, for up to 1 year	☐ ₂
Yes, for 1 year	☐ ₃
Yes, for 2 years	☐ ₄
Yes, for 3 years	☐ ₅
Yes, for 4 years or more	☐ ₆

TC014**During the past 12 months, did you participate in any of the following activities?***(Please tick one box in each row.)*

		Yes	No
TC014Q01TA	Qualification programme (e.g., a <degree programme>)	☐ ₁	☐ ₂
TC014Q02TA	A network of teachers formed specifically for the professional development of teachers	☐ ₁	☐ ₂
TC014Q03TA	Individual or collaborative research on a topic of interest to you professionally	☐ ₁	☐ ₂
TC014Q04TA	Mentoring and/or peer observation and coaching, as part of a formal school arrangement	☐ ₁	☐ ₂
TC014Q05TA	Reading professional literature (e.g., journals, evidence-based papers, thesis papers)	☐ ₁	☐ ₂
TC014Q06TA	Engaging in informal dialogue with your colleagues on how to improve your teaching	☐ ₁	☐ ₂
TC014Q07TA	Attending courses or workshops on teaching methods	☐ ₁	☐ ₂
TC014Q08TA	Attending courses or workshops relevant to your subject-matter field	☐ ₁	☐ ₂
TC014Q09TA	Visiting other schools to observe their programmes	☐ ₁	☐ ₂
TC014Q10TA	Attending training course in private companies or other organisations	☐ ₁	☐ ₂

TC015**During the past 12 months, have you had any of the following health problems?***(Please tick one box in each row.)*

		Yes	No
TC015Q01NA	A chronic disease (e.g., heart disease, lung or other respiratory problems, cancer, diabetes)	☐ ₁	☐ ₂
TC015Q02NA	An infectious disease (e.g., cholera, malaria, tuberculosis)	☐ ₁	☐ ₂
TC015Q03NA	Gastrointestinal problems (e.g., heartburn, stomach pain, constipation, diarrhoea)	☐ ₁	☐ ₂
TC015Q04NA	A cold or flu	☐ ₁	☐ ₂
TC015Q05NA	An injury that needed treatment	☐ ₁	☐ ₂
TC015Q06NA	Long-lasting or recurring pain	☐ ₁	☐ ₂
TC015Q07NA	Depression	☐ ₁	☐ ₂
TC015Q08NA	Panic and anxiety attacks	☐ ₁	☐ ₂
TC015Q09NA	Insomnia	☐ ₁	☐ ₂
TC015Q10NA	Fatigue that was long-lasting or recurring	☐ ₁	☐ ₂

TC021

TC021Q01NA

What language do you use most often at home with your family?*(Please tick only one box.)*

<Language 1>	☐ ₁
<Language 2>	☐ ₂
<Language 3>	☐ ₃
< ...etc. >	☐ ₄
Other language	☐ ₅



TC023 TC023Q01NA	What language do you use when teaching your students in this school? (Please tick only one box.)	
	Only <language of instruction>	☐ ₁
	Mostly <language of instruction> but sometimes their home language	☐ ₂
	Sometimes <language of instruction> and sometimes their home language	☐ ₃
	Mostly their home language	☐ ₄
	Always their home language	☐ ₅

TC024	When the <language of instruction> is not the home language of some of your students, how often do you speak with them using their home language? (Please tick one box in each row.)			
		<i>Never</i>	<i>Sometimes</i>	<i>Most of the time</i>
TC024Q01NA	When students are new to the school.	☐ ₁	☐ ₂	☐ ₃
TC024Q02NA	When I explain complex concepts.	☐ ₁	☐ ₂	☐ ₃
TC024Q03NA	When I try to engage them in a discussion.	☐ ₁	☐ ₂	☐ ₃
TC024Q04NA	When I want them to feel appreciated.	☐ ₁	☐ ₂	☐ ₃
TC024Q05NA	When they struggle learning the curriculum.	☐ ₁	☐ ₂	☐ ₃

TC025 TC025Q01NA	Think of a typical class that you teach. What proportion of your students lack the reading skills required to learn the curriculum? (Please tick only one box.)	
	Only a small proportion (less than 10%)	☐ ₁
	About one quarter of the class (25%)	☐ ₂
	About one-half of the class (50%)	☐ ₃
	About three quarters of the class (75%)	☐ ₄
	Most or all of the students in the class (more than 90%)	☐ ₅

TC026 TC026Q01NA	Think of a typical class that you teach. What proportion of your students lack the numeracy skills required to learn the curriculum? (Please tick only one box.)	
	Only a small proportion (less than 10%)	☐ ₁
	About one-quarter of the class (25%)	☐ ₂
	About one-half of the class (50%)	☐ ₃
	About three-quarters of the class (75%)	☐ ₄
	Most or all of the students in the class (more than 90%)	☐ ₅

TC027

In this school, what happens to students who lack the reading or numeracy skills to learn the curriculum?

(Please tick one box in each row.)

		Yes	No
TC027Q01NA	These students repeat a <grade>.	☐ ₁	☐ ₂
TC027Q02NA	These students drop out.	☐ ₁	☐ ₂
TC027Q03NA	Students are assigned to different classes based on their ability.	☐ ₁	☐ ₂
TC027Q04NA	Students are promoted to the next <grade> even if they do not have the reading and numeracy skills to succeed.	☐ ₁	☐ ₂
TC027Q05NA	These students receive extra help in <remedial classes> organised at school.	☐ ₁	☐ ₂
TC027Q06NA	These students receive extra support from staff at the school.	☐ ₁	☐ ₂
TC027Q07NA	Students' parents are asked to help students with homework and extra practice.	☐ ₁	☐ ₂

TC028

To what extent do you agree with the following statements?

(Please tick one box in each row.)

		Strongly Disagree	Disagree	Agree	Strongly Agree
TC028Q01NA	Many of the students who are behind should have been <held back>.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC028Q02NA	Students who repeat one or more <grades> will eventually leave school before they graduate.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC028Q03NA	After students have repeated a <grade>, they are still unprepared for the next grade.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC028Q04NA	Students who have repeated a <grade> learn the hard lesson that they must try harder to succeed.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC028Q05NA	Teachers should try to teach the curriculum, even to students who do not have the basic reading and numeracy skills.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC028Q06NA	Students with disabilities should be taught in <special schools>.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC028Q07NA	Teachers waste their time trying to support teen mothers to remain in school.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC028Q08NA	Teachers should adjust the curriculum to the cultural diversity in their classes	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC028Q09NA	Students who lag behind should be placed in special classes.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC028Q10NA	Teachers should be able to teach classes with students with differing levels of ability.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC028Q11NA	A student should never have to repeat more than one <grade>.	☐ ₁	☐ ₂	☐ ₃	☐ ₄

**TC029**

In your classes, what strategies do you use for teaching students who lack the reading skills to learn the curriculum?

(Please tick one box in each row.)

		Yes	No
TC029Q01NA	I give them separate lessons at school, whenever is possible.	☐ ₁	☐ ₂
TC029Q02NA	I meet them outside of school and offer them extra help.	☐ ₁	☐ ₂
TC029Q03NA	I set lower achievement goals for them.	☐ ₁	☐ ₂
TC029Q04NA	I teach to the rest of the class while they work at their desk.	☐ ₁	☐ ₂
TC029Q05NA	I pair them with stronger students to do group activities.	☐ ₁	☐ ₂
TC029Q06NA	I rely on volunteering parents or members of the community to come to class and work with them.	☐ ₁	☐ ₂
TC029Q07NA	I monitor their progress in reading skills.	☐ ₁	☐ ₂
TC029Q08NA	I vary the learning activities so that all students can learn.	☐ ₁	☐ ₂
TC029Q09NA	I try to enrol them in <remedial classes> outside of regular school hours.	☐ ₁	☐ ₂
TC029Q10NA	I give them extra homework.	☐ ₁	☐ ₂
TC029Q11NA	I suggest to their parents that they find a <private tutor>.	☐ ₁	☐ ₂
TC029Q12NA	I sit them beside stronger students who can help them.	☐ ₁	☐ ₂
TC029Q13NA	I sit them with students at their level and let them work on simpler tasks.	☐ ₁	☐ ₂

TC032

To what extent do you agree with the following statements regarding your school?

(Please tick one box in each row.)

		Strongly disagree	Disagree	Agree	Strongly agree
TC032Q01TA	The principal tries to achieve consensus with all staff when defining priorities and goals in school.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC032Q02TA	The principal is aware of my needs.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC032Q03TA	The principal inspires new ideas for my professional learning.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC032Q04TA	The principal treats teaching staff as professionals.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC032Q05TA	The principal ensures our involvement in decision making.	☐ ₁	☐ ₂	☐ ₃	☐ ₄

TC033

We would like to know how you generally feel about your job. To what extent do you agree with the following statements?

(Please tick one box in each row.)

		Strongly disagree	Disagree	Agree	Strongly agree
TC033Q01TA	The advantages of being a teacher clearly outweigh the disadvantages.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC033Q02TA	If I could decide again, I would still choose to work as a teacher.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC033Q03TA	I regret that I decided to become a teacher.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC033Q04TA	I enjoy working at this school.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC033Q05TA	I wonder whether it would have been better to choose another profession.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC033Q06TA	I would recommend my school as a good place to work.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC033Q07TA	I am satisfied with my performance in this school.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC033Q08TA	All in all, I am satisfied with my job.	☐ ₁	☐ ₂	☐ ₃	☐ ₄

TC034

We would like to know more about your satisfaction with specific aspects of your job. To what extent do you agree with the following statements?

(Please tick one box in each row.)

		<i>Strongly disagree</i>	<i>Disagree</i>	<i>Agree</i>	<i>Strongly agree</i>
TC034Q01NA	The salary that I receive is fair.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC034Q02NA	For the amount of time that I work every day, the pay is too low.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC034Q03NA	The employment benefits that I receive as a teacher meet my expectations.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC034Q04NA	The working conditions and longer holidays make up for the fact that teachers are not very well paid.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC034Q05NA	<Country specific option>	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC034Q06NA	<Country specific option>	☐ ₁	☐ ₂	☐ ₃	☐ ₄
TC034Q07NA	<Country specific option>	☐ ₁	☐ ₂	☐ ₃	☐ ₄

TC016

During the last month, did any of the following situations prevent you from going to work?

(Please tick one box in each row.)

		<i>Yes</i>	<i>No</i>
TC016Q01NA	I experienced a physical illness	☐ ₁	☐ ₂
TC016Q02NA	I experienced emotional or mental health problems	☐ ₁	☐ ₂
TC016Q03NA	Someone in my family was sick	☐ ₁	☐ ₂
TC016Q04NA	Someone in my family needed care	☐ ₁	☐ ₂
TC016Q05NA	I had to run errands	☐ ₁	☐ ₂
TC016Q06NA	I had an appointment with a doctor or dentist	☐ ₁	☐ ₂
TC016Q07NA	I was hospitalised	☐ ₁	☐ ₂
TC016Q08NA	There was a death in my family	☐ ₁	☐ ₂
TC016Q09NA	I had a conflict with the school principal	☐ ₁	☐ ₂
TC016Q10NA	I had a conflict with the school colleagues	☐ ₁	☐ ₂
TC016Q11NA	There was too much violence in the school	☐ ₁	☐ ₂
TC016Q12NA	There was a strike	☐ ₁	☐ ₂
TC016Q13NA	There was no public transportation to reach the school	☐ ₁	☐ ₂
TC016Q14NA	I did not have a mean of transportation to reach the school	☐ ₁	☐ ₂
TC016Q15NA	There was extreme weather or a hazard (e.g., heavy rain, fire)	☐ ₁	☐ ₂



OUT-OF-SCHOOL YOUTH QUESTIONNAIRE

Main survey version

Youth that are not reflected in the school-based survey, including those out-of-school and those who are in school but enrolled at Grade 6 or below, complete the out-of-school youth questionnaire. The questionnaire takes about 30 minutes to complete and covers information about the youth themselves, their well-being, educational attainment and attitudes towards learning, their homes, and their school and learning experiences.

Technical terms are given in <brackets> and are adapted to the national context by the national data collection centre of the participating country or economy.

YI001c	On what date were you born? <i>[Interviewer starts with the day, then the month, and finally the year. If the respondent does not know, ask how old he/she thinks he/she is].</i>		
Y001CQ01TA	Day	_____	Use software to calculate age. If day is unknown, round to nearest 0.1 years.
Y001CQ02TA	Month	_____	If age < 14 or age > 16, terminate interview.
Y001CQ03TA	Year	_____	If age ≥ 14 or ≤ 16 go to YI004ab
	Did not know	☐ ₉₃	Go to YI002c
	Refused to answer	☐ ₉₄	Go to YI002c

YI002c Y002CQ01TA	How old are you?		
	AGE	_____	If age < 14 or age > 16, terminate interview. If age ≥ 14 or ≤ 16 go to YI004c
	Did not know	☐ ₃	Go to YI003c
	Refused to answer	☐ ₄	Go to YI003c

YI003c Y003CQ01NA	Even though you don't know your exact age, are you 14, 15, or 16?		
	Yes	☐ ₁	Go to YI004c
	No	☐ ₂	Terminate interview
	Did not know	☐ ₃	Go to YI004c
	Refused to answer	☐ ₄	Go to YI004c

YI004c Y004CQ01NA	Are you formally enrolled in school? By formally enrolled, I mean having paid fees for the year or being on the school register. <i>[INTERVIEWER: Help clarify the meaning of school and formal enrollment if the respondent is unsure.]</i>		
	Yes	☐ ₁	Go to YI005a
	No	☐ ₂	Go to Part II (YI007b)
	Did not know	☐ ₃	Go to Part II (YI007b)
	Refused to answer	☐ ₄	Go to Part II (YI007b)

YI005a
Y005AQ01TA

What <grade> are you in?

[HELP: Being in a <grade> means attending classes or having paid fees for the year and being on the school register for a specific <grade>.]

_____ <grade>	If grade is less than 7, go to Part II (YI010c). If grade is greater or equal to 7, go to YI006a.	
School is ungraded	☐ _7	Go to YI010c
Did not know	☐ _93	Go to YI010c
Refused to answer	☐ _94	Go to YI010c

YI006a
Y006AQ01NA

Did you attend school regularly within the last month?

[HELP: By going regularly to school I mean that you have been taking courses, attending class, or attending lessons every day or almost every day of the typical school week.]

Yes	☐ _1	Terminate interview
No	☐ _2	Go to YI010c
Did not know	☐ _3	Go to YI010c
Refused to answer	☐ _4	Go to YI010c

YI007b
Y007BQ01NA

Have you ever attended school?

[HELP: Attending school means attending lessons, taking courses, or being in a class.]

Yes	☐ _1	Go to YI008b
No	☐ _2	Go to YI010c
Did not know	☐ _3	Go to YI010c
Refused to answer	☐ _4	Go to YI010c

YI008b
Y008BQ01NA

When you last attended school, what <grade> were you in?

[HELP: Being in a <grade> means attending classes or having paid fees for the year and being on the school register for a specific <grade>.]

_____ <grade>	Go to YI009b	
Did not know	☐ _93	Go to YI010c
Refused to answer	☐ _94	

YI010c
Y001CQ01TA

Are you male or female?

[INTERVIEWER: Indicate the sex of respondent. Indicate Don't Know if the sex is unclear.]

Female	☐ _1
Male	☐ _2
Did not know	☐ _3



YI011c
Y011CQ01NA

We want to know if you have you ever repeated a grade. Did you repeat a grade during <ISCED 1>? If so, was it more than once?

[INTERVIEWER: If the youth responds yes, ask "How many times did you repeat a <grade>?"]

		No, Never	Yes, once	Yes, twice or more	Did not know	Refused to answer	
	At <ISCED 1>	☐ ₁	☐ ₂	☐ ₃	☐ ₉₃	☐ ₉₄	If No, go to YI013c. If Yes, either once or twice or more, go to YI012c. If Did not know or Refused to answer go to Y0012c.

YI012c

Now I want to ask you in greater detail about your school experience during <ISCED 1>.

How many years did you spend in <Grade 1>?

How many years did you spend in <Grade 2>?

How many years did you spend in <Grade 3>?

[INTERVIEWER: Only completed years are to be counted as having been spent in a particular <grade>.]

			Did not know	Refused to answer
Y012CQ01NA	<Grade 1>	_____ Years	☐ ₉₃	☐ ₉₄
Y012CQ02NA	<Grade 2>	_____ Years	☐ ₉₃	☐ ₉₄
Y012CQ03NA	<Grade 3>	_____ Years	☐ ₉₃	☐ ₉₄

YI013c
Y013CQ01NA

We want to know if you have you ever repeated a grade. Did you repeat a grade during <ISCED 2>? If so, was it more than once?

[INTERVIEWER: If the youth responds yes, ask "How many times did you repeat a <grade>?"]

		No, Never	Yes, once	Yes, twice or more	Did not know	Refused to answer
	At <ISCED 2>	☐ ₁	☐ ₂	☐ ₃	☐ ₉₃	☐ ₉₄

YI014c
Y014CQ01NA

We want to know if you have you ever repeated a grade. Did you repeat a grade during <ISCED 3>? If so, was it more than once?

[INTERVIEWER: If the youth responds yes, ask "How many times did you repeat a <grade>?"]

		No, Never	Yes, once	Yes, twice or more	Did not know	Refused to answer
	At <ISCED 3>	☐ ₁	☐ ₂	☐ ₃	☐ ₉₃	☐ ₉₄

YI015c
Y015CQ01NA

Have you ever missed school for more than three months in a row? If so, was it more than once?

[INTERVIEWER: If the youth responds yes, ask «How many times have you missed school for more than three months in a row?»]

No	☐ ₁	Type A, Go to YI017a Type B, go to YI018b
Yes, once	☐ ₂	Go to YI016c
Yes, twice or more	☐ ₃	Go to YI016c
Did not know	☐ ₉₃	Type A, Go to YI017a Type B, go to YI018b
Refused to answer	☐ ₉₄	Type A, Go to YI017a Type B, go to YI018b

YI016c

I am now going to ask you about the reasons why you missed school for more than three months. This could have happened to you several times. Please listen as I read a list of reasons and tell me whether you missed school for more than three months for any of these reasons.

[INTERVIEWER: Read each item separately and indicate Yes or No (or Don't know or Refuse).]

		Yes	No	DK	RF
Y016CQ01NA	because you were pregnant? (enter "no" for males)	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y016CQ02NA	because you were sick?	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y016CQ03NA	because you had to take care of a family member?	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y016CQ04NA	because you had to take care of a sick parents or relative?	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y016CQ05NA	because you had to help with work at home?	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y016CQ06NA	because you had to help with seasonal work on the family land?	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y016CQ07NA	because you had to work to bring money home?	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y016CQ08NA	because you could not understand the language in which the lessons were given?	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y016CQ09NA	because you were no longer interested in school?	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y016CQ10NA	because you were not doing well at school?	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y016CQ11NA	because you did not feel safe at school?	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y016CQ12NA	because you did not have a teacher?	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y016CQ13NA	because the teachers or students were on strike?	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y016CQ14NA	because the school was closed because of a natural disaster (e.g., flood, earthquake)?	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y016CQ15NA	because you could not reach school due to transportation problems?	☐ ₁	☐ ₂	☐ ₃	☐ ₄

YI019c

We would like to know whether certain factors would enable you to continue with your schooling. Please listen as I read a list of factors and tell me whether any of these factors would enable you to continue your schooling.

Would you be more likely to continue your schooling if there were

[INTERVIEWER: Read each item separately and indicate Yes or No (or Don't know or Refuse). If required, repeat the stem, "Would you be more likely to continue your schooling if there were ..."]

		Yes	No	DK	RF
Y019CQ01NA	A school that was closer to home	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y019CQ02NA	Better quality teaching	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y019CQ03NA	Accessible for students with disabilities	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y019CQ04NA	A safe school	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y019CQ05NA	No discrimination, such as gender, racial, ethnic, religious discrimination	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y019CQ06NA	A school that was more accepting of students' differences	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y019CQ07NA	Financial incentives, such as conditional cash transfers	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y019CQ08NA	No school fees or if I did not have to pay them	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y019CQ09NA	A vocational program that would result in a job	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y019CQ10NA	Help with improving my reading skills	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y019CQ11NA	A school where I could learn at my own pace	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y019CQ12NA	Support for my children while attending school	☐ ₁	☐ ₂	☐ ₃	☐ ₄



YI020c
Y020CQ01TA

Overall, how satisfied are you with your life as a whole these days? Using this card on which 1 means you are “completely dissatisfied” and 10 means you are “completely satisfied” where would you put your satisfaction with life as a whole?

[INTERVIEWER: Present SHOW CARD 2. The respondent is to indicate the point on the scale that he or she feels best represents his or her level of satisfaction.]

_____ score on 0-10 scale	
Did not know	☐ ₉₃
Refused to answer	☐ ₉₄

YI021c

I would like to ask you about certain feelings you may have at home, at school, or at work. For each of the following statements, please use the categories on the show card and tell me how often you feel this way at home, at school, or at work.

[INTERVIEWER: Present SHOW CARD 3. Read each item one by one and indicate the response based on the SHOW CARD.]

		Never or almost never	About once a week	2 to 3 times a week	Almost every day	DK	RF
Y021CQ01NA	You are too fearful or nervous.	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₉₃	☐ ₉₄
Y021CQ02NA	You are afraid that other people will think you are stupid.	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₉₃	☐ ₉₄
Y021CQ03NA	You worry more than most people.	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₉₃	☐ ₉₄
Y021CQ04NA	You worry about what other people think of you.	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₉₃	☐ ₉₄
Y021CQ05NA	You cry without a good reason.	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₉₃	☐ ₉₄
Y021CQ06NA	You feel lonely.	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₉₃	☐ ₉₄
Y021CQ07NA	Other people seem to have more fun than you.	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₉₃	☐ ₉₄
Y021CQ08NA	You feel sad or depressed.	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₉₃	☐ ₉₄
Y021CQ09NA	You have trouble falling asleep at night.	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₉₃	☐ ₉₄
Y021CQ10NA	A lot of things seem to bother you.	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₉₃	☐ ₉₄

YI022c
Y022CQ01NA

In general, how would you rate your health? Using this card on which 1 means you have poor health and 10 means you have excellent health, how would you regard your health?

[INTERVIEWER: Present SHOW CARD 2. Explain that 0 indicates poor health and 10 is excellent health.]

_____ score on 0-10 scale	
Did not know	☐ ₉₃
Refused to answer	☐ ₉₄

YI023c

I am now going to ask you about some common health problems that youth of your age experience. During the past year, have you had any of the following health problems? Please listen while I read a list of health problems.

[INTERVIEWER: Read each item separately and indicate Yes or No (or Don't know or Refuse).]

		Yes	No	DK	RF
Y023CQ01NA	A chronic health problem, such as heart disease, lung or other respiratory problems, cancer, or diabetes.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y023CQ01NA	An infectious health problem, such as cholera, malaria, tuberculosis.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y023CQ02NA	Gastrointestinal problems, such as heartburn, stomach pain, constipation, diarrhoea.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y023CQ03NA	A cold or flu.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y023CQ04NA	An injury that needed treatment.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y023CQ05NA	Pain that was long-lasting or recurring.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y023CQ06NA	Depression.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y023CQ07NA	Panic and anxiety attacks.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y023CQ08NA	Insomnia.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y023CQ09NA	Fatigue that was long-lasting or recurring.	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y023CQ10NA	HIV/AIDS	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y023CQ11NA	<country specific>	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y023CQ12NA	<country specific>	☐ ₁	☐ ₂	☐ ₃	☐ ₄

YI024c

Y024CQ01NA

Do you have a disability or medical condition that limits your daily activities?

Yes	☐ ₁	Go to YI025c
No	☐ ₂	Go to YI026c
Did not know	☐ ₃	Go to YI026c
Refused to answer	☐ ₄	Go to YI026c

YI025c

What is the nature of this disability or medical condition? Is it a . . .

[INTERVIEWER: Read the item for each disability and indicate whether the respondent has it or not.]

		Yes	No	DK	RF
Y025CQ01NA	Physical disability?	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y023CQ02NA	Visual impairment?	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y023CQ03NA	Hearing impairment?	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y023CQ04NA	Speech impairment?	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y023CQ05NA	Learning disability?	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y023CQ06NA	Chronic medical condition (e.g., asthma, allergies, diabetes)?	☐ ₁	☐ ₂	☐ ₃	☐ ₄



YI026c
Y026CQ01TA

I would like to ask you few questions concerning the language that you speak at home and in other places, such as at school or at work. What language do you speak most of the time?

<Language of out-of-school test>	☐ ₁
<Language 2>	☐ ₂
<Language 3>	☐ ₃
<Language 4>	☐ ₄
<Language 5>	☐ ₅
<Language 6>	☐ ₆
<Language 7>	☐ ₇
Other language	☐ ₉₃
Refused to answer	☐ ₉₄

YI027c
Y027CQ01NA

When did you begin learning <language of out-of-school achievement test >? Did you learn it at home before you started school, did you learn it at school, or did you never learn it?

At home before I started school	☐ ₁
At school	☐ ₂
I never learned it	☐ ₃
Did not know	☐ ₉₃
Refused to answer	☐ ₉₄

YI028c
Y028CQ01NA

In which language did you first begin learning how to read?

<Language of out-of-school test>	☐ ₁
<Language 2>	☐ ₂
<Language 3>	☐ ₃
<Language 4>	☐ ₄
<Language 5>	☐ ₅
<Language 6>	☐ ₆
<Language 7>	☐ ₇
Other language	☐ ₉₃
Refused to answer	☐ ₉₄

YI030c
Y030CQ01NA

When you entered school for the first time, what language did most of your teachers use for instruction?

[HELP: The language used for instruction refers to the language used by the teachers to discuss topics, to test, to probe, to read books or textbooks.]

<Language of out-of-school test>	☐ ₁
<Language 2>	☐ ₂
<Language 3>	☐ ₃
<Language 4>	☐ ₄
<Language 5>	☐ ₅
<Language 6>	☐ ₆
<Language 7>	☐ ₇
Other language	☐ ₉₃
Refused to answer	☐ ₉₄

YI032c
Y032CQ01TA

I would like to ask you a few questions concerning the country where you and your parents were born. In what country were you born?

<Country of test>	☐ ₁	Go to YI034c
<Country B>	☐ ₂	Go to YI033c
<Country C>	☐ ₃	Go to YI033c
<Country D>	☐ ₄	Go to YI033c
<etc.>	☐ ₅	Go to YI033c
Other countries	☐ ₆	Go to YI033c
Did not Know	☐ ₉₃	Go to YI034c
Refused to answer	☐ ₉₄	Go to YI034c

YI034c
Y034CQ01TA

In what country was your mother born?

<Country of test>	☐ ₁
<Country B>	☐ ₂
<Country C>	☐ ₃
<Country D>	☐ ₄
<etc.>	☐ ₅
Other countries	☐ ₆
Did not Know	☐ ₉₃
Refused to answer	☐ ₉₄


YI035c
Y035CQ01TA In what country was your father born?

<Country of test>	☐ ₁
<Country B>	☐ ₂
<Country C>	☐ ₃
<Country D>	☐ ₄
<etc>	☐ ₅
Other countries	☐ ₆
Did not Know	☐ ₉₃
Refused to answer	☐ ₉₄

YI036c
Y036CQ01NA

This next question is about your daily life and whether you or your family have been able to afford the food you need. In the past 30 days, how often were you hungry because there was not enough food?

[Interviewer hands over the *SHOW CARD*]

	Never or almost never	About once a week	2 to 3 times a week	Almost every day	DK	RF
	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₉₃	☐ ₉₄

YI040c

How many of the following items are there in your home?

[INTERVIEWER: Read the list, item by item, and indicate the number of items per category.]

		Number	DK	RF
Y040CQ01TA	Televisions	____ (drop-down 0 to 5)	☐ ₉₃	☐ ₉₄
Y040CQ02TA	Cars, vans or trucks	____ (drop-down 0 to 5)	☐ ₉₃	☐ ₉₄
Y040CQ03TA	Rooms with a bath or shower	____ (drop-down 0 to 5)	☐ ₉₃	☐ ₉₄
Y040CQ04TA	<Cell phones> with internet access (e.g., smartphones)	____ (drop-down 0 to 5)	☐ ₉₃	☐ ₉₄
Y040CQ05TA	<Cell phones> without internet access	____ (drop-down 0 to 5)	☐ ₉₃	☐ ₉₄
Y040CQ06TA	Computers (desktop computer, portable laptop, or notebook)	____ (drop-down 0 to 5)	☐ ₉₃	☐ ₉₄
Y040CQ07TA	Musical instruments (e.g., guitar, piano, <country-specific example>)	____ (drop-down 0 to 5)	☐ ₉₃	☐ ₉₄

YI041c

Which of the following items are in your home?

[INTERVIEWER: Read the list, item by item, and indicate whether each item is available in the home.]

		Yes	No	DK	RF
Y041CQ01NA	A table to have meals	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y041CQ02NA	A fixed telephone (phone line)	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y041CQ03NA	A washer	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y041CQ04NA	A refrigerator or freezer	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y041CQ05NA	A vacuum cleaner	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y041CQ06NA	A stove or burner for cooking	☐ ₁	☐ ₂	☐ ₃	☐ ₄

YI042c

Y042CQ01NA

How many books are there in your home? Please, do not include magazines, newspapers, or schoolbooks.

[INTERVIEWER: To help respondents give an estimate, indicate that «a stack of 30 books is about this high» (with your hand, indicate about one metre from the floor).]

There are no books	☐ ₁
There are fewer than 10 books	☐ ₂
There are 10 - 50 books	☐ ₃
There are more than 50 books	☐ ₄
Did not know	☐ ₉₃
Refused to answer	☐ ₉₄

YI243c

Y243CQ01NA

What is the floor of your home mostly made of?

[INTERVIEWER: If the respondent does not provide a clear answer, read the following options, one by one. Stop when the respondent indicates the type of flooring.]

Earth, mud, sand or dung	☐ ₁
Wood planks, palm or bamboo	☐ ₂
Parquet, polished wood	☐ ₃
Vinyl or asphalt strips or PVC tiles	☐ ₄
Ceramic, porcelain tiles	☐ ₅
Cement	☐ ₆
Did not know	☐ ₉₃
Refused to answer	☐ ₉₄



YI044c Y044CQ01NA	Do you have access to a <flush toilet> at your home?	
	Yes	☐ ₁
	No	☐ ₂
	Did not know	☐ ₃
	Refused to answer	☐ ₄

YI045c Y045CQ01NA	Do you share a toilet facility with others who are not members of your household? <i>[HELP: A shared toilet is one that other families or people who do not live in the same household, and are unrelated to the respondent, have access to.]</i>	
	Yes	☐ ₁
	No	☐ ₂
	Did not know	☐ ₃
	Refused to answer	☐ ₄

YI049c Y049CQ01NA	Do you have children of your own? These can include biological, adoptive, or step children.	
	Yes	☐ ₁
	No	☐ ₂
	Did not know	☐ ₃
	Refused to answer	☐ ₄

The next set of questions are about your mother or those persons who are like a mother to you, such as a guardian, step-mother, or foster parent.

If you share your time with more than one set of parents or guardians, please answer the following questions for those parents or guardians you spend the most time with.

YI050c Y050CQ01NA	Where is your mother living now?		
	She is living in the same home as me	☐ ₁	Go to YI051c
	She is living alone or with another family	☐ ₂	Go to YI051c
	She is working outside of the <country of test>	☐ ₃	Go to YI052c
	She is deceased	☐ ₄	Go to YI057c
	Don't know	☐ ₉₃	Go to YI051c
	Refused to answer	☐ ₉₄	Go to YI051c

YI051c
Y051CQ01NA

What is your mother's main activity?

[INTERVIEWER: Present SHOW CARD 4. Read each option to the respondent and ask the respondent to say what the **main** activity is.]

Caring for our family	☐ ₁
Working to earn money	☐ ₂
Caring for our family and working to earn money	☐ ₃
Working on our family land	☐ ₄
Going to school	☐ ₅
Recovering from illness	☐ ₆
Looking for work	☐ ₇
Other	☐ ₈
Did not know	☐ ₉₃
Refused to answer	☐ ₉₄

YI052c
Y052CQ01TA

The following two questions concern your mother's job. If she is not working now, please tell us her last main job. What is your mother's main job? (e.g., school teacher, kitchen-hand, sales manager)

[INTERVIEWER: Please write the job title.]

YI252c
Y252CQ01TA

What does your mother do in her main job? (e.g., teaches high school students, helps the cook prepare meals in a restaurant, manages a sales team) Please use a sentence to describe the kind of work she does or did in that job.

[INTERVIEWER: Please write the description.]

YI053c
Y053CQ01NA

I am now going to ask you a few questions about your mother's reading and writing skills.

[INTERVIEWER: Present SHOW CARD 5. Read each response option separately and ask which one applies.]

She cannot read or write	☐ ₁
She can read and write her name	☐ ₂
She can read but not write	☐ ₃
She can read and write simple sentences	☐ ₄
She can read and write well	☐ ₅
Did not know	☐ ₉₃
Refused to answer	☐ ₉₄



YI055c
Y055CQ01TA

What is the <highest level of schooling> completed by your mother?

[INTERVIEWER: Present SHOW CARD 6. Read each of the response options and ask the respondent to indicate the highest level.]

<ISCED level 3A>	☐ ₁
<ISCED level 3B, 3C>	☐ ₂
<ISCED level 2>	☐ ₃
<ISCED level 1>	☐ ₄
She did not complete <ISCED level 1>	☐ ₅
Did not know	☐ ₉₃
Refused to answer	☐ ₉₄

YI056c

Does your mother have any of the following qualifications?

[INTERVIEWER: Present SHOW CARD 7. Read each of the options and ask the respondent to indicate whether or not the qualification is attained.]

		Yes	No	DK	RF
Y056CQ01TA	<ISCED level 6>	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y056CQ02TA	<ISCED level 5A> Master's degree	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y056CQ03TA	<ISCED level 5A> Bachelor's degree	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y056CQ04TA	<ISCED level 5B>	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y056CQ05TA	<ISCED level 4>	☐ ₁	☐ ₂	☐ ₃	☐ ₄

The next set of questions are about your father or those persons who are like a father to you, such as a guardian, step-father, or foster parent.

If you share your time with more than one set of parents or guardians, please answer the following questions for those parents or guardians you spend the most time with.

YI057c
Y057CQ01NA

Where is your father living now?

He is living in the same home as me	☐ ₁	Go to YI058c
He is living alone or with another family	☐ ₂	Go to YI058c
He is working outside of the <country of test>	☐ ₃	Go to YI059c
He is deceased	☐ ₄	Go to YI064c
Don't know	☐ ₉₃	Go to YI058c
Refused to answer	☐ ₉₄	Go to YI058c

YI058c
Y058CQ01NA

What is your father's main activity?

[INTERVIEWER: Present SHOW CARD 4. Read each option to the respondent and ask the respondent to say what the main activity is.]

Caring for our family	☐ ₁
Working to earn money	☐ ₂
Caring for our family and working to earn money	☐ ₃
Working on our family land	☐ ₄
Going to school	☐ ₅
Recovering from illness	☐ ₆
Looking for work	☐ ₇
Other	☐ ₈
Did not know	☐ ₉₃
Refused to answer	☐ ₉₄

YI059c
Y059CQ01TA

The following two questions concern your father's job. If she is not working now, please tell us his last main job. What is your father's main job? (e.g., school teacher, kitchen-hand, sales manager)

[INTERVIEWER: Please write the job title.]

YI259c
Y259CQ01TA

What does your father do in his main job? (e.g., teaches high school students, helps the cook prepare meals in a restaurant, manages a sales team) Please use a sentence to describe the kind of work he does or did in that job.

[INTERVIEWER: Please write the description.]

YI060c
Y060CQ01NA

I am now going to ask you a few questions about your father's reading and writing skills.

[INTERVIEWER: Present SHOW CARD 8. Read each response separately and ask which one applies.]

He cannot read or write	☐ ₁
He can read and write her name	☐ ₂
He can read but not write	☐ ₃
He can read and write simple sentences	☐ ₄
He can read and write well	☐ ₅
Did not know	☐ ₉₃
Refused to answer	☐ ₉₄



YI062c
Y062CQ01TA

What is the <highest level of schooling> completed by your father?

[INTERVIEWER: Present SHOW CARD 6. Read each of the response options and ask the respondent to indicate the highest level.]

<ISCED level 3A>	☐ ₁
<ISCED level 3B, 3C>	☐ ₂
<ISCED level 2>	☐ ₃
<ISCED level 1>	☐ ₄
He did not complete <ISCED level 1>	☐ ₅
Did not know	☐ ₉₃
Refused to answer	☐ ₉₄

YI063c

Does your father have any of the following qualifications?

[INTERVIEWER: Present SHOW CARD 7. Read each of the options and ask the respondent to indicate whether or not the qualification is attained.]

		Yes	No	DK	RF
Y063CQ01TA	<ISCED level 6>	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y063CQ02TA	<ISCED level 5A> Master's degree	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y063CQ03TA	<ISCED level 5A> Bachelor's degree	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y063CQ04TA	<ISCED level 5B>	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y063CQ05TA	<ISCED level 4>	☐ ₁	☐ ₂	☐ ₃	☐ ₄

YI066c

I am going to ask you about the work that you did last week in your household.

[INTERVIEWER: Ask each question separately.]

		Yes	No	DK	RF
Y066CQ01NA	In the past week, did you look after young children?	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y066CQ02NA	In the past week, did you look after elderly adults?	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y066CQ03NA	In the past week, did you take care of sick family members and relatives?	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y066CQ04NA	In the past week, did you cook a family meal?	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y066CQ05NA	In the past week, did you do house cleaning?	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y066CQ06NA	In the past week, did you wash clothes?	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y066CQ07NA	In the past week, did you fetch water?	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y066CQ08NA	In the past week, did you chop or collect fire wood?	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y066CQ09NA	In the past week, did you do the family grocery shopping?	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y066CQ10NA	In the past week, did you work in the family <garden>?	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y066CQ11NA	In the past week, did you take care of livestock?	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y066CQ12NA	In the past week, did you help with the family business without pay?	☐ ₁	☐ ₂	☐ ₃	☐ ₄

YI067c

In general, how often do your parents or someone in your family do the following things with you?

[INTERVIEWER: Present SHOW CARD 9.]

		Never or hardly ever	A few times a year	About once a month	Several times a month	Several times a week	DK	RF
Y067CQ01NA	Discuss political or social issues with you	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₅	☐ ₉₃	☐ ₉₄
Y067CQ01NA	Discuss books, films, or television programmes with you	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₅	☐ ₉₃	☐ ₉₄
Y067CQ02NA	Eat <the main meal> with you	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₅	☐ ₉₃	☐ ₉₄
Y067CQ03NA	Spend time just talking with you	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₅	☐ ₉₃	☐ ₉₄
Y067CQ04NA	Talk to you about the importance of completing <secondary school>	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₅	☐ ₉₃	☐ ₉₄
Y067CQ05NA	Talk to you about your future education	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₅	☐ ₉₃	☐ ₉₄

YI068c

Y068CQ01NA

Please, think about the last time you were in school. Although you may have used a variety of means of transportation, think about the one you used most often. When you went to school, how did you usually travel?

[INTERVIEWER: If the respondent does not provide a mode of transportation, present SHOW CARD 10, read each of the options and indicate one response.]

Did not need to travel as boarded at the school	☐ ₁	Go to YI070c
Walked or ran	☐ ₂	
By bicycle	☐ ₃	
By motorcycle or scooter	☐ ₄	
By boat or canoe	☐ ₅	
By car	☐ ₆	
By <school bus, truck, or van>	☐ ₇	
By <train or metro>	☐ ₈	
<Country specific>	☐ ₉	
<Country specific>	☐ ₁₀	
Other	☐ ₁₁	
Did not know	☐ ₉₃	Go to YI069c
Refused to answer	☐ ₉₄	

YI069c

Y069CQ01NA

How long would it usually take you to get from your home to school?

[INTERVIEWER: If the youth does not provide the travel time, present SHOW CARD 11, read each of the options and indicate one response.]

15 minutes or less	☐ ₁
More than 15 minutes but less than 30 minutes	☐ ₂
30 minutes or more, but less than 60 minutes	☐ ₃
60 minutes or more, but less than 90 minutes	☐ ₄
90 minutes or more	☐ ₅
Did not know	☐ ₉₃
Refused to answer	☐ ₉₄

**YI070c**

I am now going to ask you about your overall experience with being in school and learning from it. Please, tell me to what extent do you agree with the following statements?

[INTERVIEWER: Present SHOW CARD 12.]

		Strongly agree	Agree	Disagree	Strongly disagree	DK	RF
Y070CQ01NA	School has done little to prepare you for adult life	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₉₃	☐ ₉₄
Y070CQ02NA	School is a waste of time	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₉₃	☐ ₉₄
Y070CQ03NA	School gave you confidence to make decisions	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₉₃	☐ ₉₄
Y070CQ04NA	School taught you things which could be useful in a job	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₉₃	☐ ₉₄
Y070CQ05NA	Trying hard at school helps you get a good job	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₉₃	☐ ₉₄

YI071c

Still thinking about your experience at school, to what extent do you agree with the following statements?

[INTERVIEWER: Present SHOW CARD 12.]

		Strongly agree	Agree	Disagree	Strongly disagree	DK	RF
Y071CQ01NA	You felt safe at your school	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₉₃	☐ ₉₄
Y071CQ02NA	You felt safe on your way to school	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₉₃	☐ ₉₄
Y071CQ03NA	You felt safe on your way home from school	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₉₃	☐ ₉₄

YI072a

I want you to think about your last four weeks in school and whether particular events occurred?

[INTERVIEWER: Read each of the following options and indicate one response in each row.]

		Yes	No	DN	RF
Y072AQ01NA	Were you in a physical fight on school property?	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y072AQ02NA	Did you stay home from school because you felt unsafe?	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y072AQ03NA	Was your school vandalized?	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y072AQ04NA	Did you give money to someone at school because they threatened to hurt you?	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y072AQ05NA	Did you witness a fight on school property in which someone got hurt?	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y072AQ06NA	Did you see gangs in and around the school?	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y072AQ07NA	Did you hear a student threaten to hurt another student?	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y072AQ08NA	Did someone steal something of yours at school?	☐ ₁	☐ ₂	☐ ₃	☐ ₄
Y072AQ09NA	Did you see a student carrying a gun or a knife at school?	☐ ₁	☐ ₂	☐ ₃	☐ ₄

YI073a

I am going to ask you about your recent experience in school, and especially about your thoughts in relation to other students and to teachers. To what extent do you agree with the following statements?

[INTERVIEWER: Present SHOW CARD 12.]

		Strongly agree	Agree	Disagree	Strongly disagree	DK	RF
Y073AQ01NA	Students socialize in groups based on their culture.	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₉₃	☐ ₉₄
Y073AQ02NA	Rich or poor, teachers treat us all equally.	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₉₃	☐ ₉₄
Y073AQ03NA	Students who struggle in school are treated fairly by your teachers.	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₉₃	☐ ₉₄
Y073AQ04NA	Teachers treat students differently depending on their cultural background.	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₉₃	☐ ₉₄
Y073AQ05NA	Teachers care less about girls than boys.	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₉₃	☐ ₉₄

YI074b

Y074BQ01NA

Do you work for at least ten hours per week for pay in a regular job? By regular job, I mean a job in which you worked for a company or for someone outside of your family to earn money?

Yes	☐ ₁	Go to YI075b
No	☐ ₂	
Did not know	☐ ₃	Go to YI086b
Refused to answer	☐ ₄	

YI075b

Y075BQ01NA

The following two questions concern your current job. What is your main job? (e.g., school teacher, kitchen-hand, sales manager)

[INTERVIEWER: Please write the job title.]

YI275b

Y075BQ02NA

What do you do in your main job? (e.g., teach high school students, help the cook prepare meals in a restaurant, manage a sales team) Please use a sentence to describe the kind of work you do in that job.

[INTERVIEWER: Please write the description.]

YI076b

Y076BQ01NA

How many hours do you usually work per week in this job?

[INTERVIEWER: If the number of hours per week varies, ask the respondent for the average over the last 4 weeks. Present SHOW CARD 13 and indicate the response that applies.]

10 – 20 hours	☐ ₁	Go to YI077b
21 – 30 hours	☐ ₂	Go to YI077b
31 – 40 hours	☐ ₃	Go to YI077b
41 – 50 hours	☐ ₄	Go to YI077b
More than 50 hours	☐ ₅	Go to YI077b
Did not know	☐ ₉₃	Go to YI086b
Refused to answer	☐ ₉₄	Go to YI086b



YI077b
Y077BQ01NA

What is the easiest way for you to tell us how much you are paid for your current job? Would it be ...

[INTERVIEWER: Read the response options to the respondent.]

Per hour	☐ ₁	Go to YI078b
Per day	☐ ₂	Go to YI079b
Per week	☐ ₃	Go to YI080b
Per two-week period	☐ ₄	Go to YI081b
Per month	☐ ₅	Go to YI082b
Per year	☐ ₆	Go to YI083b
Piece rate	☐ ₇	Go to YI084b
I get no salary or wage at all	☐ ₈	Go to YI085b
Did not know	☐ ₉₃	Go to YI078b
Refused to answer	☐ ₉₄	Go to YI078b

YI078b
Y078BQ01NA

About how much are you paid per hour at this job? If you are unsure, please try to tell me an approximate amount.

[INTERVIEWER: Enter the amount in the <country currency>.]

_____ per hour		Go to YI086b
Did not know	☐ ₉₃	Go to YI079b
Refused to answer	☐ ₉₄	Go to YI079b

YI079b
Y079BQ01NA

About how much are you paid per day at this job? If you are unsure, please try to tell me an approximate amount.

[INTERVIEWER: Enter the amount in the <country currency>.]

_____ per day		Go to YI086b
Did not know	☐ ₉₃	Go to YI080b
Refused to answer	☐ ₉₄	Go to YI080b

YI080b
Y080BQ01NA

About how much are you paid per week at this job? If you are unsure, please try to tell me an approximate amount.

[INTERVIEWER: Enter the amount in the <country currency>.]

_____ per week		Go to YI086b
Did not know	☐ ₉₃	Go to YI081b
Refused to answer	☐ ₉₄	Go to YI081b

YI081b
Y081BQ01NA

About how much are you paid per two-week period at this job? If you are unsure, please try to tell me an approximate amount.

[INTERVIEWER: Enter the amount in the <country currency>.]

_____ per two-week period		Go to YI086b
Did not know	☐ ₉₃	Go to YI082b
Refused to answer	☐ ₉₄	Go to YI082b

YI082b
Y082BQ01NA

About how much are you paid per month at this job? If you are unsure, please try to tell me an approximate amount.

[INTERVIEWER: Enter the amount in the <country currency>.]

_____ per month		Go to YI086b
Did not know	☐ ₉₃	Go to YI083b
Refused to answer	☐ ₉₄	Go to YI083b

YI083b
Y083BQ01NA

About how much are you paid per year at this job? If you are unsure, please try to tell me an approximate amount.

[INTERVIEWER: Enter the amount in the <country currency>.]

_____ per year		Go to YI086b
Did not know	☐ ₉₃	Go to YI084b
Refused to answer	☐ ₉₄	Go to YI084b

YI084b

About how long does it take you to make one piece?

[INTERVIEWER: Fill in one category only, depending on whether the respondent replies in minutes or hours.]

Y084BQ01NA	_____ minutes		Go to YI085b
Y084BQ02NA	_____ hours		Go to YI085b
	Does not know	☐ ₉₃	Go to YI085b
	Refused to answer	☐ ₉₄	Go to YI085b

YI085b
Y085BQ01NA

About how much are you paid per piece at this job? If you are unsure, please try to tell me an approximate amount.

[INTERVIEWER: Enter the amount in the <country currency>.]

_____ per piece		
Did not know	☐ ₉₃	
Refused to answer	☐ ₉₄	

YI086b
Y086BQ01NA

During the past month, have you worked on your own to earn money?

Yes	☐ ₁	Go to YI087b
No	☐ ₂	
Did not know	☐ ₃	Go to YI093b
Refused to answer	☐ ₄	



YI087b
Y087BQ01NA

How many hours do you usually work per week in this job?

[INTERVIEWER: If the number of hours per week varies, ask the youth for the average over the last 4 weeks. Present SHOW CARD 13 and indicate the response that applies.]

10 – 20 hours	☐ ₁	Go to YI088b
21 – 30 hours	☐ ₂	Go to YI088b
31 – 40 hours	☐ ₃	Go to YI088b
41 – 50 hours	☐ ₄	Go to YI088b
More than 50 hours	☐ ₅	Go to YI088b
Does not know	☐ ₉₃	Go to YI093b
Refused to answer	☐ ₉₄	Go to YI093b

YI093b
Y093BQ01NA

During the past month, have you worked in a family business or on the family farm?

Yes	☐ ₁	Go to YI094b
No	☐ ₂	
Did not know	☐ ₃	Go to YI101b
Refused to answer	☐ ₄	

YI094b
Y094BQ01NA

How many hours do you usually work per week in this job?

[INTERVIEWER: If the number of hours per week varies, ask the youth for the average over the last 4 weeks. Present SHOW CARD 13 and indicate the response that applies.]

10 – 20 hours	☐ ₁	Go to YI095b
21 – 30 hours	☐ ₂	Go to YI095b
31 – 40 hours	☐ ₃	Go to YI095b
41 – 50 hours	☐ ₄	Go to YI095b
More than 50 hours	☐ ₅	Go to YI095b
Does not know	☐ ₉₃	Go to YI101b
Refused to answer	☐ ₉₄	Go to YI101b

YI101b

How often do you do each of the following?

[INTERVIEWER: Present SHOW CARD 3.]

		Never or almost never	About once a week	2 to 3 times a week	Almost every day	DK	RF
YI01BQ01NA	Read a newspaper or magazine	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₉₃	☐ ₉₄
YI01BQ02NA	Read a book	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₉₃	☐ ₉₄
YI01BQ03NA	Write a text (SMS) or email message	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₉₃	☐ ₉₄
YI01BQ04NA	Write a note to a family member or friend	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₉₃	☐ ₉₄

OUT-OF-SCHOOL YOUTH PARENT QUESTIONNAIRE

Main survey version

The parent or person most knowledgeable about the out-of-school youth completes the parent questionnaire. The questionnaire takes about 20 minutes to complete and covers information about the parent or caregiver's background, the youth and the family, the youth's early years and educational experience, and the parent or caregiver's educational expectations for the youth.

Technical terms are given in <brackets> and are adapted to the national context by the national data collection centre of the participating country or economy.

About the parent or person most knowledgeable about the youth

HH001 HH001Q01NA	What is your relationship to the youth we interviewed? You are his or her: (Please tick only one box.)
Mother	☐ ₁
Father	☐ ₂
Grandmother	☐ ₃
Grandfather	☐ ₄
A brother or a sister (including adoptive brother, adoptive sister)	☐ ₅
A husband or wife	☐ ₆
Other relative (e.g., <aunt, uncle>, or <cousin>)	☐ ₇
Friend	☐ ₈
Member of another family living in the same household	☐ ₉
A caregiver (e.g., <keeper or legal tutor>)	☐ ₁₀
I refuse to answer	☐ ₉₄

HH002 HH002Q01NA	What is your main activity? (Please tick only one box.)
Caring for my family	☐ ₁
Working to earn money	☐ ₂
Caring for my family <i>and</i> working to earn money	☐ ₃
Working on my family land	☐ ₄
Going to school	☐ ₅
Recovering from illness	☐ ₆
Looking for work	☐ ₇
Other (please specify) _____	☐ ₈
I refuse to answer	☐ ₉₄



HH003 HH003Q01TA	What is your main job? (e.g., school teacher, kitchen-hand, sales manager) (If you are not working now, what was your last main job?)
	Please write the job title. _____

HH004 HH004Q01TA	What do you do in your main job? (e.g., I teach high school students, I help the cook prepare meals in a restaurant, I manage a sales team)
	Please describe the kind of work you do or did in that job. _____

HH005 HH005Q01NA	What is the highest level of schooling that you have completed? (Please select one response.)																				
	<table border="1"> <tr> <td>I never went to school</td> <td>☐ ₁</td> </tr> <tr> <td>I started <ISCED 1> but never completed it</td> <td>☐ ₂</td> </tr> <tr> <td><ISCED level 1></td> <td>☐ ₃</td> </tr> <tr> <td><ISCED level 2></td> <td>☐ ₄</td> </tr> <tr> <td><ISCED level 3></td> <td>☐ ₅</td> </tr> <tr> <td><ISCED level 4></td> <td>☐ ₆</td> </tr> <tr> <td><ISCED level 5></td> <td>☐ ₇</td> </tr> <tr> <td><ISCED level 6></td> <td>☐ ₈</td> </tr> <tr> <td>I don't know</td> <td>☐ ₉₃</td> </tr> <tr> <td>I refuse to answer</td> <td>☐ ₉₄</td> </tr> </table>	I never went to school	☐ ₁	I started <ISCED 1> but never completed it	☐ ₂	<ISCED level 1>	☐ ₃	<ISCED level 2>	☐ ₄	<ISCED level 3>	☐ ₅	<ISCED level 4>	☐ ₆	<ISCED level 5>	☐ ₇	<ISCED level 6>	☐ ₈	I don't know	☐ ₉₃	I refuse to answer	☐ ₉₄
I never went to school	☐ ₁																				
I started <ISCED 1> but never completed it	☐ ₂																				
<ISCED level 1>	☐ ₃																				
<ISCED level 2>	☐ ₄																				
<ISCED level 3>	☐ ₅																				
<ISCED level 4>	☐ ₆																				
<ISCED level 5>	☐ ₇																				
<ISCED level 6>	☐ ₈																				
I don't know	☐ ₉₃																				
I refuse to answer	☐ ₉₄																				

About the youth's educational experience

HH006 HH006Q01NA	Did he or she receive <formal ECEC>? (Please tick only one box.)										
	<table border="1"> <tr> <td>No</td> <td>☐ ₁</td> </tr> <tr> <td>Yes, for one year or less</td> <td>☐ ₂</td> </tr> <tr> <td>Yes, for more than one year</td> <td>☐ ₃</td> </tr> <tr> <td>I don't know</td> <td>☐ ₉₃</td> </tr> <tr> <td>I refuse to answer</td> <td>☐ ₉₄</td> </tr> </table>	No	☐ ₁	Yes, for one year or less	☐ ₂	Yes, for more than one year	☐ ₃	I don't know	☐ ₉₃	I refuse to answer	☐ ₉₄
No	☐ ₁										
Yes, for one year or less	☐ ₂										
Yes, for more than one year	☐ ₃										
I don't know	☐ ₉₃										
I refuse to answer	☐ ₉₄										

HH007 HH007Q01NA	Did the youth we interviewed attend <ISCED 0>? (Please tick only one box.)										
	<table border="1"> <tr> <td>No</td> <td>☐ ₁</td> </tr> <tr> <td>Yes, for one year or less</td> <td>☐ ₂</td> </tr> <tr> <td>Yes, for more than one year</td> <td>☐ ₃</td> </tr> <tr> <td>I don't know</td> <td>☐ ₉₃</td> </tr> <tr> <td>I refuse to answer</td> <td>☐ ₉₄</td> </tr> </table>	No	☐ ₁	Yes, for one year or less	☐ ₂	Yes, for more than one year	☐ ₃	I don't know	☐ ₉₃	I refuse to answer	☐ ₉₄
No	☐ ₁										
Yes, for one year or less	☐ ₂										
Yes, for more than one year	☐ ₃										
I don't know	☐ ₉₃										
I refuse to answer	☐ ₉₄										

HH008

When the youth we interviewed was a child (from 0 to 6 years old), how often did you or someone else in your home do the following activities with him or her?

(Please tick one box in each row.)

		<i>Never or hardly ever</i>	<i>Once or twice a month</i>	<i>Once or twice a week</i>	<i>Every day or almost every day</i>	<i>I don't know</i>	<i>I refuse to answer</i>
HH008Q01NA	Read books	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₉₃	☐ ₉₄
HH008Q02NA	Tell stories	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₉₃	☐ ₉₄
HH008Q03NA	Sing songs	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₉₃	☐ ₉₄
HH008Q04NA	Play with alphabet toys (for example, blocks with letters of the alphabet)	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₉₃	☐ ₉₄
HH008Q06NA	Talk about what he or she had read	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₉₃	☐ ₉₄
HH008Q07NA	Play word games	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₉₃	☐ ₉₄
HH008Q08NA	Write letters or words	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₉₃	☐ ₉₄
HH008Q09NA	Read aloud signs and labels	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₉₃	☐ ₉₄
HH008Q11NA	Identify sounds or letters	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₉₃	☐ ₉₄

About the youth's early years of life

HH009

While pregnant with the youth we interviewed, did the mother experience any of the following events?

(Please tick all that apply.)

HH009Q01NA	She had pregnancy diabetes	☐ ₁
HH009Q01NB	She had high blood pressure (e.g. <preeclampsia>)	☐ ₁
HH009Q01NC	She had an infection (e.g. <toxoplasmosis, dengue>)	☐ ₁
HH009Q01ND	She had rubella	☐ ₁
HH009Q01NE	She had chickenpox	☐ ₁
HH009Q01NF	She had mental health problems	☐ ₁
HH009Q01NG	She was exposed to environmental toxins or toxic wastes	☐ ₁
HH009Q01NH	She was malnourished	☐ ₁
HH009Q01NI	She had anemia	☐ ₁
HH009Q01NJ	She had scarlet fever or seizures	☐ ₁
HH009Q01NK	She smoked	☐ ₁
HH009Q01NL	She drank alcohol daily	☐ ₁
HH009Q01NM	She used drugs	☐ ₁
HH009Q01NN	I don't know	☐ ₉₃
HH009Q01NO	I refuse to answer	☐ ₉₄



HH010 HH010Q01NA	Under what conditions was he or she born? (Please tick only one box.)
	In a hospital, under the care of a specialist (doctor or specialized nurse) ☐ ₁
	In a hospital, under the care of a midwife ☐ ₂
	In a health centre, under the care of a specialist (doctor or specialized nurse) ☐ ₃
	In a health centre, under the care of a midwife ☐ ₄
	At home, under the care of a midwife ☐ ₅
	At home, under the care of a family member or other person ☐ ₆
	At home, without any help ☐ ₇
	I don't know ☐ ₉₃
	I refuse to answer ☐ ₉₄

HH011	Was he or she... (Please tick one box in each row.)				
		Yes	No	I don't know	I refuse to answer
HH011Q01NA	Born prematurely?	☐ ₁	☐ ₂	☐ ₉₃	☐ ₉₄
HH011Q02NA	Born underweight?	☐ ₁	☐ ₂	☐ ₉₃	☐ ₉₄
HH011Q03NA	Delivered with a birthing aide <e.g. suction, clamps>?	☐ ₁	☐ ₂	☐ ₉₃	☐ ₉₄

HH013	How was he or she fed during the first six months? (Please tick one box in each row.)				
		Yes	No	I don't know	I refuse to answer
HH013Q01NA	Breast fed	☐ ₁	☐ ₂	☐ ₉₃	☐ ₉₄
HH013Q02NA	Fed with <formula>	☐ ₁	☐ ₂	☐ ₉₃	☐ ₉₄
HH013Q03NA	Fed with animal milk (e.g., cow's milk or goat milk)	☐ ₁	☐ ₂	☐ ₉₃	☐ ₉₄
HH013Q04NA	Fed with vegetal milk (e.g. soya, almond)	☐ ₁	☐ ₂	☐ ₉₃	☐ ₉₄
HH013Q05NA	Fed with alternative beverage (e.g. <corn beverage>)	☐ ₁	☐ ₂	☐ ₉₃	☐ ₉₄

HH014
HH014Q01NA

During the first two years of life, about how often was he or she hungry because there was not enough food in the house?

(Please tick only one box.)

Never	☐ _1
Once a month	☐ _2
About once a week	☐ _3
Two or three times a week	☐ _4
Almost every day	☐ _5
I don't know	☐ _93
I refuse to answer	☐ _94

HH015
HH015Q01NA

During the first two years of life, about how many times a day was he or she fed?

(Please tick only one box.)

Once	☐ _1
Twice	☐ _2
Three times	☐ _3
Four times	☐ _4
More than four times	☐ _5
I don't know	☐ _93
I refuse to answer	☐ _94

HH016

During the first five years of life, did he or she experience any of the following health problems?

(Please tick one box in each row.)

		Yes	No	I don't know	I refuse to answer
HH016Q01NA	Parasite infection (e.g. hookworm)	☐ _1	☐ _2	☐ _93	☐ _94
HH016Q02NA	Allergies	☐ _1	☐ _2	☐ _93	☐ _94
HH016Q03NA	Infectious disease (e.g., cholera, tuberculosis)	☐ _1	☐ _2	☐ _93	☐ _94
HH016Q04NA	Respiratory infection or pneumonia	☐ _1	☐ _2	☐ _93	☐ _94
HH016Q05NA	Anemia	☐ _1	☐ _2	☐ _93	☐ _94
HH016Q06NA	Viral disease such as measles, rubella, chickenpox, polio, or yellow fever	☐ _1	☐ _2	☐ _93	☐ _94
HH016Q07NA	<malaria>	☐ _1	☐ _2	☐ _93	☐ _94
HH016Q08NA	<dengue>	☐ _1	☐ _2	☐ _93	☐ _94
HH016Q09NA	<HIV/AIDS>	☐ _1	☐ _2	☐ _93	☐ _94



About the parent or caregiver's educational expectations for the youth

HH019

Which of the following factors do you think may prevent the youth we interviewed from completing <compulsory education>?

(Please tick one box in each row.)

		Yes	No	I don't know	I refuse to answer
HH019Q01NA	Distance that he or she would need to travel to school	☐ ₁	☐ ₂	☐ ₉₃	☐ ₉₄
HH019Q02NA	Lack of motivation for further studies	☐ ₁	☐ ₂	☐ ₉₃	☐ ₉₄
HH019Q03NA	Poor grades	☐ ₁	☐ ₂	☐ ₉₃	☐ ₉₄
HH019Q04NA	Difference in the language of instruction	☐ ₁	☐ ₂	☐ ₉₃	☐ ₉₄
HH019Q05NA	Being expelled from school	☐ ₁	☐ ₂	☐ ₉₃	☐ ₉₄
HH019Q06NA	Alcohol or drug addiction	☐ ₁	☐ ₂	☐ ₉₃	☐ ₉₄
HH019Q07NA	Migration to another country	☐ ₁	☐ ₂	☐ ₉₃	☐ ₉₄
HH019Q08NA	Parents relocating for work	☐ ₁	☐ ₂	☐ ₉₃	☐ ₉₄
HH019Q09NA	Not having enough money	☐ ₁	☐ ₂	☐ ₉₃	☐ ₉₄
HH019Q10NA	Wanting to start earning money in a full-time job	☐ ₁	☐ ₂	☐ ₉₃	☐ ₉₄
HH019Q11NA	Caring for his or her children	☐ ₁	☐ ₂	☐ ₉₃	☐ ₉₄
HH019Q12NA	Caring for parents or other relatives	☐ ₁	☐ ₂	☐ ₉₃	☐ ₉₄
HH019Q13NA	His or her beliefs that school will not pay off in the long run	☐ ₁	☐ ₂	☐ ₉₃	☐ ₉₄
HH019Q14NA	Not knowing what he or she wants to do later on	☐ ₁	☐ ₂	☐ ₉₃	☐ ₉₄
HH019Q15NA	Fear of violence in the school he or she would attend	☐ ₁	☐ ₂	☐ ₉₃	☐ ₉₄
HH019Q16NA	Inability to enrol in school because of <racial or ethnic discrimination>	☐ ₁	☐ ₂	☐ ₉₃	☐ ₉₄
HH019Q17NA	Inability to enrol in school because of <gender discrimination>	☐ ₁	☐ ₂	☐ ₉₃	☐ ₉₄
HH019Q18NA	Inability to enrol in school because of <religious discrimination>	☐ ₁	☐ ₂	☐ ₉₃	☐ ₉₄
HH019Q19NA	A health problem or disability	☐ ₁	☐ ₂	☐ ₉₃	☐ ₉₄

HH020

Thinking about school, to what extent do you agree with the following statements?*(Please tick one box in each row.)*

		<i>Strongly agree</i>	<i>Agree</i>	<i>Disagree</i>	<i>Strongly disagree</i>	<i>I don't know</i>	<i>I refuse to answer</i>
HH020Q01NA	School does little to prepare youth for adult life	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₉₃	☐ ₉₄
HH020Q02NA	School helps getting a job	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₉₃	☐ ₉₄
HH020Q03NA	School is a waste of time	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₉₃	☐ ₉₄
HH020Q04NA	School teaches how to be a citizen	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₉₃	☐ ₉₄
HH020Q05NA	School ignores native languages and cultures	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₉₃	☐ ₉₄
HH020Q06NA	School is a waste of money	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₉₃	☐ ₉₄
HH020Q07NA	School integrates people into society	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₉₃	☐ ₉₄
HH020Q08NA	School builds confidence to make decisions	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₉₃	☐ ₉₄
HH020Q09NA	School teaches useful work skills	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₉₃	☐ ₉₄
HH020Q10NA	Top-of-their class graduates get very good jobs	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₉₃	☐ ₉₄
HH020Q11NA	School is the only way to get better life opportunities	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₉₃	☐ ₉₄
HH020Q12NA	School helps overcoming ignorance	☐ ₁	☐ ₂	☐ ₃	☐ ₄	☐ ₉₃	☐ ₉₄

HH021

Did the youth we interviewed or his or her family, receive any of the following benefits from government or local authorities?*(Please tick one box in each row.)*

		<i>Yes</i>	<i>No</i>	<i>I don't know</i>	<i>I refuse to answer</i>
HH021Q01NA	Child or family <benefits>	☐ ₁	☐ ₂	☐ ₉₃	☐ ₉₄
HH021Q02NA	Child or family grants to attend school	☐ ₁	☐ ₂	☐ ₉₃	☐ ₉₄
HH021Q03NA	Cash transfer to attend school	☐ ₁	☐ ₂	☐ ₉₃	☐ ₉₄
HH021Q04NA	Scholarships	☐ ₁	☐ ₂	☐ ₉₃	☐ ₉₄
HH021Q05NA	School meals	☐ ₁	☐ ₂	☐ ₉₃	☐ ₉₄
HH021Q06NA	Vouchers or coupons for food	☐ ₁	☐ ₂	☐ ₉₃	☐ ₉₄
HH021Q07NA	Waiver of school fees	☐ ₁	☐ ₂	☐ ₉₃	☐ ₉₄
HH021Q08NA	School books or notebooks	☐ ₁	☐ ₂	☐ ₉₃	☐ ₉₄
HH021Q09NA	<Solidarity bags>	☐ ₁	☐ ₂	☐ ₉₃	☐ ₉₄
HH021Q10NA	<Country specific>	☐ ₁	☐ ₂	☐ ₉₃	☐ ₉₄

HH022

HH022Q01NA

Did you have help in completing this questionnaire?*(Please tick only one box.)*

No, I completed it myself.	☐ ₁
Yes, the interviewer helped me.	☐ ₂
Yes, the youth that was interviewed helped me.	☐ ₃
Yes, another person helped me.	☐ ₄



HOUSEHOLD OBSERVATION QUESTIONNAIRE

Main study version

The interviewer conducting the out-of-school youth survey completes the household observation questionnaire. The questionnaire takes about 15 minutes to complete and covers information about the location and surrounding characteristics of the youth's household.

Technical terms are given in <brackets> and are adapted to the national context by the national data collection centre of the participating country or economy.

HO004 HO004Q01NA	What is the main road to the dwelling made of? (Select the most common material.)
Tar	☐ ₁
Dirt	☐ ₂
Trail or path	☐ ₃
There is no road	☐ ₄
Other: _____	☐ ₅
Unable to determine	☐ ₉₃

HO005 HO005Q01NA	Does the area surrounding the dwelling have street lights?	Yes	No
	☐ ₁	☐ ₂	

HO006 HO006Q01NA	Are there visible signs that security is a concern for the inhabitants of the area (e.g., barriers, security guards, barbed wire, guns, etc.)?	Yes	No
	☐ ₁	☐ ₂	

HO007 HO007Q01NA	Kind of dwelling (Select the dwelling's best fitting description.)
Single family house/semi-detached house	☐ ₁
Part of a house/part of an apartment (e.g., private bedrooms with access to a common kitchen and bathroom)	☐ ₂
Apartment in a building with less than 10 dwellings	☐ ₃
School	☐ ₄
Other: _____	☐ ₅
Unable to determine	☐ ₉₃

HO008
HO008Q01NA

Main roofing material

(Select the roof's most noticeable material.)

No roof	☐ ₁
Natural roofing (e.g., thatch, palm leaf, or sod)	☐ ₂
Rudimentary roofing (e.g., rustic mat, rough wood planks, or cardboard)	☐ ₃
Finished roofing (e.g., metal, asbestos, finished wood, calamine or cement fibre, ceramic tiles, cement, or roofing shingles)	☐ ₄
Other: _____	☐ ₅
Unable to determine	☐ ₉₃

HO009
HO009Q01NA

Main material of the exterior walls

(Select the exterior walls' most noticeable material.)

No walls	☐ ₁
Natural walls (e.g., cane, palm, trunks or dirt)	☐ ₂
Rudimentary walls (e.g., bamboo with mud, stone with mud, uncovered adobe, plywood cardboard, or reused wood)	☐ ₃
Finished walls (e.g., cement, stone with lime or cement, bricks, cement blocks, covered adobe, finished wood planks, or shingles)	☐ ₄
Other: _____	☐ ₅
Unable to determine	☐ ₉₃

HO010
HO010Q01NA

Does this dwelling have electricity?

Yes	No	Unable to determine
☐ ₁	☐ ₂	☐ ₃

HO011
HO011Q01NA

Besides the respondent, was anyone else present during the interview?

Yes	☐ ₁
No	☐ ₂

HO012
HO012Q01NA

Did this/these person(s) assist the respondent in answering the background questionnaire (youth interview)?

Yes	☐ ₁
No	☐ ₂



HO013 HO013Q01NA	Did this/these person(s) assist the respondent in answering the skills assessment?	
	Yes	☐ ₁
	No	☐ ₂

HO014 HO014Q01NA	Overall, how often did you feel that the respondent understood the questions in the interview?	
	Never	☐ ₁
	Rarely	☐ ₂
	About half the time	☐ ₃
	Often	☐ ₄
	Always	☐ ₅

HO015 HO015Q01NA	Did the respondent ask for clarification on any questions while undertaking the interview?	
	Yes	No
	☐ ₁	☐ ₂

HO016 HO016Q01NA	Where did the interview mainly take place?	
	Living/dining room	☐ ₁
	Kitchen	☐ ₂
	Bedroom	☐ ₃
	Entrance	☐ ₄
	Hallway or corridor	☐ ₅
	Office	☐ ₆
	Other space in the household	☐ ₇
	Other space outside of the household (e.g., near the house, <library>, etc.)	☐ ₈

HO017 HO017Q01NA	Additional comments on the interview:	



Annex B

PISA FOR DEVELOPMENT EXPERTS AND SUPPORT STAFF

Annex B lists the experts and support staff involved in developing the PISA for Development framework for the cognitive instruments and the background questionnaires.

**PISA-D expert group**

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ORGANISATION FOR ECONOMIC CO-OPERATION AND DEVELOPMENT

The OECD is a unique forum where governments work together to address the economic, social and environmental challenges of globalisation. The OECD is also at the forefront of efforts to understand and to help governments respond to new developments and concerns, such as corporate governance, the information economy and the challenges of an ageing population. The Organisation provides a setting where governments can compare policy experiences, seek answers to common problems, identify good practice and work to co-ordinate domestic and international policies.

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PISA for Development Assessment and Analytical Framework

READING, MATHEMATICS AND SCIENCE

“What is important for citizens to know and be able to do?” The OECD Programme for International Student Assessment (PISA) seeks to answer that question through the most comprehensive and rigorous international assessment of student knowledge and skills. As more countries join its ranks, PISA is evolving to successfully cater for a larger and more diverse group of participants. The OECD launched the PISA for Development initiative in 2014 to support evidence-based policy making globally and offer universal tools in monitoring progress towards the Education Sustainable Development Goal. Spanning six years, this unique pilot project aims to make the assessment more accessible and relevant to a wider range of countries, while maintaining the overall PISA framework and accordance with PISA’s technical standards and usual practices.

The PISA for Development Assessment and Analytical Framework presents the conceptual foundations of the project, and covers reading, mathematics and science. PISA for Development has a school-based component and an out-of-school one. For the school-based component, a questionnaire about students’ background is distributed to all participating students. School principals complete a school questionnaire that describes the school, its students and teachers, and the learning environment. Teachers also complete a questionnaire about themselves, the school’s resources, their teaching practice and their students. The out-of-school respondents complete a background questionnaire, and their parent (or person most knowledgeable about them) answers a questionnaire about the youth’s background and childhood experiences. A household observation questionnaire is completed by the interviewer, and information about the location of the household is collected by PISA for Development National Centres. Nine countries participated in the PISA for Development assessment: Bhutan, Cambodia, Ecuador, Guatemala, Honduras, Panama, Paraguay, Senegal and Zambia.

Contents

- Chapter 1. What are PISA and PISA for Development?
- Chapter 2. PISA for Development reading framework
- Chapter 3. PISA for Development mathematics framework
- Chapter 4. PISA for Development science framework
- Chapter 5. PISA for Development contextual questionnaires framework

PISA for Development

PISA for Development has been designed by the OECD and its partners to enable greater PISA participation by middle- and low-income countries. It is doing this by:

- extending the PISA test instruments to a wider range of performance levels
- developing contextual questionnaires and data-collection instruments to effectively capture the diverse situations in low- and middle-income countries
- establishing methods and approaches to include out-of-school youth in the assessments.

PISA for Development also builds capacity for managing and using the results of large-scale student learning assessment.

Consult this publication on line at: <http://dx.doi.org/10.1787/9789264305274-en>

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